



EdgeFire 1014

Administrator's Guide

Version 2.2

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Data Security Warning

Before removing and disposing of this product, ensure all data and configurations stored on the device are erased. You can do this by:

- manually resetting the product to factory settings, or
- handing the product to the TXOne support team, who will help you clear all stored data and configurations.

TXOne Support Team Contact Information

- Email: support@txone.com
- Website: <https://www.txone.com/>
- Call Center: <https://help.txone.com/>
- Phone:
 - US: +1 3465867975
 - Netherland: +31 402310122
 - Taiwan: +886 277275120

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1 About EdgeFire 1014

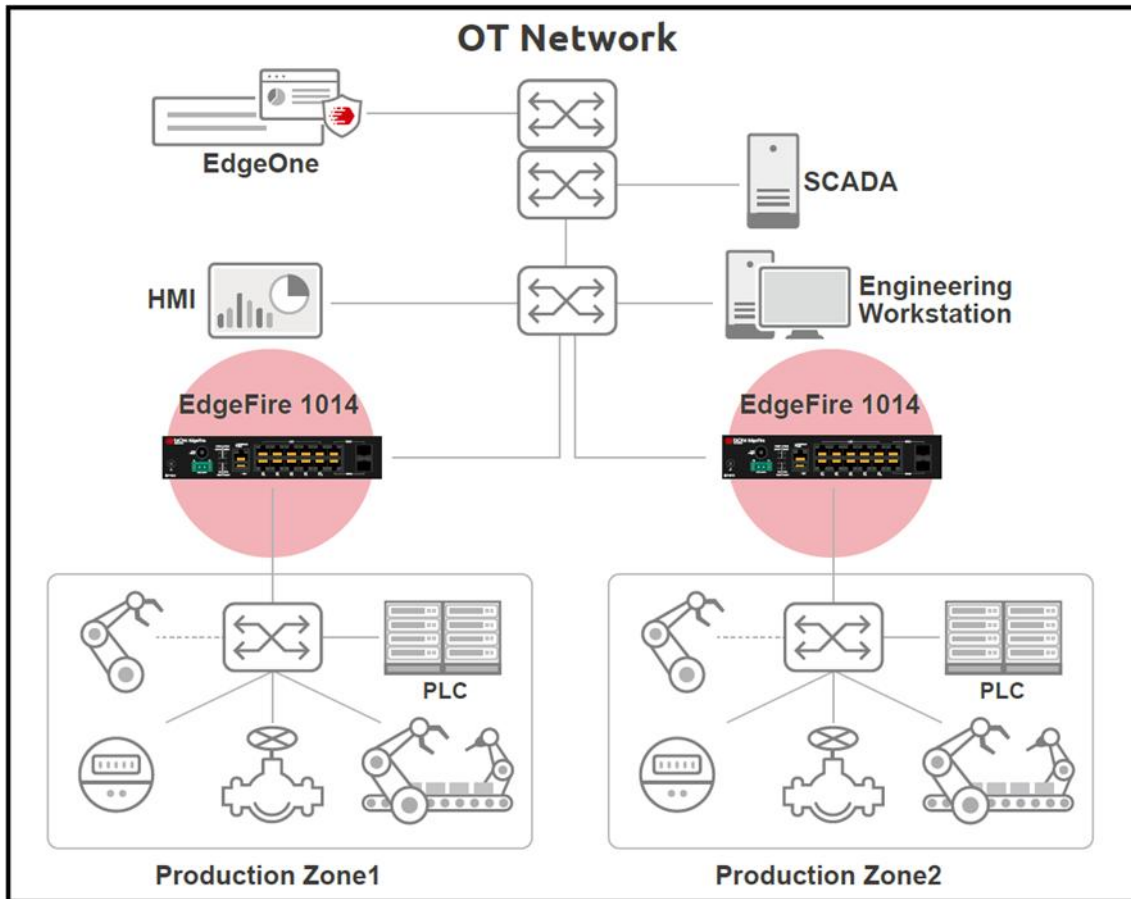
1.1 Introduction to EdgeFire 1014

EdgeFire 1014, a next-generation firewall, is a highly integrated industrial multiport secure router with firewall, NAT, and IPS functions. It is designed for Ethernet-based security applications on factory networks, and it provides an electronic security perimeter for the protection of critical cyber assets including pump-and-treat systems in water stations, DCS systems in oil and gas applications, and PLC/SCADA systems in factory automation. Users can access its web-based console that provides a graphical user interface for device configuration and security policy settings. The whole management process is designed to comply with the manufacturing SOPs of the industry.

IT and OT traditionally operate separately, each with its own network, transportation team, goals, and needs. In addition, each industrial environment is equipped with tools and devices that were not designed to connect to a corporate network; thus, provisioning timely security updates or patches can be difficult. Therefore, the need for security products that provide proper security protection and visibility is on the rise.

TXOne provides a wide range of security products that cover both your IT and OT layers. These easy-to-build solutions provide an active and immediate protection to Industrial Control System (ICS) environments with the following features:

- Certified industrial grade hardware with size, power consumption, and durability tailored for OT environments, as well as the ability to tolerate a wide range of temperature variations
- Threat detection and interception, with safeguards against the spread of worms
- Protection against Advanced Persistent Threats (APTs) and Denial of Service (DoS) attacks that target vulnerable legacy devices
- Virtual patch protection against OT device exploits



1.2 Main Functions of EdgeFire 1014

EdgeFire 1014 is a network security appliance. The main functions of this device are as follows:

Extensive Support for Industrial Protocols

The device supports the identification of a wide range of industrial control protocols, including Modbus and other protocols used by well-known international companies such as Siemens, Mitsubishi, Schneider Electric, ABB, Rockwell, Omron, and Emerson. In addition to allowing OT and IT security system administrators to work together, this feature also allows the flexibility to deploy defense measures in appropriate network segments and seamlessly connects them to existing factory networks.

Policy Enforcement for Mission-Critical Machines

The device's core technology TXODI (One-pass Deep Packet Inspection for Industrial) allows administrators to maintain a Policy Enforcement database. By analyzing Layer 2 to Layer 7 network traffic between mission-critical production machines, Policy Enforcement executes filtering of control commands within the protocols and blocks traffic that is not defined in the policy rules. This feature can help prevent unexpected operational traffic, block unknown network attacks, and block other activities that match defined policy rules.

Intelligence Learning and Rule Generation for Daily Network Traffic

The device's machine learning technology, Policy Rule Auto-Learning, allows administrators to create a learning task that will analyze the network traffic of daily operations and generate baseline policy rules as a trust list. During the learning process, it creates a rule list for rapid

deployment and filters potential cyber threats. This feature can help OT and IT security system administrators to silently deploy security solutions without affecting daily operations and efficiently generate policy rules with minimal manpower.

Intrusion Prevention and Intrusion Detection

The device provides a powerful and up-to-date first line of defense against known threats. Vulnerability filtering rules provide effective protection against exploits at the network level. Manufacturing personnel manage patching and updating, providing pre-emptive protection against critical production failures and additional protection for old or terminated software.

Cyber-Physical System Detection and Response

The device is equipped with TXOne's pioneering CPSDR (Cyber-Physical System Detection and Response) technology, designed to identify and predict anomalous network behaviors early on. With the CPSDR, your OT network can proactively stay a step ahead of cyber risks, blocking potential threats before they materialize.

Switch Between Two Flexible Modes, 'Monitor' & 'Prevention'

The device can flexibly switch between 'Monitor' and 'Prevention' modes. 'Monitor' mode will log traffic without interfering, while 'Prevention' mode will filter traffic based on policies you create. These modes work together to preserve your productivity while maximizing security.

Top Threat Intelligence and Analytics

The device provides advanced protection against unknown threats with its up-to-date threat information. With the help of the Zero Day Initiative (ZDI) vulnerability reward program, the device offers your systems exclusive protection from undisclosed and zero-day threats.

Flexible Segmentation and Isolation

The device is an ideal solution for segmenting a network into easily managed security zones. It segments networks and isolates connectivity both to and between facilities as well as production zones. The device comes equipped to make your IT and OT networks as integrated and coordinated with each other as possible, and to grant visibility of your shadow OT environment.

Virtual Private Network for Remote OT Asset Management and Secure Communication

The device features IPSec VPN and L2TP/IPSec VPN that allow for secure remote management of OT sites and assets. IPSec VPN establishes a secure VPN connection between VPN gateways, while L2TP/IPSec VPN adds an extra layer of security with a two-step authentication process. Both features enable secure remote device management from a central location, ensuring the security and reliability of the OT network.

Easily Centralized Management with Convenient, Consolidated Overview

TXOne's EdgeOne provides a graphical user interface for policy management in compliance with manufacturing SOP. It centrally monitors operations information, edits network protection policies, and sets patterns for attack behaviors.

All protections are deployed throughout the entire information technology (IT) and operational technology (OT) infrastructure. These include:

- A centralized policy deployment and reporting system
- Full visibility into assets, operations, and security threats

- IPS and policy enforcement configuration can be assigned per device group, allowing all devices in the same device group to share the same policy configuration
- Management permissions for device groups can be assigned per user account

2 Getting Started

This chapter guides you on how to get started with EdgeFire 1014 and configure the initial settings.

 For an overview of the physical hardware and characteristics, or a more condensed manual to help with initial setup of the device, see the document **EdgeFire 1014 - Quick Setup Guide**.

2.1 Getting Started Task List

This task list provides a high-level overview of all procedures required to get EdgeFire 1014 up and running as quickly as possible. Each step links to more detailed instructions found later in the document.

Procedure

1. Open the management console.

 For more information, see [Opening the Management Console](#).

2. Change the administrator account ID and password.

 For more information, see [Changing Account ID/Password](#).

3. Ensure that the link speed modes of the network ports are correct for your environment.

 For more information, see [Configuring Port Settings](#).

4. Change the default web management console IP address. The default web management console IP address is **192.168.127.254**. The IP address is bound to the **LAN1** network interface.

 For more information, see [Configuring LAN Network Interfaces](#).

5. Configure the network interfaces.

 For more information, see [Network Interfaces \(Only in Gateway Mode\)](#).

6. Configure the system time.

 For more information, see [Configuring System Time](#).

7. (Optional) Configure the Syslog settings.

 For more information, see [Configuring Syslog Settings](#).

8. Configure objects and profiles.

 For more information, see [Objects & Profiles](#).

9. Configure security policies.

 For more information, see [Security Policies](#).

10. Configure the device name and location.

 For more information, see [Configuring Device Hostname and Location](#).

11. (Optional) Configure management protocols and ports.

 For more information, see [Configuring Management Protocols and Ports](#).

12. (Optional) Configure the device access control list.

 For more information, see [Configuring Access Control List and Access from WAN Interface](#).

13. (Optional) Update the DPI (Deep Packet Inspection) pattern for the device.

 For more information, see [Updating IPS Patterns](#).

14. (Optional) Enable EdgeOne Management settings.

 For more information, see [Enabling EdgeOne Management](#).

15. (Optional) Configure password policy.

 For more information, see [Configuring Password Policy](#).

2.2 Opening the Management Console

We provide a built-in management console that you can use to configure and manage this Edge series device. View the management console using a web browser.

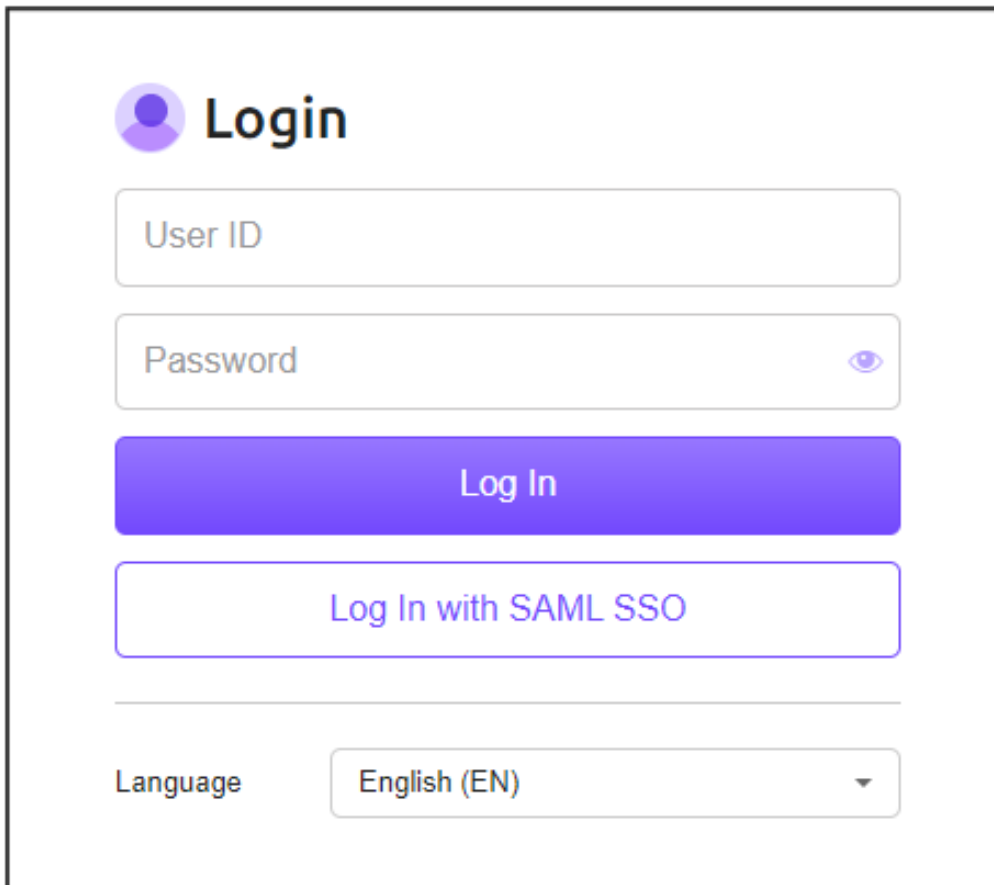
 View the management console using Google Chrome version 63 or later; Firefox version 53 or later; Safari version 10.1 or later; Edge version 15 or later.

Procedure

1. In a web browser, type the address of the device management console in the following format.

https://<target server IP address or FQDN>

The login page will appear.



The screenshot shows a login page with a purple header containing a user icon and the word "Login". Below the header are two input fields: "User ID" and "Password". The "Password" field has a toggle icon (an eye) to the right. Below these fields is a large purple "Log In" button. Underneath the button is a link that says "Log In with SAML SSO". At the bottom of the page, there is a "Language" label followed by a dropdown menu currently set to "English (EN)".



1. The default IP address of this Edge series device is 192.168.127.254 with subnet 255.255.255.0. Before you connect a PC/Laptop to the device, the PC's IP address should be set to an IP address that is able to access the default IP address of the device. After that, connect the PC and the device using an Ethernet cable.
2. TXOne products use an automatically generated self-signed SSL certificate to encrypt communications to and from the client accessing the device. Given that the certificate is self-signed, most browsers will not trust the certificate and will give a warning that the certificate being used is not signed by a known authority.

2. Input your login credentials (user ID and password). Use the default administrator credentials when logging in for the first time.

- User ID: *admin*
- Password: *txone*

3. Click [Log In].

4. When logging in for the first time or after a factory reset, you will be prompted to read and accept our End User License Agreement, and to change the default user ID and password before accessing the system.



End User License Agreement (EULA)

Please read the End User License Agreement (EULA) and accept the terms in the agreement before using this product.

[EULA\(EN\)](#) | [EULA Website Link](#)

☒ I have read and accept the terms in the EULA.

[Continue](#)



Reset Login Credentials

Change the default user ID and password to prevent unauthorized access to your account.

New User ID*



New Password*




Confirm Password*




Save

Cancel

- 
 1. You must change the default login name and password at your first login before you can access the management console.
 2. New login name cannot be “root”, “admin”, “administrator” or “auditor” (case-insensitive).
 3. For enhanced security, consider using a strong password with at least 12 characters that consists of both uppercase and lowercase letters, numbers, and symbols. Avoid using common words or personal information.
5. You will be automatically logged out of the system. The login page will appear again. Log in again using your new credentials.

3 Visibility

This chapter describes the visibility-related features designed to provide comprehensive insights into your OT network.

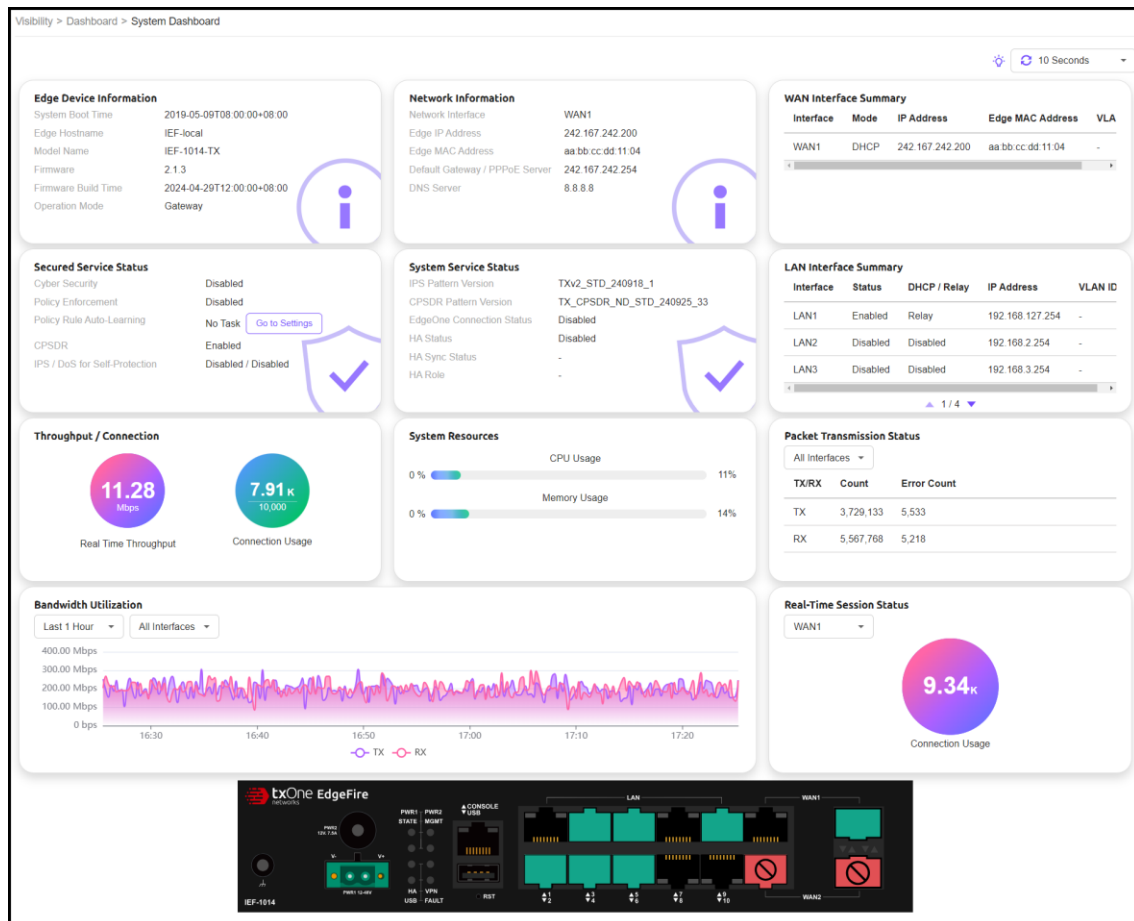
3.1 Dashboard

Monitor and analyze your assets, devices, network status, and threat detection.

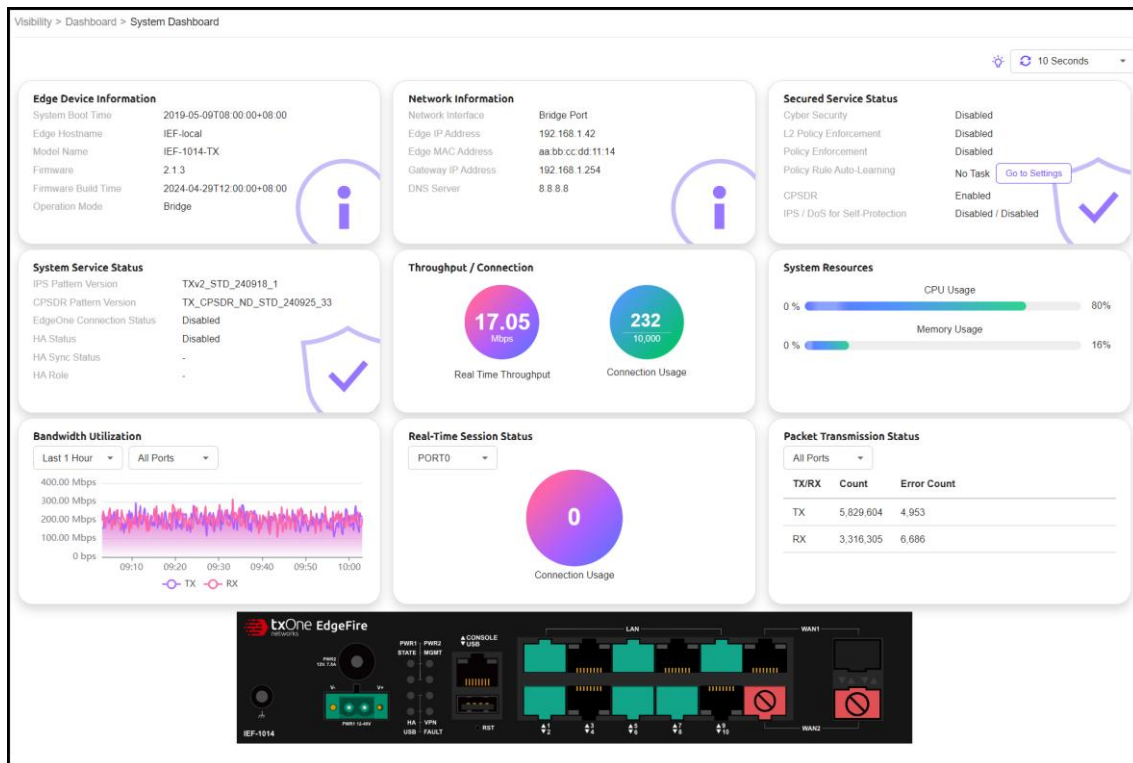
- [System Dashboard](#)

3.1.1 System Dashboard

Monitor the EdgeFire 1014 device information, network information, secured service status, system service status, throughput/connection, system resources, bandwidth utilization, real-time session status, packet transmission status, WAN interface summary, and LAN interface summary on the system dashboard.



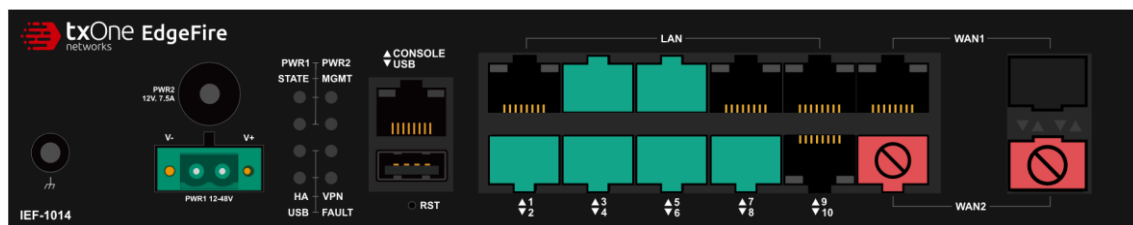
System Dashboard in Gateway Mode



System Dashboard in Bridge Mode

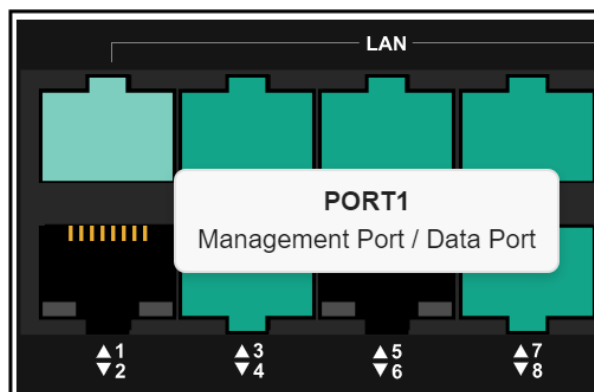
3.1.1.1 Viewing System Dashboard Device Panel

In the left area, you may view the device panel and the status of each port.



The panel view on the page shows a graphical depiction of the connected ports on the device.

Hover your mouse over a port on the device panel to view details of the port.



Port Status Information

In the upper right corner, you may view the port status information by clicking the  button.

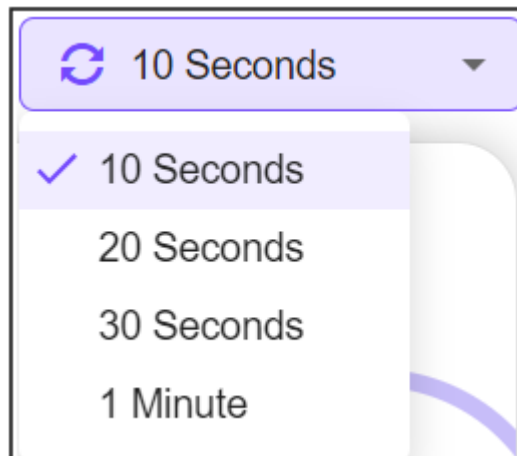
Port Status Information

Port	Status	Description
	Disconnected	Port Link Down
	Connected	Port Link Up
	Disabled	Port Disabled

Close

Refresh Interval Settings

In the upper right corner, you may select **10 seconds** (default), **20 seconds**, **30 seconds** or **1 minute** for the refresh interval.



Edge Device Information

This widget shows the system boot time, Edge hostname, model name, firmware version, firmware build time, and operation mode.

Edge Device Information

System Boot Time	2019-05-09T08:00:00+08:00
Edge Hostname	IEF-local
Model Name	IEF-1014-TX
Firmware	2.1.3
Firmware Build Time	2024-04-29T12:00:00+08:00
Operation Mode	Gateway



Network Information

This widget shows the network interface, Edge IP address, Edge MAC address, default gateway/PPPoE server, and DNS server.

Network Information

Network Interface	WAN1
Edge IP Address	222.247.146.248
Edge MAC Address	aa:bb:cc:dd:11:11
Default Gateway / PPPoE Server	222.247.146.254
DNS Server	-



Secured Service Status

The widget shows the statuses of Cyber Security, L2 Policy Enforcement (supported in Bridge Mode only), Policy Enforcement, Policy Rule Auto-Learning, CPSDR, and IPS/DoS Setting for Self-Protection.

Secured Service Status

Cyber Security	Disabled
L2 Policy Enforcement	Disabled
Policy Enforcement	Disabled
Policy Rule Auto-Learning	No Task
CPSDR	Enabled
IPS / DoS for Self-Protection	Disabled / Disabled

[Go to Settings](#)


System Service Status

This widget shows the versions of IPS and CPSDR patterns in use, EdgeOne connection status, HA status, HA sync status (when HA is enabled), and its HA role (when HA is enabled).

System Service Status	
IPS Pattern Version	TXv2_STD_240918_1
CPSDR Pattern Version	TX_CPSDR_ND_STD_240925_33
EdgeOne Connection Status	Disabled
HA Status	Disabled
HA Sync Status	-
HA Role	-



WAN Interface Summary (Only in Gateway Mode)

This widget shows a summary of the WAN interface information.

WAN Interface Summary				
Interface	Mode	IP Address	Edge MAC Address	VLAN ID
WAN1	DHCP	6.215.124.32	aa:bb:cc:dd:11:10	-

LAN Interface Summary (Only in Gateway Mode)

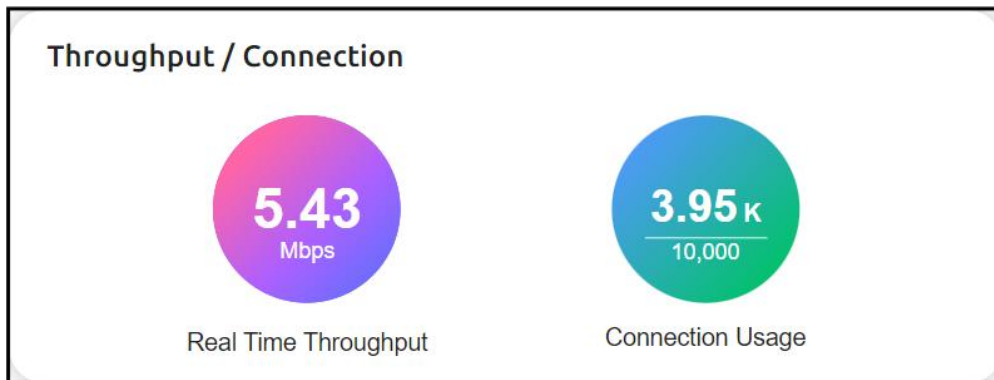
This widget shows a summary of the LAN1, LAN2 and DMZ interfaces information.

LAN Interface Summary				
Interface	Status	DHCP / Relay	IP Address	VLAN ID
LAN1	Enabled	Relay	192.168.127.254	-
LAN2	Disabled	Disabled	192.168.2.254	-
LAN3	Disabled	Disabled	192.168.3.254	-

1 / 4

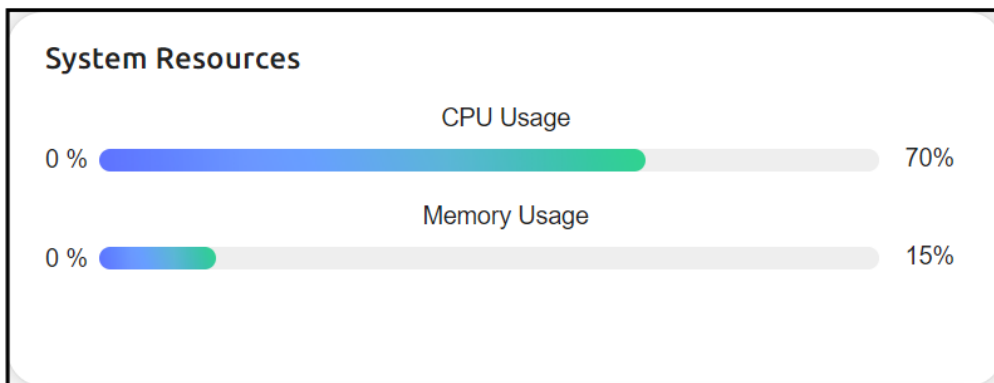
Throughput / Connection

The widget shows the current network throughput and the current network connection usage on the device.



System Resources

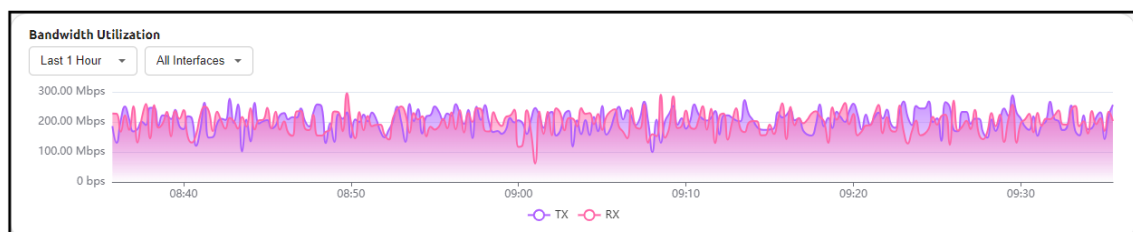
This widget shows system resources (CPU and memory) usages on the device.



Item	Description
CPU Usage	Percentage of real-time CPU utilization
Memory Usage	Percentage of real-time memory utilization

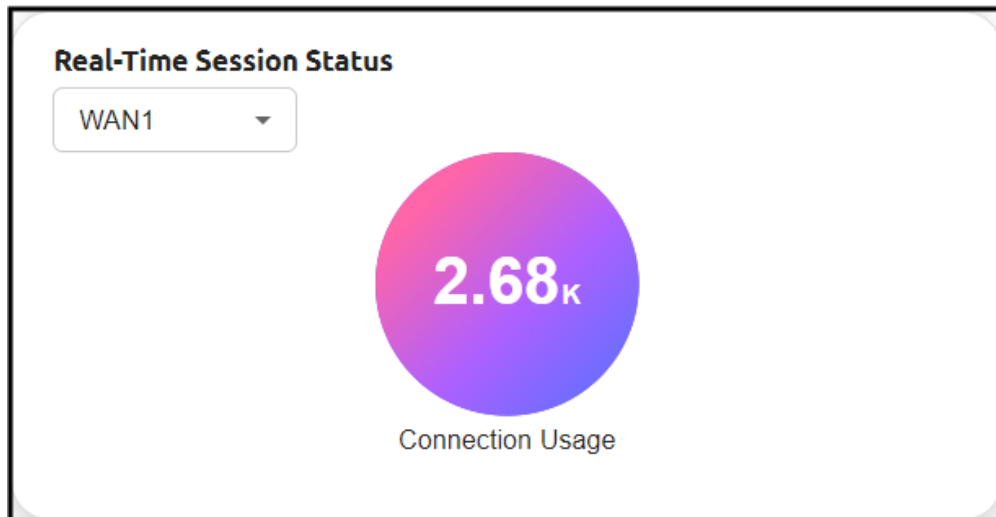
Bandwidth Utilization

This widget shows TX and RX bandwidth utilization based on the specified time interval.



Real-Time Session Status

This widget shows how many sessions are currently in use.



Packet Transmission Status

This widget shows the packet count number on each interface and error count information.

Packet Transmission Status

All Interfaces ▼

TX/RX	Count	Error Count
TX	4,243,909	5,373
RX	4,430,492	4,923

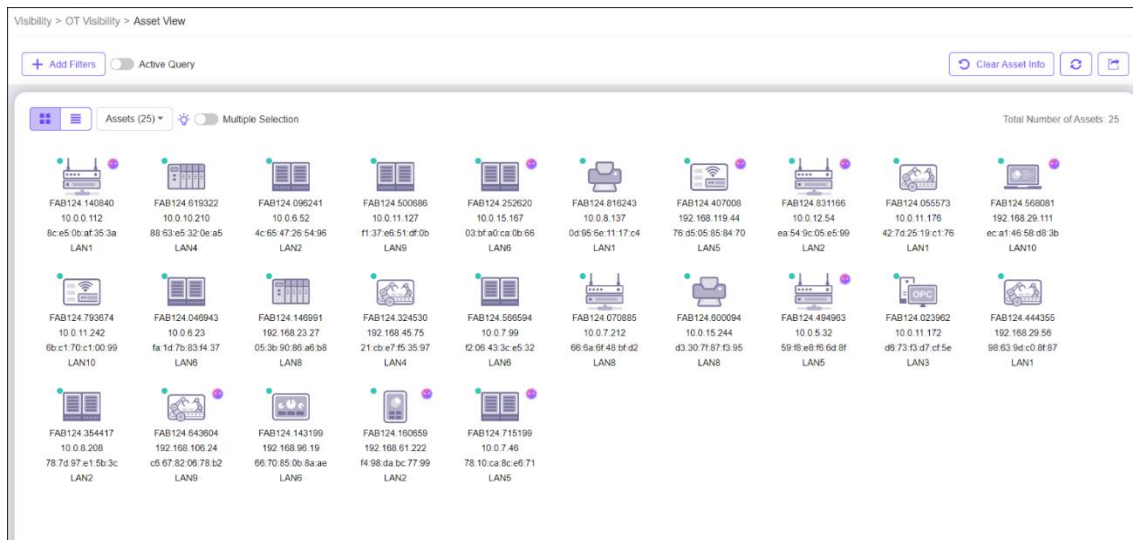
3.2 OT Visibility

Gain a comprehensive view of your OT networks from various aspects.

- [Asset View](#)

3.2.1 Asset View

The [Asset View] page gives you an overview of your managed assets. The page provides you with timely and accurate information on the assets that are managed by your Edge series device.

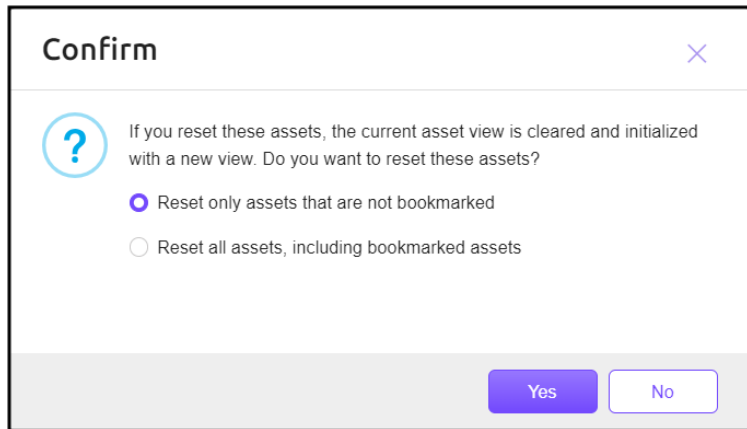


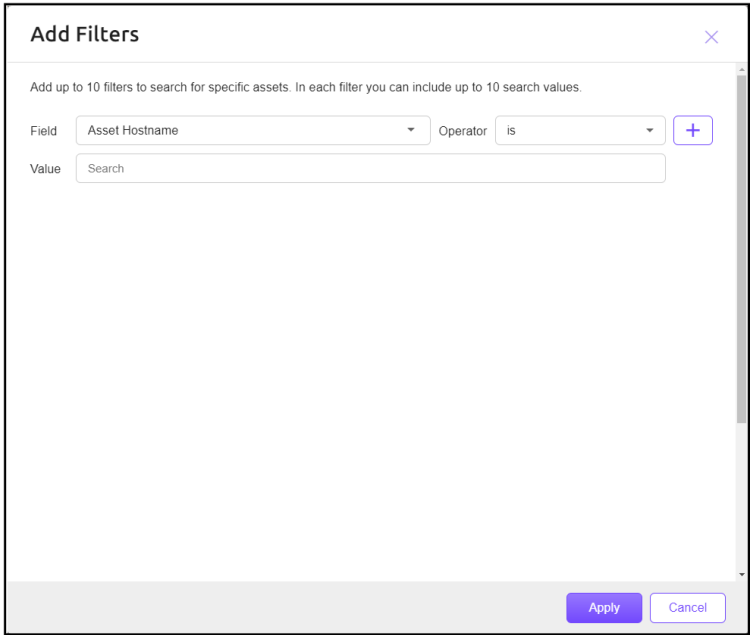
The assets listed on this page are automatically detected by the Edge series devices.



The term **asset** in this chapter refers to a device or host that is protected by an Edge device.

The following table lists the actions that you can perform on the [Asset View] page.

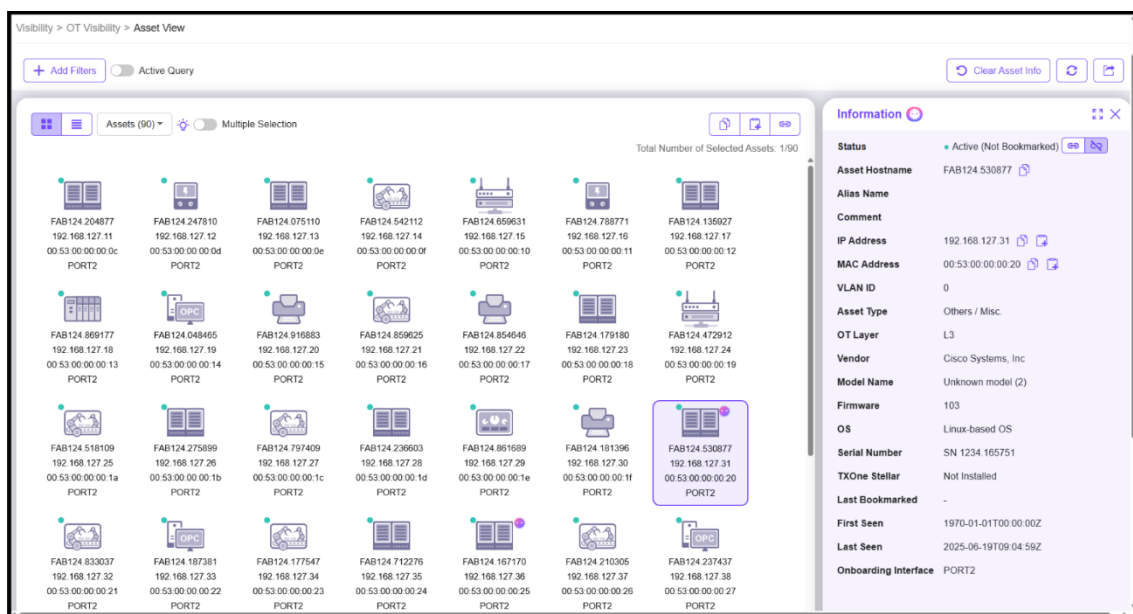
Task	Procedure
To clear the current asset information and initialize the page with a new asset view (only available for an admin role)	- Click the [Clear Asset Info] button in the upper right corner. A confirmation window will appear.  - Select an appropriate option and click [Yes].
To refresh the displayed information	Click the refresh button  in the upper right corner.
To export asset information to a CSV file	Click the export button  in the upper right corner.
To search for specific assets	Click [Add Filters] in the upper left corner. The [Add Filters] window will appear.

Task	Procedure
	 - Create up to 10 search filters. In each filter you can include up to 10 search values. - Click [Apply].
To enable the function for detecting assets that do not proactively send out OT commands in the network	Click the [Active Query] toggle in the upper left corner.  Starting from firmware v1.1, the Active Query function supports 4 protocols, including Modbus, CIP, OMRON FINS and SMB.
To switch between the grid view and table view modes	Click the grid/table view buttons  .
To select multiple assets	- Enable the [Multiple Selection] toggle. - Click on the assets you want to select. To select all assets, click the [Select All] checkbox.
To copy an asset hostname, IP address, or MAC address	- Select a specific asset. - Click the copy button . - Select an action, [Copy Asset Hostname], [Copy IP Address], or [Copy MAC Address], from the pop-up menu.
To use an asset's IP or MAC address for object creation	- Select a specific asset. - Click the create profile button . - Select an action, [Create IP Object] or [Create MAC Object], from the pop-up menu.
To bookmark or unbookmark an asset	- Select one or multiple assets. - Click the [Bookmark Asset]  or [Unbookmark Asset]  button.

Task	Procedure
	 - When an inactive asset is bookmarked, its cached information remains accessible in the Edge device GUI until the device is reset to its default settings. - The maximum number of detected assets is fixed for the Edge device. When an asset is bookmarked, it occupies a portion of the quota for detecting newly onboarded assets. This means that when the number of bookmarked assets is equal to the maximum number of detected assets, the Edge device will no longer be able to detect new assets.
To edit the alias name, comment, asset type, or OT layer of an asset	- Select a bookmarked asset. - Click the edit asset information button .

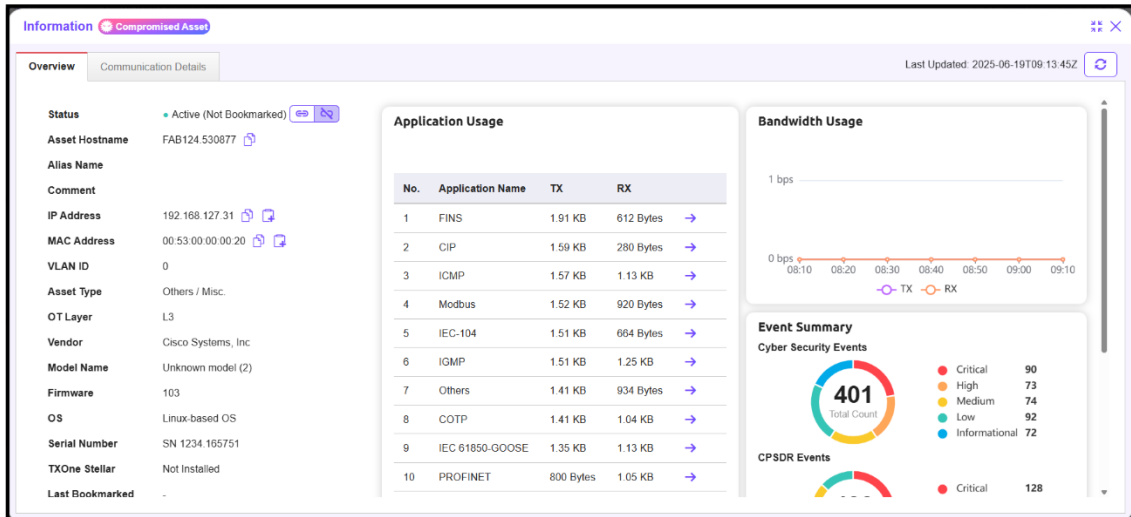
3.2.1.1 Viewing in Grid View Mode

Under the [Grid View] display mode, select a specific asset and click the expand button  to view general asset information, application usage, bandwidth usage, event summary, and communication details.



Information > Overview

On the [Information] window, switch to the [Overview] tab to view the general asset information, application usage, bandwidth usage, and event summary. If the selected asset is impacted by high- or critical-severity IPS or CPSDR events within the last 24 hours, the [Information] window will be labeled [Compromised Asset ]. If the asset has communicated with a public IP address within the last 24 hours, the window will be labeled [Internet Exposed ].



Item	Description
Status	The current status of the asset. Click the   buttons to bookmark or unbookmark an asset. Active (Not Bookmarked): An asset is marked as active when it has network traffic passing through it within the last hour. If the asset does not have network traffic for more than an hour and is not bookmarked, EdgeOne automatically removes it from the GUI. Active (Bookmarked): An asset is marked as active when it has network traffic passing through it within the last hour. If the asset does not have network traffic for more than an hour and is bookmarked, it continues to be displayed in the EdgeOne GUI. Inactive (Bookmarked): An asset is marked as inactive when no network traffic is detected within the last hour. However, if the asset is bookmarked, its cached information remains accessible in the EdgeOne GUI, despite its inactivity.
Asset Hostname	The hostname of the asset. Click the  button to copy the asset hostname.
Alias Name	User-defined name of the asset. Click the  button to edit the alias name.  You can only edit the alias name of a bookmarked asset.
Comment	User-defined comment for the asset. Click the  button to edit the comment.  You can only edit the comment for a bookmarked asset.
IP Address	The IP address of the asset. Click the  button to copy the IP address, or the  button to create an IP object.

Item	Description
	 For more information about creating an IP object, see Configuring IP Objects .
MAC Address	The MAC address of the asset. Click the  button to copy the MAC address, or the  button to create a MAC object.  For more information about creating a MAC object, see Configuring MAC Objects .
VLAN ID	The VLAN ID of the asset.
Asset Type	The type of the asset. Click the  button to edit the asset type.  You can only edit the type of a bookmarked asset.
OT Layer	The OT layer in which the asset is located. Click the  button to edit the OT layer.  You can only edit the OT layer of a bookmarked asset.
Vendor	The vendor of the asset.
Model Name	The model name of the asset.
Firmware	The firmware version of the asset.
OS	The operating system of the asset.
Serial Number	The serial number of the asset.
TXOne Stellar	The TXOne Stellar installation status of the asset.
Last Bookmarked	The date and time when the asset was last bookmarked.
First Seen	The date and time when the asset was first seen.
Last Seen	The date and time when the asset was last seen.
Onboarding Interface	The Edge device interface to which the asset is onboarded.

Information > Overview > Application Usage

In the [Application Usage] pane, you can view the statistics of the network traffic passing from/to a specific asset based on different applications.

Application Usage

No.	Application Name	TX	RX	
1	SSL/TLS	315.32 MB	486.69 MB	→
2	HTTP Protocol ov...	21.08 MB	41.25 MB	→
3	MS Remote Deskt...	15.52 MB	25.50 MB	→
4	Others	6.34 MB	8.84 MB	→
5	Microsoft.com	1.58 MB	3.29 MB	→
6	FTP Applications	1.56 MB	11.62 MB	→
7	Windows Update	768.28 KB	5.90 MB	→
8	SMB2	449.97 KB	323.36 KB	→
9	HTTP	130.90 KB	6.09 MB	→
10	IGMP	84.84 KB	0 Bytes	→
11	Microsoft Office	39.61 KB	144.40 KB	→
12	Google APIs(SSL)	20.87 KB	106.56 KB	→
13	Lets Encrypt	14.78 KB	51.71 KB	→
14	QUIC	14.21 KB	28.89 KB	→
15	Microsoft Live.com	6.38 KB	16.42 KB	→
16	Web File Transfer	5.56 KB	70.30 KB	→

Item	Description
No.	The ordinal number of the application traffic.
Application Name	The application type of the traffic.
TX	The amount of traffic transmitted for this application.

Item	Description
RX	The amount of traffic received for this application.

Click the arrow next to a specific application to view its peer IP, peer MAC, and role.

Application Usage

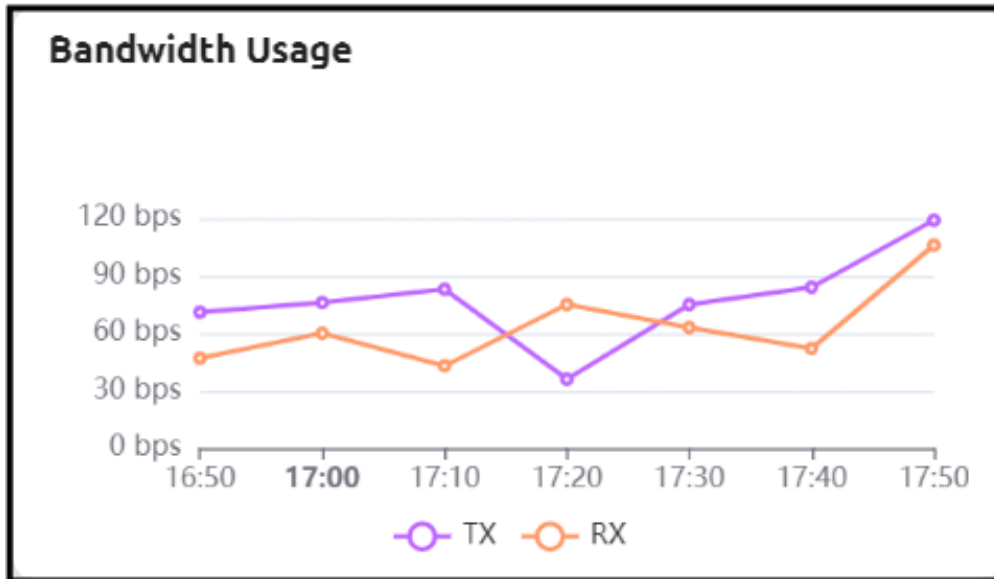
← HTTP Protocol over TLS SSL

No.	Peer IP	Peer MAC	Role
1	10.24.200.131	00:0c:29:e7:03:77	Server
2	34.107.243.93	5c:26:0a:f6:92:62	Client
3	20.90.156.32	5c:26:0a:f6:92:62	Server
4	10.24.200.193	00:0c:29:89:49:bc	Server and Client
5	10.24.7.72	00:0c:29:2a:f9:10	Server and Client
6	20.90.152.133	5c:26:0a:f6:92:62	Server

Item	Description
Peer IP	The IP address of the network element on the other end of the connection.
Peer MAC	The MAC address of the network element on the other end of the connection.
Role	Server: The network element on the other end of the connection transmits network packets to the asset. Client: The network element on the other end of the connection receives network packets from the asset.

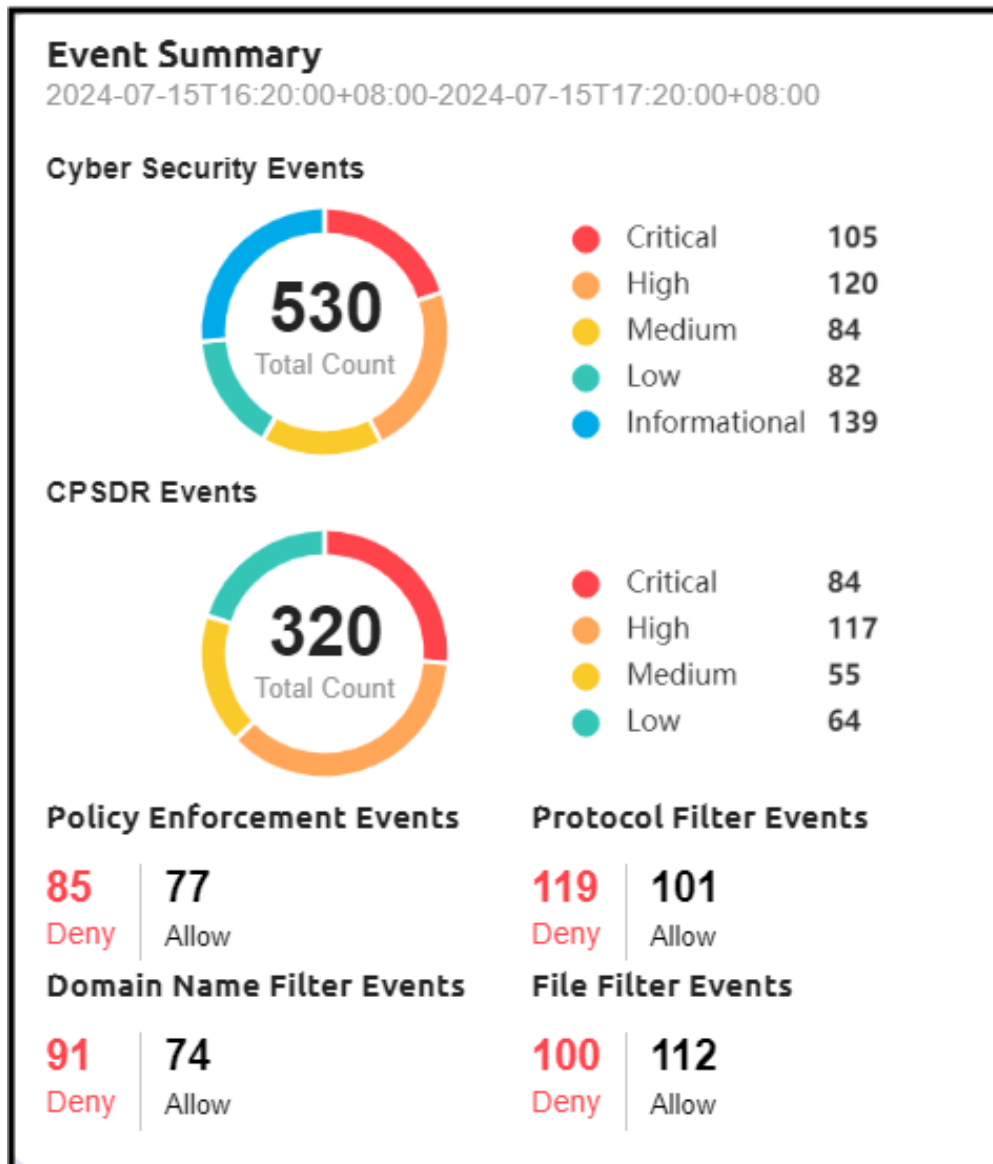
Information > Overview > Bandwidth Usage

In the [Bandwidth Usage] pane, you can monitor the bandwidth usage of a specific asset.



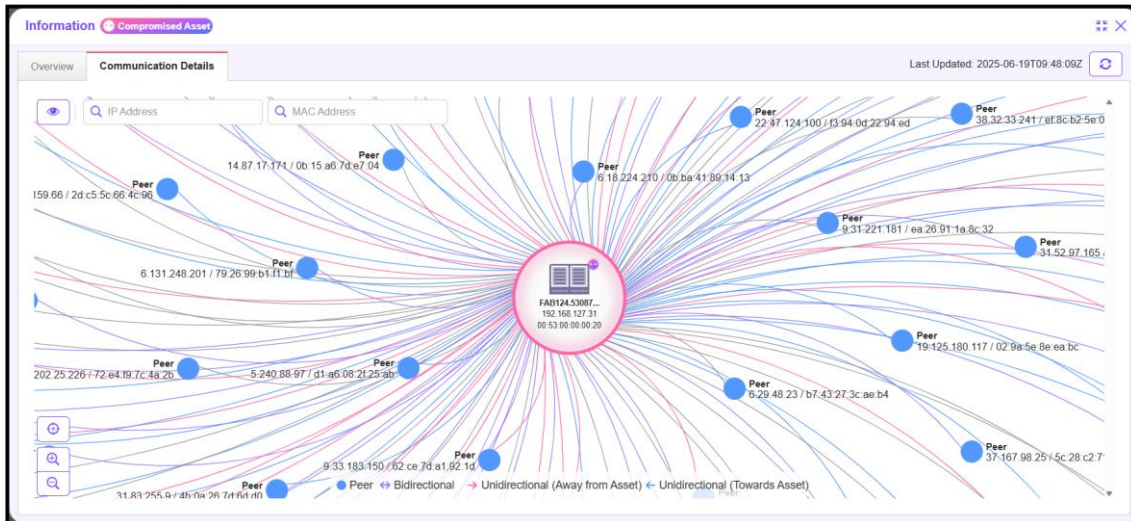
Information > Overview > Event Summary

In the [Event Summary] pane, you can monitor the counts of various types of security events, including Cyber Security, CPSDR, Policy Enforcement, Protocol Filter, Domain Name Filter, and File Filter events.



Information > Communication Details

On the [Information] window, switch to the [Communication Details] tab to view a visualized network graph showing which devices are communicating with this asset.



Task	Description
To configure display options	Click [Configure Display Options]  , then check or uncheck the display options.
To search for a specific peer device by IP address	Input the IP address of the target device in the [IP Address] field.
To search for a specific peer device by MAC address	Input the MAC address of the target device in the [MAC Address] field.
To reset and center the communication graph on the asset	Click the [Recenter]  button in the bottom-left corner of the communication graph.
To zoom in on the communication graph	Click the [Zoom In]  button in the bottom-left corner of the communication graph.
To zoom out of the communication graph	Click the [Zoom Out]  button in the bottom-left corner of the communication graph.

3.2.1.2 Viewing in Table View Mode

Under the [Table View] display mode, you can view detailed information on all assets managed by this Edge device.

Visibility > OT Visibility > Asset View

+ Add Filters ☐ Active Query Clear Asset Info

Assets (25) ☐ Multiple Selection

Total Number of Selected Assets: 1/25 1 / 1

Status	Asset Hostname	Alias Name	Comment	IP Address	MAC Address	VLAN ID	Asset Type	OT Layer	Vendor
Active (Not Bookmarked)	FAB124.140840			10.0.0.112	8c:e5:0b:af:35:3a	125	Network Appliance	L2	Netbotz
Active (Not Bookmarked)	FAB124.619322			10.0.10.210	88:63:e5:32:0e:a5	516	Industrial Controller - PLC/RTU	L1	SIEMENS AG
Active (Not Bookmarked)	FAB124.096241			10.0.6.52	4c:65:47:26:54:96	262	Others / Misc.	L3	DFI Inc.
Active (Not Bookmarked)	FAB124.500666			10.0.11.127	f1:37:a6:51:df:0b	572	Others / Misc.	L3	Syabas Technology (Amquest)
Active (Not Bookmarked)	FAB124.252620			10.0.15.167	03:bf:a0:ca:0b:66	532	Others / Misc.	L3	NATIONAL INSTRUMENTS CORP
Active (Not Bookmarked)	FAB124.816243			10.0.8.137	0d:95:6e:11:17:c4	439	Printer in OT	L3	Zebra Technologies Corp.
Active (Not Bookmarked)	FAB124.407008			192.168.119.44	76:d5:05:85:84:70	117	Wireless Access Point and Controller	L3	Buffalo Inc.
Active (Not Bookmarked)	FAB124.631166			10.0.12.54	ea:54:9c:05:e5:99	250	Network Appliance	L2	Routerboard.com
Active (Not Bookmarked)	FAB124.055573			10.0.11.176	42:7d:25:19:c1:76	586	Industrial Production Machines / Assets	L1	Fujitsu Technology Solutions GmbH
Active (Not Bookmarked)	FAB124.568081			192.168.29.111	eca1:46:58:d8:3b	374	OT Appliance (Desktop / Laptop)	L3	ASUSTek COMPUTER Inc.
Active (Not Bookmarked)	FAB124.793674			10.0.11.242	8b:c1:70:c1:00:99	106	Wireless Access Point and Controller	L3	Ruckus Wireless
Active (Not Bookmarked)	FAB124.046943			10.0.6.23	fa:1d:7b:63:f4:37	120	Others / Misc.	L3	Control4
Active (Not Bookmarked)	FAB124.146991			192.168.23.27	05:3b:90:86:a6:b8	23	Industrial Controller - PLC/RTU	L1	Mitsubishi Electric
Active (Not Bookmarked)	FAB124.324530			192.168.45.75	21:cb:e7:f5:35:97	44	Industrial Production Machines / Assets	L1	Dell Inc.
Active (Not Bookmarked)	FAB124.566594			10.0.7.99	f2:06:43:3c:e5:32	470	Others / Misc.	L3	Sophos Ltd

Click on a row and the expand button  to view general asset information, application usage, bandwidth usage, event summary, and communication details.

Information Last Updated: 2024-07-17T18:07:30+08:00

Status Active (Not Bookmarked)

Asset Hostname FAB124.560081

Alias Name

Comment

IP Address 192.168.115.9

MAC Address e9:a2:d8:79:4a:a3

VLAN ID 179

Asset Type Industrial Production Machines / Assets

OT Layer L1

Vendor Fujitsu Technology Solutions GmbH

Model Name Unknown model (1)

Firmware 104

OS Windows 7 / Server 2008, SP1, Build 7...

Serial Number SN 1234.241199

TXOne Stellar Installed

Last Bookmarked

First Seen 2024-07-11T23:18:15+08:00

Last Seen 2024-07-17T18:07:30+08:00

Onboarding Interface PORT2

App Usage

No.	Application Name	TX	RX
1	IEC-104	2.36 MB	2.37 MB
2	IEC 61850-MMS	2.36 MB	2.35 MB
3	DNP3	2.35 MB	2.34 MB
4	Modbus	2.34 MB	2.35 MB
5	SMB	2.34 MB	2.36 MB
6	ICMP	2.34 MB	2.36 MB
7	FINS	2.34 MB	2.34 MB
8	COTP	2.34 MB	2.33 MB
9	IEC 61850-GOOSE	2.33 MB	2.35 MB
10	PROFINET	2.33 MB	2.33 MB
11	CIP	2.33 MB	2.33 MB
12	Others	2.33 MB	2.34 MB

Bandwidth Usage

Event Summary
2024-07-15T16:30:00+08:00-2024-07-15T17:30:00+08:00

Cyber Security Events

Severity	Count
Critical	55
High	118
Medium	68
Low	116
Informational	107

CPSDR Events

Severity	Count
Critical	49
High	76
Medium	127

 For more information about the information under the [Overview] and [Communication Details] tabs, see [Viewing in Grid View Mode](#).

Click the  button to show/hide certain type of information and adjust the column order.

Customize Column Display

Click on the checkboxes to show/hide columns and the arrows to adjust the column order.

1.	☒ Asset Security Status	^	v
2.	☒ Asset Hostname	^	v
3.	☒ Alias Name	^	v
4.	☐ Comment	^	v
5.	☒ IP Address	^	v
6.	☒ MAC Address	^	v
7.	☐ VLAN ID	^	v
8.	☒ Asset Type	^	v
9.	☐ OT Layer	^	v
10.	☐ Internet Exposure	^	v
11.	☐ Vendor	^	v
12.	☐ Model Name	^	v

Apply
Cancel



The [Asset Security Status], [Asset Hostname], [IP Address], [MAC Address], and [Onboarding Interface] columns cannot be hidden.

Click the  button to adjust the number of rows in the table.

4 Device

This chapter describes how to configure network-related features and port settings under the [Device] tab.

4.1 Network & Ports

Configure device network and port settings.

- [Port Settings](#)
- [VLAN Bridge \(Only in Bridge Mode\)](#)
- [Port Mapping](#)
- [Network Interfaces \(Only in Gateway Mode\)](#)
- [DDNS Settings \(Only in Gateway Mode\)](#)
- [MGMT Interface Settings \(Only in Gateway Mode\)](#)
- [HA Settings](#)
- [Port Mirroring](#)
- [Operation Mode](#)

4.1.1 Port Settings

The [Port Settings] page contains setting items for port connection statuses and port link speeds.

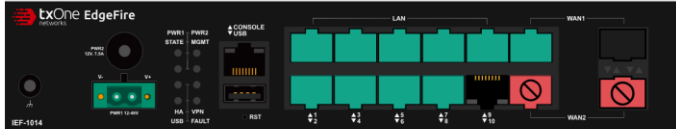
4.1.1.1 Configuring Port Settings

Procedure

1. Go to [Device] > [Network & Ports] > [Port Settings].

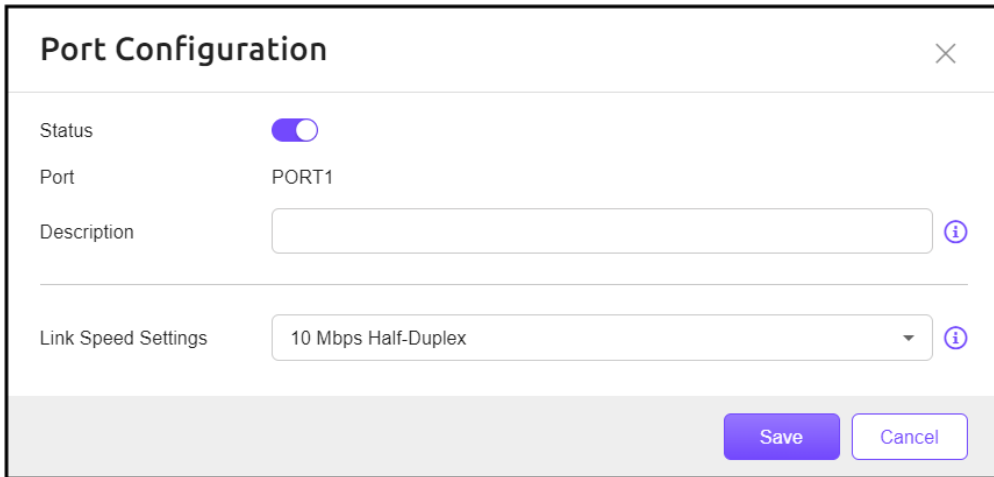
Device > Network & Ports > Port Settings
Manual Refresh

Port	Port Status	Link Speed Settings	Link Speed Status	Link Status	Description
WAN1	Enabled	100 Mbps Full-Duplex	100 Mbps Full-Duplex	Link Up	Default WAN Port
WAN2	Disabled	10 Mbps Full-Duplex	-	Link Down	Default WAN Port
PORT1	Enabled	10 Mbps Half-Duplex	10 Mbps Full-Duplex	Link Up	
PORT2	Enabled	10 Mbps Half-Duplex	100 Mbps Full-Duplex	Link Up	
PORT3	Enabled	10 Mbps Half-Duplex	1 Gbps Half-Duplex	Link Up	
PORT4	Enabled	10 Mbps Half-Duplex	10 Mbps Half-Duplex	Link Up	
PORT5	Enabled	10 Mbps Half-Duplex	0 Mbps Full-Duplex	Link Up	
PORT6	Enabled	10 Mbps Half-Duplex	0 Mbps Half-Duplex	Link Up	
PORT7	Enabled	10 Mbps Half-Duplex	1 Gbps Full-Duplex	Link Up	
PORT8	Enabled	10 Mbps Half-Duplex	10 Mbps Half-Duplex	Link Up	
PORT9	Enabled	10 Mbps Half-Duplex	100 Mbps Half-Duplex	Link Up	
PORT10	Enabled	10 Mbps Half-Duplex	-	Link Down	




The panel view on the page shows a graphical depiction of the connected ports on the device.

- Click a port in the [Port] column. The [Port Configuration] window will appear.



The Port Configuration window is a modal dialog with a title bar containing a close button (X). It contains the following fields and controls:

- Status:** A toggle switch currently turned on (blue).
- Port:** A text field containing the value "PORT1".
- Description:** An empty text input field with an information icon (i) to its right.
- Link Speed Settings:** A dropdown menu currently showing "10 Mbps Half-Duplex" with an information icon (i) to its right.
- Buttons:** "Save" and "Cancel" buttons at the bottom right.

- Configure the following port settings.

Item	Description
Status	The connection status of the port. Enable or disable the port by clicking the toggle.
Description	(Optional) Type a description for the port.
Link Speed Settings	Select a speed and negotiation method, Auto Negotiation, 10 Mbps Half-Duplex, 10 Mbps Full-Duplex, 100 Mbps Half-Duplex, or 100 Mbps Full-Duplex for the port.  Select [Auto Negotiation] if you would like the interface to automatically negotiate with the highest speed that both sides can work with. Otherwise, the link speed would be set to the configured speed value based on your selection.

- Click [Save].

4.1.2 VLAN Bridge (Only in Bridge Mode)

A VLAN bridge is used to connect multiple VLANs for traffic communications by forwarding traffic between the VLANs, even if they are on different physical interfaces. This allows operators to create complex and secure network topologies that can meet their security needs.

4.1.2.1 Configuring VLAN Bridge

Procedure

- Go to [Device] > [Network & Ports] > [VLAN Bridge].

Device > Network & Ports > VLAN Bridge

☒ Trunk Port for WAN1

Native VLAN ID for WAN1* ⓘ

PORT1 - VLAN ID* ⓘ

PORT2 - VLAN ID* ⓘ

PORT3 - VLAN ID* ⓘ

PORT4 - VLAN ID* ⓘ

PORT5 - VLAN ID* ⓘ

PORT6 - VLAN ID* ⓘ

PORT7 - VLAN ID* ⓘ

PORT8 - VLAN ID* ⓘ

PORT9 - VLAN ID* ⓘ

PORT10 - VLAN ID* ⓘ

2. Enable [Trunk Port for WAN1] by clicking the toggle.
3. Input a native VLAN ID number (default: 1) for untagged traffic on a trunk port.
4. Input port VLAN ID numbers between 1 and 4094 (except for the native VLAN ID) for each port.
5. Click [Save].

4.1.3 Port Mapping

On the [Port Mapping] page, you can view the mapping between the ports and the interfaces.

4.1.3.1 Viewing Port Mapping

Procedure

1. Go to [Device] > [Network & Ports] > [Port Mapping].
2. The [Port Mapping] page shows the mapping between the physical ports and the network interfaces/independent ports.

Device > Network & Ports > Port Mapping

	WAN1	WAN2	PORT1	PORT2	PORT3	PORT4	PORT5	PORT6	PORT7	PORT8	PORT9	PORT10
Network Interface												
WAN1	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
WAN2	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
LAN1	☐	☐	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒
LAN2	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
LAN3	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
LAN4	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
LAN5	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
LAN6	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
LAN7	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
LAN8	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
LAN9	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
LAN10	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
DMZ	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Independent Port												
MGMT Port ⓘ	☐	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
HA Port ⓘ	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Mirror Port ⓘ	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Save Cancel

Port Mapping in Gateway Mode

Device > Network & Ports > Port Mapping

	WAN1	WAN2	PORT1	PORT2	PORT3	PORT4	PORT5	PORT6	PORT7	PORT8	PORT9	PORT10
Network Interface												
Bridge Port	☒	☐	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒
Independent Port												
MGMT Port ⓘ	☐	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
HA Port ⓘ	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Mirror Port ⓘ	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Save Cancel

Port Mapping in Bridge Mode

4.1.4 Network Interfaces (Only in Gateway Mode)

On the [Network Interface] page, you can configure the following:

- Network settings of the network interfaces of the device
- DHCP settings on the LAN/DMZ network interface, including:
 - DHCP service
 - DHCP relay
- Connection type for the WAN network interface, including:
 - Static IP address settings
 - DHCP client
 - PPPoE settings



1. The term **network interface** or **interface** in this document refers to the logical interface that maps to one or multiple physical ports.
2. The default web management console IP address is **192.168.127.254**. The IP address is bound to the **LAN1** network interface.



According to RFC3927, the WAN or LAN interface cannot accept the IPv4 link-local network address range 169.254.0.0/16. It means the link-local addresses in the IPv4 169.254.0.0/16 range (commonly referred to as Automatic Private IP Addressing or APIPA addresses) are not meant to be routed outside of their local network segment. A router should never forward packets that have source or destination addresses in the 169.254.0.0/16 range, so these addresses are non-routable at the IP-layer. This ensures that these addresses remain confined within a single broadcast domain or network segment. Link-local addresses can only be used to communicate with devices that reside on the same local link. They cannot be used to reach hosts on other networks, whether those networks are private (e.g., 192.168.x.x) or public (e.g., routable, globally unique IPv4 addresses).



For more information about changing the default IP address, see [Configuring LAN Network Interfaces](#).

4.1.4.1 Configuring LAN Network Interfaces

Procedure

1. Go to [Device] > [Network & Ports] > [Network Interfaces].

Device > Network & Ports > Network Interfaces

Network Interfaces		Network Interface Advanced Settings					
Network Interface	Status	Connection Type	IP Address	MAC Address	Subnet Mask	VLAN ID	Description
WAN1	Enabled	DHCP Client	17.104.197.17	aa:bb:cc:dd:11:12	255.255.255.0	-	
WAN2	Disabled	DHCP Client	183.253.197.213	aa:bb:cc:dd:11:10	255.255.255.0	-	
LAN1	Enabled	DHCP Relay	192.168.127.254	aa:bb:cc:dd:11:11	255.255.255.0	-	
LAN2	Disabled	DHCP Service Disabled	192.168.2.254	aa:bb:cc:dd:11:09	255.255.255.0	-	
LAN3	Disabled	DHCP Service Disabled	192.168.3.254	aa:bb:cc:dd:11:06	255.255.255.0	-	
LAN4	Disabled	DHCP Service Disabled	192.168.4.254	aa:bb:cc:dd:11:07	255.255.255.0	-	
LAN5	Disabled	DHCP Service Disabled	192.168.5.254	aa:bb:cc:dd:11:13	255.255.255.0	-	
LAN6	Disabled	DHCP Service Disabled	192.168.6.254	aa:bb:cc:dd:11:01	255.255.255.0	-	
LAN7	Disabled	DHCP Service Disabled	192.168.7.254	aa:bb:cc:dd:11:02	255.255.255.0	-	
LAN8	Disabled	DHCP Service Disabled	192.168.8.254	aa:bb:cc:dd:11:03	255.255.255.0	-	
LAN9	Disabled	DHCP Service Disabled	192.168.9.254	aa:bb:cc:dd:11:14	255.255.255.0	-	
LAN10	Disabled	DHCP Service Disabled	192.168.10.254	aa:bb:cc:dd:11:08	255.255.255.0	-	
DMZ	Enabled	DHCP Service Disabled	192.168.254.254	aa:bb:cc:dd:11:05	255.255.255.0	-	Default DMZ

2. Under the [Network Interfaces] tab, click a LAN interface. The [Edit Network Interface] window will appear.

Edit Network Interface

×

Status

☒

Network Interface

LAN1

Description

i

Network Settings

IP Address*

192.168.127.254

MAC Address*

Default

▼

Subnet Mask*

255.255.255.0

☐ VLAN ID

i

NAT

☒ Enabled ☐ Disabled

DHCP Service

DHCP Service

DHCP Server

▼

Start IP Address*

192.168.127.1

End IP Address*

192.168.127.100

Gateway*

192.168.127.254

Lease Time*

86400

i

DNS Server 1

8.8.8.8

DNS Server 2

8.8.8.9

Save

Cancel

3. Configure the following network interface settings.

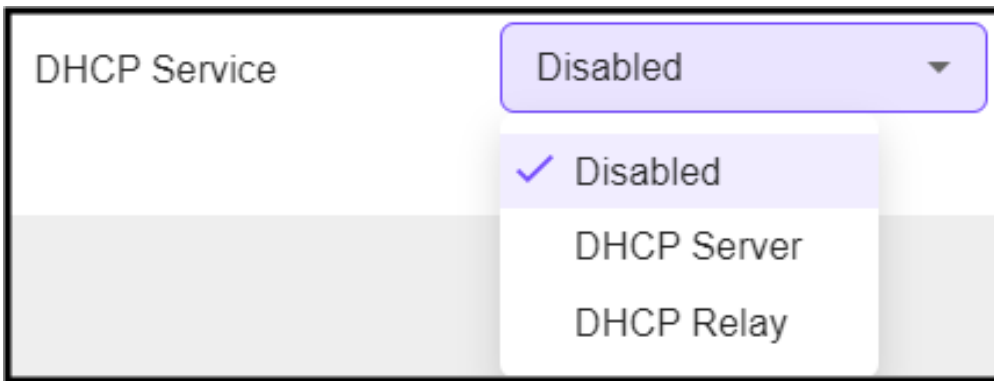
Item	Description
Status	The connection status of the LAN interface. Enable or disable the LAN interface by clicking the toggle.
Description	(Optional) Type a description for the LAN interface.

- Network Settings

Item	Description
IP Address	Input an IP Address of the device on the LAN interface.
MAC Address	Select Default or Custom . If Custom is selected, further input a MAC address intended for unicast use according to RFC-9542. The network interface cannot accept a MAC address used for multicast or broadcast purposes (e.g., a MAC address with the least significant bit (M bit) of the first byte set to 1 or an "FF:FF:FF:FF:FF:FF" address).

Item	Description
Subnet Mask	Input a subnet mask of the device.
VLAN ID	(Optional) Enable this function by clicking the toggle. If enabled, input a VLAN ID of the device to allow packets with the specified VLAN tag to access the LAN interface.
Default Gateway	Select WAN1 or WAN2 as the default gateway for the device.  This setting is configurable only when the WAN2 interface on the [Port Mapping] page is enabled.
NAT	Enable or disable the NAT (Network Address Translation) function based on the requirements for cross-network communication. Some assets on the LAN or DMZ interface may have IP addresses outside the private address space and may not require NAT. When NAT is enabled, the IP addresses of assets on the LAN or DMZ network will be translated for communication with other network interfaces. If NAT is disabled, the IP addresses will be routed directly to other network interfaces without translation.

- DHCP Service: Select **Disabled**, **DHCP Server** or **DHCP Relay** for the DHCP Service.



- Disabled: No DHCP service will be provided on this interface.
- DHCP Server: This interface will provide DHCP service to the devices that connect to the interface. If this is selected, provide the following information.

DHCP Service

DHCP Service	DHCP Server ▼
Start IP Address*	192.168.127.1
End IP Address*	192.168.127.100
Gateway*	192.168.127.254
Lease Time*	86400 i
DNS Server 1	8.8.8.8
DNS Server 2	8.8.8.9

Item	Description
DHCP Service	DHCP Server
Start IP Address	Input a start IP address of the DHCP service.
End IP Address	Input an end IP address of the DHCP service.
Gateway	Input a gateway IP address that will be assigned to the clients.
Lease Time	Input a specific amount of time ranging from 300 to 86400 seconds (default: 86400) that a client device can use the IP address settings assigned by the DHCP server.
DNS Server 1	Input a DNS server IP address that will be assigned to the clients.
DNS Server 2	Input another DNS server IP address that will be assigned to the clients.

• DHCP Relay: This interface will relay the traffic from the clients to a relayed server for DHCP service. If this is selected, provide the following information.

DHCP Service

DHCP Service

DHCP Relay ▼

Interface

WAN1 ▼

Relay Server Address*

192.168.50.1

Item	Description
DHCP Service	DHCP Relay
Interface	When the WAN2 interface on the [Port Mapping] page is enabled, select either WAN1 or WAN2. If WAN2 is disabled, this setting is set to WAN1.
Relay Server Address	Input the IP address of the server that will provide DHCP service.

4. Click [OK].

4.1.4.2 Configuring DMZ Network Interface

Procedure

1. Go to [Device] > [Network & Ports] > [Network Interfaces].

Device > Network & Ports > Network Interfaces

Network Interfaces

Network Interface Advanced Settings

Network Interface	Status	Connection Type	IP Address	MAC Address	Subnet Mask	VLAN ID	Description
WAN1	Enabled	DHCP Client	17.104.197.17	aa bb cc dd 11 12	255.255.255.0	-	
WAN2	Disabled	DHCP Client	183.253.197.213	aa bb cc dd 11 10	255.255.255.0	-	
LAN1	Enabled	DHCP Relay	192.168.127.254	aa bb cc dd 11 11	255.255.255.0	-	
LAN2	Disabled	DHCP Service Disabled	192.168.2.254	aa bb cc dd 11 09	255.255.255.0	-	
LAN3	Disabled	DHCP Service Disabled	192.168.3.254	aa bb cc dd 11 06	255.255.255.0	-	
LAN4	Disabled	DHCP Service Disabled	192.168.4.254	aa bb cc dd 11 07	255.255.255.0	-	
LAN5	Disabled	DHCP Service Disabled	192.168.5.254	aa bb cc dd 11 13	255.255.255.0	-	
LAN6	Disabled	DHCP Service Disabled	192.168.6.254	aa bb cc dd 11 01	255.255.255.0	-	
LAN7	Disabled	DHCP Service Disabled	192.168.7.254	aa bb cc dd 11 02	255.255.255.0	-	
LAN8	Disabled	DHCP Service Disabled	192.168.8.254	aa bb cc dd 11 03	255.255.255.0	-	
LAN9	Disabled	DHCP Service Disabled	192.168.9.254	aa bb cc dd 11 14	255.255.255.0	-	
LAN10	Disabled	DHCP Service Disabled	192.168.10.254	aa bb cc dd 11 08	255.255.255.0	-	
DMZ	Enabled	DHCP Service Disabled	192.168.254.254	aa bb cc dd 11 05	255.255.255.0	-	Default DMZ

2. Under the [Network Interfaces] tab, click the DMZ interface. The [Edit Network Interface] window will appear.

Edit Network Interface

Status

☒

Network Interface

DMZ

Description

Network Settings

IP Address*

MAC Address*

Default

Subnet Mask*

☐ VLAN ID

NAT

☒ Enabled
☐ Disabled

DHCP Service

DHCP Service

DHCP Server

Start IP Address*

End IP Address*

Gateway*

Lease Time*

DNS Server 1

DNS Server 2

Save

Cancel

3. Configure the following network interface settings.

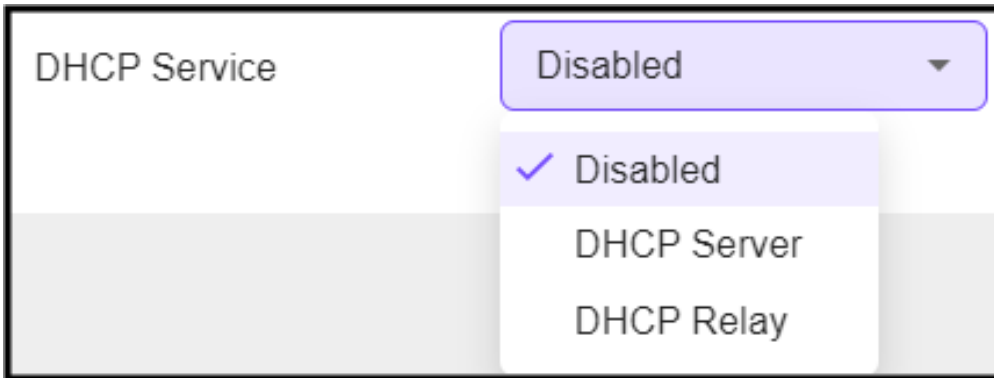
Item	Description
Status	The connection status of the DMZ interface. Enable or disable the DMZ interface by clicking the toggle.
Description	(Optional) Type a description for the DMZ interface.

- Network Settings

Item	Description
IP Address	Input an IP Address of the device on the DMZ interface.
MAC Address	Select Default or Custom . If Custom is selected, further input a MAC address intended for unicast use according to RFC-9542. The network interface cannot accept a MAC address used for multicast or broadcast purposes (e.g., a MAC address with the least significant bit (M bit) of the first byte set to 1 or an "FF:FF:FF:FF:FF:FF" address).
Subnet Mask	Input a subnet mask of the device.

Item	Description
VLAN ID	(Optional) Enable this function by clicking the toggle. If enabled, input a VLAN ID of the device to allow packets with the specified VLAN tag to access the DMZ interface.
Default Gateway	Select WAN1 or WAN2 as the default gateway for the device.  This setting is configurable only when the WAN2 interface on the [Port Mapping] page is enabled.
NAT	Enable or disable the NAT (Network Address Translation) function based on the requirements for cross-network communication. Some assets on the LAN or DMZ interface may have IP addresses outside the private address space and may not require NAT. When NAT is enabled, the IP addresses of assets on the LAN or DMZ network will be translated for communication with other network interfaces. If NAT is disabled, the IP addresses will be routed directly to other network interfaces without translation.

- DHCP Service: Select **Disabled**, **DHCP Server** or **DHCP Relay** for the DHCP Service.



The screenshot shows a configuration interface for the DHCP Service. On the left, the text "DHCP Service" is displayed. To its right is a dropdown menu currently set to "Disabled". The dropdown menu is open, showing three options: "Disabled" (which is selected and has a blue checkmark), "DHCP Server", and "DHCP Relay".

- Disabled: No DHCP service will be provided on this interface.
- DHCP Server: This interface will provide DHCP service to the devices that connect to the interface. If this is selected, provide the following information.

DHCP Service

DHCP Service

DHCP Server ▼

Start IP Address*

192.168.127.1

End IP Address*

192.168.127.100

Gateway*

192.168.127.254

Lease Time*

86400

i

DNS Server 1

8.8.8.8

DNS Server 2

8.8.8.9

Item	Description
DHCP Service	DHCP Server
Start IP Address	Input a start IP address of the DHCP service.
End IP Address	Input an end IP address of the DHCP service.
Gateway	Input a gateway IP address that will be assigned to the clients.
Lease Time	Input a specific amount of time ranging from 300 to 86400 seconds (default: 86400) that a client device can use the IP address settings assigned by the DHCP server.
DNS Server 1	Input a DNS server IP address that will be assigned to the clients.
DNS Server 2	Input another DNS server IP address that will be assigned to the clients.

- DHCP Relay: This interface will relay the traffic from the clients to a relayed server for DHCP service. If this is selected, provide the following information.

Device – 44

DHCP Service

DHCP Service

DHCP Relay

Interface

WAN1

Relay Server Address*

192.168.50.1

Item	Description
DHCP Service	DHCP Relay
Interface	When the WAN2 interface on the [Port Mapping] page is enabled, select either WAN1 or WAN2. If WAN2 is disabled, this setting is set to WAN1.
Relay Server Address	Input the IP address of the server that will provide DHCP service.

4. Click [OK].

4.1.4.3 Configuring WAN Network Interface

Procedure

1. Go to [Device] > [Network & Ports] > [Network Interfaces].

Device > Network & Ports > Network Interfaces

Network Interfaces

Network Interface Advanced Settings

Network Interface	Status	Connection Type	IP Address	MAC Address	Subnet Mask	VLAN ID	Description
WAN1	Enabled	DHCP Client	17.104.197.17	aa:bb:cc:dd:11:12	255.255.255.0	-	
WAN2	Disabled	DHCP Client	183.253.197.213	aa:bb:cc:dd:11:10	255.255.255.0	-	
LAN1	Enabled	DHCP Relay	192.168.127.254	aa:bb:cc:dd:11:11	255.255.255.0	-	
LAN2	Disabled	DHCP Service Disabled	192.168.2.254	aa:bb:cc:dd:11:09	255.255.255.0	-	
LAN3	Disabled	DHCP Service Disabled	192.168.3.254	aa:bb:cc:dd:11:06	255.255.255.0	-	
LAN4	Disabled	DHCP Service Disabled	192.168.4.254	aa:bb:cc:dd:11:07	255.255.255.0	-	
LAN5	Disabled	DHCP Service Disabled	192.168.5.254	aa:bb:cc:dd:11:13	255.255.255.0	-	
LAN6	Disabled	DHCP Service Disabled	192.168.6.254	aa:bb:cc:dd:11:01	255.255.255.0	-	
LAN7	Disabled	DHCP Service Disabled	192.168.7.254	aa:bb:cc:dd:11:02	255.255.255.0	-	
LAN8	Disabled	DHCP Service Disabled	192.168.8.254	aa:bb:cc:dd:11:03	255.255.255.0	-	
LAN9	Disabled	DHCP Service Disabled	192.168.9.254	aa:bb:cc:dd:11:14	255.255.255.0	-	
LAN10	Disabled	DHCP Service Disabled	192.168.10.254	aa:bb:cc:dd:11:08	255.255.255.0	-	
DMZ	Enabled	DHCP Service Disabled	192.168.254.254	aa:bb:cc:dd:11:05	255.255.255.0	-	Default DMZ

2. Under the [Network Interfaces] tab, click the WAN interface. The [Edit Network Interface] window will appear.

Edit Network Interface

Status
☒

Network Interface
WAN1

Description

Network Settings

Connection Type
Static IP

IP Address*
10.100.100.1

Subnet Mask*
255.255.255.0

Gateway*
10.100.100.254

DNS Server 1
8.8.8.8

DNS Server 2
8.8.8.9

☐ VLAN ID

Save

Cancel

3. Configure the following network interface settings.

Item	Description
Status	The connection status of the WAN interface. Enable or disable the LAN interface by clicking the toggle.
Description	(Optional) Type a description for the WAN interface.

- Network Settings: Select **Static IP**, **DHCP Client** or **PPPoE** for the Connection Type.

Connection Type
Static IP

IP Address*

Subnet Mask*

Static IP

DHCP Client

PPPoE

- Static IP: This device uses a static IP address for this interface. If this is selected, provide the following information.

Network Settings

Connection Type

Static IP

IP Address*

10.100.100.1

Subnet Mask*

255.255.255.0

Gateway*

10.100.100.254

DNS Server 1

8.8.8.8

DNS Server 2

8.8.8.9

VLAN ID

Item	Description
Connection Type	Static IP
IP Address	Input an IP Address of the device on the WAN interface.
Subnet Mask	Input a subnet mask of the device.
Gateway	Input a gateway IP address of the device on the WAN interface.
DNS Server 1	Input a DNS server IP address.
DNS Server 2	Input another DNS server IP address.
VLAN ID	(Optional) Enable this function by clicking the toggle. If enabled, input a VLAN ID of the device to allow packets with the specified VLAN tag to access the WAN interface.

- DHCP Client: The interface functions as a DHCP client and automatically get an IP address from a DHCP server. If this is selected, provide the following information.

Network Settings

Connection Type

DHCP Client

 EdgeFire gets IP address automatically.

VLAN ID

Item	Description
Connection Type	DHCP Client
VLAN ID	(Optional) Enable this function by clicking the toggle. If enabled, input a VLAN ID of the device to allow packets with the specified VLAN tag to access the WAN interface.

- PPPoE: The interface functions as a PPPoE client. If this is selected, provide the following information.

Network Settings

Connection Type

PPPoE

Authentication Type

Auto

Username*

i

Password*

.....

i

DNS Server 1

8.8.8.8

DNS Server 2

8.8.8.9

MTU*

1480

i

Reconnection Mode

Always On

Retry Interval*

30

(1-60 seconds)

Item	Description
Connection Type	PPPoE
Authentication Type	Select an authentication protocol, MSCHAPv2 , MSCHAPv1 , CHAP , PAP , or Auto to include all the protocols on the list.
Username	Input the username for authenticating the connection.
Password	Input the password for authenticating the connection.
DNS Server 1	Input a DNS server IP address.
DNS Server 2	Input another DNS server IP address.
MTU	Input a number for the Maximum Transmission Unit (MTU) that defines the largest size packet (default: 1480) to be transmitted over the network.

Item	Description
Reconnection Mode	Select a reconnection mode, Always On, On Demand or Manual for the PPPoE connection establishment and maintenance. • Always On: Select this option for the PPPoE connection to be always active, and the Edge device will automatically reconnect if the connection is lost. • On Demand: Select this option for the PPPoE connection to be only established when needed. • Manual: Select this option for the PPPoE connection to be manually established and disconnected.
Retry Interval	If the reconnection mode is Always On, then input a number between 1 and 60 (seconds) (default: 30) to specify how long the PPPoE client should wait before attempting to re-establish the PPPoE connection after a connection failure or disconnection.
Idle Timeout	If the reconnection mode is On Demand, then input a number between 10 and 3600 (seconds) (default: 300) to specify how long the PPPoE connection should remain active when there is no network traffic passing through it.

4. Click [OK].

4.1.4.4 Configuring Network Interface Advanced Settings

Procedure

1. Go to [Device] > [Network & Ports] > [Network Interfaces].

Device > Network & Ports > Network Interfaces

Network Interfaces

Network Interface Advanced Settings

System Default Gateway Setting

System Default Gateway  ☒ WAN1 ☐ WAN2

Failover Settings

☐ Failover 

IP Address for WAN1 Ping

IP Address for WAN2 Ping

Failover Check Interval (1-60 seconds)

Failback Trigger Level

Failover Trigger Level

Save

Cancel

 Make sure the WAN2 port on the [Port Mapping] page is active before configuring the network interface advanced settings.

2. Configure the following settings.

- System Default Gateway Setting

Item	Description
System Default Gateway	Select the WAN1 or WAN2 interface. The system default gateway enables EdgeFire's local services to communicate with external networks. Changing the system default gateway may require the EdgeFire local services to reconnect to external networks.

- Failover Settings

Item	Description
Failover	Enable the failover function by clicking the toggle. Failover ensures backup communication for EdgeFire when the primary connection is unavailable. To determine if failover should activate, EdgeFire performs ping tests on specified IP addresses in the WAN1 or WAN2 network. Once the primary connection is restored, EdgeFire automatically switches back from the backup to the primary connection.

Item	Description
IP Address for WAN1 Ping	Input an IP address in the WAN1 network for EdgeFire to perform ping tests to check if the primary connection of EdgeFire is unavailable.
IP Address for WAN2 Ping	Input an IP address in the WAN2 network for EdgeFire to perform ping tests to check if the primary connection of EdgeFire is unavailable.
Failover Check Interval	Input a number between 1 and 60 to specify how often (default: 5 seconds) EdgeFire performs ping tests to check if the primary connection of EdgeFire is unavailable.
Failback Trigger Level	Select how many times (default: 3 times) the primary connection of EdgeFire can be pinged before EdgeFire switches back from the backup to the primary connection.
Failover Trigger Level	Select how many times (default: 3 times) the primary connection of EdgeFire fails to be pinged before EdgeFire switches from the primary to the backup connection.

3. Click [Save].

4.1.5 DDNS Settings (Only in Gateway Mode)

On the [DDNS Settings] page, you can assign a domain name to your network's public IP address, which can change over time. This allows you to access your network remotely using a domain name instead of a changing IP address.

4.1.5.1 Configuring DDNS Settings

Procedure

1. Go to [Device] > [Network & Ports] > [DDNS Settings].



No.	Status	Name	Registration Status	Last Updated	Interface	Hostname	Service Provider	IP Address	Description
1	Enabled	DDNS1	Successful	2024-10-30T16:46:27+08:00	WAN1	abc.ddns.net	No-IP	-	

2. Do one of the following.

- Click [Create] to create an item. The [Create DDNS] window will appear.
- Click an item name to edit settings. The [Edit DDNS] window will appear.

Create DDNS

Status

Name*

Description

Host Settings

Interface

WAN1

Hostname*

IP Address

Use Port IP

IP Edit Checking Interval*

20

(4-1440 Minutes)

Service Provider Settings

Service Provider

DynDNS

Login Name*

Password*

.....

Save

Cancel

3. Configure the following settings.

Item	Description
Status	Enable or disable the DDNS settings by clicking the toggle.
Name	Type a name for this DDNS setting object.
Description	Type a description.

- Host Settings

Item	Description
Interface	This is set to the WAN1 interface.
Hostname	Input a domain name that you want to use for your network.
IP Address	Select either Use Port IP or Use NATed Public IP for the public IP address associated with your hostname.

Item	Description
	- Use Port IP: Use the WAN1 interface's IP address as the public IP address for your network. - Use NATed Public IP: Use the public IP address assigned to the WAN1 interface after NAT is applied.  For more information, see NAT Rules.
IP Edit Checking Interval	Input a number between 4 and 1440 (minutes) (default: 20) to specify how often the Edge device should check for changes in the IP address.

- Service Provider Settings

Item	Description
Service Provider	Select DynDNS , No-IP or DynuDNS as your service provider.
Login Name	Input the username that you use to log in to your DDNS service provider account.
Password	Input the password that you use to log in to your DDNS service provider account.

4. Click [OK].

4.1.6 MGMT Interface Settings (Only in Gateway Mode)

The MGMT port (or management interface, which is mapped to WAN2) is a reserved out-of-band management port that serves as an independent interface for the remote system management and network device configuration on EdgeFire when the device is in Gateway mode. The MGMT port is logically separate from the data ports. Traffic passing through these two types of ports is isolated and does not interchange (except for transferring management data stored in EdgeFire).

- MGMT Port:** This port connects EdgeFire to EdgeOne for device management.
- Data Ports:** These ports connect EdgeFire to target assets for asset protection.

4.1.6.1 Configuring MGMT Interface Settings

Procedure

- Go to [Device] > [Network & Ports] > [MGMT Interface Settings].

Device > Network & Ports > MGMT Interface Settings

☒ Independent MGMT

Connection Type

Static IP ▼

Edge IP Address*

192.168.1.42

Subnet Mask*

255.255.255.0

Gateway*

192.168.1.254

DNS

8.8.8.8

☒ VLAN ID

42 ⓘ

Save

Cancel

2. Use the toggle to enable or disable the [Independent MGMT] function.
3. If Independent MGMT is enabled, select **Static IP** or **DHCP Client** for the Connection Type.
 - **Static IP:** This device uses a static IP address for this interface. If this option is selected, provide the following information.

Item	Description
Edge IP Address	Input an IP Address of the device.
Subnet Mask	Input a subnet mask of the device.
Gateway	Input an IP address of the gateway that will send packets to a different network.
DNS	(Optional) Input a DNS address of the device.
VLAN ID	(Optional) Enable this function by clicking the toggle. If enabled, input a VLAN ID of the device to allow packets with the specified VLAN tag to access the management interface.

- **DHCP Client:** The interface functions as a DHCP client and automatically get an IP address from a DHCP server. If this option is selected, provide the following information.

Device > Network & Ports > MGMT Interface Settings

☒ Independent MGMT

Connection Type

DHCP Client

⚠ The Edge device gets IP address automatically.

☒ VLAN ID

42

ℹ

Save

Cancel

Item	Description
Connection Type	DHCP Client
VLAN ID	(Optional) Enable this function by clicking the toggle. If enabled, input a VLAN ID for the device to allow packets with the specified VLAN tag to access the management interface.

4. Click [Save].

4.1.7 HA Settings

Using a single device for network security and traffic flow can create a single point of failure. This Edge series device provides a High Availability (HA) feature that enables redundancy by way of the ability to add and synchronize configurations with a secondary backup device. This eliminates the single point of failure and allows for a seamless switchover from primary device to secondary backup device in case of primary device failure. Thus, it allows for the planning and building of a fault-tolerant, resilient, and secure network to minimize any OT operational downtime.

The HA feature allows for the grouping of two devices to form an HA group where configurations of the devices in the group are synchronized to support full fail-over redundancy. This section describes how to configure HA.

4.1.7.1 Configuring HA Settings

Procedure

1. Go to [Device] > [Network & Ports] > [HA Settings].

Device > Network & Ports > HA Settings

HA Status

High Availability

Enabled

Connection Status

Connected

Peer Sync Status

Not Synced - Ready for first-time sync

Sync

Device Role Status

Primary

⚙ Edit HA Settings

Item	Description
High Availability	The status of HA, either Enabled or Disabled .

Item	Description
	If High Availability is enabled, the following status information will be shown.
Connection Status	The status of connection, either Connected or Disconnected .
Peer Sync Status	The status of peer synchronization, either Synced or Not Synced .
Device Role Status	The status of device role, either Primary or Secondary .



You cannot configure the HA settings if the HA port is inactive on the [Port Mapping] page, or the WAN interface connection type is set to DHCP Client or PPPoE.

2. Click the [Edit HA Settings] button. The [Edit HA Settings] window will appear.

Edit HA Settings

Status
☒

HA Settings

Mode
Active / Standby

HA Interface IP Address*

HA Interface Subnet Mask*

☐ VLAN ID
i

Unicast Heartbeat Peer IP*

Shared Secret Key*
i

Heartbeat Interval

Failover Trigger Level

Management IP List *i*

Management IP*

3. Configure the following HA settings and management IP list.

Item	Description
Status	The HA sync status between two Edge devices in the same HA group. This also displays the sync progress. Enable the HA function by clicking the toggle.

- HA Settings

Item	Description
Mode	Active/Standby: Only one device in the HA group is active and online. The secondary/standby device is only brought online in case of primary device failure and traffic only flows via the active device. In this mode, the switching of traffic between devices is handled internally on the devices via HA protocol logic set up between the devices.
HA Interface IP Address	Input an IP Address of the HA interface that sends/receives HA heartbeats and data messages to/from its peer.
HA Interface Subnet Mask	Input a subnet mask of the HA interface.
VLAN ID	(Optional) Enable this function by clicking the toggle. If enabled, input a VLAN ID of the device to allow packets with the specified VLAN tag to access the HA interface.
Unicast Heartbeat Peer IP	Input an IP address of the HA peer.
Shared Secret key	Input a shared secret key to allow for two HA peers to authenticate and communicate with each other
Heartbeat Interval	Select an interval between sent heartbeats ranging from 1 to 10 seconds (default: 1).
Failover Trigger Level	Select a maximum number of consecutive missed heartbeats before secondary to primary switch is triggered ranging from 1 to 10 times (default: 3).

- Management IP List

Item	Description
Management IP	When HA is enabled and the configuration is synced up, the IP settings for all the interfaces will be synced to the active device. To access the management interface of the active device or the standby device, create at least a management IP to bridge to a designated management interface. Click the add button  to add another entry, or the remove button  to remove an entry.

4. Click [Save].

4.1.8 Port Mirroring

To monitor the network, all the traffic from any of the protected segments will be mirrored to the reserved port, i.e., the Mirror Port.



Instead of pair-based, the Port Mirroring feature is port-based.

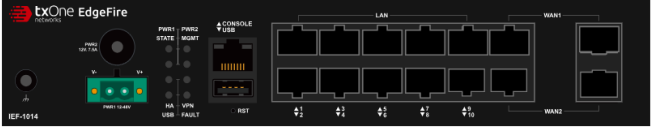
4.1.8.1 Configuring Port Mirroring Settings

Procedure

1. Go to [Device] > [Network & Ports] > [Port Mirroring].

Device > Network & Ports > Port Mirroring

☐ Enabled

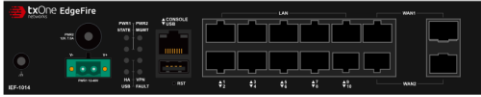


2. Enable the Port Mirroring function by clicking the toggle. You will then be able to configure the port mirroring settings on this page.

Device > Network & Ports > Port Mirroring

☒ Enabled

Click on a toggle to enable or disable the Port Mirroring function of a specific port.

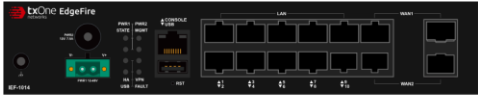


Status	Interface	Port	TX / RX
☐	WAN1	WAN1	TX / RX
☐	WAN2	-	TX / RX
☐	LAN1	PORT1, PORT2, PORT3, PORT4, PORT5, PORT6, PORT7, PORT8, PORT9	TX / RX
☐	LAN2	-	TX / RX
☐	LAN3	-	TX / RX
☐	LAN4	-	TX / RX
☐	LAN5	-	TX / RX
☐	LAN6	-	TX / RX
☐	LAN7	-	TX / RX
☐	LAN8	-	TX / RX
☐	LAN9	-	TX / RX
☐	LAN10	-	TX / RX
☐	DMZ	-	TX / RX

Port Mirroring in Gateway Mode

Device > Network & Ports > Port Mirroring

☒ Enabled
Click on a toggle to enable or disable the Port Mirroring function of a specific port.



Status	Port	TX / RX
☐	WAN1	TX / RX
☐	PORT1	TX / RX
☐	PORT2	TX / RX
☐	PORT3	TX / RX
☐	PORT4	TX / RX
☐	PORT5	TX / RX
☐	PORT6	TX / RX
☐	PORT7	TX / RX
☐	PORT8	TX / RX
☐	PORT9	TX / RX

Port Mirroring in Bridge Mode

3. Click a port on the front panel to configure the following port mirror settings.

Status	Interface	Port	TX / RX
☒	WAN1	WAN1	TX / RX TX Only RX Only ✓ TX / RX

Port Mirroring in Gateway Mode

Status	Port	TX / RX
☒	WAN1	TX / RX TX Only RX Only ✓ TX / RX

Port Mirroring in Bridge Mode

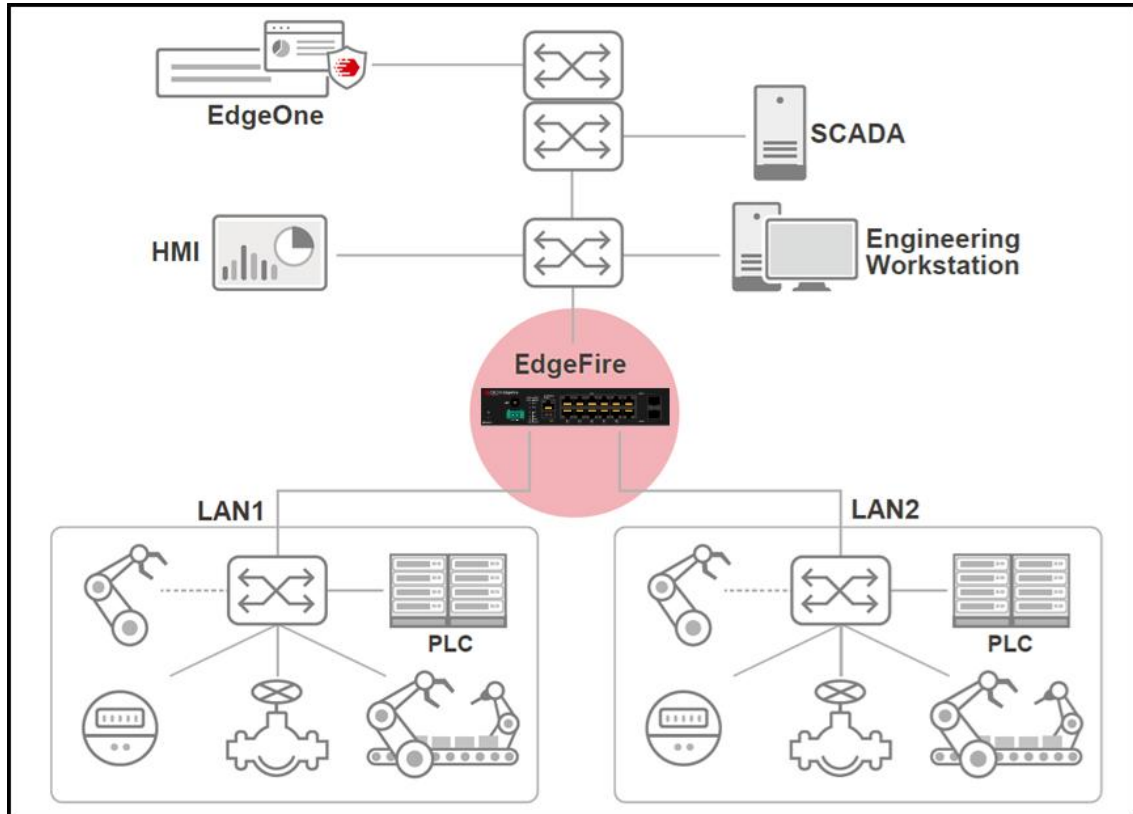
Item	Description
Status	The connection status of the mirror port. Enable or disable the port by clicking the toggle.
TX/RX	Select a mirror traffic direction, TX Only , RX Only or TX/RX for the port.

4. Click [Save].

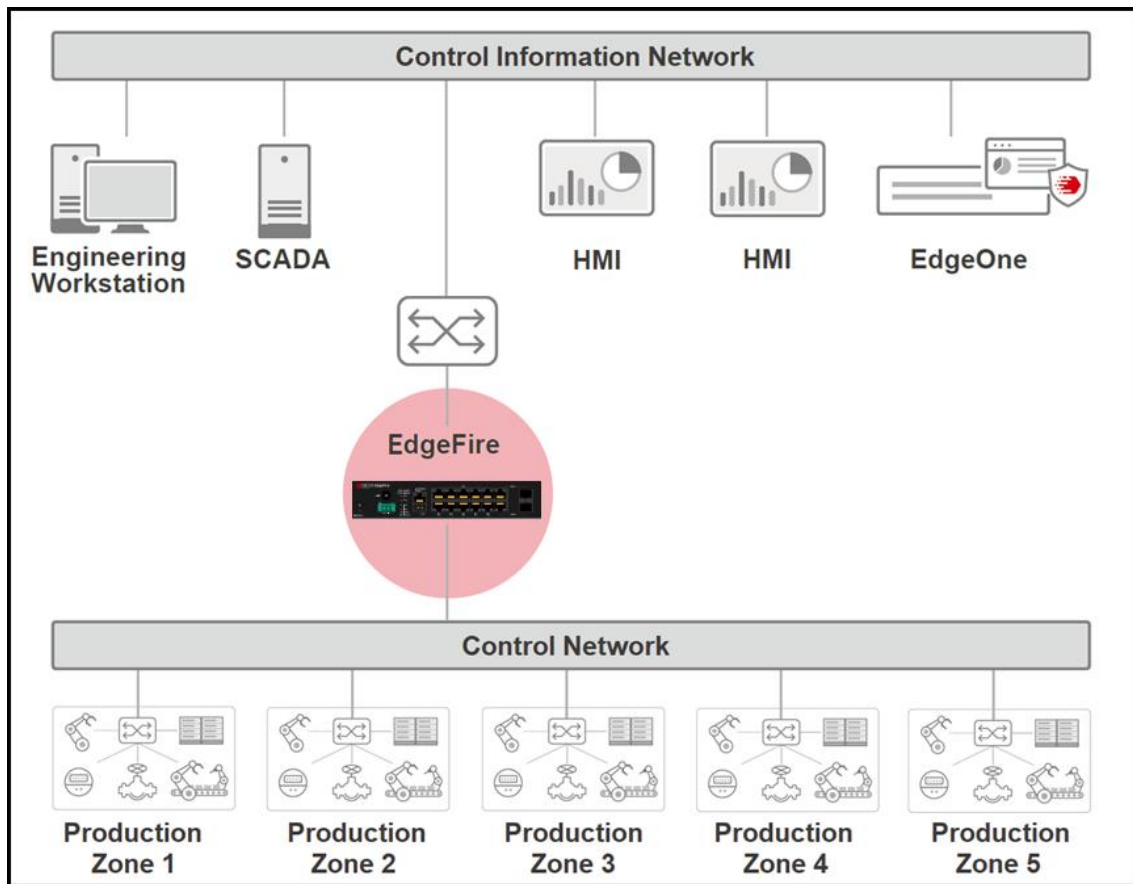
4.1.9 Operation Mode

EdgeFire offers two system operation modes:

- **Gateway Mode:** When EdgeFire is in gateway mode, it operates as a gateway with NAT and routing main features between multiple different network segments to actively analyze, filter, and take actions on all traffic that passes through it.



- **Bridge Mode:** When EdgeFire is in bridge mode, it sits in the direct communication path between source and destination to actively analyze, filter and take actions on all traffic that passes through it.



4.1.9.1 Switching to Gateway Mode

Procedure

1. Go to [Device] > [Network & Ports] > [Operation Modes].

Device > Network & Ports > Operation Modes

Operation Modes

☐ Gateway Mode
 ☒ Bridge Mode

Network Settings

Management Interface	Bridge Port
Connection Type	Static IP
Edge IP Address*	192.168.1.42
Subnet Mask*	255.255.255.0
Gateway*	192.168.1.254
DNS	8.8.8.8
☒ VLAN ID	42 
☐ STP 	

LLDP Setting

Transmit LLDP ☒

2. Click the [Gateway Mode] radio button. The [LAN1 Network Settings] and [LAN1 DHCP Service] panes for the gateway mode will appear. The information here is for viewing only.

Device > Network & Ports > Operation Modes

Operation Modes

☒ Gateway Mode
 ☐ Bridge Mode

LAN1 Network Settings

IP Address	192.168.127.254
Subnet Mask	255.255.255.0

LAN1 DHCP Service

DHCP Service	Relay
Start IP Address	192.168.127.1
End IP Address	192.168.127.100
Gateway	192.168.127.254
Lease Time	86400

LLDP Setting

Transmit LLDP ☒

 For more information about configuring the LAN network settings and DHCP service, see [Configuring LAN Network Interfaces](#).

3. Configure the LLDP setting.

- LLDP Setting

Item	Description
Transmit LLDP	(Optional) Enable the LLDP function by clicking the toggle.  When the transmission of LLDP (Link Layer Discovery Protocol) for discovery and configuration is enabled, it allows a network device to advertise its identity and capabilities on the network.

4. Click [Save].

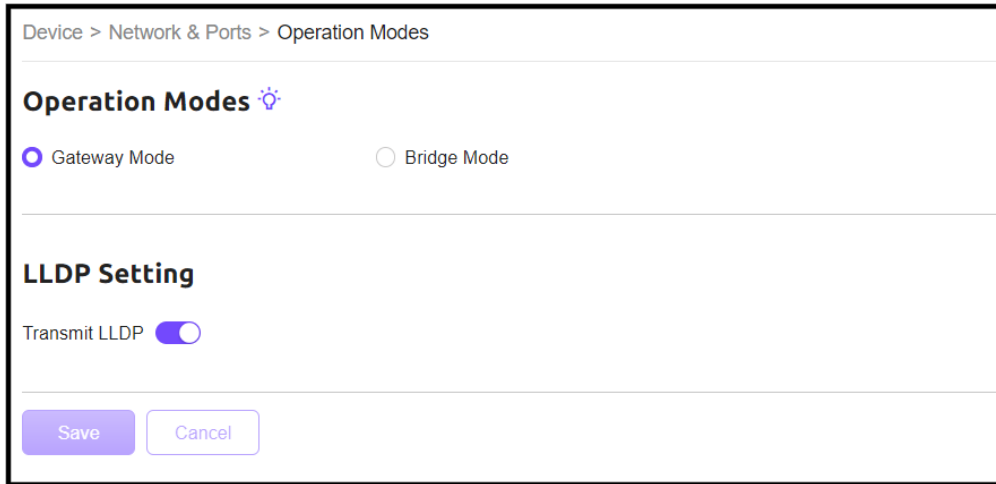


The configuration of the policy enforcement rule is not compatible between gateway mode and bridge mode. Therefore, the policy enforcement rule needs to be reconfigured after the device is switched from bridge mode to gateway mode.

4.1.9.2 Switching to Bridge Mode

Procedure

1. Go to [Device] > [Network & Ports] > [Operation Modes].



2. Click the [Bridge Mode] radio button. The [Network Settings] pane for the bridge mode will appear.

Device > Network & Ports > Operation Modes

Operation Modes

☐ Gateway Mode
 ☒ Bridge Mode

Network Settings

Management Interface	Bridge Port
Connection Type	Static IP
Edge IP Address*	192.168.1.42
Subnet Mask*	255.255.255.0
Gateway*	192.168.1.254
DNS	8.8.8.8
☒ VLAN ID	42 
☐ STP 	

LLDP Setting

Transmit LLDP ☒

3. Configure the following settings.

- Network Settings: Select either **Static IP** or **DHCP Client** for the Connection Type.

Connection Type	Static IP
Edge IP Address*	☒ Static IP ☐ DHCP Client

- Static IP:** This device uses a static IP address for this interface. If this option is selected, provide the following information.

Network Settings

Management Interface

Bridge Port

Connection Type

Static IP

Edge IP Address*

192.168.1.42

Subnet Mask*

255.255.255.0

Gateway*

192.168.1.254

DNS

8.8.8.8

☒
VLAN ID

42

☐
STP

Item	Description
Management Interface	Select either Bridge Port or Independent MGMT Port for the management interface.
Connection Type	Static IP
Edge IP Address	Input an IP Address of the device.
Subnet Mask	Input a subnet mask of the device.
Gateway	Input an IP address of the gateway that will route the management connection to a different network segment.
DNS	(Optional) Input a DNS address of the device.
VLAN ID	(Optional) Enable this function by clicking the toggle. If enabled, input a VLAN ID of the device to allow packets with the specified VLAN tag to access the interface.
STP	(Optional) Enable the STP (Spanning Tree Protocol) function by clicking the toggle for preventing bridge loops or broadcast radiation.

• **DHCP Client:** The interface functions as a DHCP client and automatically get an IP address from a DHCP server. If this option is selected, provide the following information.

Network Settings

Management Interface

Bridge Port

Connection Type

DHCP Client

 The Edge device gets IP address automatically.

VLAN ID

42



STP



Item	Description
Management Interface	Select either Bridge Port or Independent MGMT Port for the management interface.
Connection Type	DHCP Client
VLAN ID	(Optional) Enable this function by clicking the toggle. If enabled, input a VLAN ID of the device to allow packets with the specified VLAN tag to access the interface.
STP	(Optional) Enable the STP (Spanning Tree Protocol) function by clicking the toggle for preventing bridge loops or broadcast radiation.

- LLDP Setting

Item	Description
Transmit LLDP	(Optional) Enable the LLDP function by clicking the toggle.  When the transmission of LLDP (Link Layer Discovery Protocol) for discovery and configuration is enabled, it allows a network device to advertise its identity and capabilities on the network.

4. Click [Save].

	1. When EdgeFire is in bridge mode, the [Port Mapping], [Network Interfaces], [NAT Rules], [ALG Settings], and [Static Routes] features will be non-operational and non-configurable. 2. The configuration of the [Policy Enforcement] is not compatible between gateway mode and bridge mode. Therefore, the Policy Enforcement rules need to be reconfigured after the device is switched from gateway mode to bridge mode.
---	--

4.2 NAT (Only in Gateway Mode)

Configure NAT rules and enable/disable application layer gateways (ALGs).

- [NAT Rules](#)

- [ALG Settings](#)

4.2.1 NAT Rules

On the [NAT Rules] page, you can configure the following:

- **1-to-1 Network Address Translation Rules:** A 1-to-1 NAT rule allows you to map a destination IP address in the incoming traffic to another IP address located in a specific network.
- **Multiple 1-to-1 Network Address Translation Rules (CIDR or IP Range):** A multiple 1-to-1 NAT rule allows you to map multiple destination IP addresses in the incoming traffic to different IP addresses located in a specific network as shown in the following example.

Original Destination IP Addresses

172.1.1.5
172.1.1.20
172.1.1.50
172.1.1.69

... Are Mapped to These Destination IP Addresses

192.168.100.5
192.168.100.20
192.168.100.50
192.168.100.69

- **Port Forwarding Network Address Translation Rules:** A port forwarding NAT rule allows you to map a host IP address in the incoming traffic to another IP address located in your local network.

The following table describes the tasks you can perform on the [NAT Rules] page.

Task	Description
To create a NAT rule	Click the [Create] button to create a new NAT rule.
To edit a NAT rule	Click a NAT rule name to edit the rule settings.
To delete a NAT rule	Click the checkbox of a NAT rule and click the [Delete] button.
To duplicate a NAT rule	Click the checkbox of a NAT rule and click the [Duplicate] button.
To change rule priority	Click the checkbox of a NAT rule and click the [Change Priority] button.

4.2.1.1 Configuring NAT Rules

Procedure

1. Go to [Device] > [NAT] > [NAT Rules].

Device > NAT > NAT Rules

[+ Create](#) Total Number of Items: 5 (Max: 64) 1 / 1 [<](#) [>](#) [📄](#)

☐	No.	Status	Rule Name	NAT Type	Incoming Interface	Original IP	Mapped IP	Outgoing Interface	Description
☐	1	☒	NAT_1	1-to-1 NAT	WAN1	10.1.1.1	192.168.27.1	-	
☐	2	☒	NAT_2	Multiple 1-to-1 NAT (CIDR)	WAN1	172.1.1.1/24	192.168.100.1/24	-	
☐	3	☒	NAT_3	Multiple 1-to-1 NAT (IP Range)	WAN1	15.1.1.10-15.1.1.30	192.168.5.10-192.168.5.30	-	
☐	4	☒	NAT_4	Port Forwarding	WAN1	Current IP (1-1)	192.168.30.1 (1-1)	-	
☐	5	☒	NAT_5	Source NAT	WAN1	Any	-	LAN1	

[Save](#) [Cancel](#)

2. Do one of the following.

- Click [\[Create\]](#) to create a NAT rule. The [\[Create NAT Rule\]](#) window will appear.
- Click a rule name to edit settings. The [\[Edit NAT Rule\]](#) window will appear.

Create NAT Rule

×

Status ☒

NAT Type 1-to-1 NAT

Rule Name*

Description

Source and Destination Selection

Incoming Interface WAN1

Original IP*

Mapped IP*

NAT Loopback ☐

[Save](#) [Cancel](#)

3. Configure the following settings.

Item	Description
Status	Enable or disable the rule by clicking the toggle.
NAT Type	Select 1-to-1 NAT , Multiple 1-to-1 NAT (CIDR) , Multiple 1-to-1 NAT (IP Range) , Port Forwarding , or Source NAT .
Rule Name	Type a rule name for this NAT rule.
Description	Type a description.

- Source and Destination Selection

Item	Description
Incoming Interface	Select an interface that will process the incoming traffic for this rule.
Protocol	Select a protocol of incoming traffic that this rule applies to.  This setting is applicable only when the NAT type is [Port Forwarding].
Original IP	• 1-to-1 NAT: Input the destination IP address that will be translated. When the device receives an incoming packet going through the interface specified in the [Incoming Interface], the destination IP address in the packet that matches this [Original IP] field will be mapped to another IP address. • Multiple 1-to-1 NAT (CIDR): Input a CIDR (Classless Inter-Domain Routing) format of the IP addresses that will be translated; for example, 172.1.1.0 / 24. When the device receives an incoming packet going through the interface specified in the [Incoming Interface], the destination IP address in the packet that matches this [Original IP] field will be mapped to another IP address. These IP addresses are usually the ones assigned by an ISP (Internet Service Provider). • Multiple 1-to-1 NAT (IP Range): Input an IP range format of the IP addresses that will be translated; for example, 172.1.1.1 – 172.1.1.254. When the device receives an incoming packet going through the interface specified in the [Incoming Interface], the destination IP address in the packet that matches this [Original IP] field will be mapped to another IP address. These IP addresses are usually the ones assigned by an ISP (Internet Service Provider). • Port Forwarding: Input a port range that will be translated. When the device receives an incoming packet going through the interface specified in the [Incoming Interface], the destination IP address in the packet that matches this [Original IP] field will be mapped to another IP address and port. • Source NAT: Select Any to forward packets from any incoming interface IP address to the selected outgoing interface, or select Custom to forward packets from the specified incoming interface IP address to the selected outgoing interface.
Mapped IP	• 1-to-1 NAT: Input the IP address you will map to. This IP address is usually a private IP address in your local network. • Multiple 1-to-1 NAT (CIDR): Input a CIDR (Classless Inter-Domain Routing) format of the IP addresses that will be mapped to; for example, 192.168.100.0 / 24. • Multiple 1-to-1 NAT (IP Range): Input an IP range format of the IP addresses that will be mapped to; for example, 192.168.100.1 – 192.168.100.254. • Port Forwarding: Input an IP address and port range that will be mapped to. This IP address is usually a private IP address in your local network.

Item	Description
Outgoing Interface	Select an interface that will forward the incoming traffic to the destination.
NAT Loopback	Enable or disable NAT loopback by clicking the toggle.

4. On the [Create NAT Rule] or [Edit NAT Rule] window, click [Save].

5. On the [NAT Rules] page, click [Save].

4.2.2 ALG Settings

An ALG or Application Layer Gateway allows the client applications to communicate with the server applications when the server ports are dynamically opened to client applications. These ports are usually dynamically assigned in the application protocol. An ALG understands application protocols, recognizes application specific commands, and helps open the ports dynamically on the device for communication. Without ALG, client applications like FTP would not be able to transfer files when the FTP client is in an NAT network.

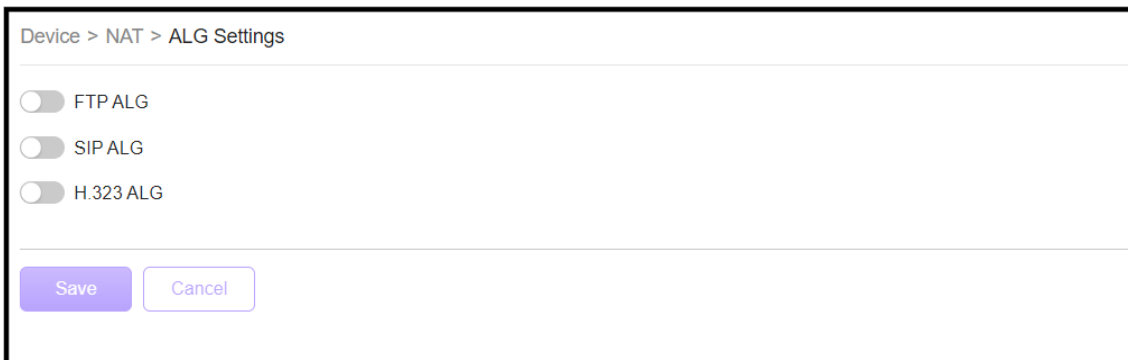
On the [ALG Settings] page, you can configure the following:

- FTP ALG
- SIP ALG
- H.323 ALG

4.2.2.1 Configuring ALG Settings

Procedure

1. Go to [Device] > [NAT] > [ALG Settings].



2. Enable or disable FTP ALG, SIP ALG and/or H.323 ALG by clicking the toggles.

3. Click [Save].

4.3 Routing (Only in Gateway Mode)

Create and manage device static routes.

- [Static Routes](#)

4.3.1 Static Routes

Static routes are generally used when no appropriate dynamic route is present, or when you want the traffic to follow the static route you specify as opposed to following the dynamic route that is automatically learned and generated by the device.

On the [Static Routes] page, you can view a list of current static routes on the device and configure their settings.

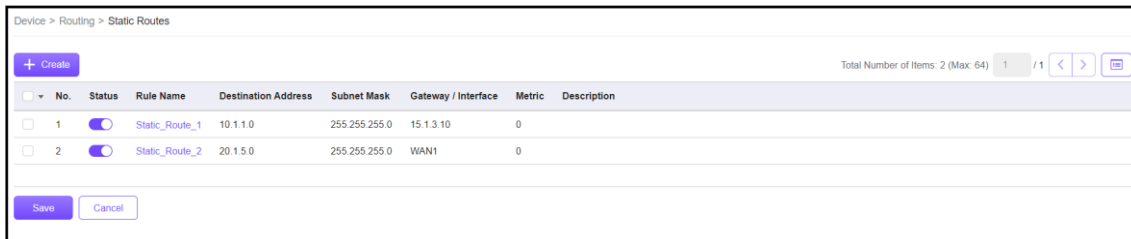
The following table describes the tasks you can perform on the [Static Routes] page.

Task	Description
To create a static route	Click the [Create] button to create a new static route.
To edit a static route	Click a static route name to edit the settings.
To delete a static route	Click the checkbox of a static route and click the [Delete] button.
To duplicate a static route	Click the checkbox of a static route and click the [Duplicate] button.

4.3.1.1 Configuring a Static Route

Procedure

1. Go to [Device] > [Routing] > [Static Routes].



Device > Routing > Static Routes

[+ Create](#) Total Number of Items: 2 (Max: 64) 1 / 1

☐	No.	Status	Rule Name	Destination Address	Subnet Mask	Gateway / Interface	Metric	Description
☐	1	☒	Static_Route_1	10.1.1.0	255.255.255.0	15.1.3.10	0	
☐	2	☒	Static_Route_2	20.1.5.0	255.255.255.0	WAN1	0	

[Save](#) [Cancel](#)

2. Do one of the following.

- Click [Create] to create a static route. The [Create Static Route] window will appear.
- Click a rule name to edit settings. The [Edit Static Route] window will appear.

Create Static Route

Status ☒

Rule Name*

i

Description

i

Static Route Settings

Destination Address*

Subnet Mask*

Next Hop Type

☒ Gateway*

☐ Network Interface

WAN1

Metric

0

i

OK

Cancel

3. Configure the following settings.

Item	Description
Status	Enable or disable the rule by clicking the toggle.
Rule Name	Type a rule name for this NAT rule.
Description	Type a description.

- Static Route Settings

Item	Description
Destination Address	Input a destination IP address.
Subnet Mask	Input a subnet mask.  If the destination is a single IP address, then input 255.255.255.255 in the [Subnet Mask] field. If the destination is a subnet of IP addresses, then input, for example, 255.255.255.0, in the [Subnet Mask] to present the destination IP address range.

- Next Hop Type

Item	Description
Gateway/ Network Interface	Select either Gateway or Network Interface. • Gateway: Select [Gateway] and input an IP address if the next hop is a gateway. The gateway needs to be on the same network as the interface. • Network Interface: Select [Network Interface] and choose a network interface from the drop-down menu if the next hop is a network interface on the device.
Metric	Input a metric value for this static route. The device determines which static route to use based on the metric value. The lower number represents higher priority.

4. On the [Create Static Route] or [Edit Static Route] window, click [OK].

5. On the [Static Route] page, click [Save].

5 VPN (Only in Gateway Mode)

This chapter describes how to configure VPN-related settings under the [VPN] tab.

5.1 VPN

Create and manage VPN connections for secure communication.

- [IPSec VPN](#)
- [L2TP/IPSec VPN](#)
- [VPN Management](#)

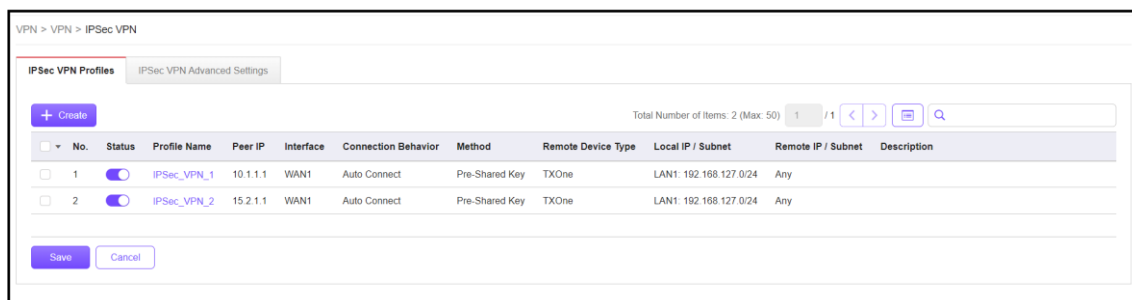
5.1.1 IPSec VPN

IPSec (Internet Protocol Security) VPN is a type of virtual private network that uses the IPSec protocol to secure and encrypt communications between two devices for establishing secure, site-to-site or remote access VPN connections. With IPSec VPN, data is encrypted before it is transmitted. This ensures that any data exchanged between the two VPN gateways remains private and secure from unauthorized access or interception.

5.1.1.1 Configuring IPSec VPN Profiles

Procedure

1. Go to [VPN] > [VPN] > [IPSec VPN].



VPN > VPN > IPSec VPN

IPSec VPN Profiles | IPSec VPN Advanced Settings

+ Create

Total Number of Items: 2 (Max: 50) 1 / 1

No.	Status	Profile Name	Peer IP	Interface	Connection Behavior	Method	Remote Device Type	Local IP / Subnet	Remote IP / Subnet	Description
1	☒	IPSec_VPN_1	10.1.1.1	WAN1	Auto Connect	Pre-Shared Key	TXOne	LAN1: 192.168.127.0/24	Any	
2	☒	IPSec_VPN_2	15.2.1.1	WAN1	Auto Connect	Pre-Shared Key	TXOne	LAN1: 192.168.127.0/24	Any	

Save Cancel

2. Under the [IPSec VPN Profiles] tab, do one of the following.

- Click [Create] to create a VPN profile. The [Create IPSec VPN Profile] window will appear.
- Click a profile name to edit settings. The [Edit IPSec VPN Profile] window will appear.

Create IPsec VPN

Status

Profile Name*

Description

Network Settings

Peer IP*

Interface

WAN1

Connection Behavior*

Auto Connect

30

(Retry Interval: 10-3600 seconds)

Authentication Settings

Method*

Pre-Shared Key

Remote Device Type

Remote Device Type

TXOne

View and Edit Additional Settings

Local Interface*

LAN1 (192.168.127.0/24)

Remote IP / Subnet*

Any

OK

Cancel

3. Configure the following settings.

Item	Description
Status	Enable or disable the IPsec VPN profile by clicking the toggle.
Name	Type a name for this IPsec VPN profile.
Description	Type a description.

- Network Settings

Item	Description
Peer IP	Input the IP address of the remote device that you connect to.
Interface	This is set to the WAN1 interface.
Connection Behavior	Select a connection behavior, Auto Connect or On-Demand .

Item	Description
	- Auto Connect: Select this option for the local device (the EdgeFire device you are managing) to automatically connect to the remote device. If the connection behavior is Auto Connect, further specify the retry interval. - Retry Interval: Input a number between 10 and 3600 (seconds) (default: 30) to specify how often the local device should retry the connection. - On-Demand: Select this option for the local device to establish a connection when it detects communications traffic. If the connection behavior is On-Demand, further specify the idle timeout. - Idle Timeout: Input a number between 10 and 3600 (seconds) (default: 600) to specify after how long the connection will be terminated if no communications traffic is detected.

- Authentication Settings

Item	Description
Method	This is set to Pre-Shared Key. Input a secret key shared between the local and remote devices.

- Remote Device Type

Item	Description
Remote Device Type	Select a brand of the remote device, or Custom .

4. (Optional) Enable the [View and Edit Additional Settings] option by clicking the toggle and modify the following settings if the remote device does not use the default VPN configuration of the specified brand and you need to modify the configuration.

☒ View and Edit Additional Settings

IKE
Version 2
Peer Option

Phase1

Proposal 1	Encryption	AES128	Authentication / PRF	SHA256	Diffie-Hellman Groups	14	-
Proposal 2	Encryption	AES256	Authentication / PRF	SHA256	Diffie-Hellman Groups	14	- +

Key Lifetime* 86400 120-172800 Seconds

Phase2

Local Interface* LAN1 (192.168.127.0/24) +

Remote IP / Subnet* Any +

Proposal 1	Encryption	AES128	Authentication	SHA1	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 14	-
Proposal 2	Encryption	AES256	Authentication	SHA1	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 14	-
Proposal 3	Encryption	AES128	Authentication	SHA256	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 14	-

Key Lifetime* 86400 120-172800 Seconds



After you select a brand of the remote device from the [Remote Device Type] drop-down list, the default VPN configuration of the corresponding type will be automatically set in this section. If the remote device does not use the default VPN configuration and you need to modify the configuration here, the [Remote Device Type] will then be automatically adjusted to **Custom**.

Item	Description
IKE	Select a version, either Version 1 or Version 2 , for the IKE protocol.
Peer Option	Click the [Peer Option] button and select either Any or Specific ID for the [Local Peer ID] and [Remote Peer ID]. Peer Option Local Peer ID* Any Remote Peer ID* Any OK Cancel Ensure that the local peer ID and the remote peer ID on both the local and remote devices match. For example, if the local device's remote peer ID is set to Any, then the remote

Item	Description
	device's local peer ID should also be set to Any, as the local device's remote peer is equivalent to the remote device's local peer.
Negotiation	If the IKE is Version 1 , then select either Main Mode or Aggressive Mode for the negotiation mode between the local and remote devices.

- Phase 1: Click the add button :ODC_add_tab_button: to add another proposal, or the remove button :remove_entry_button: to remove a proposal.

Item	Description
Encryption	Select an encryption algorithm, or Mixed to include all the algorithms on the drop-down list.  The option "Mixed" is available only when the IKE is set to Version 2.
Authentication/PRF	Select an authentication algorithm from the drop-down list.
Diffie-Hellman Groups	Select a number, or Mixed to include all the numbers on the drop-down list.  The option "Mixed" is available only when the IKE is set to Version 2.
Key Lifetime	Input a number between 120 and 172800 (seconds) to specify the rekey interval.



Ensure that the remote device has a corresponding proposal with the same encryption algorithm, authentication algorithm, and Diffie-Hellman group number in phase 1.

- Phase 2: Click the add button :ODC_add_tab_button: to add another proposal, or the remove button :remove_entry_button: to remove a proposal.

Item	Description
Local Interface	Select a network interface for the local device.
Remote IP/Subnet	Select Any , Single IP or IP Subnet for the remote device. You may click the add button  to add another entry, or the remove button  to remove an entry. If Single IP or IP Subnet is selected, further specify the IP address or the IP subnet. You may click the add button  to add another entry, or the remove button  to remove an entry.
Encryption	Select an encryption algorithm, or Mixed to include all the algorithms on the drop-down list.  The option "Mixed" is available only when the IKE is set to Version 2.

Item	Description
Authentication	Select an authentication algorithm, or Mixed to include all the algorithms on the drop-down list.  The option "Mixed" is available only when the IKE is set to Version 2.
Perfect Forward Secrecy (PFS)	Select a Diffie-Hellman group, or Mixed to include all the groups on the drop-down list, or NULL to select none.  The option "Mixed" is available only when the IKE is set to Version 2.
Key Lifetime	Input a number between 120 and 172800 (seconds) to specify the rekey interval.

 Ensure that the remote device has a corresponding proposal with the same encryption algorithm, authentication algorithm, and PFS setting in phase 2.

5. If the [View and Edit Additional Settings] option is disabled, configure the following settings.

☐ View and Edit Additional Settings

Local Interface*

LAN1 (192.168.127.0/24)
+

Remote IP / Subnet*

Any
+

Item	Description
Local Interface	Select a network interface for the local device.
Remote IP/Subnet	Select Any, Single IP or IP Subnet for the remote device. You may click the add button  to add another entry, or the remove button  to remove an entry. If Single IP or IP Subnet is selected, further specify the IP address or the IP subnet. You may click the add button  to add another entry, or the remove button  to remove an entry.

6. Click [OK] on the [Create IPsec VPN Profile] or [Edit IPsec VPN Profile] window.

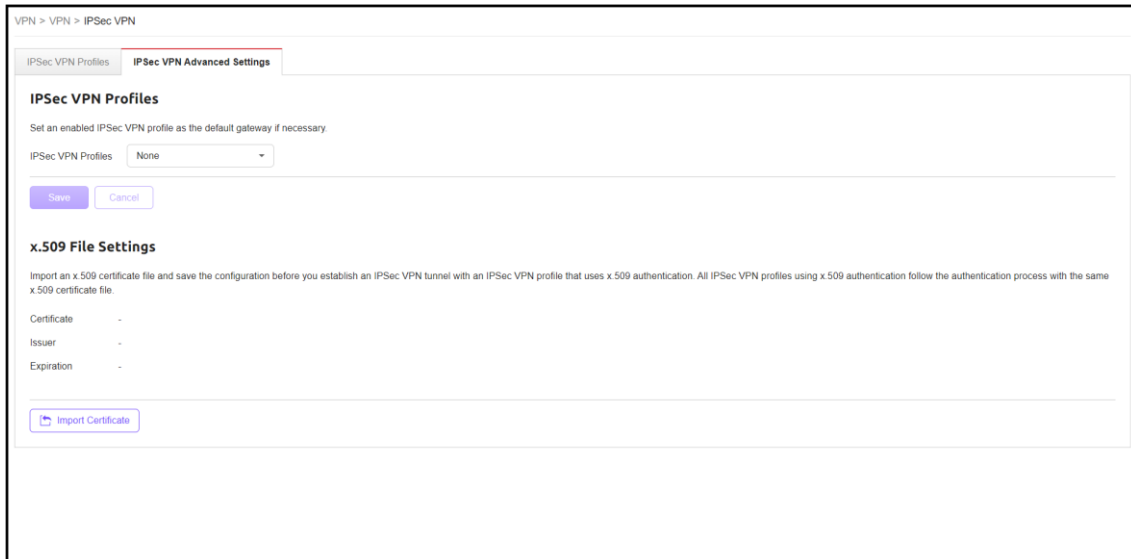
7. Click [Save] on the [IPsec VPN] page.

 You cannot create or enable a VPN profile while HA function is enabled. Similarly, you cannot enable HA function while one of the VPN profiles is enabled.

5.1.1.2 Setting Default Gateway for IPsec VPN Tunnel

Procedure

- Go to [VPN] > [VPN] > [IPsec VPN].



VPN > VPN > IPsec VPN

IPSec VPN Profiles **IPSec VPN Advanced Settings**

IPSec VPN Profiles

Set an enabled IPsec VPN profile as the default gateway if necessary.

IPSec VPN Profiles None

Save Cancel

x.509 File Settings

Import an x.509 certificate file and save the configuration before you establish an IPsec VPN tunnel with an IPsec VPN profile that uses x.509 authentication. All IPsec VPN profiles using x.509 authentication follow the authentication process with the same x.509 certificate file.

Certificate -

Issuer -

Expiration -

Import Certificate

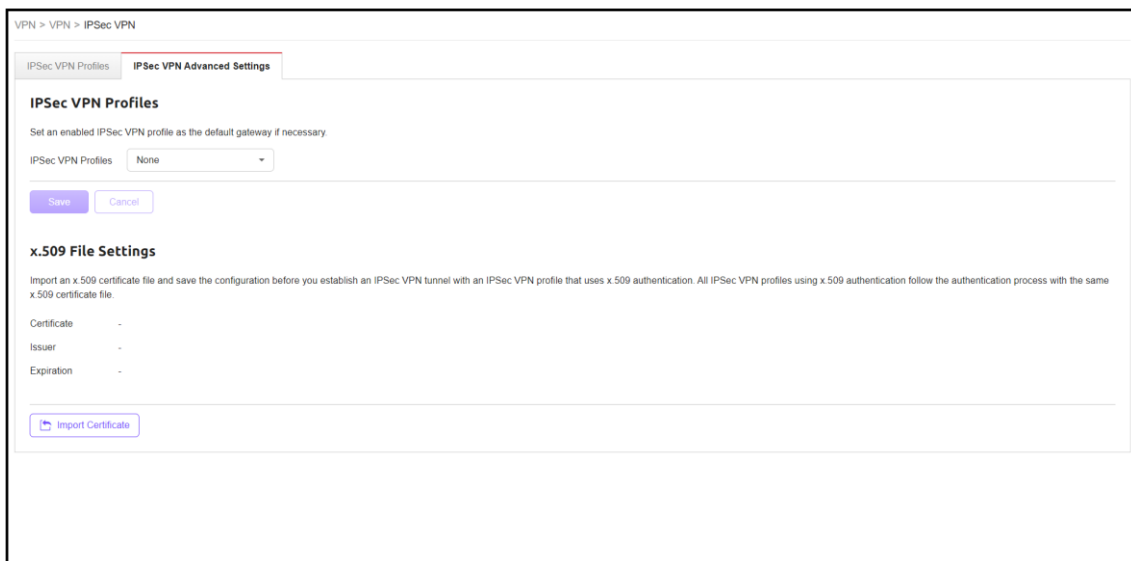
2. Under the [IPSec VPN Advanced Settings] tab, select an enabled IPsec VPN profile as the VPN tunnel default gateway of EdgeFire.

3. Click [Save].

5.1.1.3 Importing x.509 Certificate File

Procedure

1. Go to [VPN] > [VPN] > [IPsec VPN].



VPN > VPN > IPsec VPN

IPSec VPN Profiles **IPSec VPN Advanced Settings**

IPSec VPN Profiles

Set an enabled IPsec VPN profile as the default gateway if necessary.

IPSec VPN Profiles None

Save Cancel

x.509 File Settings

Import an x.509 certificate file and save the configuration before you establish an IPsec VPN tunnel with an IPsec VPN profile that uses x.509 authentication. All IPsec VPN profiles using x.509 authentication follow the authentication process with the same x.509 certificate file.

Certificate -

Issuer -

Expiration -

Import Certificate

2. Under the [IPSec VPN Advanced Settings] tab, click [Import Certificate]. The [Import Certificate] window will appear.

Import Certificate



After importing a certificate, you must save the configuration on the IPsec VPN page before you enable an IPsec VPN profile for it to take effect.

Certificate*

Select File

Private Key*

Select File

Passphrase

Import

Cancel

3. Import your certificate file by clicking the [Select File] button next to the [Certificate] field.
4. Import the private key (PEM-encoded PKCS#1 format) for the certificate file by clicking the [Select File] button next to the [Private Key] field.
5. Input the passphrase of the private key if the private key is encrypted.
6. Click [Import].

5.1.2 L2TP/IPSec VPN

L2TP/IPSec VPN is a type of virtual private network that combines the Layer 2 Tunneling Protocol (L2TP) with the Internet Protocol Security (IPSec) protocol to provide secure and encrypted connections between the VPN gateways and the VPN clients.

5.1.2.1 Configuring L2TP/IPSec VPN Profiles

Procedure

1. Go to [VPN] > [VPN] > [L2TP/IPSec VPN].

VPN > VPN > L2TP / IPsec VPN

L2TP/IPSec VPN Profiles

L2TP/IPSec VPN Settings

Create

Total Number of Items: 2 (Max: 50) 1 / 1

No.	Status	Profile Name	Assigned IP Address	Username	Description
1		L2TP_IPSec_VPN_1	192.168.127.1	test	
2		L2TP_IPSec_VPN_2	192.168.254.1	123	

Save

Cancel

2. Under the [L2TP/IPSec VPN Profiles] tab, do one of the following.
 - Click [Create] to create a VPN profile. The [Create L2TP/IPSec VPN Profile] window will appear.
 - Click a profile name to edit settings. The [Edit L2TP/IPSec VPN Profile] window will appear.

Create L2TP/IPSec VPN Profile

Status

☒

Profile Name*

Description

Assigned IP Settings

Assigned IP Address*

Interface

LAN1

User Account Settings

Username*

Password*

OK

Cancel

3. Configure the following settings.

Item	Description
Status	Enable or disable the L2TP/IPSec VPN profile by clicking the toggle.
Name	Type a name for this L2TP/IPSec VPN profile.
Description	Type a description.

- Assigned IP Settings

Item	Description
Assigned IP Address	Input an IP address assigned to the remote device.
Interface	Select a network interface for the IP subnet of the remote device.

- User Account Settings

Item	Description
Username	Input the username for authenticating the connection.
Password	Input the password for authenticating the connection.

4. Click [OK] on the [Create L2TP/IPSec VPN Profile] or [Edit L2TP/IPSec VPN Profile] window.

5. Click [Save] on the [L2TP/IPSec VPN] page.



You cannot create or enable a VPN profile while HA function is enabled. Similarly, you cannot enable HA function while one of the VPN profiles is enabled.

5.1.2.2 Configuring L2TP/IPSec VPN Settings

Procedure

1. Go to [VPN] > [VPN] > [L2TP/IPSec VPN].

VPN > VPN > L2TP / IPSec VPN

L2TP/IPSec VPN Profiles

L2TP/IPSec VPN Settings

☐ L2TP/IPSec VPN

L2TP General Settings

Authentication Protocol

MS-CHAPv2

User Authentication Type

Local Authentication

DNS Server

IPSec General Settings

Interface

WAN1

Connection Idle Timeout*

30

(Range: 10-3600 seconds, Default: 30)

Authentication Settings

Method*

Pre-Shared Key

Dial-in Client Type

Dial-in Client Type

Windows Native

☐ View and Edit Additional Settings

Save

Cancel

2. Under the [L2TP/IPSec VPN Settings] tab, enable [L2TP/IPSec VPN] by clicking the toggle.

3. Configuring the following settings.

- L2TP General Settings

Item	Description
Authentication Protocol	Select an authentication protocol, MS-CHAPv2 , MS-CHAPv1 , CHAP , PAP , or Auto to include all the protocols on the drop-down list.
User Authentication Type	This is set to Local Authentication.
DNS Server	Input a DNS server IP address of the interface.

- IPsec General Settings

Item	Description
Interface	This is set to the WAN1 interface.
Connection Idle Timeout	Input a number between 10 and 3600 (seconds) (default: 30) to specify after how long the connection will be terminated if no communications traffic is detected.

- Authentication Settings

Item	Description
Method	This is set to Pre-Shared Key. Input a secret key shared between the local and remote devices.

- Dial-in Client Type

Item	Description
Dial-in Client Type	Select a type of the remote device, or Mixed to include macOS Native and Windows Native, or Custom .

4. (Optional) Enable the [View and Edit Additional Settings] option by clicking the toggle and modify the following settings if the VPN client device does not use the default VPN configuration of the specified brand and you need to modify the configuration.

☒ View and Edit Additional Settings

IKE
Version 1
Peer Option

Negotiation
Main Mode

Phase1

Proposal 1
Encryption
AES256
Authentication / PRF
SHA1
Diffie-Hellman Groups
20

Proposal 2
Encryption
AES128
Authentication / PRF
SHA1
Diffie-Hellman Groups
19

Proposal 3
Encryption
AES256
Authentication / PRF
SHA1
Diffie-Hellman Groups
14

Key Lifetime*
28800
120-172800 Seconds

Phase2

Proposal 1
Encryption
AES256
Authentication
SHA1
Perfect Forward Secrecy (PFS)
Null

Proposal 2
Encryption
AES128
Authentication
SHA1
Perfect Forward Secrecy (PFS)
Null

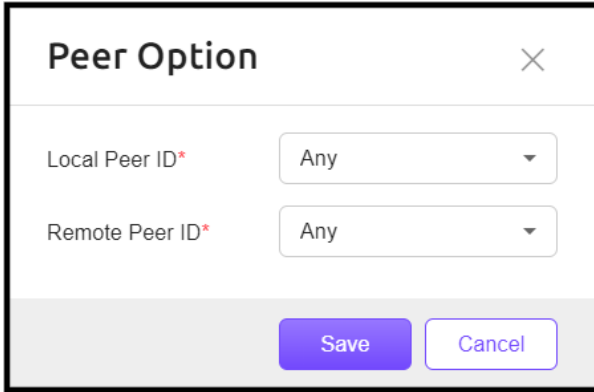
Proposal 3
Encryption
3DES
Authentication
SHA1
Perfect Forward Secrecy (PFS)
Null

Key Lifetime*
3600
120-172800 Seconds



After you select a type of the remote device from the [Dial-in Client Type] drop-down list, the default VPN configuration of the corresponding type will be automatically set in this

section. If the remote device does not use the default VPN configuration and you need to modify the configuration here, the [Dial-in Client Type] will then be automatically adjusted to **Custom**.

Item	Description
IKE	Select a version, either Version 1 or Version 2 , for the IKE protocol.
Peer Option	Click the [Peer Option] button and select either Any or Specific ID for the [Local Peer ID] and [Remote Peer ID].   Ensure that the local peer ID and the remote peer ID on both the local and remote devices match. For example, if the local device's remote peer ID is set to Any, then the remote device's local peer ID should also be set to Any, as the local device's remote peer is equivalent to the remote device's local peer.
Negotiation	If the IKE is Version 1 , then select either Main Mode or Aggressive Mode for the negotiation mode between the local and remote devices.

- Phase 1: Click the add button  to add another proposal, or the remove button  to remove a proposal.

Item	Description
Encryption	Select an encryption algorithm, or Mixed to include all the algorithms on the drop-down list.  The option "Mixed" is available only when the IKE is set to Version 2.
Authentication/PRF	Select an authentication algorithm from the drop-down list.
Diffie-Hellman Groups	Select a number, or Mixed to include all the numbers on the drop-down list.  The option "Mixed" is available only when the IKE is set to Version 2.
Key Lifetime	Input a number between 120 and 172800 (seconds) to specify the rekey interval.



Ensure that the remote device has a corresponding proposal with the same encryption algorithm, authentication algorithm, and Diffie-Hellman group number in phase 1.

- Phase 2: Click the add button  to add another proposal, or the remove button  to remove a proposal.

Item	Description
Encryption	Select an encryption algorithm, or Mixed to include all the algorithms on the drop-down list.  The option “Mixed” is available only when the IKE is set to Version 2.
Authentication	Select an authentication algorithm, or Mixed to include all the algorithms on the drop-down list.  The option “Mixed” is available only when the IKE is set to Version 2.
Perfect Forward Secrecy (PFS)	Select a Diffie-Hellman group, or Mixed to include all the groups on the drop-down list, or NULL to select none.  The option “Mixed” is available only when the IKE is set to Version 2.
Key Lifetime	Input a number between 120 and 172800 (seconds) to specify the rekey interval.

5. Click [Save].

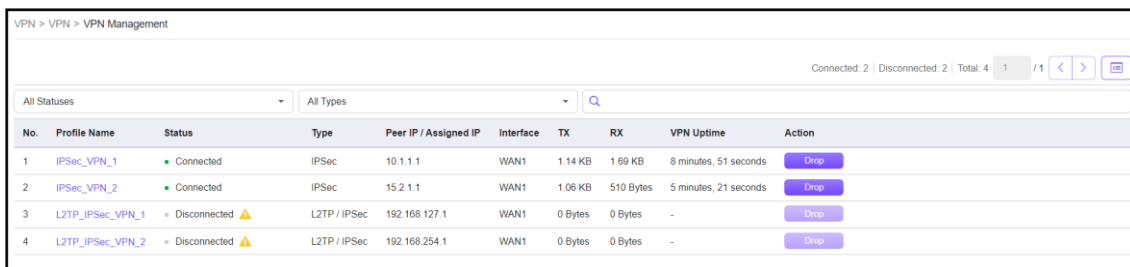
5.1.3 VPN Management

On the [VPN Management] page, you may view the connection information of VPN profiles created and enabled on the [IPSec VPN] and [L2TP/IPSec VPN] pages and activate/terminate VPN connection.

5.1.3.1 Viewing VPN Profile Information

Procedure

1. Go to [VPN] > [VPN] > [VPN Management].



No.	Profile Name	Status	Type	Peer IP / Assigned IP	Interface	TX	RX	VPN Uptime	Action
1	IPSec_VPN_1	Connected	IPSec	10.1.1.1	WAN1	1.14 KB	1.09 KB	8 minutes, 51 seconds	Drop
2	IPSec_VPN_2	Connected	IPSec	15.2.1.1	WAN1	1.06 KB	510 Bytes	5 minutes, 21 seconds	Drop
3	L2TP_IPSec_VPN_1	Disconnected	L2TP / IPSec	192.168.127.1	WAN1	0 Bytes	0 Bytes	-	Drop
4	L2TP_IPSec_VPN_2	Disconnected	L2TP / IPSec	192.168.254.1	WAN1	0 Bytes	0 Bytes	-	Drop

The following table describes the VPN Profile Information.

Field	Description
Profile Name	The name of the VPN profile.
Status	The status of the VPN connection.

Field	Description
Type	The type of the VPN profile, either IPSec or L2TP/IPSec.
Peer IP / Assigned IP	The IP address of the remote device.
Interface	The network interface of the VPN connection.
TX	The amount of traffic transmitted via the VPN connection.
RX	The amount of traffic received via the VPN connection.
VPN Uptime	The duration of the VPN connection up to the present time.

5.1.3.2 Establishing VPN Connection

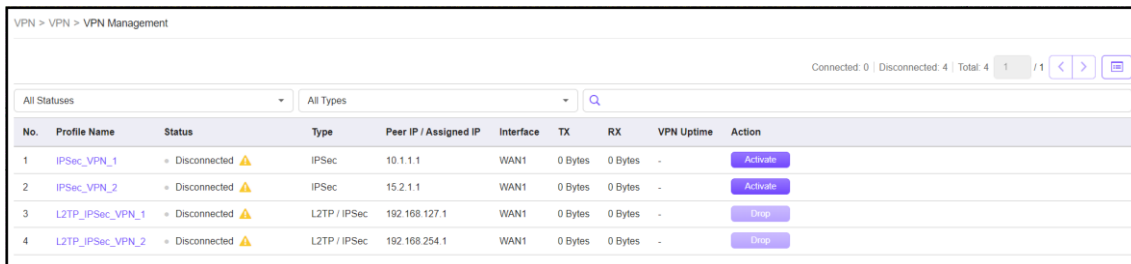
Procedure

1. Configure the required VPN profiles on both local and remote devices.



For more information, see [Configuring IPSec VPN Profiles](#) and [Configuring L2TP/IPSec VPN Profiles](#).

2. Go to [VPN] > [VPN] > [VPN Management] on both local and remote EdgeFire consoles.



No.	Profile Name	Status	Type	Peer IP / Assigned IP	Interface	TX	RX	VPN Uptime	Action
1	IPSec_VPN_1	Disconnected	IPSec	10.1.1.1	WAN1	0 Bytes	0 Bytes	-	Activate
2	IPSec_VPN_2	Disconnected	IPSec	15.2.1.1	WAN1	0 Bytes	0 Bytes	-	Activate
3	L2TP_IPSec_VPN_1	Disconnected	L2TP / IPSec	192.168.127.1	WAN1	0 Bytes	0 Bytes	-	Drop
4	L2TP_IPSec_VPN_2	Disconnected	L2TP / IPSec	192.168.254.1	WAN1	0 Bytes	0 Bytes	-	Drop



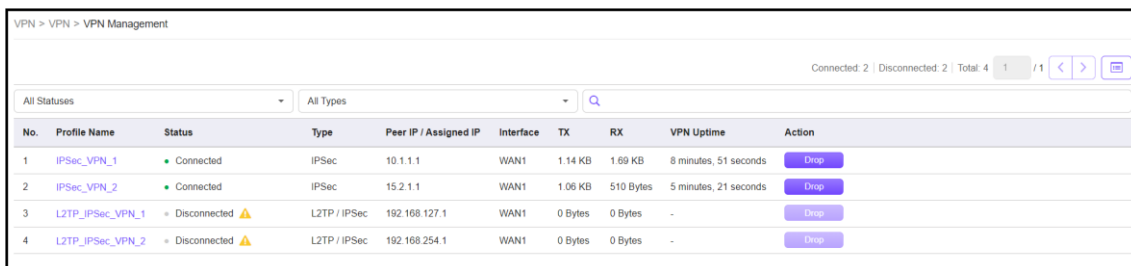
For devices from other brands, you need to initiate the VPN connection in a similar manner.

3. Click the [Activate] button on the rightmost column of the table on both local and remote EdgeFire consoles to establish the VPN connection between the two devices.

5.1.3.3 Terminating VPN Connection

Procedure

1. Go to [VPN] > [VPN] > [VPN Management].



No.	Profile Name	Status	Type	Peer IP / Assigned IP	Interface	TX	RX	VPN Uptime	Action
1	IPSec_VPN_1	Connected	IPSec	10.1.1.1	WAN1	1.14 KB	1.69 KB	8 minutes, 51 seconds	Drop
2	IPSec_VPN_2	Connected	IPSec	15.2.1.1	WAN1	1.06 KB	510 Bytes	5 minutes, 21 seconds	Drop
3	L2TP_IPSec_VPN_1	Disconnected	L2TP / IPSec	192.168.127.1	WAN1	0 Bytes	0 Bytes	-	Drop
4	L2TP_IPSec_VPN_2	Disconnected	L2TP / IPSec	192.168.254.1	WAN1	0 Bytes	0 Bytes	-	Drop

2. Click the [Drop] button on the rightmost column of the table to terminate the VPN connection between the two devices.

6 Objects & Profiles

This chapter describes how to configure different types of objects and profiles to simplify policy management by storing configurations that can be used by the Edge series device under the [Objects & Profiles] tab.

Each type of object and profile (except for MAC objects) has the following types of sub-profile lists for this device:

- **Device Profiles:** Contains objects and profiles created and configured by users with Edge device local account privileges to be applied to policy rules.
- **EdgeOne Profiles:** Contains objects and profiles synced from EdgeOne. Users with EdgeOne local account privileges can apply them to policy rules. Users with only EdgeIPS local account privileges can only view and copy objects/profiles from the [EdgeOne Profiles] list to the [Device Profiles] list.



MAC objects can only be configured on the Edge device management console.

You can perform the following tasks when you view a profile list.

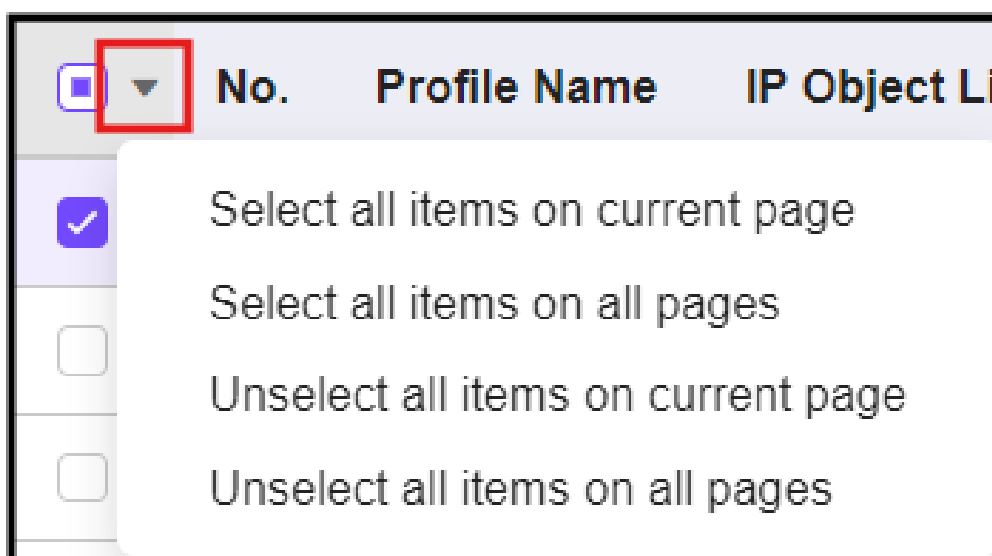
- **Device Profiles:**

Task	Description
Create object/profile	Click [Create] to create a new object/profile.
Edit object/profile	Click a(n) object/profile name to edit the settings.
Delete object/profile	Select one or multiple objects/profiles and click [Delete] to delete them.
Duplicate object/profile	Select a(n) object/profile and click [duplicate] to duplicate it.

- **EdgeOne Profiles:**

Task	Description
Copy object/profile to [Device Profiles]	Select one or multiple objects/profiles and click [Copy EdgeOne Profiles to Device Profiles] to copy the objects/profiles on the [EdgeOne Profiles] list and paste them to the [Device Profiles] list.

On the table header row, click the down arrow icon next to the checkbox, and a drop-down menu for selecting objects/profiles will be shown.



6.1 IP/MAC (Only in Bridge Mode)/Service Objects

Create and manage groups of MAC/IP addresses and service objects shared among multiple Policy Enforcement rules.

- [IP Objects](#)
- [MAC Objects \(Only in Bridge Mode\)](#)
- [Service Objects](#)

6.1.1 IP Objects

IP objects contain IP addresses which you can apply to policy rules. The types of IP addresses you can assign are:

- **Single IP address**

For example: 192.168.1.1

- **IP range**

For example: from 192.168.1.1 to 192.168.1.20

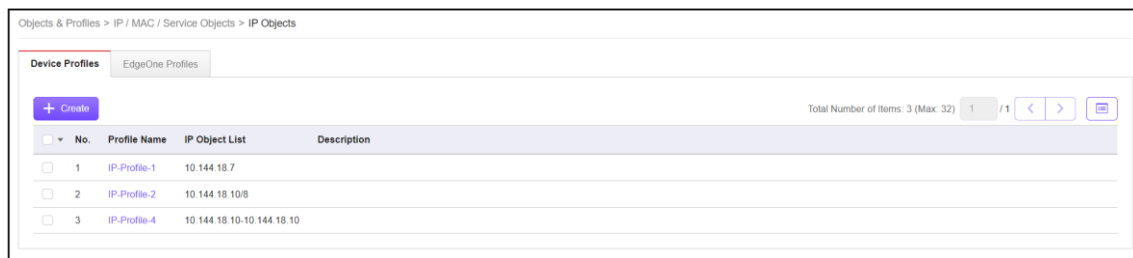
- **IP subnet**

For example: 192.168.1.0/24

6.1.1.1 Configuring IP Objects

Procedure

1. Go to [Objects & Profiles] > [IP/MAC/Service Objects] > [IP Objects].



No.	Profile Name	IP Object List	Description
1	IP-Profile-1	10.144.18.7	
2	IP-Profile-2	10.144.18.10/8	
3	IP-Profile-4	10.144.18.10-10.144.18.10	

2. Do one of the following.

- Click [Create] to create an object. The [Create IP Object] window will appear.
- Click a profile name to edit settings. The [Edit IP Object] window will appear.

Create IP Object

Profile Name*

Description

IP Object List

(Max: 32)

No. 1*

Single IP

Memo

Save

Cancel

3. Configure the following IP object settings.

Item	Description
Profile Name	Type a profile name for this IP object.
Description	(Optional) Type a description.
IP Object List	Specify an IP address, an IP range, or an IP subnet. If needed, fill out the [Memo] field. Click the [Add Entry] button  to add another entry, or the [Remove Entry] button  to remove an entry.

4. Click [Save].

6.1.2 MAC Objects (Only in Bridge Mode)

MAC objects contain MAC addresses which you can apply to policy rules.

6.1.2.1 Configuring MAC Objects

Procedure

1. Go to [Objects & Profiles] > [IP/MAC/Service Objects] > [MAC Objects].

Objects & Profiles > IP / MAC / Service Objects > MAC Objects

+ Create

Total Number of Items: 2 (Max: 32) 1 / 1

No.	Profile Name	MAC Object List	Description
1	MAC_Object_1	00 b0 d0 63 c2 26	
2	MAC_Object_2	a1 b4 c5 c1 dd 3e	

2. Do one of the following.

- Click [Create] to create an object. The [Create MAC Object] window will appear.
- Click a profile name to edit settings. The [Edit MAC Object] window will appear.

Create MAC Object

Profile Name*

Description

MAC Object List

(Max: 32)

No.1*

:

:

:

:

:

+

Save

Cancel

3. Configure the following MAC object settings.

Item	Description
Profile Name	Type a profile name for this MAC object.
Description	(Optional) Type a description.
MAC Object List	Specify a MAC address. Click the [Add Entry] button  to add another entry, or the [Remove Entry] button  to remove an entry.

4. Click [Save].

6.1.3 Service Objects

Service objects contain service definitions, including TCP port ranges, UDP port ranges, ICMP, and custom protocol numbers, which you can apply to policy rules.

- **TCP protocol port range**

For example: TCP ports and service ports ranging from 100 to 120

- **UDP protocol port range**

For example: UDP ports and service ports ranging from 100 to 120

- **ICMP protocol type and code**

For example: ICMP type 8 code 0

- Custom protocol with specified protocol number

For example: protocol number 6 and service ports ranging from 100 to 120

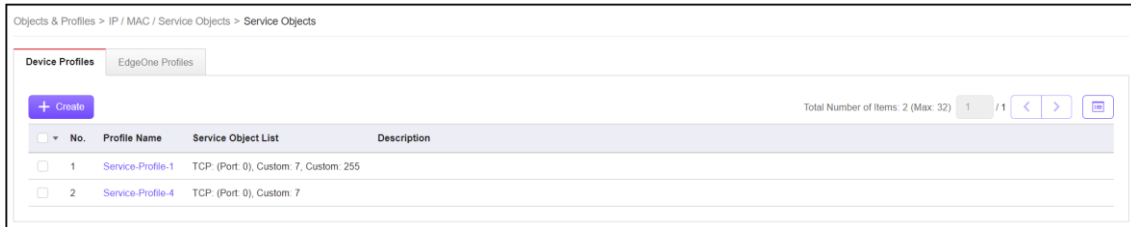


The term “protocol number” refers to the protocol number defined in the Internet protocol suite.

6.1.3.1 Configuring Service Objects

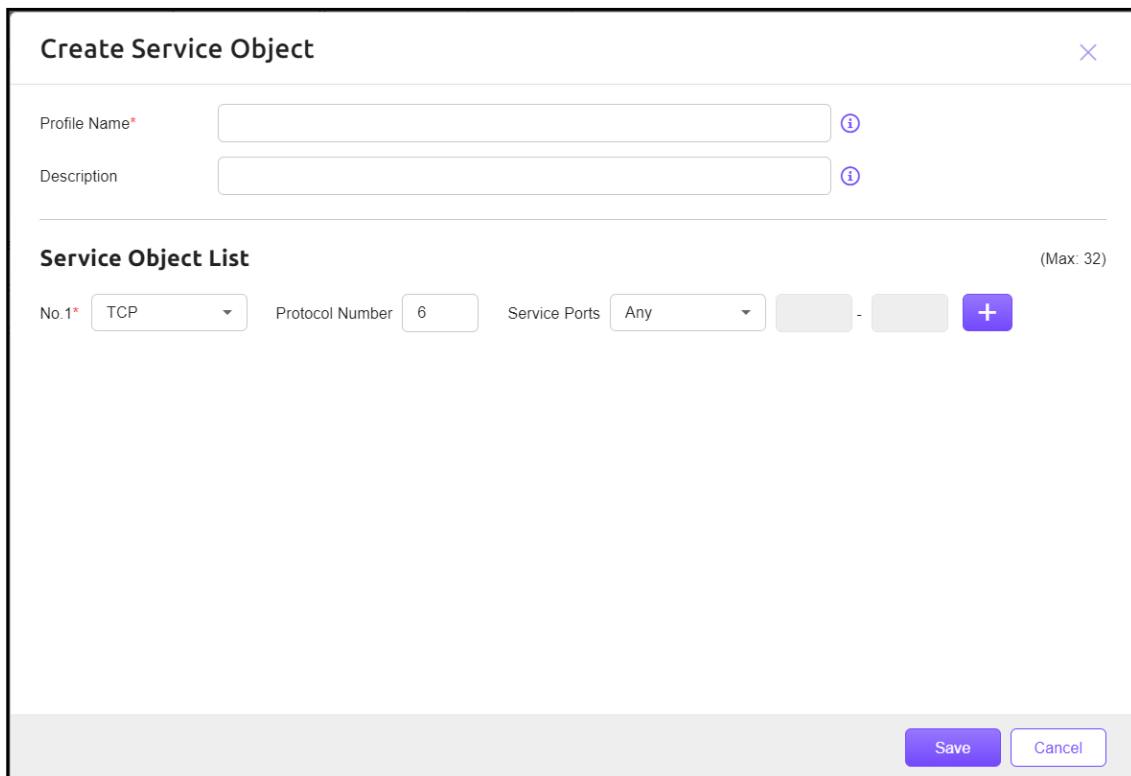
Procedure

1. Go to [Objects & Profiles] > [IP/MAC/Service Objects] > [Service Objects].



2. Do one of the following:

- Click [Create] to create an object. The [Create Service Object] window will appear.
- Click a profile name to edit settings. The [Edit Service Object] window will appear.



3. Configure the following service object settings.

Item	Description
Profile Name	Type a profile name for this service object.
Description	(Optional) Type a description.
Service Object List	Provide one of the following definitions. • TCP protocol number and its port range

Item	Description
	• UDP protocol number and its port range • ICMP protocol number and its type and code • Custom protocol with specified protocol number and port range Click the [Add Entry] button  to add another entry, or the [Remove Entry] button  to remove an entry.

4. Click [Save].

6.2 Security Profiles

Create and manage security profiles to control communications across your network.

- [Protocol Filter Profiles](#)
- [IPS Profiles](#)
- [File Filter Profiles](#)
- [Domain Name Profiles](#)

6.2.1 Protocol Filter Profiles

Protocol Filter profiles contain sophisticated and advanced ICS (Industrial Control System) protocol settings which you can apply to policy rules. The types of ICS protocols you can include in a Protocol Filter profile are:

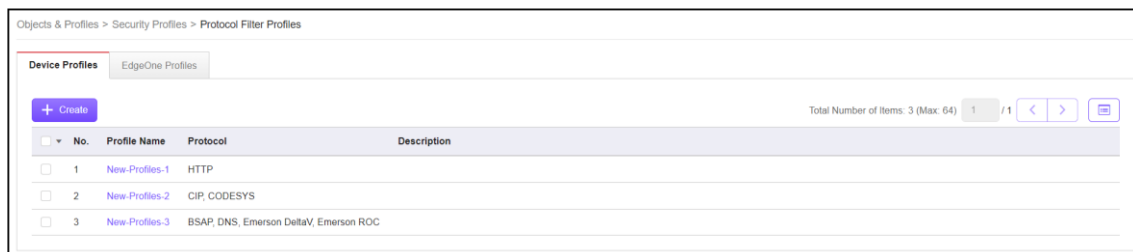
Factory Automation	• CIP • CODESYS • FINS • GE CMP • GE SRTP • MELSOFT • Modbus • OPC CLASSIC • OPC UA • PROFINET • S7COMM • S7COMM PLUS • SECS/GEM	• SLMP • TOYOPUC • TSAA • BSAP • CC-Link IE Field Basic • Emerson DeltaV • Emerson ROC • EtherCAT • FANUC FOCAS • FL-net • GE EGD (HTTP) • GE EGD (UDP)	• GE SDI • GE iFIX • HART-IP • ION • Panasonic MEWTOCOL • PTP • ROS SimpleMessage • TriStation • Valmet DNA
Building Automation	• BACnet • CN/IP	• KNX • LonWorks/LonTalk	• Niagara Fox

General OT Protocol	- CoAP - Ether-S-Bus - EtherSIO	- MDLC - Moxa Protocol - PCWorx	- RTPS/DDS - Wonderware SuiteLink
HealthCare	DICOM	HL7	
Power and Electricity	IEC 61850-GOOSE IEC 61850-R-GOOSE DNP3	ICCP TASE.2 IEC61850-MMS IEC-104 IEEE C37.118	OCPP 1.6J OCPP 2.0.1 - IEC 61850-R-SV - IEC 62056
Common	SMB - DNS - FTP - HTTP - HTTPS - MQTT	- MS RPC - PAgP - RDP - SIP/RTP - SMTP - SNMP	- SSH - TELNET - TFTP - TLS - VNC - Windows Update

6.2.1.1 Configuring Protocol Filter Profiles

Procedure

1. Go to [Objects & Profiles] > [Security Profiles] > [Protocol Filter Profiles].



2. Do one of the following.

- Click [Create] to create a profile. The [Create Protocol Filter Profile] window will appear.
- Click a profile name to edit settings. The [Edit Protocol Filter Profile] window will appear.

Create Protocol Filter Profile

Profile Name*

Description

Enabled: 0 | Disabled: 78 | Total: 78

1 / 4

< > [icon]

All Statuses

All Verticals

All Protocols

[Search]

☐	Status	Vertical	Protocol	Advanced Settings	Deny Malformed Packets
☐	☐	Building Automation	BACnet	Any	[edit]
☐	☐	Factory Automation	BSAP	-	
☐	☐	Factory Automation	CC-Link IE Field Basic	-	
☐	☐	Factory Automation	CIP	Any	[edit] ☐
☐	☐	Building Automation	CN/IP	-	
☐	☐	Factory Automation	CODESYS	Any	[edit]
☐	☐	General OT Protocol	CoAP	-	
☐	☐	HealthCare	DICOM	-	
☐	☐	Power and Electricity	DNP3	Any	[edit] ☐

Save

Cancel

3. Configure the following Protocol Filter Profile settings.

Item	Description
Profile Name	Type a profile name for this Protocol Filter Profile.
Description	(Optional) Type a description.
Status	Enable the operational statuses of one or multiple protocols you want to include in this Protocol Filter Profile by clicking the toggles. (Default status: Disabled)  If you enable Windows Update protocol, you need to enable DNS protocol as well in order for the Windows Update protocol to function properly.
Advanced Settings	Configure advanced settings of each configurable protocol by clicking the edit button .  For more information about configuring the advanced settings of each protocol, see: Configuring CIP Advanced Settings Configuring CODESYS Advanced Settings Configuring FINS and PROFINET Advanced Settings Configuring GE CMP Advanced Settings Configuring GE SRTMP Advanced Settings Configuring MELSOFT Advanced Settings Configuring Modbus Advanced Settings Configuring OPC CLASSIC Advanced Settings Configuring OPC UA Advanced Settings Configuring S7COMM Advanced Settings

Item	Description
	• Configuring S7COMM PLUS Advanced Settings • Configuring SECS/GEM Advanced Settings • Configuring SLMP Advanced Settings • Configuring TOYOPUC Advanced Settings • Configuring TSAA Advanced Settings • Configuring BACnet Advanced Settings • Configuring IEC 61850-GOOSE and IEC 61850-R-GOOSE Advanced Settings • Configuring DNP3 Advanced Settings • Configuring ICCP TASE.2 and IEC61850-MMS Advanced Settings • Configuring IEC-104 Advanced Settings • Configuring IEEE C37.118 Advanced Settings • Configuring OCPP 1.6J and OCPP 2.0.1 Advanced Settings • Configuring SMB Advanced Settings
Deny Malformed Packets	(Optional) Enable the [Deny Malformed Packets] function by clicking the toggles.  1. When the [Deny Malformed Packets] function is enabled, the Edge device will strictly check the packet format of the specified ICS protocol. If the packet format is incorrect, the device will drop the packets of the ICS protocol. 2. This feature is only applicable to the following nine protocols: IEC 61850-GOOSE, IEC 61850-R-GOOSE, CIP, FINS, Modbus, TOYOPUC, IEC-104, DNP3, and ION (firmware version v2.1 or later).

4. Click [Save].

6.2.1.2 Configuring CIP Advanced Settings

Through the [CIP Advanced Settings] window, you can further configure the command and function category access permission settings, including object class IDs and service codes based on which the protocol will work.

CIP Advanced Settings

Command and Function Category Access Permission

☒ Any
☐ Basic

☐ Read Only ☐ Read / Write ☐ Admin Config ☐ Others

☐ Advanced Matching Criteria

Object Class List: Any

Object Class ID*: Any

☐ Any Service Code
☒ Preset Service Codes*

Available: 28 Selected: 0

(0x01) Get_Attribute_All
(0x02) Set_Attribute_All
(0x03) Get_Attribute_List
(0x04) Set_Attribute_List
(0x05) Reset
(0x06) Start

>>
>
<
<<

☐ Custom Codes

Add

Total Number of Items: 0 (Max: 64) 1 / 1

No.	Object Class ID	Service Code	Command Code	Function Code
No data to display				

OK Cancel

After you access the [CIP Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands

- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].

2. If you select [Advanced Matching Criteria], follow the steps below to configure the object class IDs and service codes.

- a. From the [Object Class List] drop-down list, select an object class of this protocol. If you want to specify an object class by yourself, then select [Custom] and input an object class ID in the [Object Class ID] field.
- b. Select [Any Service Code], [Preset Service Codes] or [Custom Codes] based on your needs.

- **Any Service Code:** Select this option if you want any of the service codes within the function you specified to be applied.



The [Any Service Code] option is not available if the object class is [Any].

- **Preset Service Codes:** Select this option if you want to specify one or multiple service codes, then move the service codes from the [Available] field to the [Selected] field.

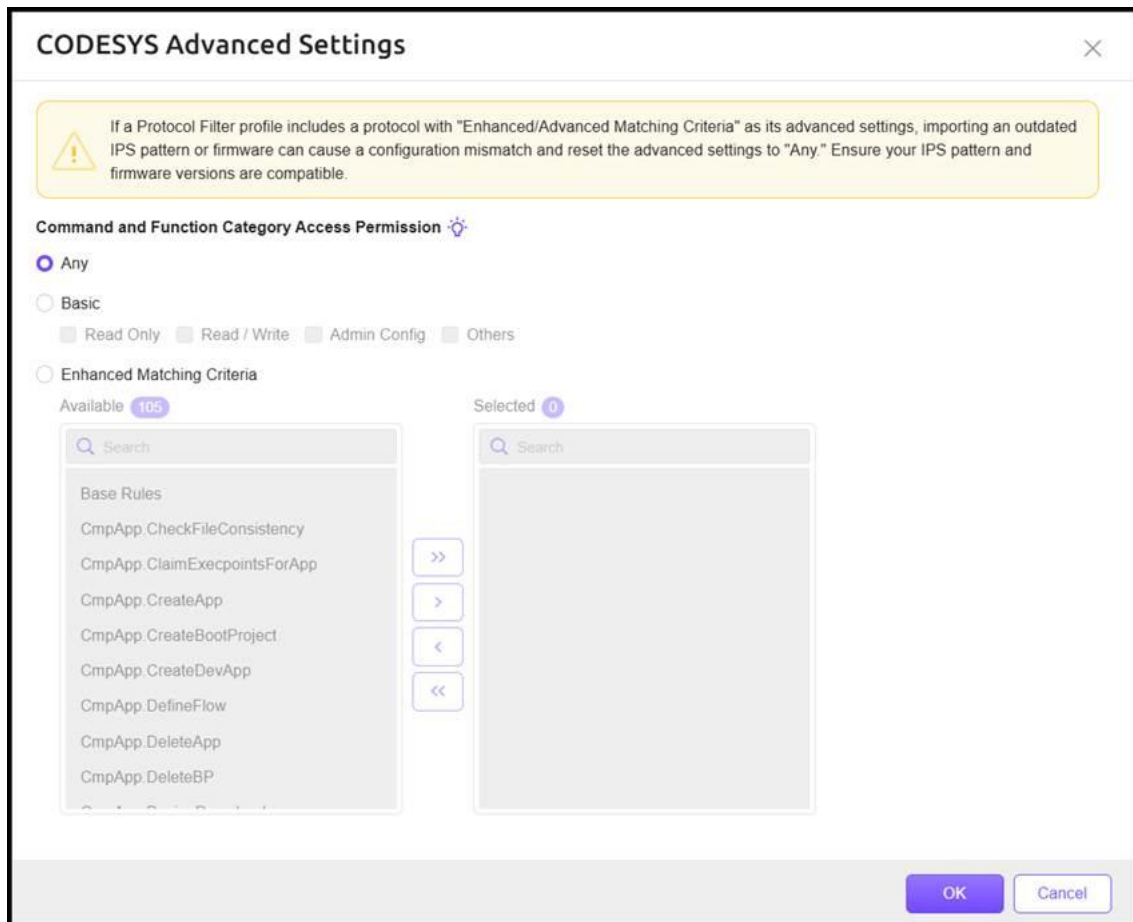
- **Custom Codes:** Select this option if you want to specify one or multiple service codes by yourself, then input service codes in the [Custom Codes] field.

- c. Click [Add].
- d. Repeat the above steps if you want to add more protocol definition entries.

3. Click [OK].

6.2.1.3 Configuring CODESYS Advanced Settings

Through the [CODESYS Advanced Settings] window, you can further configure the command and function category access permission settings, including the enhanced matching criteria based on which the protocol will work.



After you access the [CODESYS Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
- **Enhanced Matching Criteria:** Select this option to further configure enhanced matching criteria for the [Command and Function Category Access Permission].

2. If you select [Enhanced Matching Criteria], select one or multiple criteria in the [Available] field and move them to the [Selected] field.

3. Click [OK].

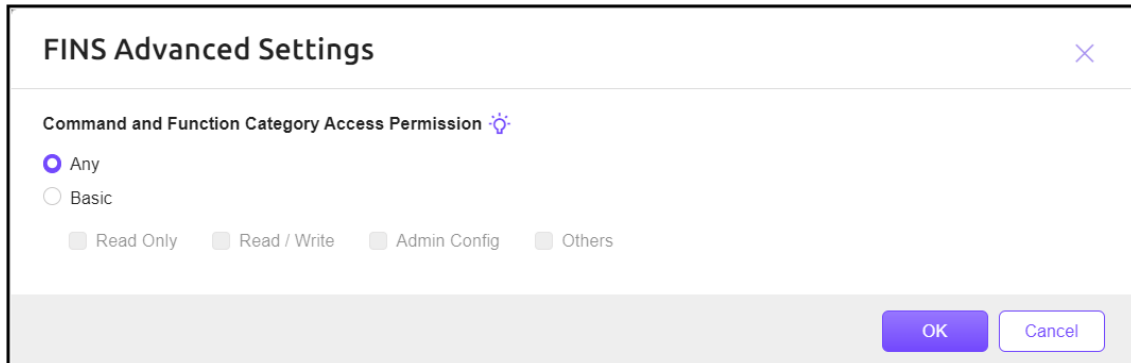


If a Protocol Filter profile includes a protocol with "Enhanced/Advanced Matching Criteria" as its advanced settings, importing an outdated IPS pattern or firmware can cause a configuration mismatch and reset the advanced settings to "Any." Ensure your IPS pattern and firmware versions are compatible.

6.2.1.4 Configuring FINS and PROFINET Advanced Settings

Through the [FINS/PROFINET Advanced Settings] window, you can further specify the command and function category access permission based on which the protocol will work.

In this section, we will demonstrate how to configure the FINS advanced settings. Please take the same approach when configuring the PROFINET advanced settings.



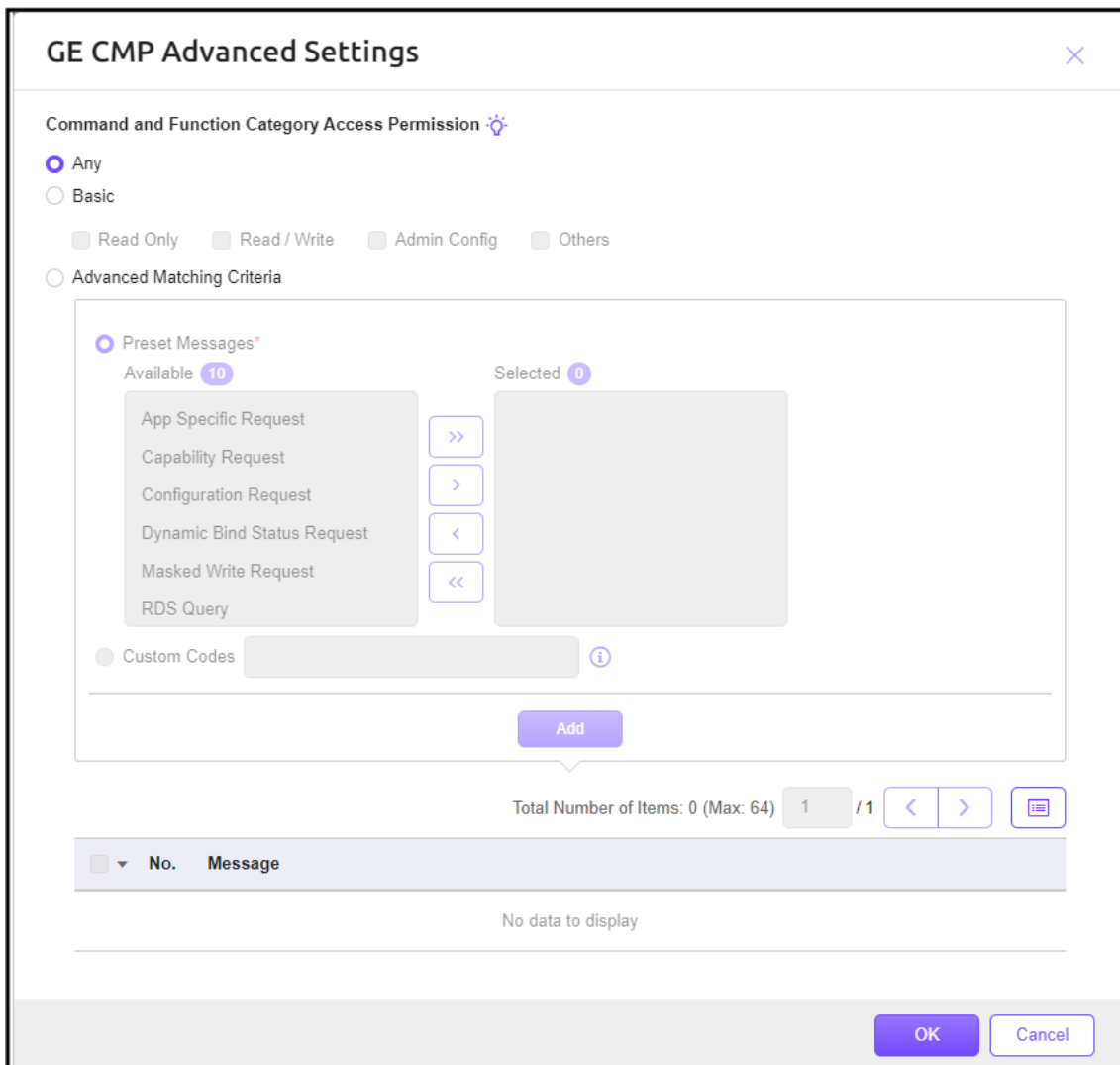
After you access the [FINS Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.
 - **Any:** Select this option to choose any of the commands or function access in this protocol.
 - **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
2. Click [OK].

6.2.1.5 Configuring GE CMP Advanced Settings

Through the [GE CMP Advanced Settings] window, you can further configure the command and function category access permission settings, including message types based on which the protocol will work.



After you access the [GE CMP Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].

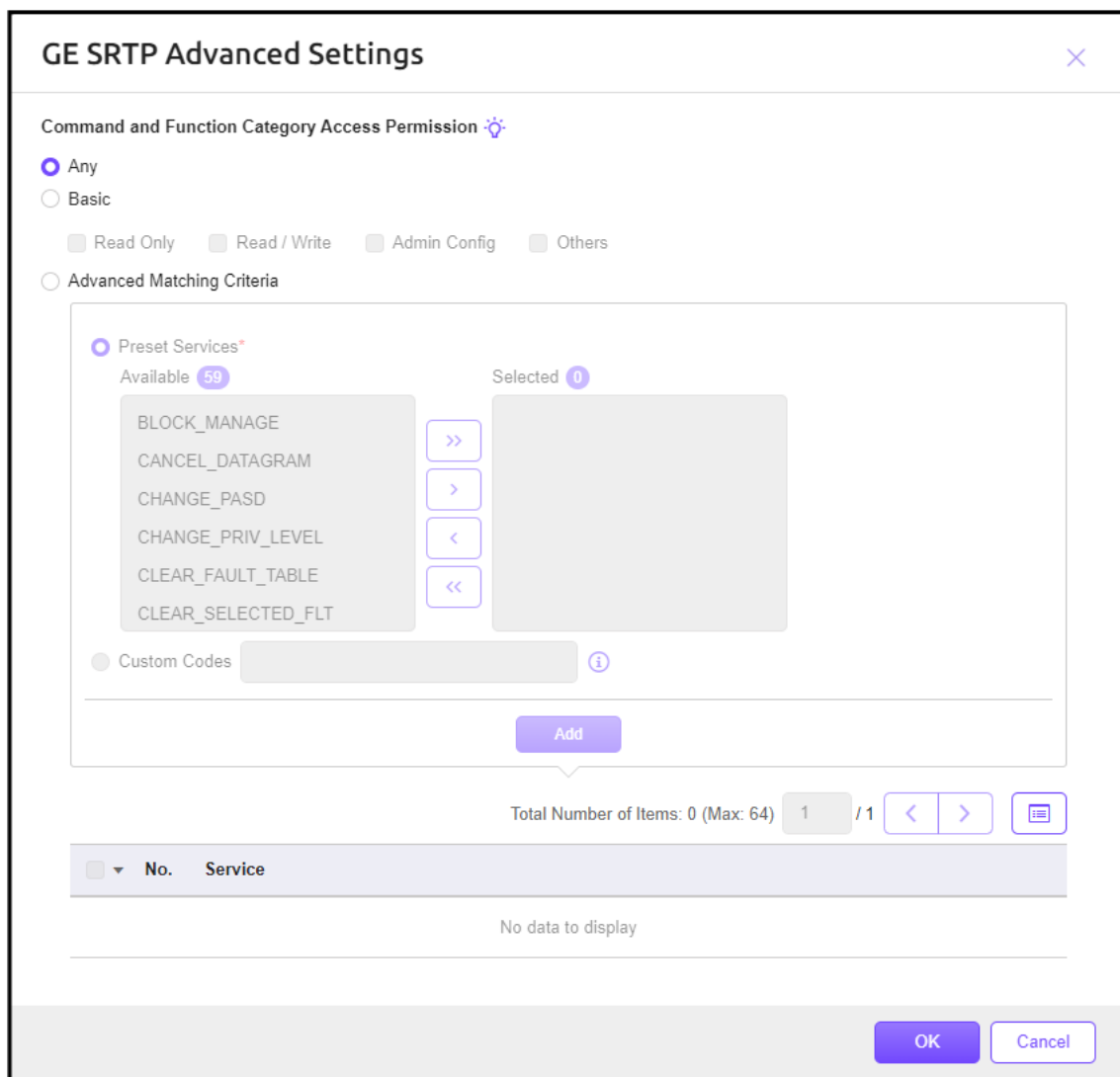
2. If you select [Advanced Matching Criteria], follow the steps below to configure the message types.

- a. Select [Preset Messages] or [Custom Codes] based on your needs.
 - **Preset Messages:** Select this option if you want to specify one or multiple types of messages, then move the message types from the [Available] field to the [Selected] field.
 - **Custom Codes:** Select this option if you want to specify one or multiple codes by yourself, then input codes in the [Custom Codes] field.
- b. Click [Add].
- c. Repeat the above steps if you want to add more protocol definition entries.

3. Click [OK].

6.2.1.6 Configuring GE SRTP Advanced Settings

Through the [GE SRTP Advanced Settings] window, you can further configure the command and function category access permission settings, including service types based on which the protocol will work.



After you access the [GE SRTP Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].

2. If you select [Advanced Matching Criteria], follow the steps below to configure the service types.

a. Select [Preset Services] or [Custom Codes] based on your needs.

- **Preset Services:** Select this option if you want to specify one or multiple types of services, then move the service types from the [Available] field to the [Selected] field.
- **Custom Codes:** Select this option if you want to specify one or multiple codes by yourself, then input codes in the [Custom Codes] field.

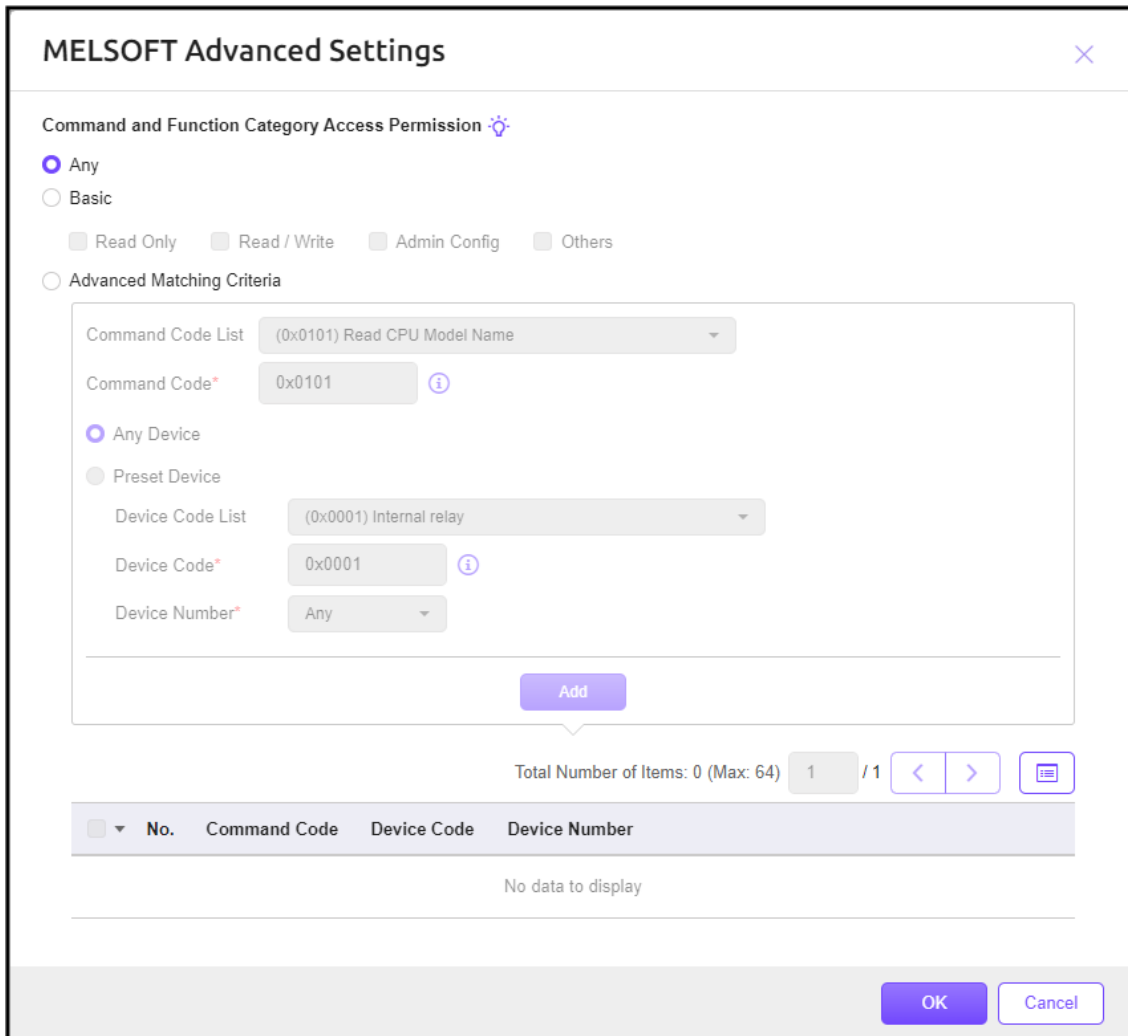
b. Click [Add].

c. Repeat the above steps if you want to add more protocol definition entries.

3. Click [OK].

6.2.1.7 Configuring MELSOFT Advanced Settings

Through the [MELSOFT Advanced Settings] window, you can further configure the command and function category access permission, including command codes, device codes, and device numbers based on which the protocol will work.



After you access the [MELSOFT Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].

2. If you select [Advanced Matching Criteria], follow the steps below to configure the command codes, device codes and device numbers.

a. From the [Command Code List] drop-down list, select a command code of this protocol. If you want to specify a command code by yourself, then select [Custom] and input a command code in the [Command Code] field.

For the command codes [(0x1401) Device Batch Write], [(0x1402) Device Random Write], [(0x1410) Device Memory Write] and [(0x1411) Device Random Write], further select [Any Device] or [Preset Device] based on your needs.

- **Any Device:** Select this option if you want any of the device codes within the function you specified to be applied.
- **Preset Device:** Select this option if you want to specify a device code, then further configure the following settings.
 - From the [Device Code List] drop-down list, select a device code. If you want to specify a device code by yourself, then select [Custom] and input a device code in the [Device Code] field.
 - From the [Device Number] drop-down list, select one of the following options based on your needs.
 - **Any:** Select this option if you don't want to include a device number as a filter criterion.
 - **Single:** Select this option if you want to specify a device number.
 - **Range:** Select this option if you want to specify a range of device numbers.

b. Click [Add].

c. Repeat the above steps if you want to add more protocol definition entries.

3. Click [OK].

6.2.1.8 Configuring Modbus Advanced Settings

Through the [Modbus Advanced Settings] window, you can further configure the command and function category access permission settings, including function codes, unit IDs, and addresses based on which the protocol will work.

Modbus Advanced Settings

Command and Function Category Access Permission

☒ Any
 ☐ Basic

☐ Read Only
 ☐ Read / Write
 ☐ Admin Config
 ☐ Others

☐ Advanced Matching Criteria

Function List

(0x01) Read Coils

Function Code*

0x01

Unit ID*

0

Address*

Any

Add

Total Number of Items: 0 (Max: 64)

1 / 1

	No.	Function Code	Unit ID	Address	UMAS Function Code
No data to display					

OK

Cancel

After you access the [Modbus Advanced Settings] window, follow the procedure below.

Procedure

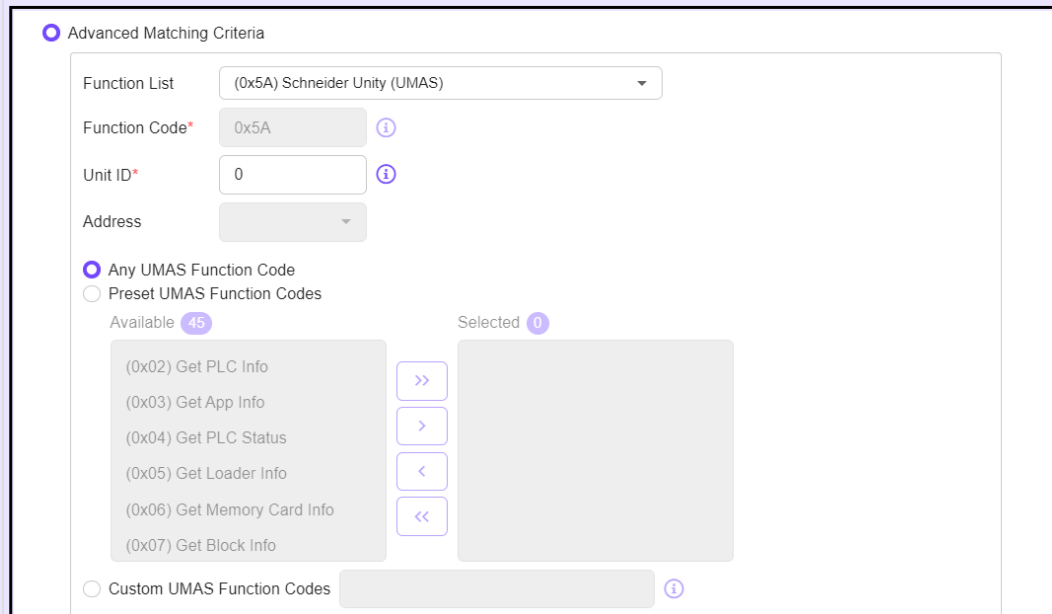
1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].

2. If you select [Advanced Matching Criteria], follow the steps below to configure the function codes, unit IDs, and addresses.

- a. From the [Function List] drop-down list, select a function of this protocol. If you want to specify a function by yourself, then select [Custom] and input a function code in the [Function Code] field.

If you select [(0x5A) Schneider Unity (UMAS)], further select [Any UMAS Function Code], [Preset UMAS Function Codes] or [Custom UMAS Function Codes] based on your needs.



- **Any UMAS Function Code:** Select this option if you want any of the UMAS function codes to be applied.
- **Preset UMAS Function Codes:** Select this option if you want to specify one or multiple UMAS function codes, and then move the UMAS function codes from the [Available] to the [Selected] field.
- **Custom UMAS Function Codes:** Select this option if you want to specify one or multiple UMAS codes by yourself, and then input UMAS codes in the [Custom UMAS Function Codes] field.

b. Type a unit ID in the [Unit ID] field.

c. For some function codes, select **Any**, **Single** or **Range** from the [Address] drop-down list. If you select Single or Range, further specify an address or range of addresses based on which the function will operate.

d. Click [Add].

e. (Optional) Repeat the above steps to add more protocol definition entries.

3. Click [OK].

6.2.1.9 Configuring OPC CLASSIC Advanced Settings

Through the [OPC CLASSIC Advanced Settings] window, you can further configure the command and function category access permission settings, including classic types, interfaces, and method types based on which the protocol will work.

OPC CLASSIC Advanced Settings

Command and Function Category Access Permission

Any

Basic

Read Only

Read / Write

Admin Config

Others

Advanced Matching Criteria

Classic Type

OPC AE

Interface

IOPCEventAreaBrowser (65168857-5783-11D1-84A0-0060...

Custom Interface Value

Any Method

Preset Methods

Available

Selected

BrowseOPCAreas

ChangeBrowsePosition

GetQualifiedAreaName

GetQualifiedSourceName

>>

>

<

<<

Custom Methods

Add

Total Number of Items: 0 (Max: 64)

1 / 1

<

>

No.	Classic Type	Interface	Method
No data to display			

OK

Cancel

After you access the [OPC CLASSIC Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands

- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].
2. If you select [Advanced Matching Criteria], follow the steps below to configure the classic types, interfaces, and method types.
- a. From the [Classic Type] drop-down list, select a classic type of this protocol, then select an interface from the [Interface] drop-down list for your selected classic type.
 - b. If your classic type and interface are [Custom], further input a UUID (Universally Unique Identifier) in the [Custom Interface Value] field.
 - c. Select [Any Method], [Preset Methods] or [Custom Methods] based on your needs.
 - **Any Method:** Select this option if you want any of the method types within the function you specified to be applied.
 - **Preset Methods:** Select this option if you want to specify one or multiple method types, then move the method types from the [Available] field to the [Selected] field.
-  The [Preset Methods] option is not available if the classic type and interface are [Custom].
- **Custom Methods:** Select this option if you want to specify one or multiple method types by yourself, then input method values in the [Custom Methods] field.
 - d. Click [Add].
 - e. Repeat the above steps if you want to add more protocol definition entries.
3. Click [OK].

6.2.1.10 Configuring OPC UA Advanced Settings

Through the [OPC UA Advanced Settings] window, you can further configure the command and function category access permission settings, including classic types, interfaces, and method types based on which the protocol will work.

OPC UA Advanced Settings

Command and Function Category Access Permission

Any

Basic

Read Only

Read / Write

Admin Config

Others

Advanced Matching Criteria

Message Name

MSG

Service Set

Attribute Service Set

Preset Methods*

Available 4

(631) ReadRequest

(664) HistoryReadRequest

(673) WriteRequest

(700) HistoryUpdateRequest

>>

>

<

<<

Selected 0

Custom Node IDs

Add

Total Number of Items: 0 (Max: 64)

1

/ 1

<

>

No.	Message Name	Service Set	Node ID
No data to display			

OK

Cancel

After you access the [OPC UA Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands

Objects & Profiles – 111

- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].
2. If you select [Advanced Matching Criteria], follow the steps below to configure the service sets and method types.
 - a. From the [Service Set] drop-down list, select a service set of this protocol and its method types, then move the method types from the [Available] field to the [Selected] field.
 - b. If your service set is [Custom], input one or multiple node IDs in the [Custom Node IDs] field.
 - c. Click [Add].
 - d. Repeat the above steps if you want to add more protocol definition entries.
 3. Click [OK].

6.2.1.11 Configuring S7COMM Advanced Settings

Through the [S7COMM Advanced Settings] window, you can further configure the command and function category access permission settings, including function codes/function group codes and sub-function codes based on which the protocol will work.

S7COMM Advanced Settings

Command and Function Category Access Permission

☒ Any
 ☐ Basic

☐ Read Only
 ☐ Read / Write
 ☐ Admin Config
 ☐ Others

☐ Advanced Matching Criteria

☒ Job

Function List: Any
 Function Code*: Any

☐ User Data

Function Group List: Any
 Function Group Code*: Any

☒ Any Sub-function Code
 ☐ Preset Sub-function Codes

Available: 0
 Selected: 0

>>

>

<

<<

☐ Custom Codes

Add

Total Number of Items: 0 (Max: 64)
 1 / 1

☐	No.	Job Function Code	Function Group Code	Sub-function Code
No data to display				

OK

Cancel

After you access the [S7COMM Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)

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- **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
- **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
- **Others:** Proprietary, customized, or undocumented commands
- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].

2. If you select [Advanced Matching Criteria], follow the steps below to configure the function codes/function group codes and sub-function codes.

a. Select [Job] or [User Data] based on your needs.

- **Job:** Select this option to specify one function code from the [Function list] drop-down list. If you want to specify a function by yourself, then select [Custom] and input a function code in the [Function Code] field.

- **User Data:** Select this option to specify one function group code from the [Function Group List] drop-down list. If you want to specify a function group by yourself, then select [Custom] and input a function group code in the [Function Group Code] field.

If you select [User Data] as the advanced matching criteria, further select [Any Sub-function Code], [Preset Sub-function Code] or [Custom Sub-function Code] based on your needs.

- **Any Sub-function Code:** Select this option if you want any of the sub-function codes within the function group code you specified to be applied.

- **Preset Sub-function Codes:** Select this option if you want to specify one or multiple sub-function codes, and then move the sub-function codes from the [Available] field to the [Selected] field.



The [Preset Sub-function Codes] option is not available if the function group is [Any] or [Custom].

- **Custom Codes:** Select this option if you want to specify one or multiple sub-function codes by yourself, and then input sub-function codes in the [Custom Codes] field.



The [Custom Codes] option is not available if the function group is [Any].

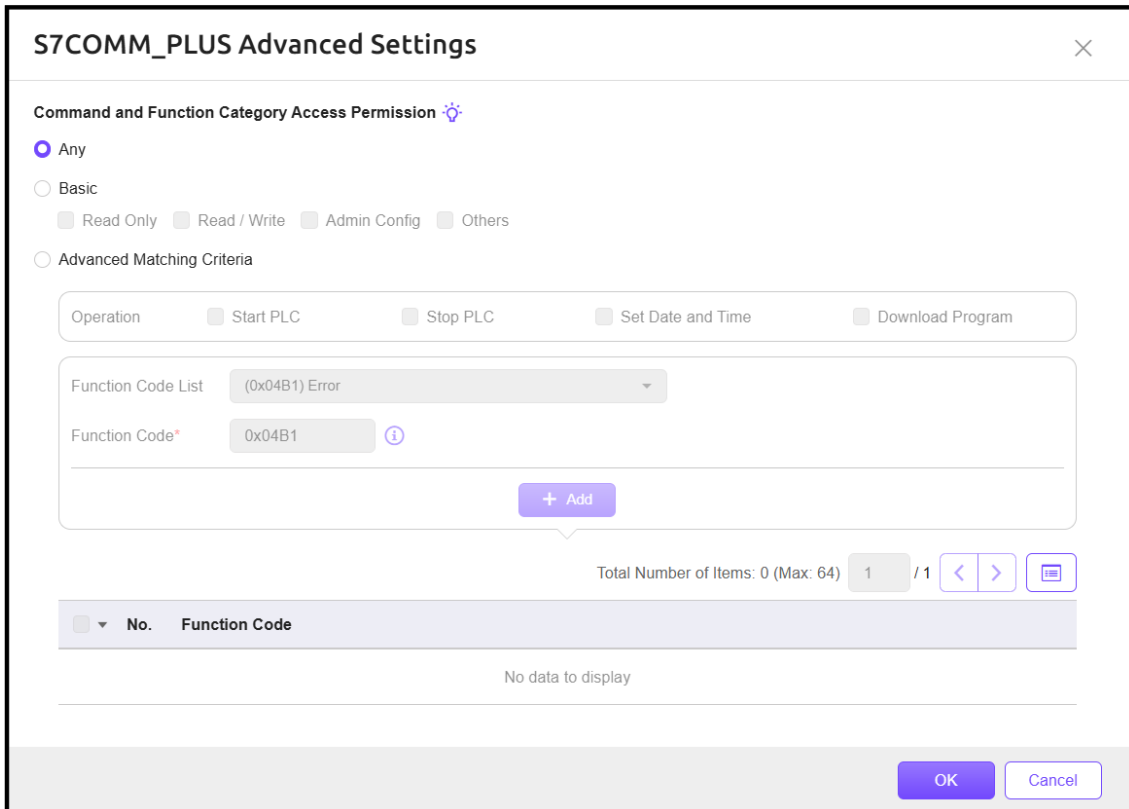
b. Click [Add].

c. Repeat the above steps if you want to add more protocol definition entries.

3. Click [OK].

6.2.1.12 Configuring S7COMM_PLUS Advanced Settings

Through the [S7COMM_PLUS Advanced Settings] window, you can further configure the command and function category access permission settings, including function codes based on which the protocol will work.



After you access the [S7COMM_PLUS Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].

2. If you select [Advanced Matching Criteria], follow the steps below to configure operations and function codes.

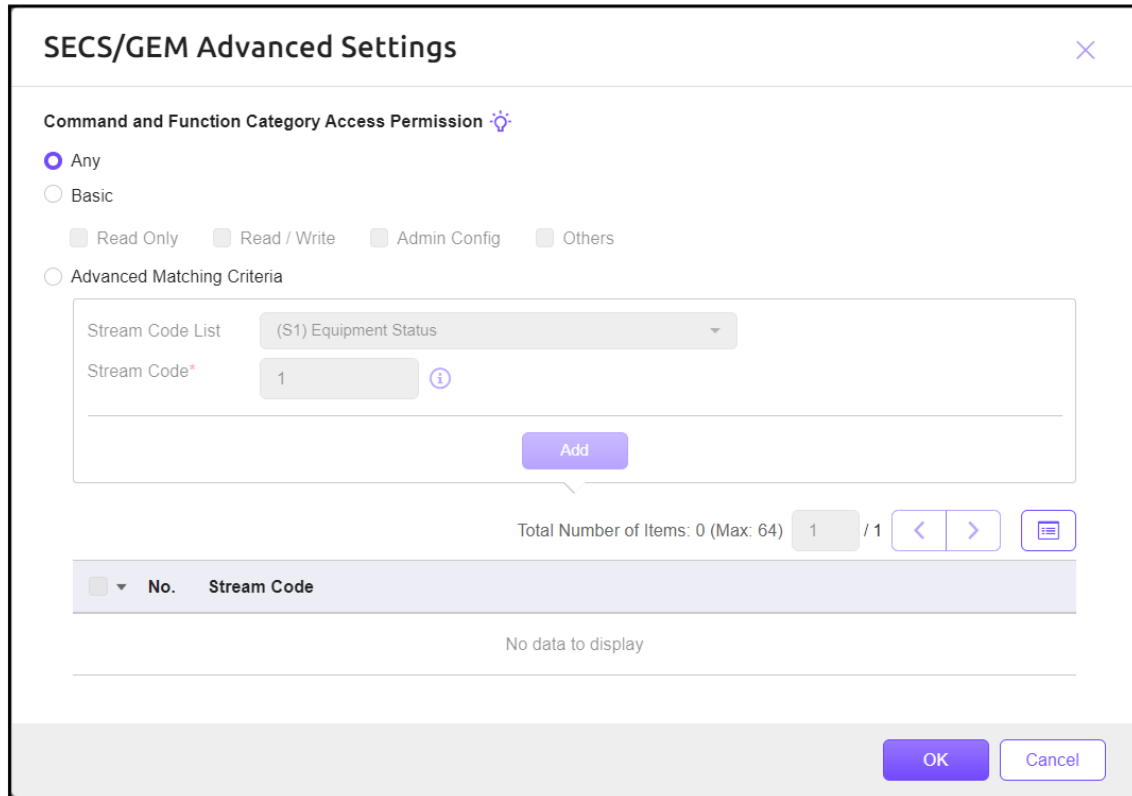
- a. (Optional) In the [Operation] section, select one or multiple operation options.
- b. From the [Function Code List] drop-down list, select a function of this protocol. If you want to specify a function by yourself, then select [Custom] and input a function code in the [Function Code] field.
- c. Click [Add].

d. Repeat the above steps if you want to add more protocol definition entries.

3. Click [OK].

6.2.1.13 Configuring SECS/GEM Advanced Settings

Through the [SECS/GEM Advanced Settings] window, you can further configure the command and function category access permission settings, including stream codes based on which the protocol will work.



After you access the [SECS/GEM Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].

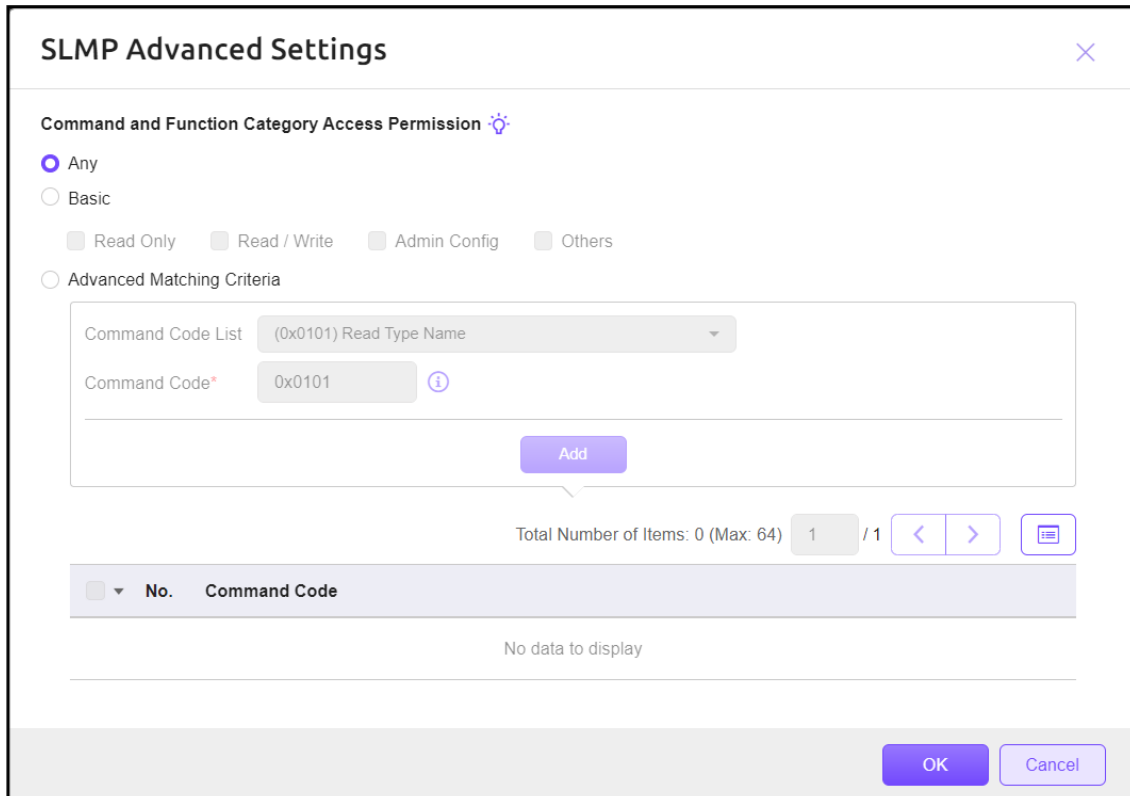
2. If you select [Advanced Matching Criteria], follow the steps below to configure the stream codes.

- a. From the [Stream Code List] drop-down list, select a function of this protocol. If you want to specify a function by yourself, then select [Custom] and input a stream code in the [Stream Code] field.
- b. Click [Add].
- c. Repeat the above steps if you want to add more protocol definition entries.

3. Click [OK].

6.2.1.14 Configuring SLMP Advanced Settings

Through the [SLMP Advanced Settings] window, you can further configure the command and function category access permission, including command codes based on which the protocol will work.



After you access the [SLMP Advanced Settings] window, follow the procedure below.

Procedure

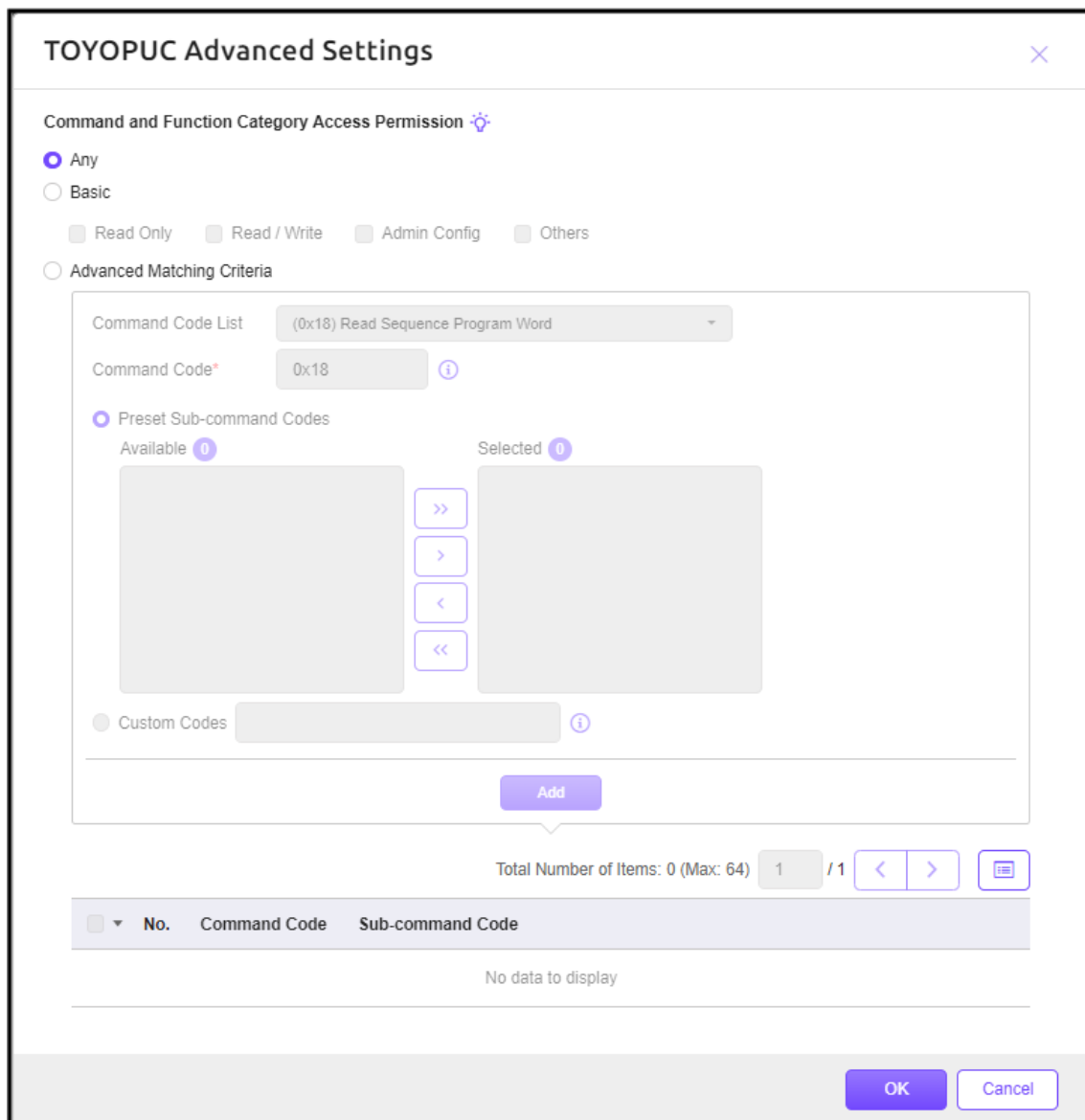
1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands

- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].
2. If you select [Advanced Matching Criteria], follow the steps below to configure the command codes.
 - a. From the [Command Code List] drop-down list, select a command code of this protocol. If you want to specify a command code by yourself, then select [Custom] and input a command code in the [Command Code] field.
 - b. Click [Add].
 - c. Repeat the above steps if you want to add more protocol definition entries.
 3. Click [OK].

6.2.1.15 Configuring TOYOPUC Advanced Settings

Through the [TOYOPUC Advanced Settings] window, you can further configure the command and function category access permission settings, including command codes and sub-command codes based on which the protocol will work.



After you access the [TOYOPUC Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].

2. If you select [Advanced Matching Criteria], follow the steps below to configure the command codes and sub-command codes.

a. From the [Command Code List] drop-down list, select a command code of this protocol. If you want to specify a command code by yourself, then select [Custom] and input a command code in the [Command Code] field.

b. For the command codes [(0x32) Function Call] and [(0xA0) Expansion Function Call], further select [Preset Sub-command Codes] or [Custom Codes] based on your needs.

- **Preset Sub-command Codes:** Select this option if you want to specify one or multiple sub-command codes, and then move the sub-command codes from the [Available] field to the [Selected] field.

- **Custom Codes:** Select this option if you want to specify one or multiple sub-command codes by yourself, then input sub-command codes in the [Custom Codes] field.

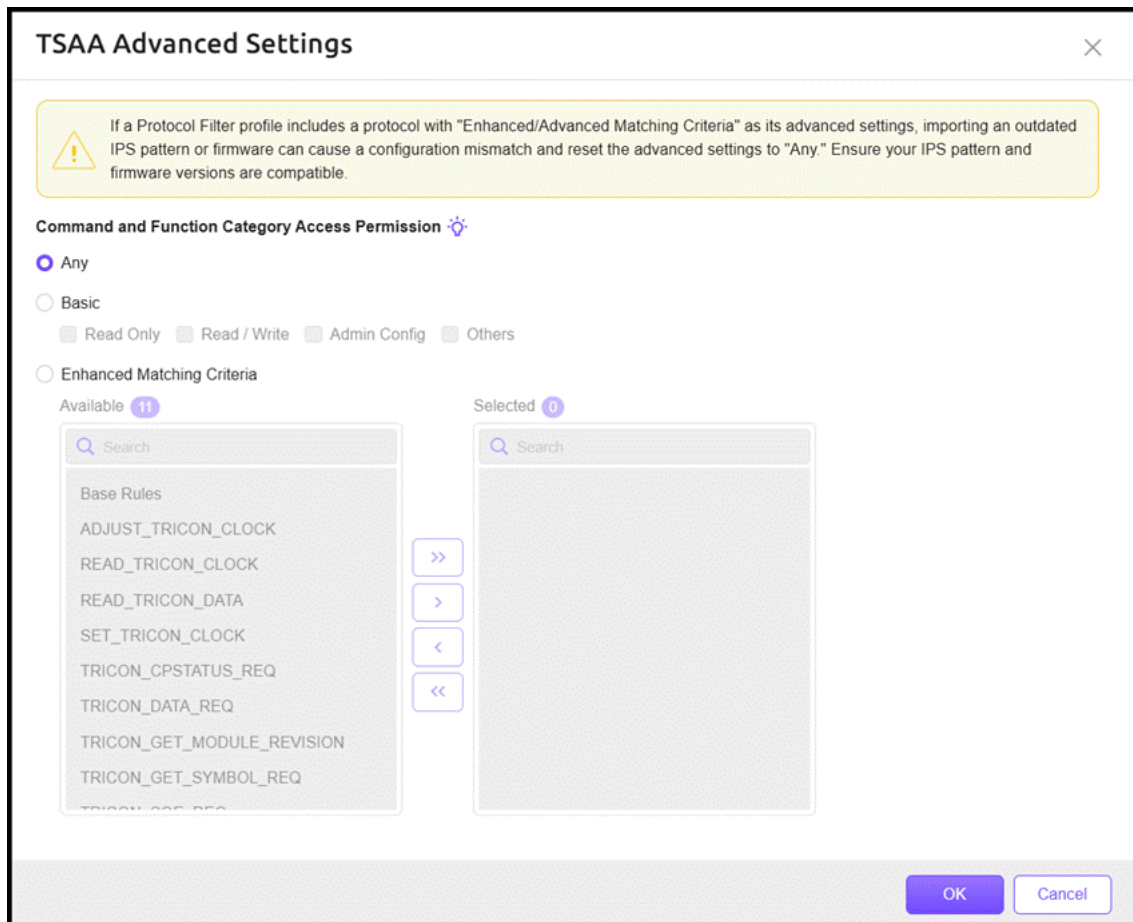
c. Click [Add].

d. Repeat the above steps if you want to add more protocol definition entries.

3. Click [OK].

6.2.1.16 Configuring TSAA Advanced Settings

Through the [TSAA Advanced Settings] window, you can further configure the command and function category access permission settings, including the enhanced matching criteria based on which the protocol will work.



After you access the [TSAA Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
- **Enhanced Matching Criteria:** Select this option to further configure enhanced matching criteria for the [Command and Function Category Access Permission].

2. If you select [Enhanced Matching Criteria], select one or multiple criteria in the [Available] field and move them to the [Selected] field.

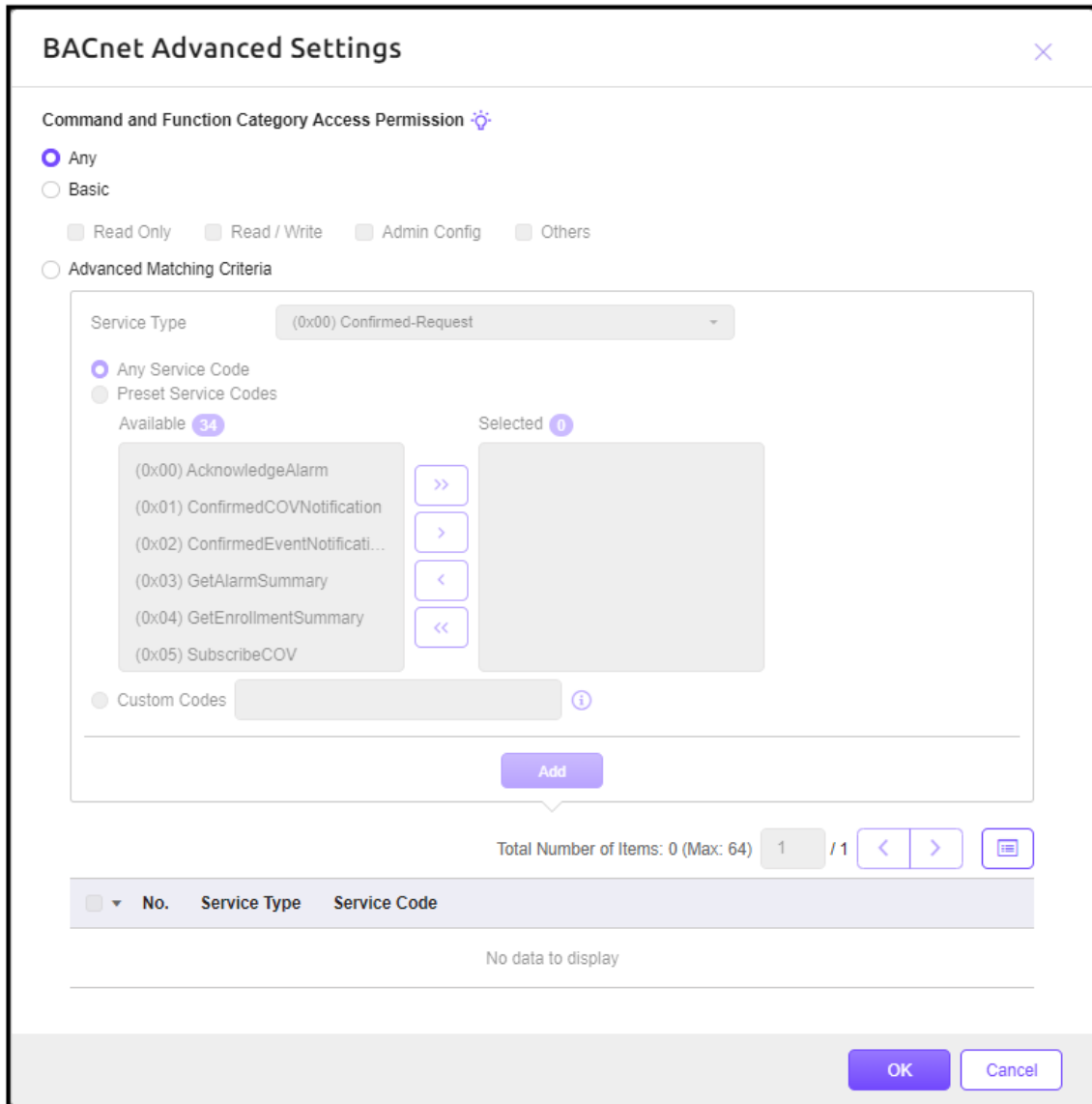
3. Click [OK].



If a Protocol Filter profile includes a protocol with "Enhanced/Advanced Matching Criteria" as its advanced settings, importing an outdated IPS pattern or firmware can cause a configuration mismatch and reset the advanced settings to "Any." Ensure your IPS pattern and firmware versions are compatible.

6.2.1.17 Configuring BACnet Advanced Settings

Through the [BACnet Advanced Settings] window, you can further configure the command and function category access permission settings, including service types and service codes based on which the protocol will work.



BACnet Advanced Settings

Command and Function Category Access Permission 

☒ Any

☐ Basic

☐ Read Only ☐ Read / Write ☐ Admin Config ☐ Others

☐ Advanced Matching Criteria

Service Type: (0x00) Confirmed-Request

☒ Any Service Code

☐ Preset Service Codes

Available: 34

Selected: 0

(0x00) AcknowledgeAlarm

(0x01) ConfirmedCOVNotification

(0x02) ConfirmedEventNotificati...

(0x03) GetAlarmSummary

(0x04) GetEnrollmentSummary

(0x05) SubscribeCOV

☐ Custom Codes

Add

Total Number of Items: 0 (Max: 64) 1 / 1

No.	Service Type	Service Code
No data to display		

OK Cancel

After you access the [BACnet Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

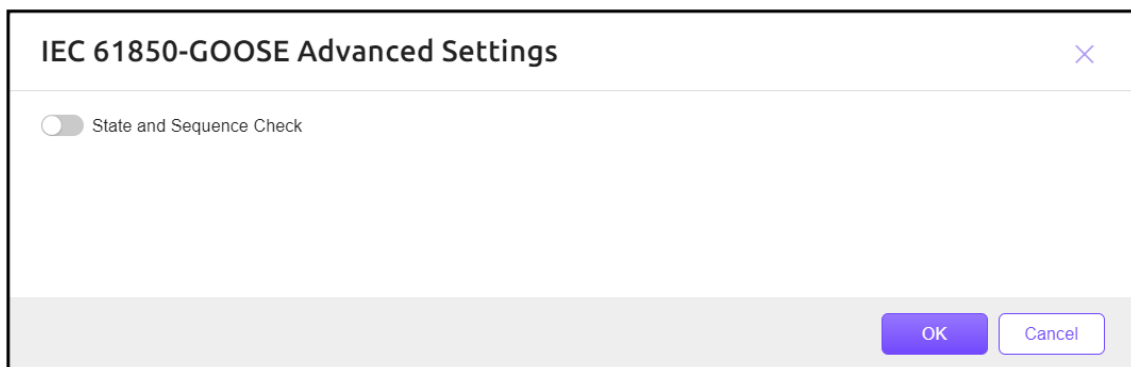
- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:

- **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
 - **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].
2. If you select [Advanced Matching Criteria], follow the steps below to configure the service types and service codes.
- a. From the [Service Type] drop-down list, select a service type of this protocol.
 - b. Select [Any Service Code], [Preset Service Codes] or [Custom Codes] based on your needs.
 - **Any Service Code:** Select this option if you want any of the service codes within the function you specified to be applied.
 - **Preset Service Codes:** Select this option if you want to specify one or multiple service codes, then move the service codes from the [Available] field to the [Selected] field.
 - **Custom Codes:** Select this option if you want to specify one or multiple service codes by yourself, then input service codes in the [Custom Codes] field.
 - c. Click [Add].
 - d. Repeat the above steps if you want to add more protocol definition entries.
3. Click [OK].

6.2.1.18 Configuring IEC 61850-GOOSE and IEC 61850-R-GOOSE Advanced Settings

Through the [IEC 61850-GOOSE/IEC 61850-R-GOOSE Advanced Settings] window, you can further enable or disable the [State and Sequence Check] function.

In this section, we will demonstrate how to configure the IEC 61850-GOOSE advanced settings. Please take the same approach when configuring the IEC 61850-R-GOOSE advanced settings.



The image shows a dialog box titled "IEC 61850-GOOSE Advanced Settings". Inside the dialog, there is a toggle switch labeled "State and Sequence Check" which is currently turned off. At the bottom right of the dialog, there are two buttons: "OK" and "Cancel".

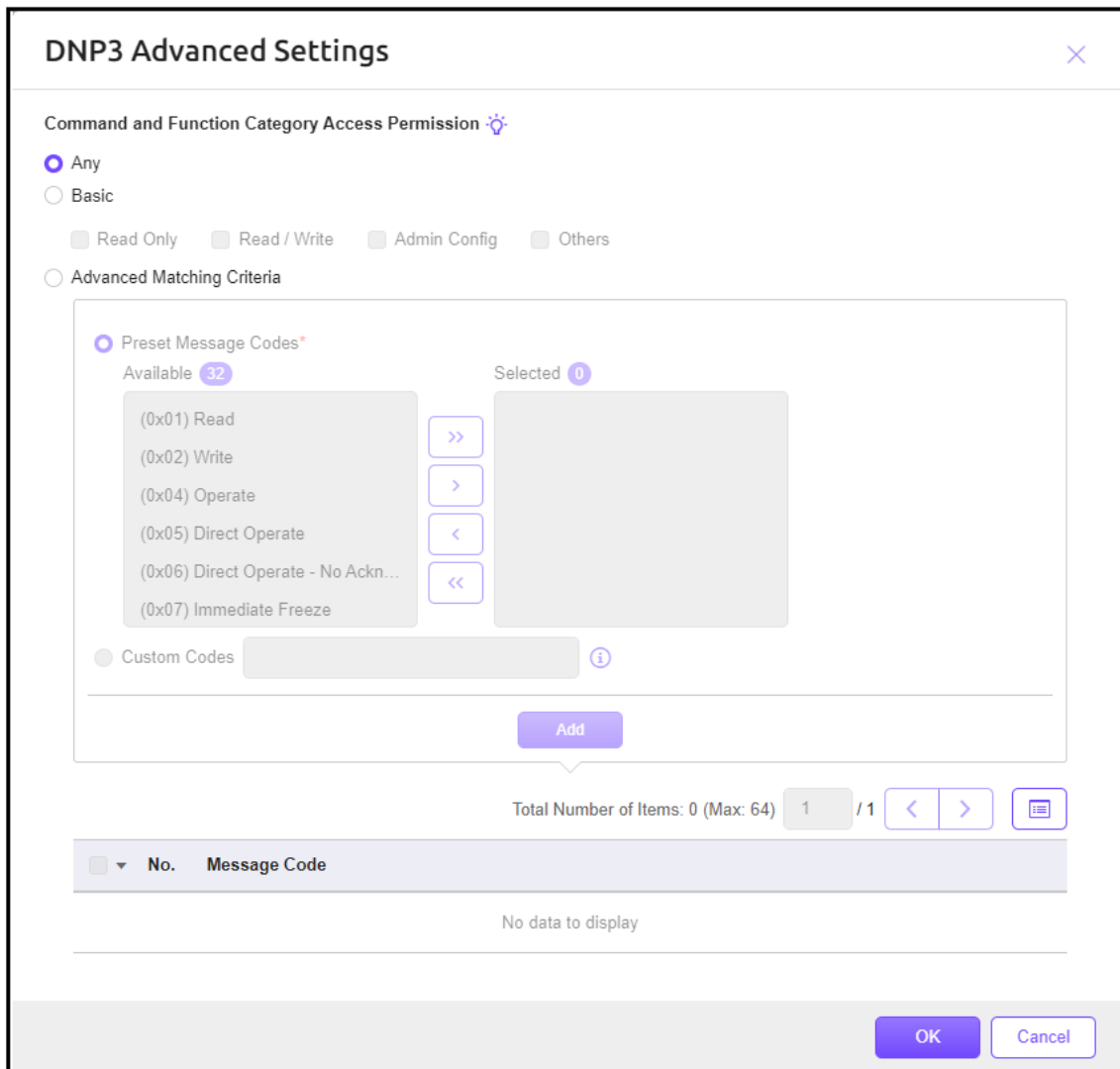
After you access the [IEC 61850-GOOSE Advanced Settings] window, follow the procedure below.

Procedure

1. Enable or disable the [State and Sequence Check] function by clicking the toggle.
2. Click [OK].

6.2.1.19 Configuring DNP3 Advanced Settings

Through the [DNP3 Advanced Settings] window, you can further configure the command and function category access permission settings, including message codes based on which the protocol will work.



After you access the [DNP3 Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system

- **Others:** Proprietary, customized, or undocumented commands
 - **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].
2. If you select [Advanced Matching Criteria], follow the steps below to configure the message codes.
- a. Select [Preset Message Codes] or [Custom Codes] based on your needs.
 - **Preset Message Codes:** Select this option if you want to specify one or multiple message codes, then move the message codes from the [Available] field to the [Selected] field.
 - **Custom Codes:** Select this option if you want to specify one or multiple message codes by yourself, then input message codes in the [Custom Codes] field.
 - b. Click [Add].
 - c. Repeat the above steps if you want to add more protocol definition entries.
3. Click [OK].

6.2.1.20 Configuring ICCP TASE.2 and IEC61850-MMS Advanced Settings

Through the [ICCP TASE.2/IEC61850-MMS Advanced Settings] window, you can further configure the command and function category access permission settings, including message types and service codes based on which the protocol will work.

In this section, we will demonstrate how to configure the ICCP TASE.2 advanced settings. Please take the same approach when configuring the IEC61850-MMS advanced settings.

ICCP TASE.2 Advanced Settings

Command and Function Category Access Permission

☒ Any
 ☐ Basic

☐ Read Only
 ☐ Read / Write
 ☐ Admin Config
 ☐ Others

☐ Advanced Matching Criteria

Message Type

(0x00) Confirmed-Request

☒ Any Service Code
 ☐ Preset Service Codes

Available 76

Selected 0

(0x00) Status

(0x01) GetNameList

(0x02) Identify

(0x03) Rename

(0x04) Read

(0x05) Write

>>

>

<

<<

☐ Custom Codes

Add

Total Number of Items: 0 (Max: 64)

1 / 1

<

>

No.	Message Type	Service Code
No data to display		

OK

Cancel

After you access the [ICCP TASE.2 Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands

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- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].
2. If you select [Advanced Matching Criteria], follow the steps below to configure the message types and service codes.
- a. From the [Message Type] drop-down list, select a message type of this protocol.
 - b. Select [Any Service Code], [Preset Service Codes] or [Custom Codes] based on your needs.
 - **Any Service Code:** Select this option if you want any of the service codes within the function you specified to be applied.
 - **Preset Service Codes:** Select this option if you want to specify one or multiple service codes, then move the service codes from the [Available] field to the [Selected] field.
 - **Custom Codes:** Select this option if you want to specify one or multiple service codes by yourself, then input service codes in the [Custom Codes] field.
 - c. Click [Add].
 - d. Repeat the above steps if you want to add more protocol definition entries.
3. Click [OK].

6.2.1.21 Configuring IEC-104 Advanced Settings

Through the [IEC-104 Advanced Settings] window, you can further configure the command and function category access permission settings, including command codes based on which the protocol will work.

IEC-104 Advanced Settings

Command and Function Category Access Permission

☒ Any
 ☐ Basic

☐ Read Only
 ☐ Read / Write
 ☐ Admin Config
 ☐ Others

☐ Advanced Matching Criteria

☒ Preset Command Codes*

Available 67

Selected 0

(0x01) M_SP_NA_1 : Single poi...

(0x02) M_SP_TA_1 : Single poi...

(0x03) M_DP_NA_1 : Double p...

(0x04) M_DP_TA_1 : Double po...

(0x05) M_ST_NA_1 : Step posit...

(0x06) M_ST_TA_1 : Step positi...

>>

>

<

<<

☐ Custom Codes

Add

Total Number of Items: 0 (Max: 64)

1 / 1

<

>

No. Command Code

No data to display

OK

Cancel

After you access the [IEC-104 Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].

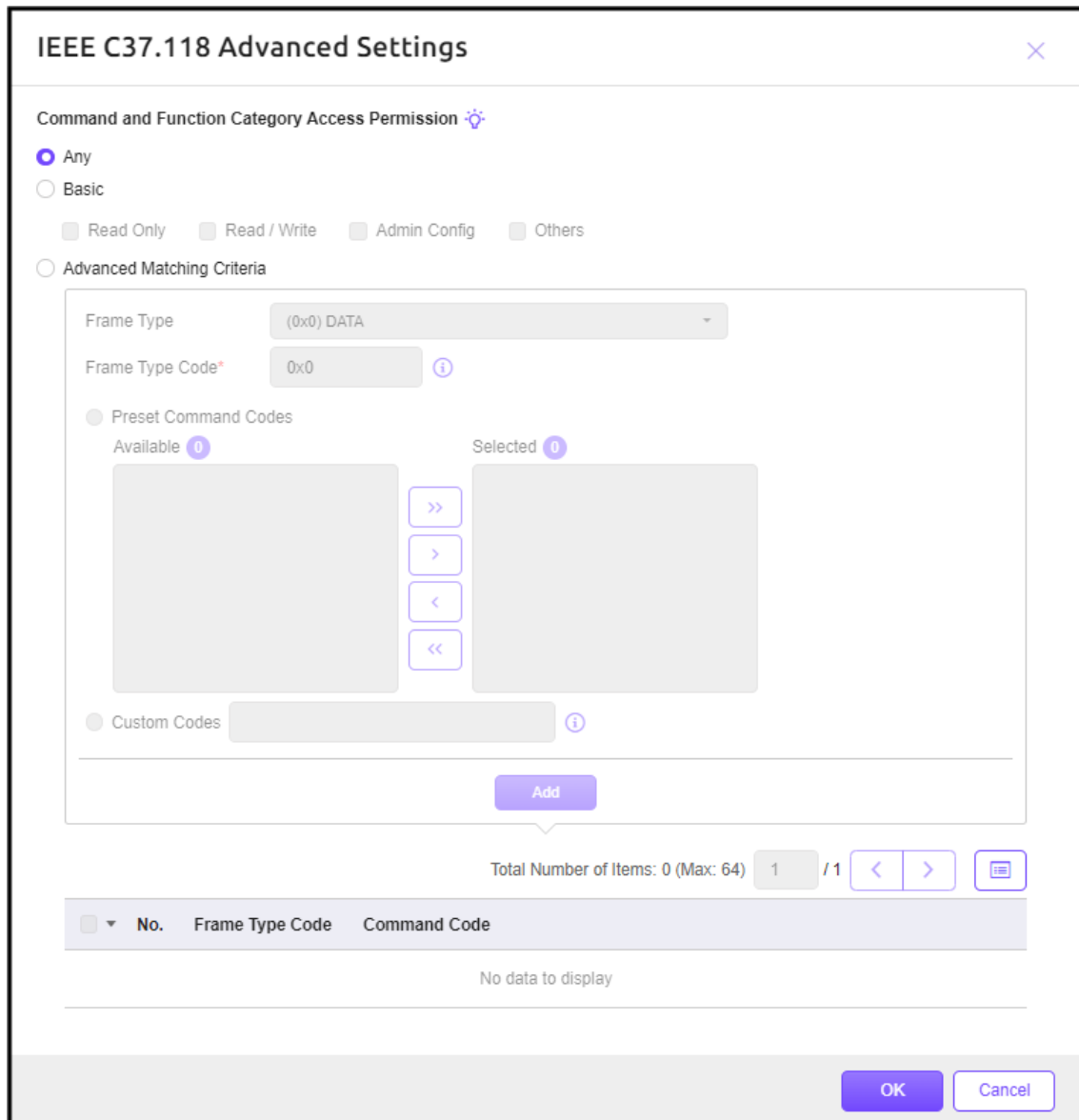
2. If you select [Advanced Matching Criteria], follow the steps below to configure the command codes.

- a. Select [Preset Command Codes] or [Custom Codes] based on your needs.
 - **Preset Command Codes:** Select this option if you want to specify one or multiple command codes, then move the command codes from the [Available] field to the [Selected] field.
 - **Custom Codes:** Select this option if you want to specify one or multiple command codes by yourself, then input command codes in the [Custom Codes] field.
- b. Click [Add].
- c. Repeat the above steps if you want to add more protocol definition entries.

3. Click [OK].

6.2.1.22 Configuring IEEE C37.118 Advanced Settings

Through the [IEEE C37.118 Advanced Settings] window, you can further configure the command and function category access permission settings, including frame types and command codes based on which the protocol will work.



After you access the [IEEE C37.118 Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.

- **Any:** Select this option to choose any of the commands or function access in this protocol.
- **Basic:** Multiple selections as follows:
 - **Read Only:** Read commands between assets (such as HMI, EWS, SCADA, and PLC)
 - **Read/Write:** Read and write commands from one asset (such as HMI, EWS, and SCADA) to another (such as PLC)
 - **Admin Config:** Configuration commands, including PLC firmware updates, admin configuration commands, and other upgrades, from an asset (such as EWS) to a programmable control system
 - **Others:** Proprietary, customized, or undocumented commands
- **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].

2. If you select [Advanced Matching Criteria], follow the steps below to configure the frame types and command codes.

a. From the [Frame Type] drop-down list, select a frame type of this protocol. If you want to specify a frame type by yourself, then select [Custom] and input a frame type code in the [Frame Type Code] field.

For the frame type code [(0x4) CMD], further select [Preset Command Codes] or [Custom Codes] based on your needs.

- **Preset Command Codes:** Select this option if you want to specify one or multiple command codes, then move the command codes from the [Available] field to the [Selected] field.
- **Custom Codes:** Select this option if you want to specify one or multiple command codes by yourself, then input command codes in the [Custom Codes] field.

b. Click [Add].

c. Repeat the above steps if you want to add more protocol definition entries.

3. Click [OK].

6.2.1.23 Configuring OCPP 1.6J and OCPP 2.0.1 Advanced Settings

Through the [OCPP 1.6J/OCPP 2.0.1 Advanced Settings] window, you can further configure the command and function category access permission settings, including message types based on which the protocol will work.

In this section, we will demonstrate how to configure the OCPP 1.6J advanced settings. Please take the same approach when configuring the OCPP 2.0.1 advanced settings.

OCPP 1.6J Advanced Settings

Command and Function Category Access Permission

☒ Any
☐ Advanced Matching Criteria

Preset Messages

Available 28

Selected 0

Authorize

BootNotification

CancelReservation

ChangeAvailability

ChangeConfiguration

ClearCache

>>

>

<

<<

Add

Total Number of Items: 0 (Max: 64) 1 / 1

No.	Message
No data to display	

OK Cancel

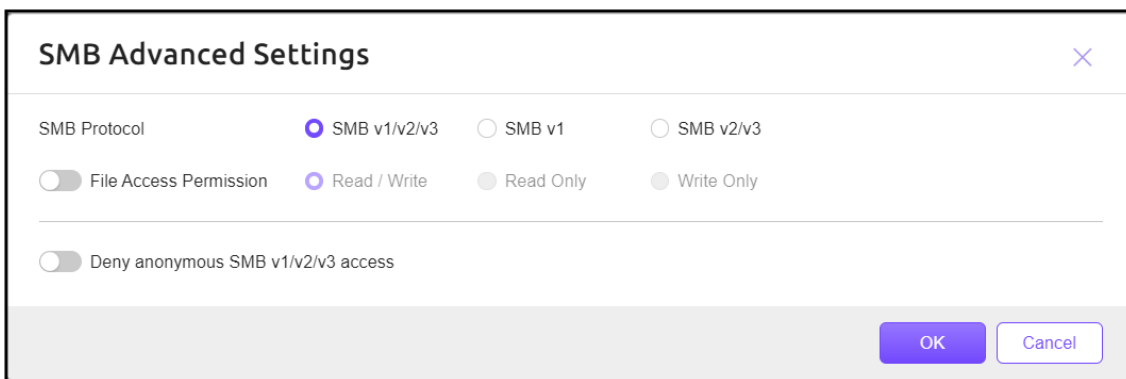
After you access the [OCPP 1.6J Advanced Settings] window, follow the procedure below.

Procedure

1. Configure [Command and Function Category Access Permission] by selecting one of the following options.
 - **Any:** Select this option to choose any of the commands or function access in this protocol.
 - **Advanced Matching Criteria:** Select this option to further configure advanced matching criteria for the [Command and Function Category Access Permission].
2. If you select [Advanced Matching Criteria], further select one or multiple types of messages, then move the message types from the [Available] field to the [Selected] field. Repeat this step if you want to add more protocol definition entries.
3. Click [OK].

6.2.1.24 Configuring SMB Advanced Settings

Through the [SMB Advanced Settings] window, you can further specify the SMB protocol version combination, file access permission and enable or disable the [Deny anonymous SMB v1/v2/v3 access] setting.



The image shows a dialog box titled "SMB Advanced Settings" with a close button (X) in the top right corner. Inside the dialog, there are three rows of settings. The first row is "SMB Protocol" with three radio buttons: "SMB v1/v2/v3" (selected), "SMB v1", and "SMB v2/v3". The second row is "File Access Permission" with a toggle switch (off) and four radio buttons: "Read / Write" (selected), "Read Only", "Write Only", and an unlabeled one. The third row is "Deny anonymous SMB v1/v2/v3 access" with a toggle switch (off). At the bottom right, there are "OK" and "Cancel" buttons.

After you access the [SMB Advanced Settings] window, follow the procedure below.

Procedure

1. Configure the following SMB advanced settings.

- **SMB Protocol:** Select an SMB protocol version combination, **SMB v1/v2/v3**, **SMB v1** or **SMB v2/v3**.
- **File Access Permission:** If needed, enable this function and select one of the following file access behaviors.
 - **Read / Write:** File access for reading and writing.
 - **Read Only:** File access for reading only.
 - **Write Only:** File access for writing only.
- **Deny anonymous SMB v1/v2/v3 access:** If needed, enable this setting to deny access over SMB v1/v2/v3 for anonymous accounts.

2. Click [OK].

6.2.2 IPS Profiles

IPS profiles contain sophisticated IPS pattern rules which you can apply to policy rules for granular control. In IPS profiles, you can configure:

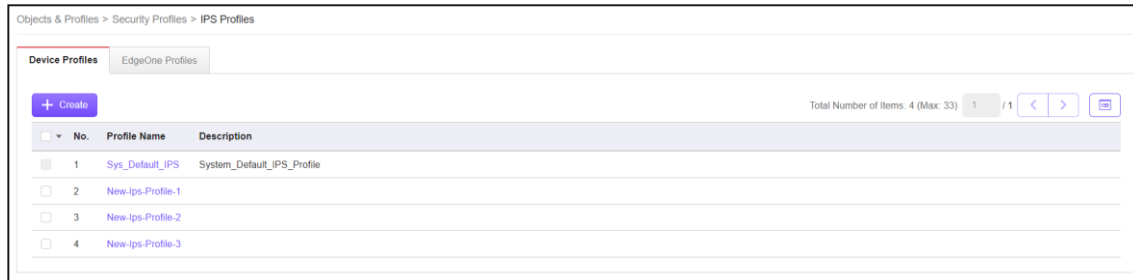
- **IPS pattern rule category**, including:
 - File Vulnerabilities
 - Buffer Overflow
 - DoS Attacks
 - Exploits
 - Malware Traffic
 - Reconnaissance
 - Web Threats
 - ICS Threats
 - Others
 - Misc
- **IPS pattern rule risk level**, including:
 - Information
 - Low
 - Medium

- High
- Critical
- **IPS pattern rule default actions**, including:
 - Allow and Log
 - Deny and Log

6.2.2.1 Configuring IPS Profiles

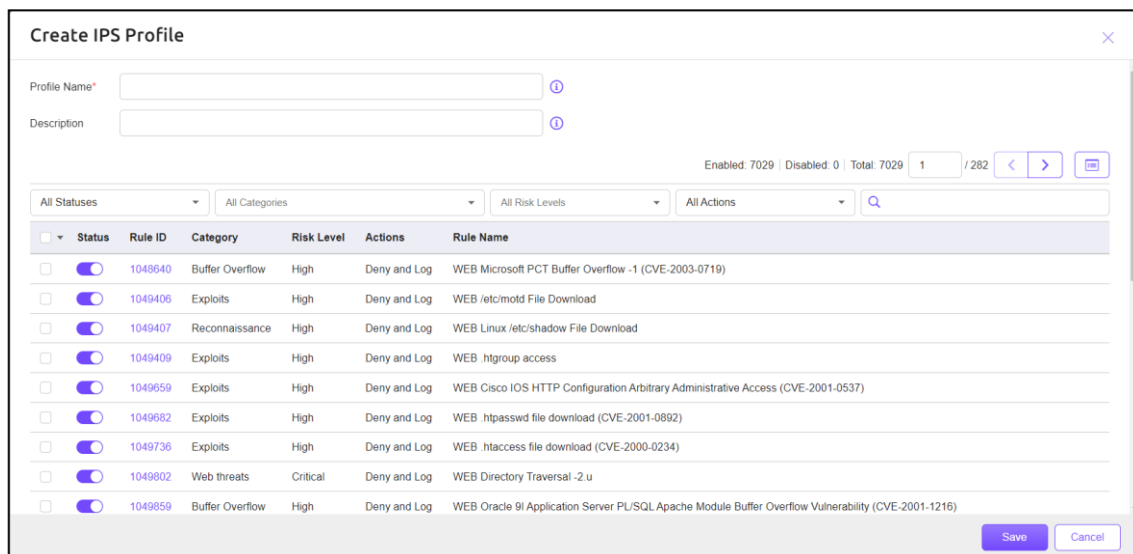
Procedure

1. Go to [Objects & Profiles] > [Security Profiles] > [IPS Profiles].



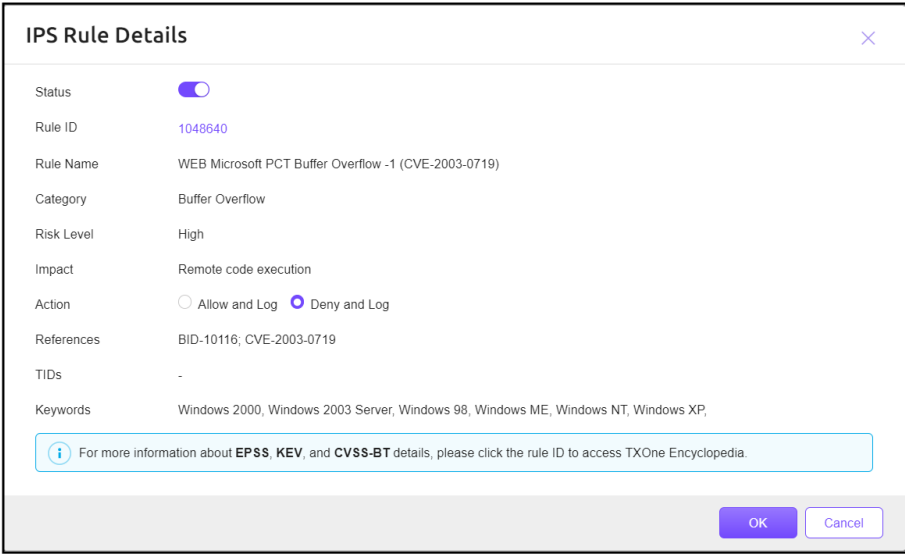
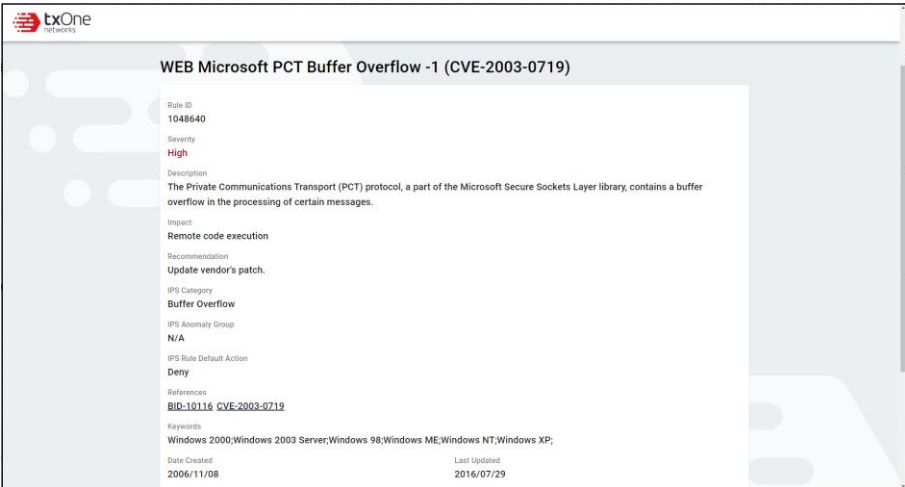
2. Do one of the following.

- Click [Create] to create a profile. The [Create IPS Profile] window will appear.
- Click a profile name to edit settings. The [Edit IPS Profile] window will appear.



3. Configure the following IPS profile settings.

Item	Description
Profile Name	Type a profile name for this IPS profile.
Description	(Optional) Type a description.
Status	Disable the operational statuses of one or multiple IPS pattern rules you want to exclude from this IPS profile by clicking the toggles. (Default status: Enabled)
Rule ID	Click the rule ID in the [Rule ID] column to configure and view details of each IPS pattern rule. The [IPS Rule Details] window will appear.

Item	Description
(IPS Rule Granular Control Details)	 • Status: The operational status of this single IPS pattern rule. Use the toggle to enable/disable this rule. • Rule ID: The ID number of this IPS pattern rule. You can view the threat information regarding this IPS pattern rule in detail by clicking the rule ID number, and you will then be redirected to the TXOne Threat Encyclopedia as shown below.  • Rule Name: The name of this IPS pattern rule regarding the cyber attack. • Category: The threat category of this IPS pattern rule regarding the cyber attack, including File Vulnerabilities, Buffer Overflow, DoS Attacks, Exploits, Malware Traffic, Reconnaissance, Web Threats, ICS Threats, Others and Misc. • Risk Level: The suggested security level regarding the cyber attack, including Information, Low, Medium, High and Critical. • Impact: The damage that will be caused to the target network device if the cyber attack succeeds. • Action: The preset action for the cyber attack. Select either [Allow and Log] or [Deny and Log]. (Default action: Deny and Log)

Item	Description
	○ Allow and Log: When the attack is detected by the Edge series device, the attack will be bypassed and logged for monitoring. ○ Deny and Log: When the attack is detected by the Edge series device, the attack will be blocked and logged for monitoring. • References: The vulnerability IDs of the cyber attack. • TIDs: MITRE ID information. • Keywords: The keywords for searching this IPS pattern rule. Click [OK] to save the configurations of the IPS rule details.

4. Click [Save].



The built-in IPS profile [Sys_Default_IPS] is view-only and cannot be deleted or configured.

6.2.3 File Filter Profiles

File Filter profiles contain detailed access protocols as well as settings for executable file types and Active Directory (AD) GPO dispatch which you can apply to policy rules. In File Filter profiles, you can configure:

- **Protocol for File Filter function**, including:
 - HTTP
 - FTP
 - SMB
- **Executable file types to be denied**, including:
 - Windows (PE files)
 - Linux (ELF files)
- **Filtering of Active Directory (AD) GPO**
 - Enable or disable AD GPO filters

On the [File Filter Profiles] page, you can further configure the **File Exclusions** function to exclude certain files from the file filter profiles you're enabling. The Edge device skips the file type scanning based on your File Exclusion list. This can increase the flexibility of configuration and obtain real control for your OT network.

We provide two methods for configuring the File Exclusion settings, either adding file items one by one or importing an edited CSV file with multiple **filename** items.

6.2.3.1 Configuring File Filter Profiles

Procedure

1. Go to [Objects & Profiles] > [Security Profiles] > [File Filter Profiles].

Objects & Profiles > Security Profiles > File Filter Profiles

File Exclusions

The system skips the file type scanning based on the file exclusion list.
To import file exclusions from a CSV file into the list, please download the CSV template and fill in full filenames and descriptions.

Total Number of File Exclusions: 0

[Edit File Exclusion List](#) [Download CSV Template](#)

File Filter Profile

Device Profiles **EdgeOne Profiles**

[+ Create](#) Total Number of Items: 3 (Max: 32) 1 / 1

No.	Profile Name	File Filter Profile Settings	AD GPO Filter	Description
1	New-Profiles-1	Enabled: HTTP, SMB, Windows (PE Files)	Disabled	
2	New-Profiles-2	Enabled: HTTP, SMB, Windows (PE Files)	Disabled	
3	New-Profiles-3	Enabled: HTTP, SMB, Windows (PE Files)	Disabled	

2. Do one of the following:

- Click [Create] to create a profile. The [Create File Filter Profile] window will appear.
- Click a profile name to edit settings. The [Edit File Filter Profile] window will appear.

Create File Filter Profile

Profile Name*

Description

File Filter Profile Settings

☐ Filter files by protocol

Protocol* ☐ HTTP ☐ FTP ☐ SMB

Deny executable files* ☐ Windows (PE Files) ☐ Linux (ELF files)

AD GPO Filter

☐ Deny Active Directory (AD) GPO dispatch

[Save](#) [Cancel](#)

3. Configure the following File Filter profile settings.

Item	Description
Profile Name	Type a profile name for this File Filter profile.
Description	(Optional) Type a description.

- File Filter Profile Settings

Item	Description
Filter files by protocol	Enable the File Filter function by clicking the toggle.
Protocol	Select one or multiple protocols for File Filter function, including HTTP , FTP and SMB .
Deny executable files	Select one or multiple packed executable file types, including Windows (PE files) and Linux (ELF files) .

- AD GPO Filter

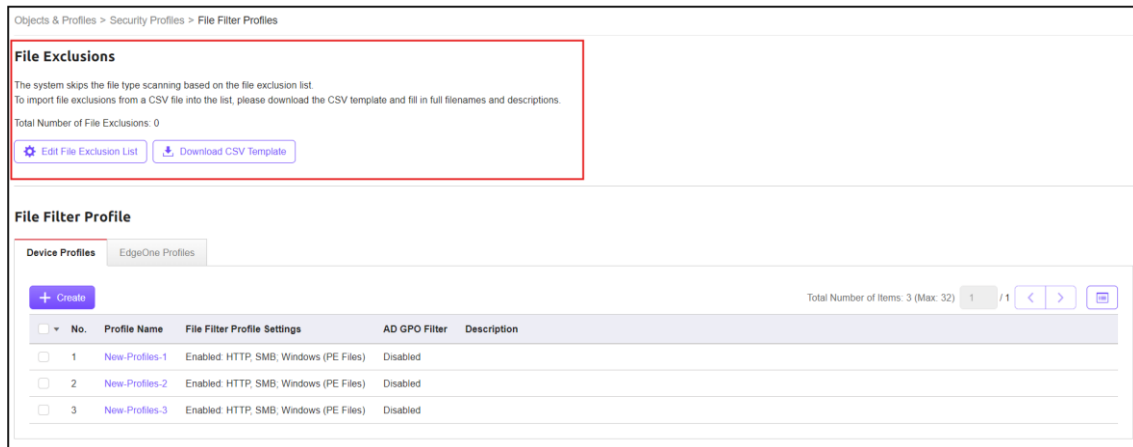
Item	Description
Deny Active Directory (AD) GPO dispatch	(Optional) Enable this function by clicking the toggle if you want to filter AD GPOs.

4. Click [Save].

6.2.3.2 Configuring File Exclusions

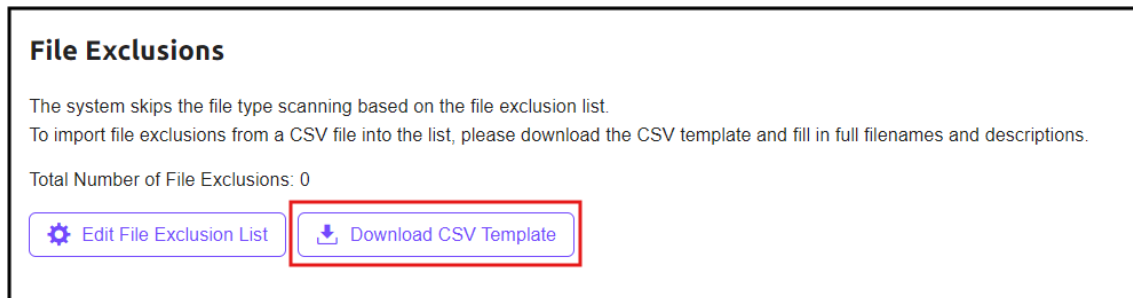
Procedure

1. Go to [Objects & Profiles] > [Security Profiles] > [File Filter Profiles].



- **Method 1 (Steps 2-5): Importing a File List Using the Built-in CSV File Template**

2. Click the [Download CSV Template] button to download the CSV template.




To avoid CVS file compatibility issue, we recommend using the CSV template provided on the page for importing full filenames into the console.

3. Open the downloaded CSV file and input the full filenames (mandatory) and descriptions (optional). For example:

Full Filename	Description
abc.exe	Windows PE File
def.bat	Windows PE File
ghi.o	Linux ELF File



1. You do not need to unzi pp zip/gz file(s) for scanning each compressed file, as the system can identify zip/gz file(s) and determine whether the filename list

contains the file items for exclusions. Notice that zip/gz files larger than 100MB will be ignored.

2. The filename format can be <filename>+<file extension> (e.g. abc.exe), or just <filename> (e.g. abc). We recommend inputting full filenames (with file extensions).
3. The file system is not case sensitive. Filenames in uppercase or lowercase are identified as the same file. In addition, Whitespace character(s) cannot be included in the beginning and the end of a filename. Special characters (<, >, :, ", /, \, |, and ?) are not allowed.
4. Filenames that start or end with whitespace characters or contain only one asterisk (*) are not valid. To use wildcard search in a file exclusion list, you can insert an asterisk at any point within the filename.
5. The maximum full filename length (<filename>+<file extension>) is 128 characters (UTF-8, English characters). If user inputs any non-English characters, the full filename length will be shortened to 32 characters.

4. Click the [Edit File Exclusion List] button to edit a File Exclusion list.

File Exclusions

The system skips the file type scanning based on the file exclusion list.
To import file exclusions from a CSV file into the list, please download the CSV template and fill in full filenames and descriptions.

Total Number of File Exclusions: 0

 Edit File Exclusion List
  Download CSV Template

5. Click the [Import] button to import file exclusion items from your CSV file.

Edit File Exclusion List

Filenames in the file exclusion list are case-insensitive. Uppercase and lowercase filenames are recognized as the same.

 Create
  Import

Total Number of Items: 0 (Max: 1000) 1 / 1 < > 

☐	No.	Full Filename	Description
No data to display			

 Save
  Cancel

- **Method 2 (Steps 2-5): Manually Adding Full Filenames to the File Exclusion List**

2. Click the [Edit File Exclusion List] button to edit a File Exclusion list.

File Exclusions

The system skips the file type scanning based on the file exclusion list.
To import file exclusions from a CSV file into the list, please download the CSV template and fill in full filenames and descriptions.

Total Number of File Exclusions: 0

[Edit File Exclusion List](#)
[Download CSV Template](#)

3. Click the [Create] button to access the [Create File Exclusion] window.

Edit File Exclusion List

Filenames in the file exclusion list are case-insensitive. Uppercase and lowercase filenames are recognized as the same.

[+ Create](#)
[Import](#)

Total Number of Items: 0 (Max: 1000) 1 / 1

No.	Full Filename	Description
No data to display		

[Save](#)
[Cancel](#)

4. Input a full filename. The description is optional.

Create File Exclusion

Full Filename*

Note: Filenames that begin or end with whitespace characters or consist only of asterisks (*) are invalid. To perform a wildcard search in the file exclusion list, insert an asterisk at any position within a filename.

Description

[OK](#)
[Cancel](#)

5. Click [OK]. Repeat the above steps 3-4 if you want to add multiple file exclusion items.

6. After you have completed the file exclusion settings by either of the methods, click [Save].

7. Use the search bar to search for the information you need if necessary.

6.2.4 Domain Name Profiles

Domain Name profiles contain lists of domain names and their types (domains or subdomains) which you can apply to Domain Name Filter rules.

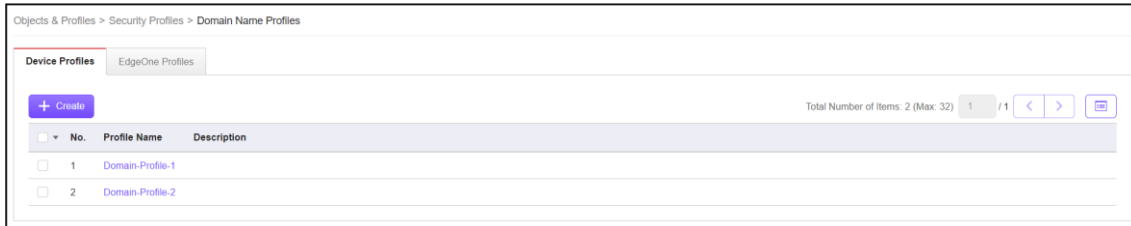


For more information about the filtering mechanisms of Domain Name Filter rules, see *Domain Name Filter Rules*.

6.2.4.1 Configuring Domain Name Profiles

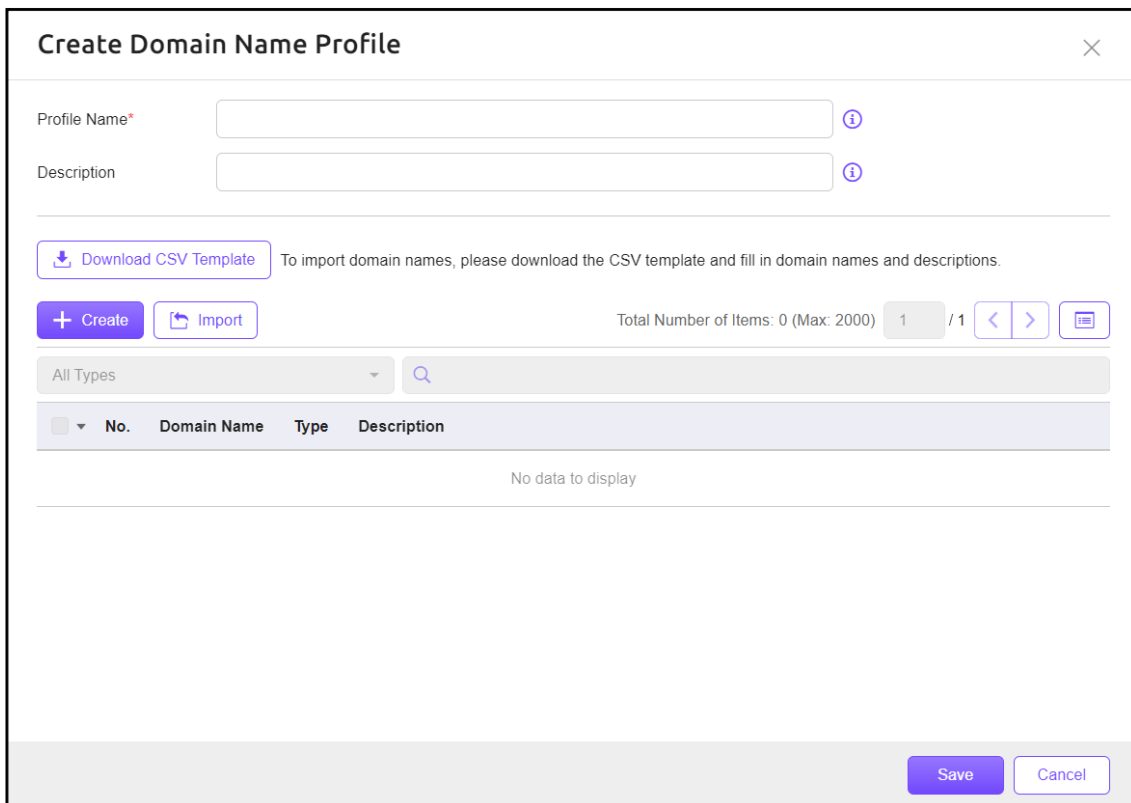
Procedure

1. Go to [Objects & Profiles] > [Security Profiles] > [Domain Name Profiles].



2. Do one of the following:

- Click [Create] to create a profile. The [Create Domain Name Profile] window will appear.
- Click a profile name to edit settings. The [Edit Domain Name Profile] window will appear.



3. Configure the following Domain Name profile settings.

Item	Description
Profile Name	Type a profile name for this Domain Name profile.
Description	(Optional) Type a description.

- **Method 1 (Steps 4-6): Importing a Domain Name List Using the Built-in CSV File Template**

4. Click the [Download CSV Template] button to download the CSV template.

5. Open the downloaded CSV file and input the domain names (mandatory), their types (mandatory), and descriptions (optional). For example:

Domain Name	Type	Description
myhost.abc.com	domain	Domain 1
www.def.com	subdomain	Domain 2

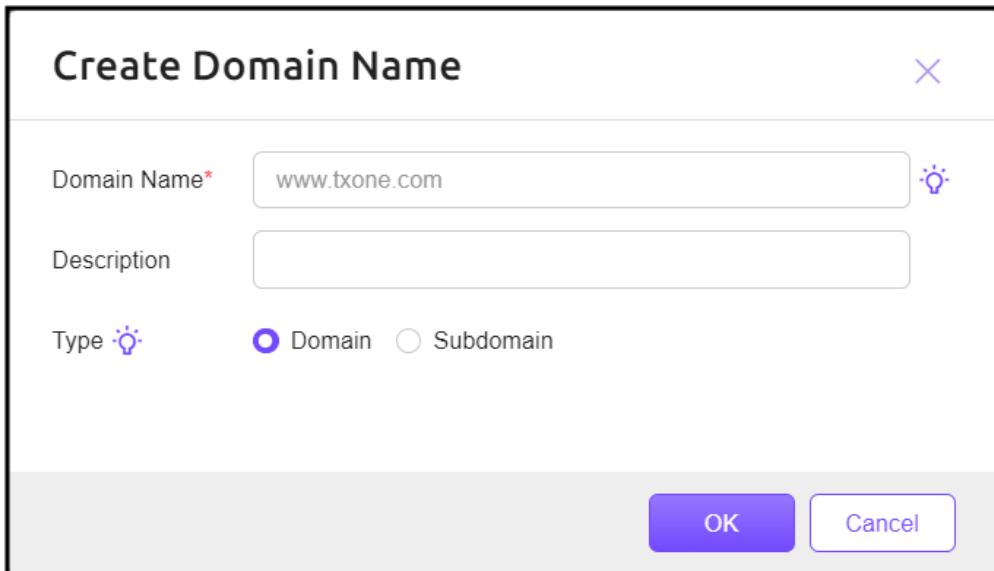


When inputting the domain name types, please input either “domain” or “subdomain” with all the letters in lowercase.

6. Click the [Import] button to import the edited CSV file.

- **Method 2 (Steps 4-6): Manually Adding Domain Names to the Domain Name List**

4. Click the [Create] button to open the [Create Domain Name] window.



The dialog box titled "Create Domain Name" has a close button (X) in the top right corner. It contains three input fields: "Domain Name*" with a lightbulb icon, "Description", and "Type" with a lightbulb icon. The "Domain Name*" field contains the text "www.txone.com". The "Type" field has two radio buttons: "Domain" (selected) and "Subdomain". At the bottom right, there are "OK" and "Cancel" buttons.

5. Input a domain name and select its type, either **Domain** or **Subdomain**. The description is optional.

6. Click [OK]. Repeat the above steps 3-4 if you want to add multiple domain names.

7. After you have completed the Domain Name profile settings by either of the methods, click [Save].

8. Use the search bar to search for the information you need if necessary.

6.3 Edge DPI Engine

Customize settings for the supported communication protocols used in your network.

- [Protocol Service Port Profiles](#)

6.3.1 Protocol Service Port Profiles

Protocol Service Port profiles contain customized ports or port ranges used by each OT protocol. By specifying which protocols use which ports, these settings improve the traffic examination accuracy of OT protocol filters, enabling the Edge device to precisely identify the protocols in use on each port.



- The [Protocol Service Port Profiles] feature is only applicable to Edge devices running firmware version 2.2 or later.
- You can customize ports or port ranges only for designated OT protocols. IT protocols such as SMB, HTTPS, and SSH are not supported.

6.3.1.1 Configuring Protocol Service Port Profiles

Procedure

1. Go to [Objects & Profiles] > [Edge DPI Engine] > [Protocol Service Port Profiles].

Objects & Profiles > Edge DPI Engine > Protocol Service Port Profiles

Custom TCP / UDP Ports

512/512

Remaining Quota

Total Number of Items: 38 1 / 2

Vertical	Protocol	Default Ports	Custom TCP / UDP Ports
Building Automation	BACnet	UDP: 47808	-
Factory Automation	BSAP	UDP: 1234-1235	-
Factory Automation	CC-Link IE Field Basic	UDP: 61450-61451	-
Factory Automation	CIP	TCP: 44818, UDP: 2222, 44818	-
Building Automation	CN/IP	UDP: 1628	-
Factory Automation	CODESYS	TCP: 11740-11743, UDP: 1740-1743	-
General OT Protocol	CoAP	UDP: 5683	-
Factory Automation	Emerson DeltaV	UDP: 18507	-
Factory Automation	Emerson ROC	TCP: 4000, UDP: 4000	-
General OT Protocol	Ether-S-Bus	UDP: 5050	-
General OT Protocol	EtherSIO	UDP: 6060	-
Factory Automation	FANUC FOCAS	TCP: 5001-65535	-
Factory Automation	FL-net	TCP: 55004, UDP: 55000-55004	-
Factory Automation	GE CMP	UDP: 7937	-
Factory Automation	GE EGD(UDP)	UDP: 18246	-

2. Click  to edit TCP/UDP ports of a protocol. The [Edit Custom TCP/UDP Port Settings] window will appear.

Edit Custom TCP / UDP Port Settings

Status ☒

Protocol BACnet

Default Ports UDP: 47808

Custom Port List (Max: 10)

You can configure up to 10 sets of ports or port ranges. Each set can be a single port or a port range within the range of 1 to 65535. The total number of ports across all sets cannot exceed the remaining quota (512).

No. 1* TCP Service Ports Single Port

3. Select **TCP** or **UDP** from the drop-down list.
4. Select **Single Port** or **Port Range** from the drop-down list.

5. Input a port number or a port range within the range of 1 to 65535 in the field.
6. Click the [Add Entry] button  to add another entry, or the [Remove Entry] button  to remove an entry.



You can create up to 10 sets of ports or port ranges for each protocol. The total number of ports for all protocols cannot exceed 512.

7. Click [Save].

7 Security

This chapter describes how to configure security features on your Edge device to protect your assets.

7.1 General Settings

Configure device operational settings.

- [Deny Action Setting](#)

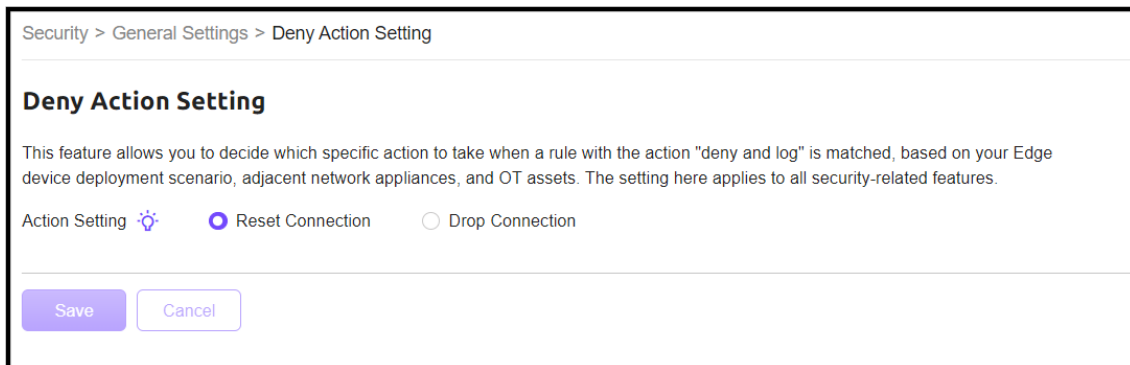
7.1.1 Deny Action Setting

On the [Deny Action Setting] page, you can decide which specific action to take when a rule with the action "deny and log" is matched, based on your Edge device deployment scenario, adjacent network appliances, and OT assets. The setting on this page applies to all security-related features across the entire console, except for [L2 Policy Enforcement].

7.1.1.1 Configuring Deny Action Setting

Procedure

1. Go to [Security] > [General Settings] > [Deny Action Setting].



2. Configure the deny action setting.

Deny Action	Description
Reset Connection	When the deny action is [Reset Connection], the Edge device sends a notification when it rejects any packets associated with an unwanted connection.
Drop Connection	When the deny action is [Drop Connection], the Edge device simply discards any packets without sending a notification.



The factory default deny action setting in EdgeFire version 2.1.5 or later has been changed from "reset connection" to "drop connection." Upgrading EdgeFire from version 1.x or 2.0.x to version 2.1.5 causes the device to use this new default setting "drop connection," as the old firmware doesn't support configuring the deny action setting.

3. Click [Save].

7.2 Security Policies

Create and manage various types of policy rules to safeguard your OT networks and assets.

- [Cyber Security](#)
- [L2 Policy Enforcement \(Only in Bridge Mode\)](#)
- [Policy Enforcement](#)
- [Policy Rule Auto-Learning](#)
- [Domain Name Filter Rules](#)
- [Suspicious Objects](#)

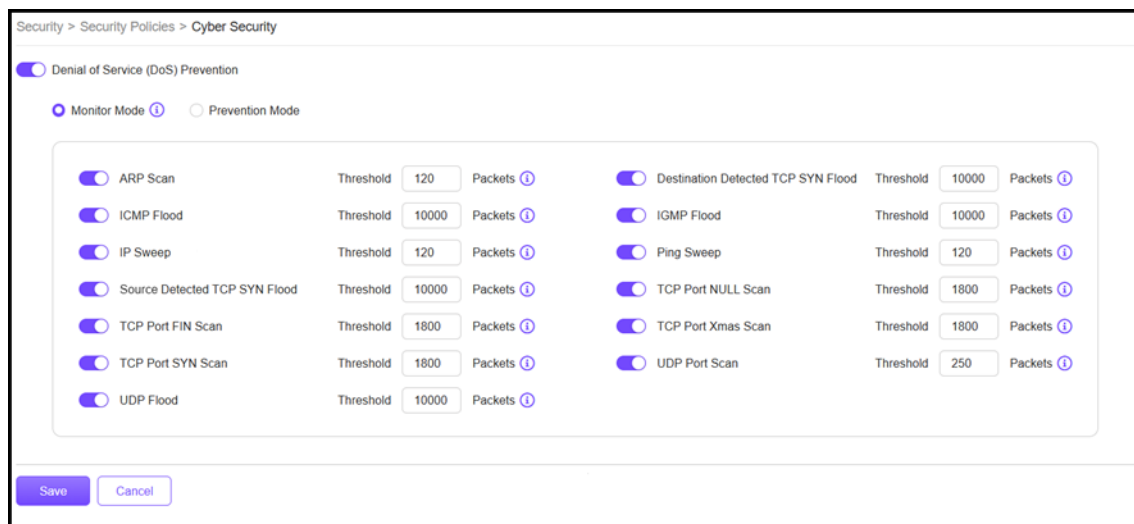
7.2.1 Cyber Security

The Cyber Security feature is used to protect OT assets against denial of service attacks. The signature rules of intrusion prevention are called “TXOne DPI (Deep Packet Inspection) Pattern.” This pattern is provided by TXOne and can be regularly updated through EdgeOne or through manual import via the device's web management console.

7.2.1.1 Configuring DoS Prevention for Asset Protection

Procedure

1. Go to [Security] > [Security Policies] > [Cyber Security].



2. Enable the [Denial of Service (DoS) Prevention] function by clicking the toggle.
3. Select a mode, **Monitor Mode** or **Prevention Mode**, for the function.
4. (Optional) Enable and configure the thresholds of the denial of service rules.

 The [Source Detected TCP SYN Flood] option is applicable to only Edge devices running firmware 2.2 or later.

5. Click [Save].

The following table summarizes the settings:

DoS Mode	Action Performed
Monitor Mode	• Detect and monitor abnormal protocol access attempts to the OT assets, without blocking network attacks.

DoS Mode	Action Performed
	Generate logs.
Prevention Mode	- Block abnormal protocol access attempts to OT assets. - Generate logs.



When the Edge device is managed by EdgeOne, the Cyber Security settings here are synchronized with EdgeOne.

7.2.2 L2 Policy Enforcement (Only in Bridge Mode)

The Edge solution is an OT-native network security solution, specifically designed for legacy production tools and environments without an IP-based production network. Our solution provides L2 Policy Enforcement feature, allowing you to set up network access policies for Layer 2 network protection by defining MAC address-related configurations while still enabling security policy configurations. By applying user-defined Layer 2 access rules with various traffic filtering conditions, the system determines whether specific network activities should be allowed or blocked based on the adopted rule settings.



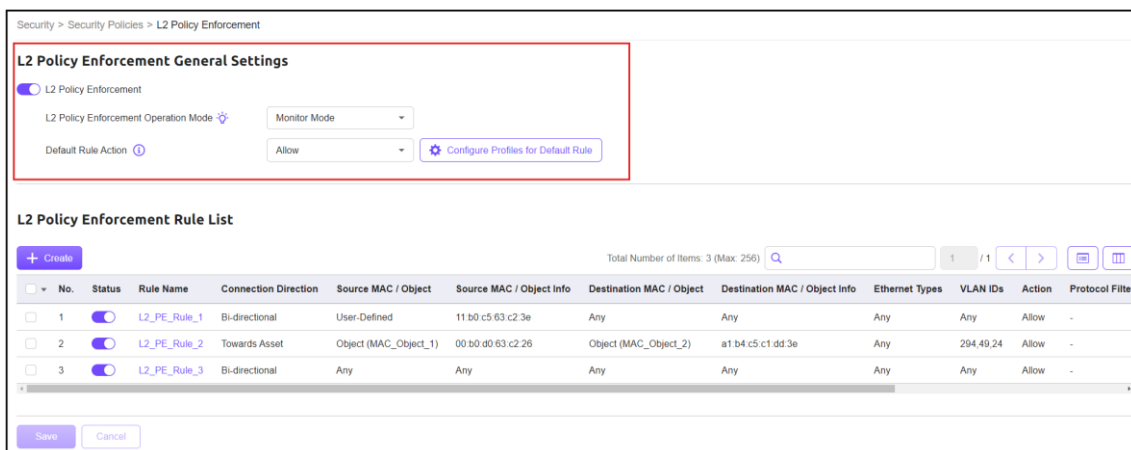
Comparing L2 and L3 Policy Enforcement Rules in Edge Device

- L2 Policy Enforcement:** L2 policies are designed to control non-IP traffic—specifically, Ethernet frames that do not encapsulate IPv4 or IPv6 packets. These are typically used in OT environments where communication is based on MAC addresses, often due to legacy protocols. For example, blocking a specific MAC address without reference to higher-layer protocols.
- L3 Policy Enforcement:** L3 policies apply to all IPv4 traffic, including ICMP, TCP, and UDP. Although IPv4 packets (like ping/ICMP) are carried within Ethernet frames, they are handled by L3 rules, not L2. Therefore, any enforcement for such traffic must be done via L3 policies. For more information about L3 Policy Enforcement, see [Policy Enforcement](#).

7.2.2.1 Configuring L2 Policy Enforcement General Settings

Procedure

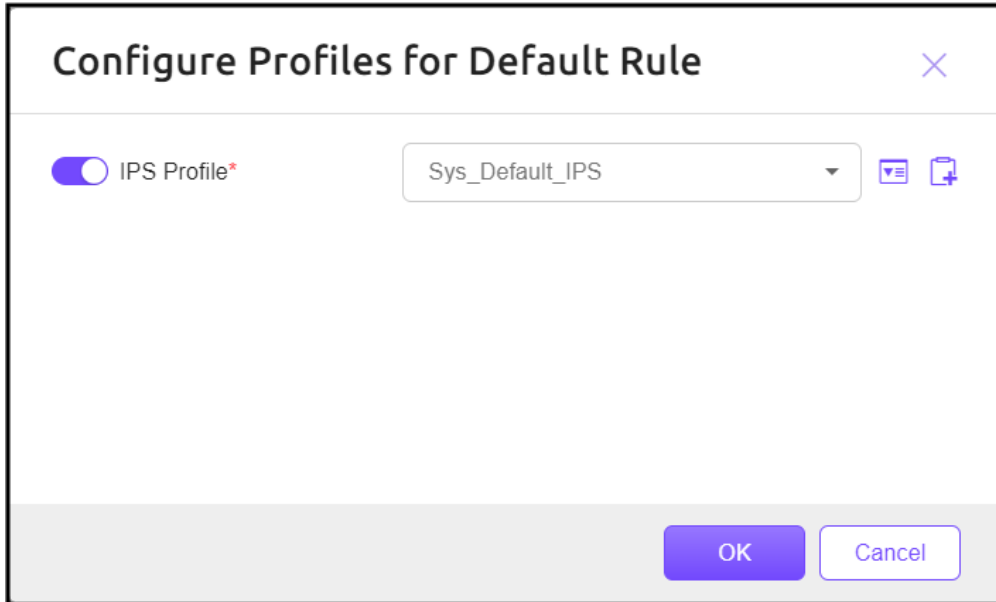
- Go to [Security] > [Security Policies] > [L2 Policy Enforcement].



The screenshot shows the 'Security > Security Policies > L2 Policy Enforcement' configuration page. The 'L2 Policy Enforcement General Settings' section is highlighted with a red box. It includes a toggle for 'L2 Policy Enforcement' (currently on), a dropdown for 'L2 Policy Enforcement Operation Mode' (set to 'Monitor Mode'), and a dropdown for 'Default Rule Action' (set to 'Allow'). There is a 'Configure Profiles for Default Rule' button. Below this is the 'L2 Policy Enforcement Rule List' table, which shows three rules: L2_PE_Rule_1, L2_PE_Rule_2, and L2_PE_Rule_3. The table has columns for No., Status, Rule Name, Connection Direction, Source MAC / Object, Source MAC / Object Info, Destination MAC / Object, Destination MAC / Object Info, Ethernet Types, VLAN IDs, Action, and Protocol Filter. The 'Save' and 'Cancel' buttons are at the bottom.

- Enable or disable the L2 Policy Enforcement function by clicking the toggle.

3. From the [L2 Policy Enforcement Operation Mode] drop-down list, select a mode, **Monitor Mode** or **Prevention Mode**, for the function.
4. From the [Default Rule Action] drop-down list, select a default action, **Allow**, **Allow and Log** or **Deny and Log**, for when no L2 Policy Enforcement rule is matched.
5. (Optional) If the default rule action is **Allow** or **Allow and Log**, click the [Configure Profiles for Default Rule] button to specify which IPS profile to use for traffic examination when no L2 Policy Enforcement rule is matched.



The dialog box titled "Configure Profiles for Default Rule" features a close button (X) in the top right corner. It contains a toggle switch labeled "IPS Profile*" which is currently turned on. To the right of the toggle is a dropdown menu showing "Sys_Default_IPS". Further right are two icons: a document with a checkmark and a document with a plus sign. At the bottom right, there are two buttons: "OK" and "Cancel".

- a. Click the [IPS Profile] toggle to enable the setting.
- b. Select a pre-defined IPS profile from the drop-down list or use the system default IPS profile (Sys_Default_IPS).
- c. (Optional) To view/edit detailed IPS profile settings, click the  button.
- d. (Optional) To create an IPS profile, click the  button.
- e. Click [OK].
6. Click [Save].

The following table summarizes the settings:

L2 Policy Enforcement Operation Mode	Action Performed
Monitor Mode	- Detect and monitor abnormal protocol access attempts to the OT assets, without blocking network attacks. - Generate logs. (If the default rule action is [Allow and Log] or [Deny and Log].)
Prevention Mode	- Block abnormal protocol access attempts to OT assets. - Generate logs. (If the default rule action is [Allow and Log] or [Deny and Log].)

7.2.2.2 Configuring L2 Policy Enforcement Rules

Procedure

1. Configure the required objects.

- MAC Objects



For more information, see [Configuring MAC Objects](#).

- Protocol Filter Profiles



For more information, see [Configuring Protocol Filter Profiles](#).

- IPS Profiles



For more information, see [Configuring IPS Profiles](#).

2. Go to [Security] > [Security Policies] > [L2 Policy Enforcement].

Security > Security Policies > L2 Policy Enforcement

L2 Policy Enforcement General Settings

☒ L2 Policy Enforcement

L2 Policy Enforcement Operation Mode  Monitor Mode

Default Rule Action  Allow [Configure Profiles for Default Rule](#)

L2 Policy Enforcement Rule List

[+ Create](#) Total Number of Items: 3 (Max: 256) 1 / 1    

No.	Status	Rule Name	Connection Direction	Source MAC / Object	Source MAC / Object Info	Destination MAC / Object	Destination MAC / Object Info	Ethernet Types	VLAN IDs	Action	Protocol Filter
1	☒	L2_PE_Rule_1	Bi-directional	User-Defined	11 b0 c5 63 c2 3e	Any	Any	Any	Any	Allow	-
2	☒	L2_PE_Rule_2	Towards Asset	Object (MAC_Object_1)	00 b0 d0 63 c2 26	Object (MAC_Object_2)	a1 b4 c5 c1 dd 3e	Any	294,49,24	Allow	-
3	☒	L2_PE_Rule_3	Bi-directional	Any	Any	Any	Any	Any	Any	Allow	-

[Save](#) [Cancel](#)

3. Do one of the following.

- Click [Create] to create a new L2 Policy Enforcement rule. The [Create L2 Policy Enforcement Rule] window will appear.
- Click a rule name to edit settings. The [Edit L2 Policy Enforcement Rule] window will appear.

Create L2 Policy Enforcement Rule

Status

Rule Name*

Description

Basic Filters

Connection Direction

Source MAC / Object

Destination MAC / Object

Ethernet Types

VLAN IDs

Action

Advanced Filters

Protocol Filter Profile

Action

IPS Profile

OK

Cancel

4. Configure the following settings.

Item	Description
Status	Enable the operational status of this L2 Policy Enforcement rule by clicking the toggle.
Rule Name	Type a rule name for this L2 Policy Enforcement rule.
Description	(Optional) Type a description.

- Basic Filter

Item	Description
Connection Direction	Select one of the following direction: Bi-directional: The Edge device doesn't consider the direction of a detected connection as a rule matching criterion in the basic filter. Towards Asset: The Edge device considers the direction of a detected connection as a rule matching criterion in the basic filter. When the connection direction is set to Towards Asset and the detected connection is initiated from a non-asset towards an asset, this criterion is met, and the Edge device takes the specified rule action accordingly. If the detected connection does not meet this criterion, the Edge

Item	Description
	device proceeds to check the next specified criterion in the basic filter. • Away from Asset: The Edge device considers the direction of a detected connection as a rule matching criterion in the basic filter. When the connection direction is set to Away from Asset and the detected connection is initiated from an asset towards a non-asset, this criterion is met, and the Edge device takes the specified rule action accordingly. If the detected connection does not meet this criterion, the Edge device proceeds to check the next specified criterion in the basic filter.
Source MAC/Object	Select one of the following for the source MAC addresses: • Any • MAC Address • MAC Object  If you select [MAC Object], you can select a MAC object that has been created beforehand on the [MAC Objects] page or click the  button to create a new MAC object.
Destination MAC/Object	Select one of the following for the destination MAC addresses: • Any • MAC Address • MAC Object
Ethernet Types	Select either one of the following for the Ethernet types: • Any • Custom: Specify at least one Ethernet type.  A maximum of five Ethernet types can be applied to one policy rule. For more information about Ethernet type, visit IANA (Internet Assigned Number Authority).
VLAN IDs	Select either one of the following for the VLAN IDs: • Any • Custom: Specify at least one VLAN ID for this protocol.  A maximum of five VLAN IDs can be applied to one policy rule.
Action	Select one of the following actions: • Allow: Select this option to allow network traffic that matches this rule. • Allow and Log: Select this option to allow network traffic that matches this rule and output a log.

Item	Description
	• Deny and Log: Select this option to block network traffic that matches this rule and output a log.

- **Advanced Filters:** If you select [Allow] or [Allow and Log] for [Action], you can set advanced filters (Protocol Filter profile and IPS profile) to conduct further actions.

Item	Description
Protocol Filter Profile	Select a Protocol Filter profile you have defined beforehand or click the  button to create a new profile. From the [Action] drop-down menu, select whether to allow or deny network traffic that matches the protocol filter.
IPS Profile	Select an IPS profile you have defined beforehand or click the  button to create a new profile.

5. Click [OK].

6. Click [Save] on the [L2 Policy Enforcement] page.



You may click the  button in the upper right corner of the table to show/hide certain type of information and adjust the column order. For more information, see [Viewing in Table View Mode](#).

7.2.2.3 Managing L2 Policy Enforcement Rules

The following table lists the common tasks for managing L2 Policy Enforcement rules.

Task	Action
To create an L2 Policy Enforcement rule	Click the [Create] button to create a new L2 Policy Enforcement rule.
To delete an L2 Policy Enforcement rule	Click the checkbox of a rule and click the [Delete] button.
To duplicate an L2 Policy Enforcement rule	Click the checkbox of a rule and click the [Duplicate] button.
To edit an L2 Policy Enforcement rule	Click the name of a rule, and an [Edit L2 Policy Enforcement Rule] window will appear.
To change the priority of an L2 Policy Enforcement rule	Click the checkbox of a rule, click the [Change Priority] button, and specify a new priority for this rule.



When more than one L2 Policy Enforcement rule is matched, the device takes action based on the rule with the highest priority and ignores the rest of the rules. The rules listed in the table on the [L2 Policy Enforcement] page are ordered by priority, with the highest priority rule listed on the first row of the table.

7.2.3 Policy Enforcement

Policy Enforcement allows you to define access rules with various conditions, including the well-known OT protocols for your own industry. Under which conditions the activity will be allowed or blocked is based on the rule settings in the adopted rules.

There are two types of Policy Enforcement rule lists for this device:

- **Device Rule List:** Contains the policy rules created and configured by users with the Edge device local account privileges.
- **EdgeOne Rule List:** Contains the policy rules synced from EdgeOne. Users with EdgeOne local account privileges can apply them to the devices managed by EdgeOne. Users with the Edge device local account privileges can only view these EdgeOne policy rules.



The device rules take higher priorities than the EdgeOne rules.



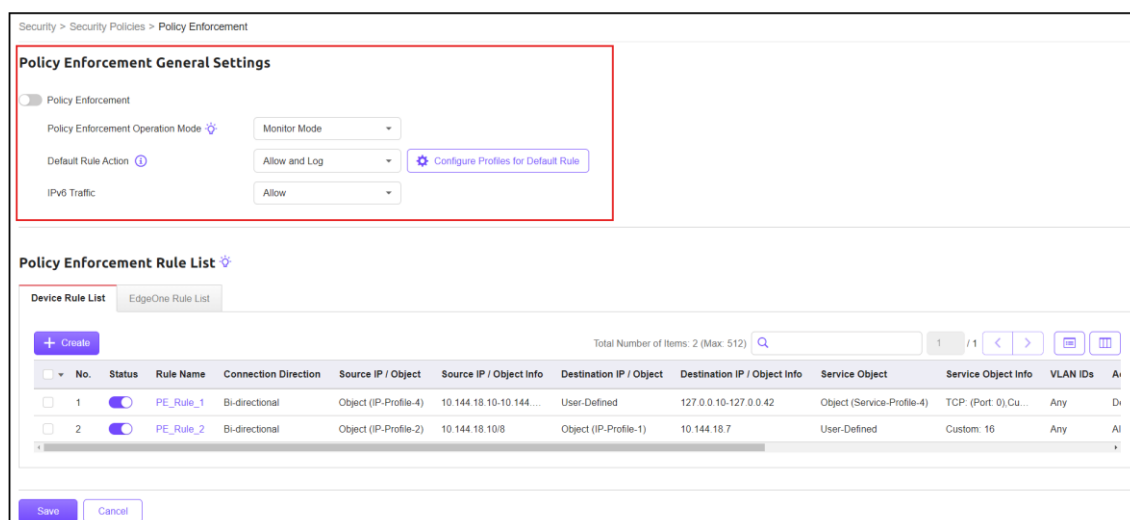
Comparing L2 and L3 Policy Enforcement Rules in Edge Device

- **L2 Policy Enforcement:** L2 policies are designed to control non-IP traffic—specifically, Ethernet frames that do not encapsulate IPv4 or IPv6 packets. These are typically used in OT environments where communication is based on MAC addresses, often due to legacy protocols. For example, blocking a specific MAC address without reference to higher-layer protocols. For more information about L2 Policy Enforcement, see [L2 Policy Enforcement \(Only in Bridge Mode\)](#).
- **L3 Policy Enforcement:** L3 policies apply to all IPv4 traffic, including ICMP, TCP, and UDP. Although IPv4 packets (like ping/ICMP) are carried within Ethernet frames, they are handled by L3 rules, not L2. Therefore, any enforcement for such traffic must be done via L3 policies.

7.2.3.1 Configuring Policy Enforcement General Settings

Procedure

1. Go to [Security] > [Security Policies] > [Policy Enforcement].

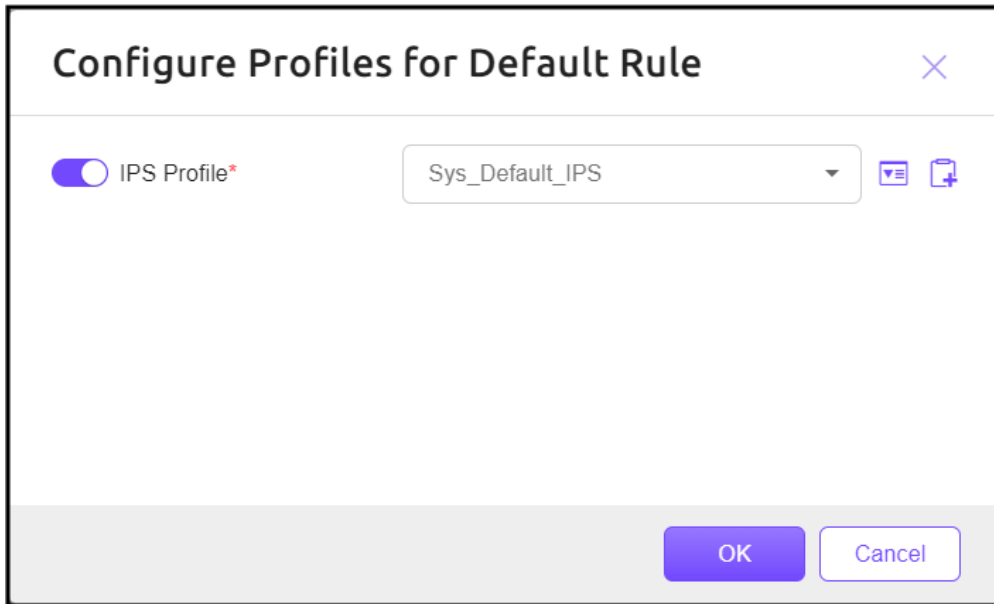


The screenshot shows the 'Policy Enforcement General Settings' and 'Policy Enforcement Rule List' configuration page. The 'Policy Enforcement General Settings' section includes a toggle for 'Policy Enforcement', a dropdown for 'Policy Enforcement Operation Mode' (set to 'Monitor Mode'), a dropdown for 'Default Rule Action' (set to 'Allow and Log'), and a dropdown for 'IPv6 Traffic' (set to 'Allow'). The 'Policy Enforcement Rule List' section shows a table with two rules, 'PE_Rule_1' and 'PE_Rule_2', both in 'On' status. The table columns include No., Status, Rule Name, Connection Direction, Source IP / Object, Source IP / Object Info, Destination IP / Object, Destination IP / Object Info, Service Object, Service Object Info, VLAN IDs, and Action.

No.	Status	Rule Name	Connection Direction	Source IP / Object	Source IP / Object Info	Destination IP / Object	Destination IP / Object Info	Service Object	Service Object Info	VLAN IDs	Action
1	On	PE_Rule_1	Bi-directional	Object (IP-Profile-4)	10.144.18.10-10.144.18.10	User-Defined	127.0.0.10-127.0.0.42	Object (Service-Profile-4)	TCP (Port: 0), Cu...	Any	Deny
2	On	PE_Rule_2	Bi-directional	Object (IP-Profile-2)	10.144.18.10/8	Object (IP-Profile-1)	10.144.18.7	User-Defined	Custom: 16	Any	Allow

2. Enable or disable the Policy Enforcement function by clicking the toggle.
3. From the [Policy Enforcement Operation Mode] drop-down list, select a mode, **Monitor Mode** or **Prevention Mode**, for the function.
4. From the [Default Rule Action] drop-down list, select a default action, **Allow**, **Allow and Log** or **Deny and Log**, for when no Policy Enforcement rule is matched.

5. (Optional) If the default rule action is **Allow** or **Allow and Log**, click the [Configure Profiles for Default Rule] button to specify which IPS profile to use for traffic examination when no Policy Enforcement rule is matched.



The dialog box titled "Configure Profiles for Default Rule" has a close button (X) in the top right corner. It contains a toggle switch labeled "IPS Profile*" which is currently turned on. To the right of the toggle is a dropdown menu showing "Sys_Default_IPS". There are two icons to the right of the dropdown: a list icon and a plus icon. At the bottom right, there are two buttons: "OK" and "Cancel".

- Click the [IPS Profile] toggle to enable the setting.
 - Select a pre-defined IPS profile from the drop-down list or use the system default IPS profile (Sys_Default_IPS).
 - (Optional) To view/edit detailed IPS profile settings, click the  button.
 - (Optional) To create an IPS profile, click the  button.
 - Click [OK].
6. From the [IPv6 Traffic] drop-down list, select an action, **Allow** or **Deny**, for the device to perform when it detects IPv6 packets.

 The [IPv6 Traffic] setting item is available only when EdgeFire is in Bridge mode.

7. Click [Save].

The following table summarizes the settings:

Policy Enforcement Operation Mode	Action Performed
Monitor Mode	- Detect and monitor abnormal protocol access attempts to the OT assets, without blocking network attacks. - Generate logs. (If the default rule action is [Allow and Log] or [Deny and Log].)
Prevention Mode	- Block abnormal protocol access attempts to OT assets. - Generate logs. (If the default rule action is [Allow and Log] or [Deny and Log].)

7.2.3.2 Configuring Policy Enforcement Rules (In Gateway Mode)

Procedure

- Configure the required objects and profiles.

- IP Objects

For more information, see [Configuring IP Objects](#).

- Service Objects

For more information, see [Configuring Service Objects](#).

- Protocol Filter Profiles

For more information, see [Configuring Protocol Filter Profiles](#).

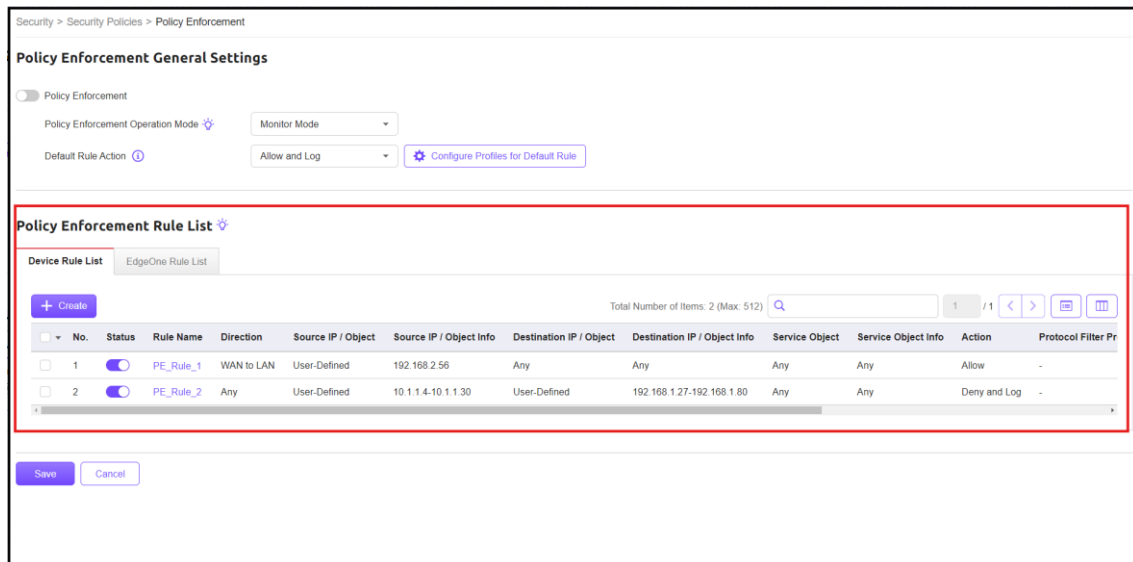
- IPS Profiles

For more information, see [Configuring IPS Profiles](#).

- File Filter Profiles

For more information, see [Configuring File Filter Profiles](#).

2. Go to [Security] > [Security Policies] > [Policy Enforcement].



Security > Security Policies > Policy Enforcement

Policy Enforcement General Settings

☐ Policy Enforcement

Policy Enforcement Operation Mode: Monitor Mode

Default Rule Action: Allow and Log [Configure Profiles for Default Rule](#)

Policy Enforcement Rule List

Device Rule List | EdgeOne Rule List

[+ Create](#) Total Number of Items: 2 (Max: 512) 1 / 1

No.	Status	Rule Name	Direction	Source IP / Object	Source IP / Object Info	Destination IP / Object	Destination IP / Object Info	Service Object	Service Object Info	Action	Protocol Filter Pr
1	☒	PE_Rule_1	WAN to LAN	User-Defined	192.168.2.56	Any	Any	Any	Any	Allow	-
2	☒	PE_Rule_2	Any	User-Defined	10.1.1.4-10.1.1.30	User-Defined	192.168.1.27-192.168.1.80	Any	Any	Deny and Log	-

[Save](#) [Cancel](#)

3. Do one of the following.

- Click [Create] to create a new Policy Enforcement rule. The [Create Policy Enforcement Rule] window will appear.
- Click a rule name to edit settings. The [Edit Policy Enforcement Rule] window will appear.

Create Policy Enforcement Rule

Status ☒

Rule Name*

Description

Basic Filters

Direction

Any

Source IP / Object

Any

Destination IP / Object

Any

Service Object

Any

Action

Allow

Advanced Filters

Protocol Filter Profile

- Select -

Action

Allow

IPS Profile

- Select -

File Filter Profile

- Select -

OK

Cancel

4. Configure the following settings.

Item	Description
Status	Enable the operational status of this Policy Enforcement rule by clicking the toggle.
Rule Name	Type a rule name for this Policy Enforcement rule.
Description	(Optional) Type a description.

- Basic Filters

Item	Description
Direction	Select one of the following for the network traffic direction: - Any - WAN to LAN - LAN to WAN - WAN to DMZ - DMZ to WAN - LAN to DMZ - DMZ to LAN - LAN to LAN

Item	Description
	• VPN to WAN • LAN to VPN • VPN to LAN • DMZ to VPN • VPN to DMZ • VPN to VPN  1. The network interface in the drop-down menu does not specify which exact network interface, but two or more network interfaces of a kind from the broad view. For example, if you select [WAN to LAN], then the policy enforcement rule will be effective on the traffic from WAN1 interface to LAN1 interface or WAN1 interface to LAN2 interface. If you select [LAN to LAN], then the policy enforcement rule will be effective on the traffic from LAN1 interface to LAN2 interface or LAN2 interface to LAN1 interface. 2. If you select [Any], then the policy enforcement rule will be effective on the traffic from all network interfaces.
Source IP/Object	Select one of the following for the source IP addresses: • Any • Single IP • IP Range • IP Subnet • IP Object  If you select [IP Object], you can select an IP object that has been created beforehand on the [IP Objects] page or click the  button to create a new IP object.
Destination IP/Object	Select one of the following for the destination IP addresses: • Any • Single IP • IP Range • IP Subnet • IP Object
Service Object	Select one of the following for the layer 4 criteria: • Any • TCP: You can further specify the port range for this protocol. • UDP: You can further specify the port range for this protocol.

Item	Description
	- ICMP: You can further specify the Type and Code for this protocol. - Custom: You can further specify the protocol number for this protocol. The term “protocol number” refers to the one defined in the Internet protocol suite. - Service Object  If you select [Service Object], you can select a service object that has been created beforehand on the [Service Objects] page or click the  button to create a new service object.
Action	Select one of the following action: Allow: Select this option to allow network traffic that matches this rule. Allow and Log: Select this option to allow network traffic that matches this rule and output a log. Deny and Log: Select this option to block network traffic that matches this rule and output a log.

- Advanced Filters: If you select [Allow] or [Allow and Log] for [Action], you can set advanced filters (Protocol Filter profile, IPS profile, and File Filter profile) to conduct further actions.

Item	Description
Protocol Filter Profile	Select a Protocol Filter profile you have defined beforehand or click the  button to create a new profile. From the [Action] drop-down menu, select whether to allow or deny network traffic that matches the protocol filter.
IPS Profile	Select an IPS profile you have defined beforehand or click the  button to create a new profile.
File Filter Profile	Select a File Filter profile you have defined beforehand or click the  button to create a new profile.

5. Click [OK].

6. Click [Save] on the [Policy Enforcement] page.

 You may click the  button in the upper right corner of the table to show/hide certain type of information and adjust the column order. For more information, see [Viewing in Table View Mode](#).

7.2.3.3 Configuring Policy Enforcement Rules (In Bridge Mode)

Procedure

1. Configure the required objects and profiles.

- IP Objects

 For more information, see [Configuring IP Objects](#).

- Service Objects

 For more information, see [Configuring Service Objects](#).

- Protocol Filter Profiles

For more information, see [Configuring Protocol Filter Profiles](#).

- IPS Profiles

For more information, see [Configuring IPS Profiles](#).

- File Filter Profiles

For more information, see [Configuring File Filter Profiles](#).

2. Go to [Security] > [Security Policies] > [Policy Enforcement].

Security > Security Policies > Policy Enforcement

Policy Enforcement General Settings

☐ Policy Enforcement

Policy Enforcement Operation Mode: Monitor Mode

Default Rule Action: Allow and Log [Configure Profiles for Default Rule](#)

IPv6 Traffic: Allow

Policy Enforcement Rule List

Device Rule List | EdgeOne Rule List

[+ Create](#) Total Number of Items: 2 (Max: 512) 1 / 1

No.	Status	Rule Name	Connection Direction	Source IP / Object	Source IP / Object Info	Destination IP / Object	Destination IP / Object Info	Service Object	Service Object Info	VLAN IDs	Action
1	☒	PE_Rule_1	Bi-directional	Object (IP-Profile-4)	10.144.18.10-10.144.18.10	User-Defined	127.0.0.10-127.0.0.42	Object (Service-Profile-4)	TCP: (Port: 0) Cu...	Any	Deny
2	☒	PE_Rule_2	Bi-directional	Object (IP-Profile-2)	10.144.18.10/8	Object (IP-Profile-1)	10.144.18.7	User-Defined	Custom: 16	Any	Allow

[Save](#) [Cancel](#)

3. Do one of the following.

- Click [Create] to create a new Policy Enforcement rule. The [Create Policy Enforcement Rule] window will appear.
- Click a rule name to edit settings. The [Edit Policy Enforcement Rule] window will appear.

Create Policy Enforcement Rule

Status

Rule Name*

Description

Basic Filters

Connection Direction

Source IP / Object

Destination IP / Object

Service Object

VLAN IDs

Action

Advanced Filters

Protocol Filter Profile

Action

IPS Profile

File Filter Profile

OK

Cancel

4. Configure the following settings.

Item	Description
Status	Enable the operational status of this Policy Enforcement rule by clicking the toggle.
Rule Name	Type a rule name for this Policy Enforcement rule.
Description	(Optional) Type a description.

- Basic Filters

Item	Description
Connection Direction	Select one of the following direction: Bi-directional: The Edge device doesn't consider the direction of a detected connection as a rule matching criterion in the basic filter. Towards Asset: The Edge device considers the direction of a detected connection as a rule matching criterion in the basic filter. When the connection direction is set to Towards Asset and the detected connection is initiated from a non-asset towards an asset, this criterion is met, and the Edge device takes the specified rule action accordingly. If the detected connection does not meet this

Item	Description
	criterion, the Edge device proceeds to check the next specified criterion in the basic filter. • Away from Asset: The Edge device considers the direction of a detected connection as a rule matching criterion in the basic filter. When the connection direction is set to Away from Asset and the detected connection is initiated from an asset towards a non-asset, this criterion is met, and the Edge device takes the specified rule action accordingly. If the detected connection does not meet this criterion, the Edge device proceeds to check the next specified criterion in the basic filter.
Source IP/Object	Select one of the following for the source IP addresses: • Any • Single IP • IP Range • IP Subnet • IP Object  If you select [IP Object], you can select an IP object that has been created beforehand on the [IP Objects] page or click the  button to create a new IP object.
Destination IP/Object	Select one of the following for the destination IP addresses: • Any • Single IP • IP Range • IP Subnet • IP Object
Service Object	Select one of the following for the layer 4 criteria: • Any • TCP: You can further specify the port range for this protocol. • UDP: You can further specify the port range for this protocol. • ICMP: You can further specify the Type and Code for this protocol. • Custom: You can further specify the protocol number for this protocol. The term “protocol number” refers to the one defined in the Internet protocol suite. • Service Object  If you select [Service Object], you can select a service object that has been created beforehand on the [Service

Item	Description
	Objects] page or click the  button to create a new service object.
VLAN IDs	Select either one of the following for the VLAN IDs: - Any - Custom: Specify at least one VLAN ID for this protocol.  A maximum of five VLAN IDs can be applied to one policy rule.
Action	Select one of the following actions: Allow: Select this option to allow network traffic that matches this rule. Allow and Log: Select this option to allow network traffic that matches this rule and output a log. Deny and Log: Select this option to block network traffic that matches this rule and output a log.

- Advanced Filters: If you select [Allow] or [Allow and Log] for [Action], you can set advanced filters (Protocol Filter profile, IPS profile, and File Filter profile) to conduct further actions.

Item	Description
Protocol Filter Profile	Select a Protocol Filter profile you have defined beforehand or click the  button to create a new profile. From the [Action] drop-down menu, select whether to allow or deny network traffic that matches the protocol filter.
IPS Profile	Select an IPS profile you have defined beforehand or click the  button to create a new profile.
File Filter Profile	Select a File Filter profile you have defined beforehand or click the  button to create a new profile.

5. Click [OK].

6. Click [Save] on the [Policy Enforcement] page.

 You may click the  button in the upper right corner of the table to show/hide certain type of information and adjust the column order. For more information, see [Viewing in Table View Mode](#).

7.2.3.4 Managing Policy Enforcement Rules

The following tables list the common tasks for managing Policy Enforcement rules.

- Device Rule List:**

Task	Action
To create a Policy Enforcement rule	Click the [Create] button to create a new Policy Enforcement rule.
To delete a Policy Enforcement rule	Click the checkbox of a rule and click the [Delete] button.
To duplicate a Policy Enforcement rule	Click the checkbox of a rule and click the [Duplicate] button.
To edit a Policy Enforcement rule	Click the name of a rule, and an [Edit Policy Enforcement Rule] window will appear.
To change the priority of a Policy Enforcement rule	Click the checkbox of a rule, click the [Change Priority] button, and specify a new priority for this rule.



When more than one Policy Enforcement rule is matched, the device takes action based on the rule with the highest priority and ignores the rest of the rules. The rules listed in the table on the [Policy Enforcement] page are ordered by priority, with the highest priority rule listed on the first row of the table.

- **EdgeOne Rule List:**

Task	Description
N/A	The Policy Enforcement rules on the EdgeOne rule list are view-only.

7.2.4 Policy Rule Auto-Learning

Policy Rule Auto-Learning allows you to create a learning task within a scheduled period of time to collect information about legitimate network traffic, and then use it to generate a trust list of Policy Enforcement baseline rules for your network environment.



1. Before you create a policy rule auto-learning task, ensure that IPS profiles are properly configured. For more information about IPS profiles, see [IPS Profiles](#).
2. L2 policy rule auto-learning is not supported when EdgeFire is in Gateway mode.

7.2.4.1 Configuring Policy Rule Auto-Learning Tasks

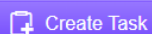
Procedure

1. Go to [Security] > [Security Policies] > [Policy Rule Auto-Learning].

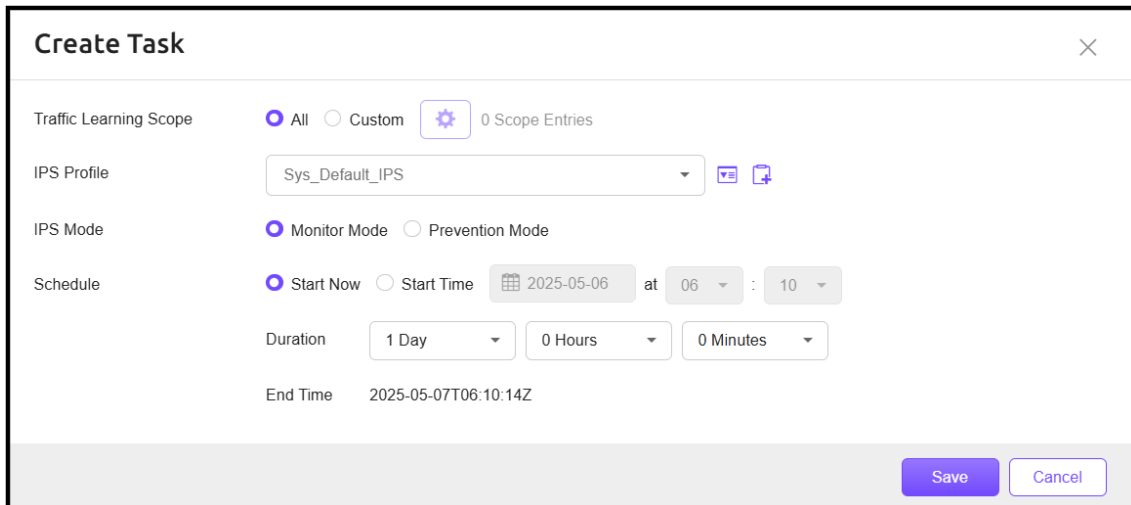
Security > Security Policies > Policy Rule Auto-Learning

Policy Rule Auto-Learning Task

This feature allows the Edge device to learn the traffic passing through it and automatically build up Policy Enforcement and L2 Policy Enforcement rules. All generated rules are allowlists and are added to the device rule list on the [Policy Enforcement] and [L2 Policy Enforcement] pages. We recommend that you configure the NTP settings to sync the system time before you create an auto-learning task. During the task, the Policy Enforcement and L2 Policy Enforcement default rule actions are skipped.

 Create Task

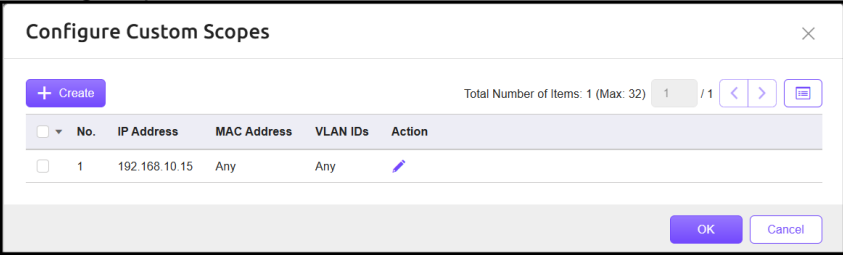
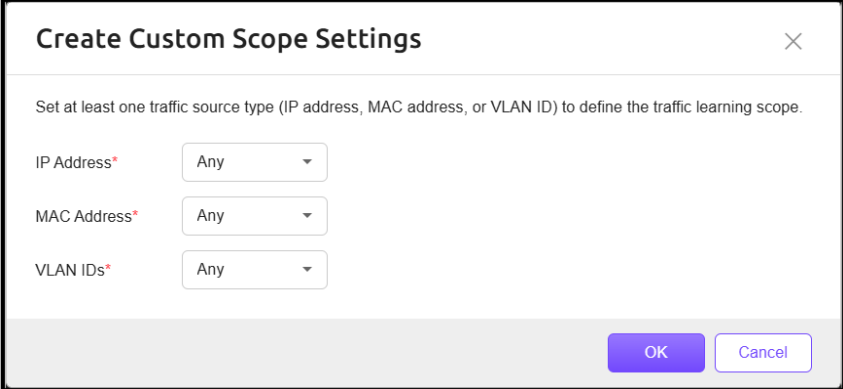
2. Click the [Create Task] button to create an auto-learning task. The [Create Task] window will appear.



The 'Create Task' window is shown with the following settings:

- Traffic Learning Scope:** All (selected), Custom (disabled), 0 Scope Entries
- IPS Profile:** Sys_Default_IPS
- IPS Mode:** Monitor Mode (selected), Prevention Mode
- Schedule:** Start Now (selected), Start Time (disabled), 2025-05-06 at 06 : 10
- Duration:** 1 Day, 0 Hours, 0 Minutes
- End Time:** 2025-05-07T06:10:14Z
- Buttons:** Save, Cancel

3. Configure the following settings.

Item	Description
Traffic Learning Scope	Select All to perform the learning task based on all traffic flowing through the device, or Custom to define the specific types of traffic the device should use for learning. If you select Custom, click the  button to further configure traffic learning scopes.  1. On the [Configure Custom Scopes] window, do one of the following. - Click [Create] to create a scope entry. The [Create Custom Scope Settings] window will appear. - Click  to edit scope settings. The [Edit Custom Scope Settings] window will appear.  2. Set at least one traffic source type, including IP address, MAC address, and VLAN ID to define the traffic learning scope.

Item	Description
	- IP Address: Select one of the following for the source IP addresses: a. Any b. Single IP: Specify an IP address. c. IP Range: Specify an IP range. d. IP Subnet: Specify an IP subnet. - MAC Address: Select either one of the following for the source MAC addresses: - o Any - o Custom: Specify a MAC address. - VLAN IDs: Select either one of the following for the source VLAN IDs: - o Any - o Custom: Specify one to five VLAN IDs.  VLAN IDs cannot be configured when EdgeFire is in Gateway mode. 3. Click [OK] on the [Create Custom Scope Settings] or [Edit Custom Scope Settings] window, then click [OK] on the [Configure Custom Scopes] window.
IPS Profile	Select an IPS profile to apply to the learning task.
IPS Mode	Select Monitor Mode or Prevention Mode to apply to the learning task.
Schedule	Select Start Now to start the task once you complete the task settings, or Start Time and specify a specific time to schedule the task. - Duration: Specify the duration of the task. - End Time: The system will calculate the end time based on your task start time and duration.

4. Click [Save].



When a learning task is running, you cannot upload a configuration file to restore settings.

7.2.4.2 Stopping Policy Rule Auto-Learning Tasks

Procedure

1. When a learning task is in progress, go to [Security] > [Security Policies] > [Policy Rule Auto-Learning].

Security > Security Policies > Policy Rule Auto-Learning

Policy Rule Auto-Learning Task Details

Learning Status	• Task Ongoing
Traffic Learning Scope	Custom 
IPS Profile for Task	Sys_Default_IPS
IPS Mode for Task	Monitor Mode
Start Time	2025-05-06T07:08:18Z
End Time	2025-05-07T07:08:18Z

 Stop Task
  Delete Task

2. Click the [Stop Task] button to stop the auto-learning task. After you stop the task, a learning result will still be generated for your review.

7.2.4.3 Deleting Policy Rule Auto-Learning Tasks

Procedure

1. When a learning task is in progress, go to [Security] > [Security Policies] > [Policy Rule Auto-Learning].

Security > Security Policies > Policy Rule Auto-Learning

Policy Rule Auto-Learning Task Details

Learning Status	• Task Ongoing
Traffic Learning Scope	Custom 
IPS Profile for Task	Sys_Default_IPS
IPS Mode for Task	Monitor Mode
Start Time	2025-05-06T07:08:18Z
End Time	2025-05-07T07:08:18Z

 Stop Task
  Delete Task

2. Click the [Delete Task] button to delete the auto-learning task. Note that if you delete the task, the learning result will be erased.

7.2.4.4 Restarting Policy Rule Auto-Learning Tasks

Procedure

1. After a learning task is completed/stopped, go to [Security] > [Security Policies] > [Policy Rule Auto-Learning].

Security > Security Policies > Policy Rule Auto-Learning

Policy Rule Auto-Learning Task Details

Learning Status	● Task Completed		
Traffic Learning Scope	Custom 		
IPS Profile for Task	Sys_Default_IPS		
IPS Mode for Task	Monitor Mode		
Start Time	2025-05-06T07:08:18Z		
End Time	2025-05-07T07:08:18Z		
Number of Learned Protocol Filter Profiles / Remaining Quota	32 / 64		
Number of Learned Policy Rules / Remaining Quota	152 / 512	 Apply All Policy Rules	 Generate
Number of Learned L2 Policy Rules / Remaining Quota	44 / 256	 Apply All L2 Policy Rules	 Generate

 Restart Task
  Delete Learning Result
  Download Learning Result

2. Click the [Restart Task] button to restart the auto-learning task. Note that if you restart the task, the previous learning result will be erased.

7.2.4.5 Generating and Applying Learned Policy Enforcement Rules

Procedure

1. After a learning task is completed/stopped, go to [Security] > [Security Policies] > [Policy Rule Auto-Learning].

Security > Security Policies > Policy Rule Auto-Learning

Policy Rule Auto-Learning Task Details

Learning Status	● Task Completed		
Traffic Learning Scope	Custom 		
IPS Profile for Task	Sys_Default_IPS		
IPS Mode for Task	Monitor Mode		
Start Time	2025-05-06T07:08:18Z		
End Time	2025-05-07T07:08:18Z		
Number of Learned Protocol Filter Profiles / Remaining Quota	32 / 64		
Number of Learned Policy Rules / Remaining Quota	152 / 512	 Apply All Policy Rules	 Generate
Number of Learned L2 Policy Rules / Remaining Quota	44 / 256	 Apply All L2 Policy Rules	 Generate

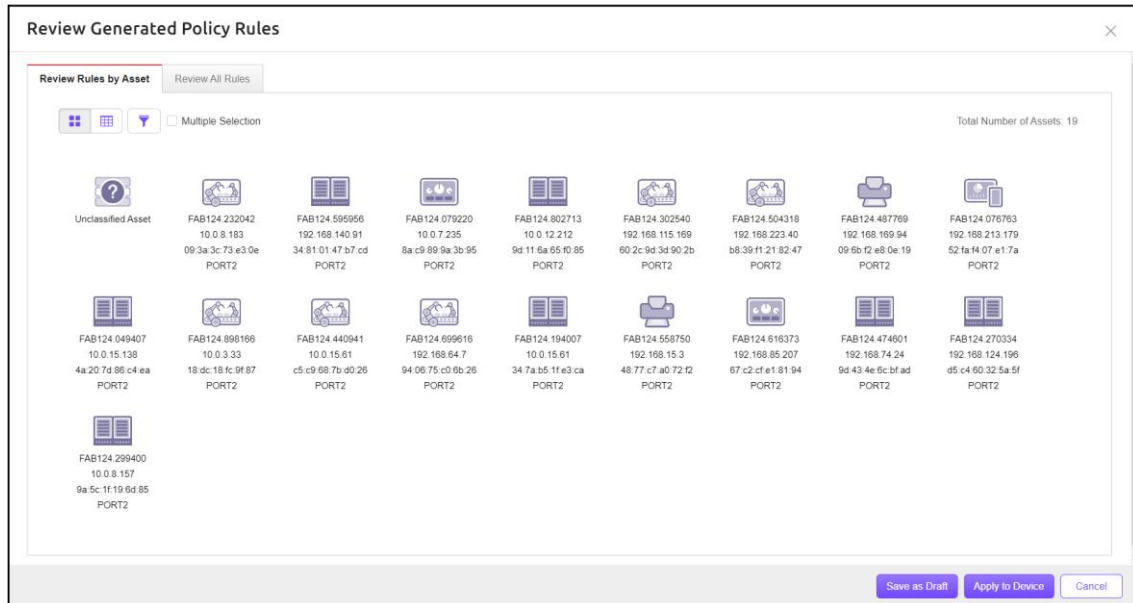
 Restart Task
  Delete Learning Result
  Download Learning Result

2. To apply all the learned policy rules to the device, click [Apply All Policy Rules] or [Apply All L2 Policy Rules].

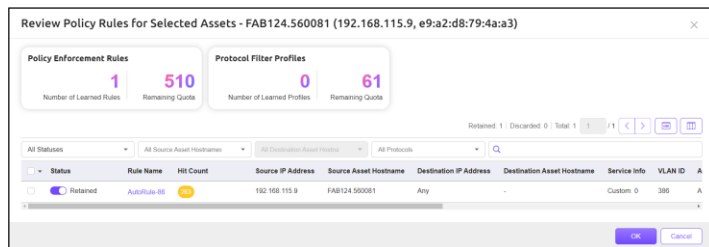


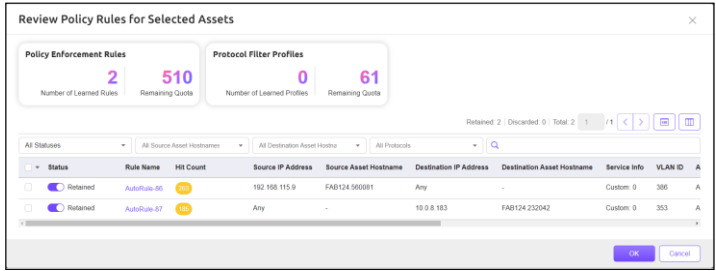
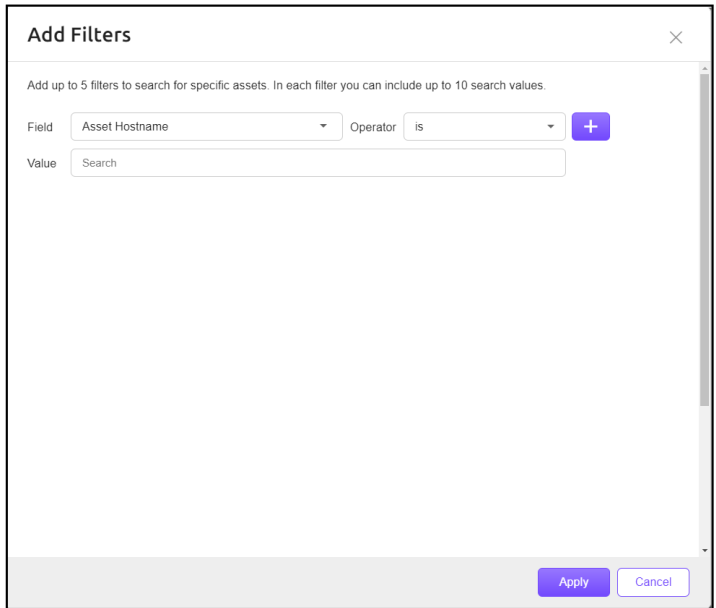
L2 policy rules are available only when EdgeFire is in Bridge mode.

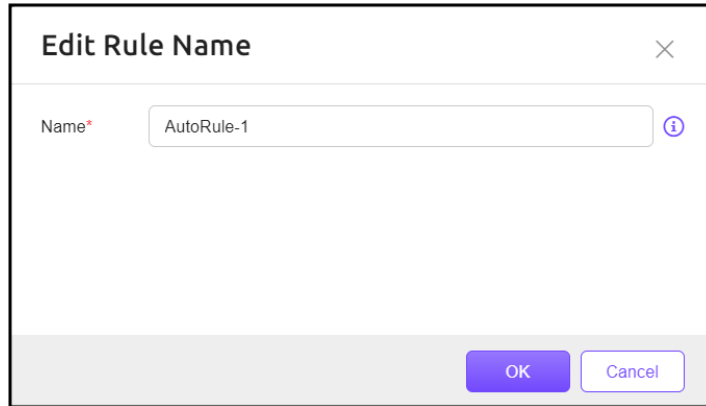
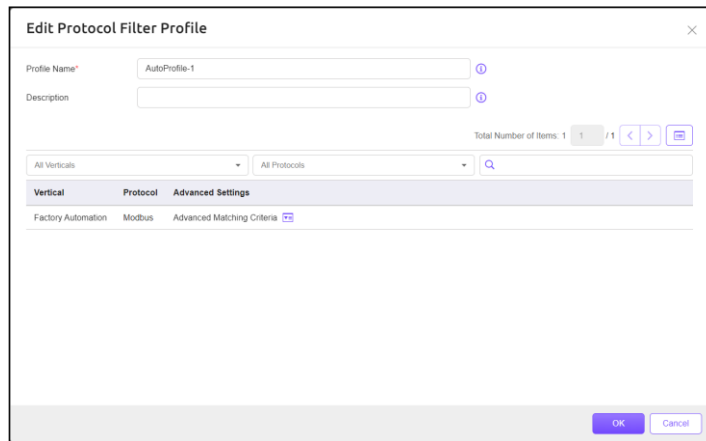
3. To review the learned policy rules and decide which rules to apply, click [Generate]. The [Review Generated Policy Rules] or [Review Generated L2 Policy Rules] window will appear.

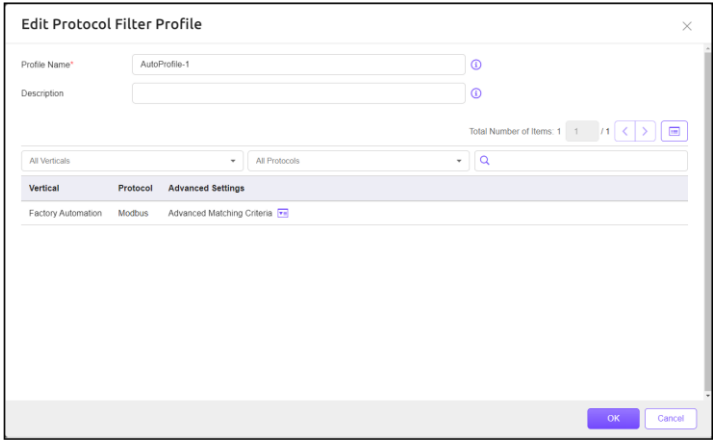
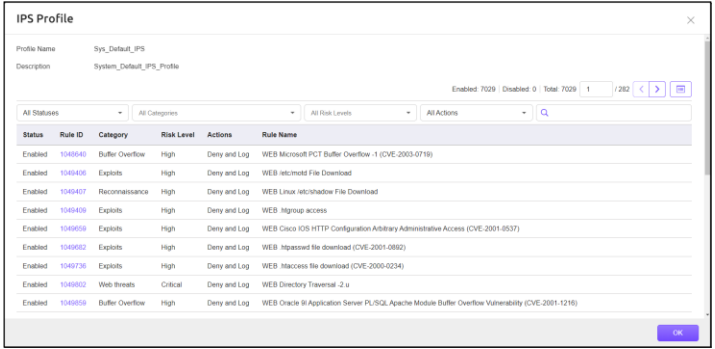


The following table lists the tasks that you can perform on the [Review Generated Policy Rules] and [Review Generated L2 Policy Rules] windows.

Task	Procedure
To review policy rules based on network traffic from one asset	- Go to the [Review Rules by Asset] tab. - Select an asset. The following window will appear.  - All learned policy rules are retained by default. To discard unwanted rules, click the toggles in the [Status] column. - Click [OK].  The number of retained policy rules and Protocol Filter profiles must not exceed their respective remaining quotas.
To review policy rules based on network traffic from multiple assets	- Go to the [Review Rules by Asset] tab. - Check the [Multiple Selection] checkbox. - Select multiple assets and click [Review Rules for Selected Assets]. The following window will appear.

Task	Procedure
	 4. All learned policy rules are retained by default. To discard unwanted rules, click the toggles in the [Status] column. 5. Click [OK].  The number of retained policy rules and Protocol Filter profiles must not exceed their respective remaining quotas.
To review all learned policy rules	1. Go to the [Review All Rules] tab. 2. All learned policy rules are retained by default. To discard unwanted rules, click the toggles in the [Status] column.  The number of retained policy rules and Protocol Filter profiles must not exceed their respective remaining quotas.
To switch between the grid view and table view modes under the [Review Rules by Asset] tab	1. Click the grid/table view buttons  .
To search for specific assets under the [Review Rules by Asset] tab	1. Click the  button. The [Add Filters] window will appear.  2. Create up to five search filters. In each filter you can include up to 10 search values.

Task	Procedure
	Click [Apply].
To edit a policy rule name	- Go to the generated policy rule list under either tab. - Click a policy rule name in the [Rule Name] column. The [Edit Rule Name] window will appear.  - Input a new name. - Click [OK].
To edit a Protocol Filter profile name and description	- Go to the generated policy rule list under either tab. - Click a Protocol Filter profile name in the [Protocol Filter Profile] column. The [Edit Protocol Filter Profile] window will appear.  - Input a new name and/or description. - Click [OK].
To view details of a learned Protocol Filter profile	- Go to the generated policy rule list under either tab. - Click a Protocol Filter profile name in the [Protocol Filter Profile] column. The [Edit Protocol Filter Profile] window will appear.

Task	Procedure
	 3. Click the [View Detailed Information] button .
To view details of the applied IPS profile	- Go to the generated policy rule list under either tab. - Click the name of the applied IPS profile in the [IPS Profile] column. The following window will appear. 
To save a draft of the generated policy rule list	Click [Save as Draft]. To edit a saved draft, click [Generate] on the [Policy Rule Auto-Learning].
To apply retained policy rules and Protocol Filter profiles to the Edge device	Click [Apply to Device]. The applied policy rules and Protocol Filter profiles will be added to the [Policy Enforcement], [L2 Policy Enforcement], and [Protocol Filter Profiles] pages.

7.2.4.6 Deleting Learning Results

Procedure

- After a learning task is completed/stopped, go to [Security] > [Security Policies] > [Policy Rule Auto-Learning].

Security > Security Policies > Policy Rule Auto-Learning

Policy Rule Auto-Learning Task Details

Learning Status	● Task Completed		
Traffic Learning Scope	Custom 		
IPS Profile for Task	Sys_Default_IPS		
IPS Mode for Task	Monitor Mode		
Start Time	2025-05-06T07:08:18Z		
End Time	2025-05-07T07:08:18Z		
Number of Learned Protocol Filter Profiles / Remaining Quota	32 / 64		
Number of Learned Policy Rules / Remaining Quota	152 / 512	 Apply All Policy Rules	 Generate
Number of Learned L2 Policy Rules / Remaining Quota	44 / 256	 Apply All L2 Policy Rules	 Generate

 Restart Task
  Delete Learning Result
  Download Learning Result

2. Click the [Delete Learning Result] button.

7.2.4.7 Downloading Learning Results

Procedure

1. After a learning task is completed/stopped, go to [Security] > [Security Policies] > [Policy Rule Auto-Learning].

Security > Security Policies > Policy Rule Auto-Learning

Policy Rule Auto-Learning Task Details

Learning Status	● Task Completed		
Traffic Learning Scope	Custom 		
IPS Profile for Task	Sys_Default_IPS		
IPS Mode for Task	Monitor Mode		
Start Time	2025-05-06T07:08:18Z		
End Time	2025-05-07T07:08:18Z		
Number of Learned Protocol Filter Profiles / Remaining Quota	32 / 64		
Number of Learned Policy Rules / Remaining Quota	152 / 512	 Apply All Policy Rules	 Generate
Number of Learned L2 Policy Rules / Remaining Quota	44 / 256	 Apply All L2 Policy Rules	 Generate

 Restart Task
  Delete Learning Result
  Download Learning Result

2. Click the [Download Learning Result] button.

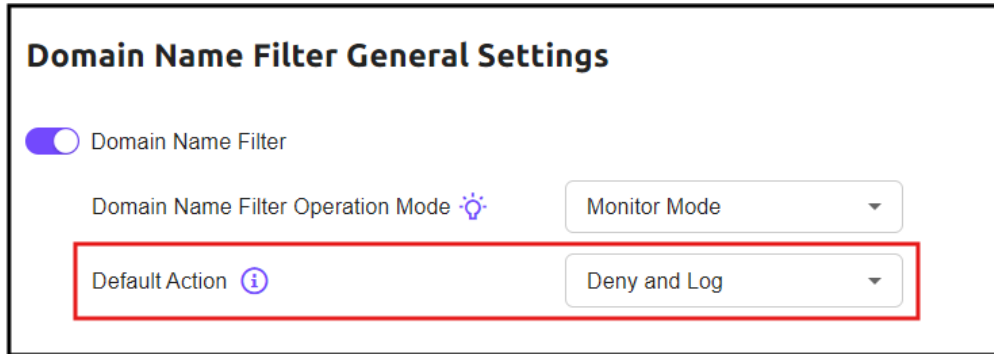
7.2.5 Domain Name Filter Rules

Domain Name Filter rules allow users to define rules to block malicious or allow reliable traffic by identifying the DNS queries sent from the OT assets. This provides users with access controls, ensuring that all traffic passing through their network is secure.

When an OT asset sends a DNS query to a DNS server, the DNS query will be examined by the asset's uplink Edge device, before it arrives at the DNS server. If the Edge device allows the DNS query, it will then be sent to the DNS server. Otherwise, it will be blocked, terminating the DNS query process.

To examine these DNS queries coming from assets, we must define Domain Name Filter rules for the Edge device to determine whether or not a DNS query should be sent out. The Domain Name Filter rule's functioning depends on how it's configured.

Let's assume the default action for the Domain Name Filter rule is "Deny and Log."



Domain Name Filter General Settings

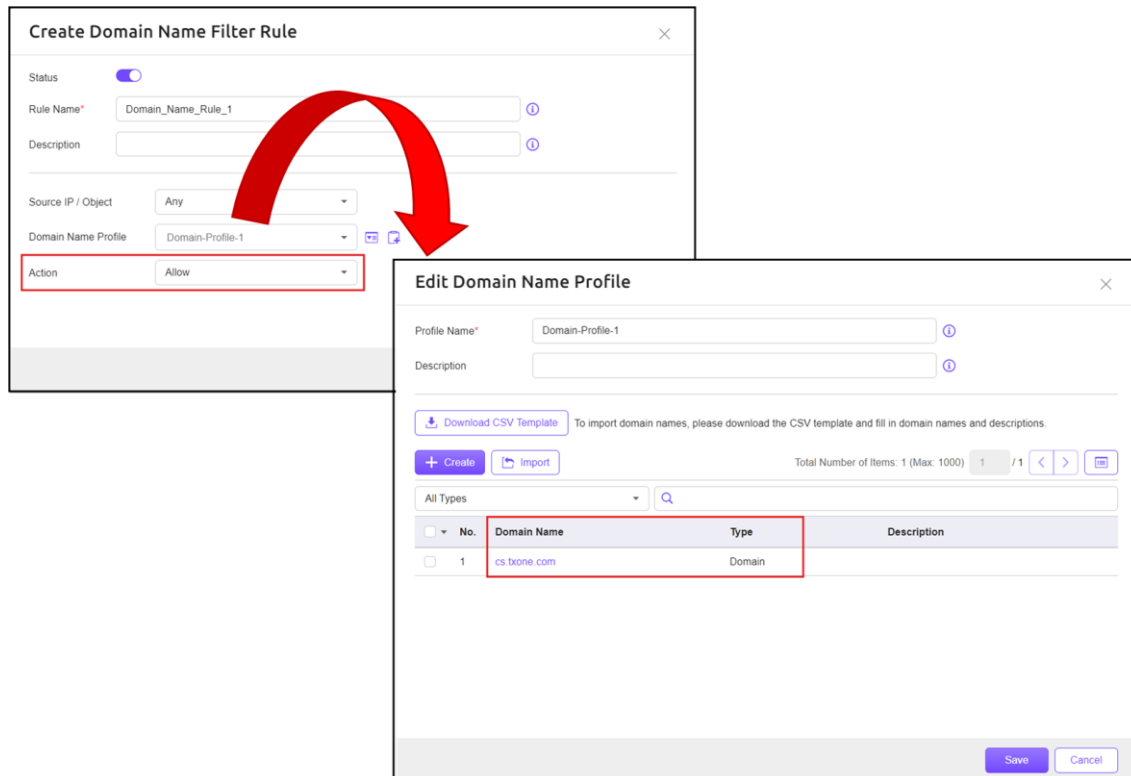
☒ Domain Name Filter

Domain Name Filter Operation Mode  Monitor Mode

Default Action  Deny and Log

We have two Domain Name Filter rules, each containing a Domain Name Object with its action set to "Allow."

- Domain_Name_Filter_Rule_1:
 - Domain Name Object:
 - Domain Name: cs.txone.com
 - Type: **Domain**
 - Rule Action if Matched: Allow



Create Domain Name Filter Rule

Status ☒

Rule Name* Domain_Name_Rule_1 

Description 

Source IP / Object Any

Domain Name Profile Domain-Profile-1 

Action Allow

Edit Domain Name Profile

Profile Name* Domain-Profile-1 

Description 

[Download CSV Template](#) To import domain names, please download the CSV template and fill in domain names and descriptions.

[+ Create](#) [Import](#) Total Number of Items: 1 (Max: 1000) 1 / 1 

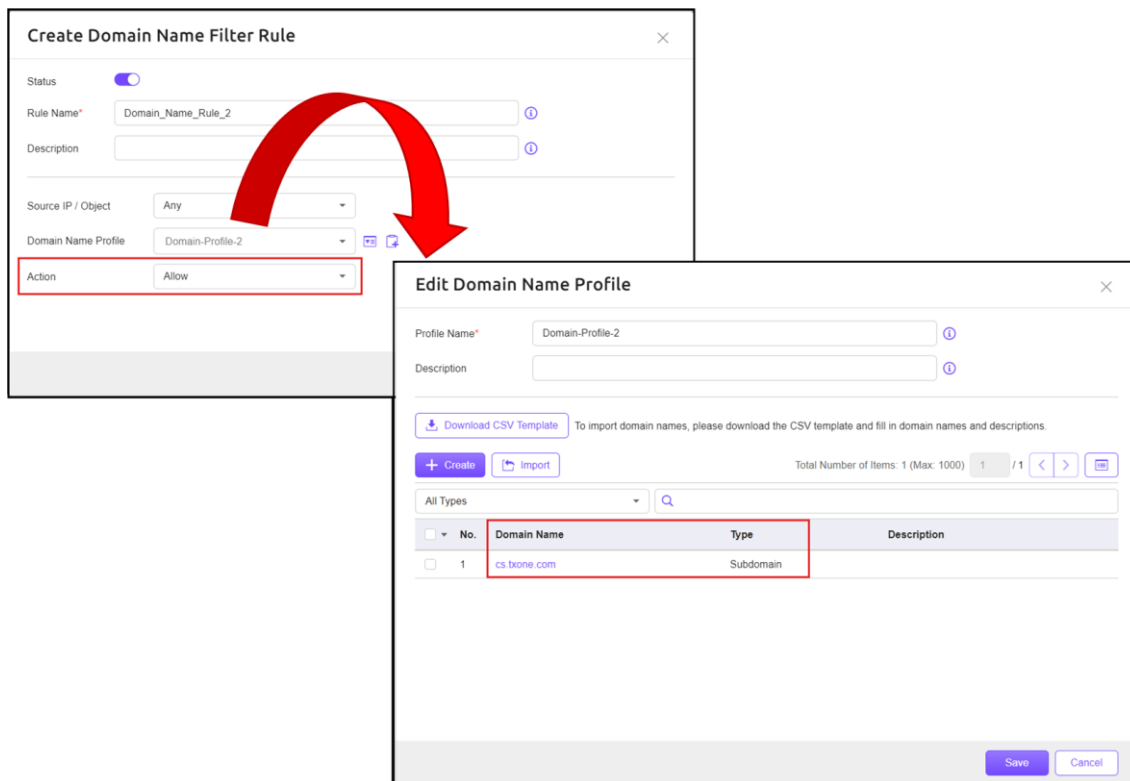
All Types 

No.	Domain Name	Type	Description
1	cs.txone.com	Domain	

[Save](#) [Cancel](#)

- Domain_Name_Filter_Rule_2:
 - Domain Name Object:

- Domain Name: cs.txone.com
- Type: **Subdomain**
- Rule Action if Matched: Allow



Create Domain Name Filter Rule

Status: ☒

Rule Name*: Domain_Name_Rule_2 ⓘ

Description: ⓘ

Source IP / Object: Any

Domain Name Profile: Domain-Profile-2 ⓘ

Action: Allow

Edit Domain Name Profile

Profile Name*: Domain-Profile-2 ⓘ

Description: ⓘ

[Download CSV Template](#) To import domain names, please download the CSV template and fill in domain names and descriptions.

[+ Create](#) [Import](#) Total Number of Items: 1 (Max: 1000) 1 / 1 < > ⓘ

All Types

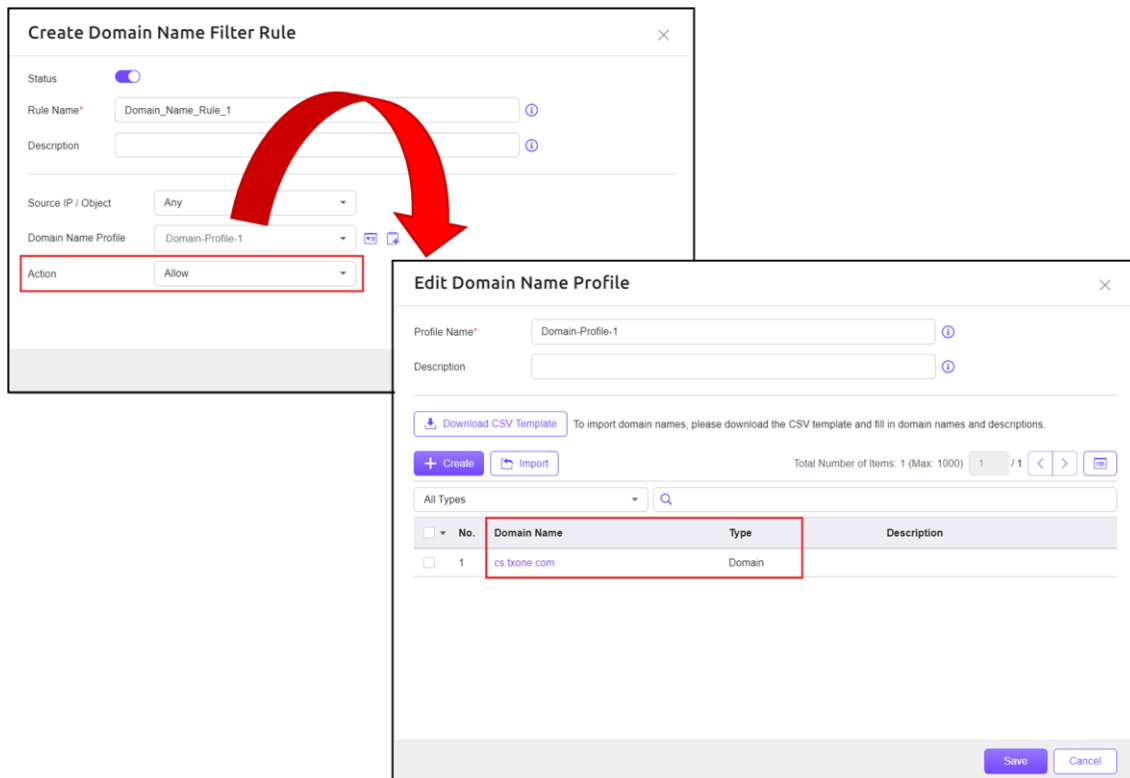
No.	Domain Name	Type	Description
1	cs.txone.com	Subdomain	

Save Cancel

Now, consider DNS queries of these three domains:

- cs.txone.com
- download-center.cs.txone.com
- txone.com

With "Domain_Name_Filter_Rule_1", DNS queries are filtered as follows:



Create Domain Name Filter Rule

Status: ☒

Rule Name*: Domain_Name_Rule_1 ⓘ

Description: ⓘ

Source IP / Object: Any

Domain Name Profile: Domain-Profile-1 ⓘ

Action: Allow

Edit Domain Name Profile

Profile Name*: Domain-Profile-1 ⓘ

Description: ⓘ

[Download CSV Template](#) To import domain names, please download the CSV template and fill in domain names and descriptions.

[+ Create](#) [Import](#) Total Number of Items: 1 (Max: 1000) 1 / 1 < > ⓘ

All Types

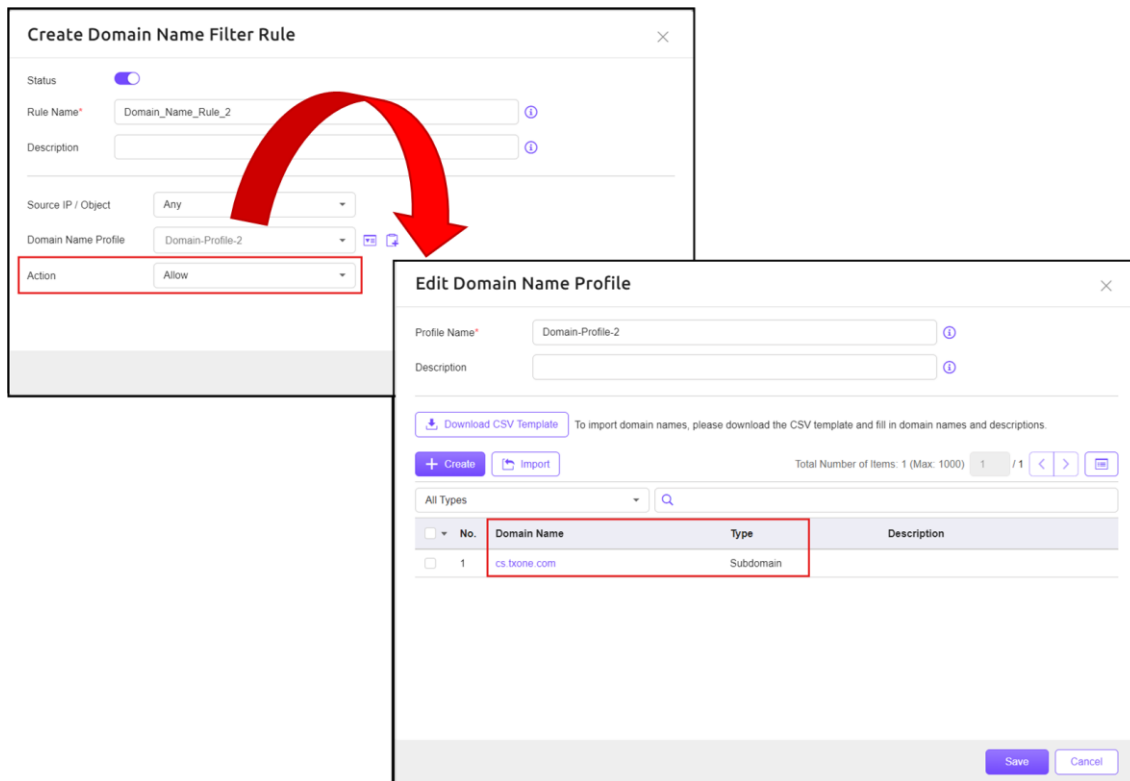
No.	Domain Name	Type	Description
1	cs.txone.com	Domain	

[Save](#) [Cancel](#)

This rule allows the DNS query of "cs.txone.com" and denies queries of all other domains.

- cs.txone.com: This DNS query is **allowed**, as it matches the domain specified.
- download-center.cs.txone.com: This DNS query is **denied**, as it doesn't match the domain specified.
- txone.com: This DNS query is **denied**, as it doesn't match the domain specified.

With "Domain_Name_Filter_Rule_2", DNS queries are filtered differently:



Create Domain Name Filter Rule

Status: ☒

Rule Name*: Domain_Name_Rule_2 ⓘ

Description: ⓘ

Source IP / Object: Any

Domain Name Profile: Domain-Profile-2 ⓘ

Action: Allow

Edit Domain Name Profile

Profile Name*: Domain-Profile-2 ⓘ

Description: ⓘ

[Download CSV Template](#) To import domain names, please download the CSV template and fill in domain names and descriptions.

[+ Create](#) [Import](#) Total Number of Items: 1 (Max: 1000) 1 / 1 < > ⓘ

All Types

No.	Domain Name	Type	Description
1	cs.txone.com	Subdomain	

Save Cancel

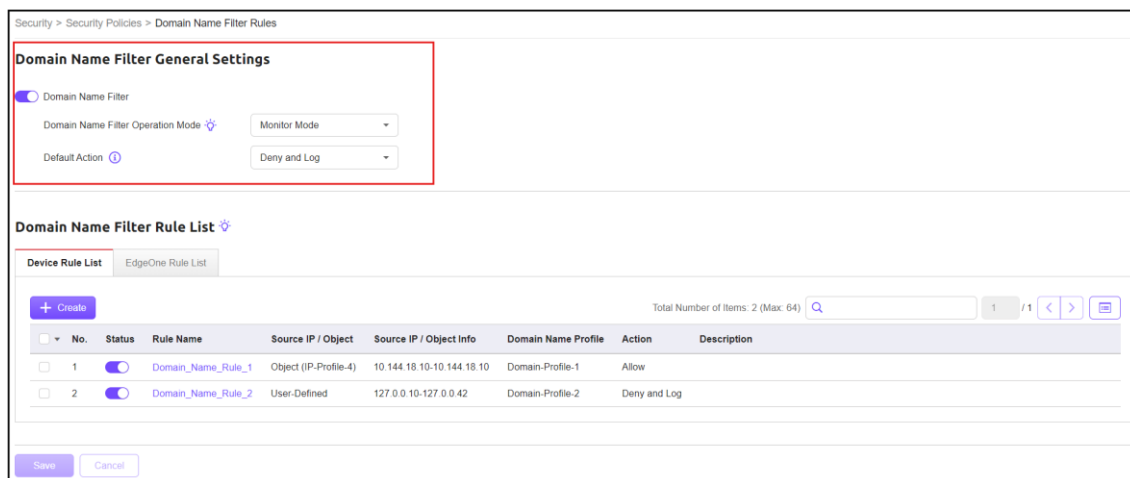
This rule allows DNS queries of any subdomain under "cs.txone.com" and denies all other queries.

- cs.txone.com: This DNS query is **denied**, as it is not a query of a subdomain under "cs.txone.com".
- download-center.cs.txone.com: This DNS query is **allowed**, as it is a query of a subdomain under "cs.txone.com".
- txone.com: This DNS query is **denied**, as it is not a query of a subdomain under "cs.txone.com".

7.2.5.1 Configuring Domain Name Filter General Settings

Procedure

1. Go to [Security] > [Security Policies] > [Domain Name Filter Rules].



Security > Security Policies > Domain Name Filter Rules

Domain Name Filter General Settings

☒ Domain Name Filter

Domain Name Filter Operation Mode ⓘ Monitor Mode

Default Action ⓘ Deny and Log

Domain Name Filter Rule List ⓘ

Device Rule List EdgeOne Rule List

[+ Create](#) Total Number of Items: 2 (Max: 64) 1 / 1 < > ⓘ

No.	Status	Rule Name	Source IP / Object	Source IP / Object Info	Domain Name Profile	Action	Description
1	☒	Domain_Name_Rule_1	Object (IP-Profile-4)	10.144.18.10-10.144.18.10	Domain-Profile-1	Allow	
2	☒	Domain_Name_Rule_2	User-Defined	127.0.0.10-127.0.0.42	Domain-Profile-2	Deny and Log	

Save Cancel

2. Enable or disable the [Domain Name Filter] function by clicking the toggle.

3. From the [Domain Name Filter Operation Mode] drop-down list, select a mode, **Monitor Mode** or **Prevention Mode**, for the function.

4. From the [Default Action] drop-down list, select a default action, **Allow**, **Allow and Log** or **Deny and Log**, for when no Domain Name Filter rule is matched.
5. Click [Save].

The following table summarizes the settings:

Domain Name Filter Operation Mode	Action Performed
Monitor Mode	- Detect and monitor abnormal protocol access attempts to the OT assets, without blocking network attacks. - Generate logs. (If the default rule action is [Allow and Log] or [Deny and Log].)
Prevention Mode	- Block abnormal protocol access attempts to OT assets. - Generate logs. (If the default rule action is [Allow and Log] or [Deny and Log].)

7.2.5.2 Configuring Domain Name Filter Rules

Procedure

1. Configure the required objects and profiles.
 - IP Objects



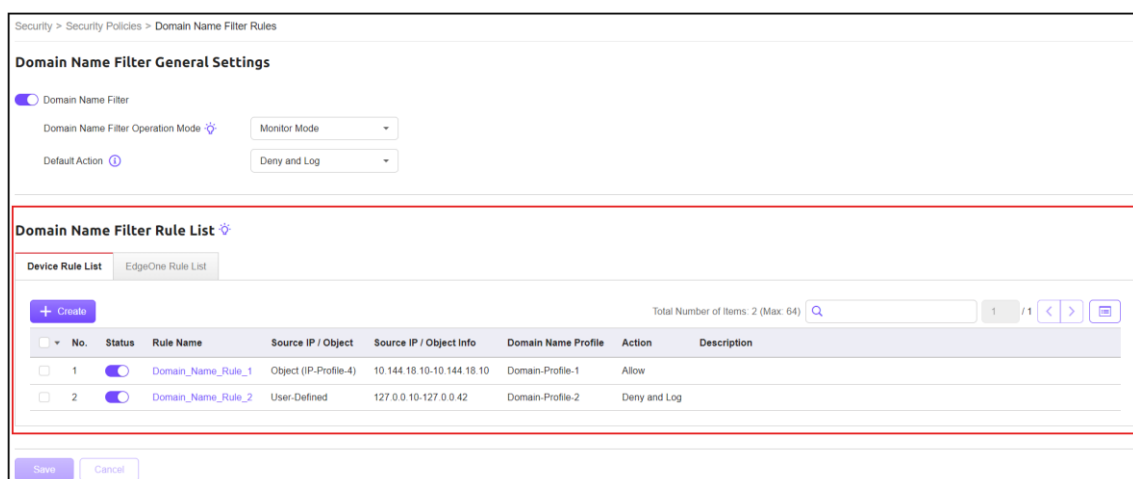
For more information, see [Configuring IP Objects](#).

- Domain Name Profiles



For more information, see [Configuring Domain Name Profiles](#).

2. Go to [Security] > [Security Policies] > [Domain Name Filter Rules].



3. Do one of the following.

- Click [Create] to create a new Domain Name Filter rule. The [Create Domain Name Filter Rule] window will appear.
- Click a rule name to edit settings. The [Edit Domain Name Filter Rule] window will appear.

Create Domain Name Filter Rule

Status ☒

Rule Name*

Description

Source IP / Object

Any

Domain Name Profile

- Select -

Action

Allow

OK

Cancel

4. Configure the following settings.

Item	Description
Status	Enable the operational status of this Domain Name Filter rule by clicking the toggle.
Rule Name	Type a rule name for this Domain Name Filter rule.
Description	(Optional) Type a description.
Source IP / Object	Select one of the following for the source IP addresses: - Any - Single IP - IP Range - IP Subnet - IP Object  If you select [IP Object], you can select an IP object that has been created beforehand on the [IP Objects] page or click the  button to create a new IP object.
Domain Name Profile	Select a Domain Name profile you have defined beforehand or click the  button to create a new profile.
Action	Select one of the following actions: Allow: Select this option to allow network traffic that matches this rule. Allow and Log: Select this option to allow network traffic that matches this rule and output a log. Deny and Log: Select this option to block network traffic that matches this rule and output a log.

5. Click [OK].

6. Click [Save] on the [Domain Name Filter Rules] page.

7.2.5.3 Managing Domain Name Filter Rules

The following tables list the common tasks for managing Domain Name Filter rules.

- **Device Rule List:**

Task	Action
To create a Domain Name Filter rule	Click the [Create] button to create a new Domain Name Filter rule.
To delete a Domain Name Filter rule	Click the checkbox of a rule and click the [Delete] button.
To duplicate a Domain Name Filter rule	Click the checkbox of a rule and click the [Duplicate] button.
To edit a Domain Name Filter rule	Click the name of a rule, and an [Edit Domain Name Filter Rule] window will appear.
To change the priority of a Domain Name Filter rule	Click the checkbox of a rule, click the [Change Priority] button, and specify a new priority for this rule.



When more than one Domain Name Filter rule is matched, the device takes action based on the rule with the highest priority and ignores the rest of the rules. The rules listed in the table on the [Domain Name Filter Rules] page are ordered by priority, with the highest priority rule listed on the first row of the table.

- **EdgeOne Rule List:**

Task	Description
N/A	The Domain Name Filter rules on the EdgeOne rule list are view-only.

7.2.6 Suspicious Objects

The [Suspicious Objects] feature allows you to define a custom node-based/link-based Suspicious Object list from EdgeOne. If any identical IP address or MAC address on the node level, or identical IP address, protocol and port numbers on the link level are detected, the activity will be allowed or blocked according to your pre-defined action for the suspicious objects.



Before you enable the [Suspicious Objects] function, ensure that the Edge device authentication settings ([Admin] > [System] > [Connection Settings]) are properly configured and that EdgeOne is connected with the third-party API from the source of the suspicious objects.

7.2.6.1 Configuring Suspicious Objects

Procedure

1. Go to [Security] > [Security Policies] > [Suspicious Objects].

Security > Security Policies > Suspicious Objects

Suspicious Object General Settings

Unable to enable Suspicious Object because the Edge device is not connected to EdgeOne. To connect the device, log in to EdgeOne and go to the [Connection Settings] page.

☒ Suspicious Object

Suspicious Object Operation Mode: ☒ Monitor Mode ☐ Prevention Mode

Suspicious Object Rule List

Suspicious Object is prioritized over Policy Enforcement during traffic examination if this feature is enabled.

Deny: 52 | Allow: 48 | Total: 100 | 1 / 4

All Types	All Sources	All Risk Levels	All Actions						
No.	ID	Type	Source	Object Contents	Risk Level	Last Updated	Expiration Time	Action	
☐	1	dtSYGGWeh	Link	NOZOMI1	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	Medium	2024-07-19T15:14:06+08:00	Never Expires	Deny and Log
☐	2	yYYapdqZn	Link	DDI2	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	High	2024-07-19T15:14:06+08:00	2050-02-10T05:18:44+08:00	Deny and Log
☐	3	vZQlyLzqpl	Link	NOZOMI1	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	Low	2024-07-19T15:14:06+08:00	2047-05-31T06:24:48+08:00	Allow and Log
☐	4	XphZcqORag	Link	DDI1	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	Medium	2024-07-19T15:14:06+08:00	2039-08-20T15:09:54+08:00	Deny and Log
☐	5	AlJxPnjIMdS	Link	NOZOMI1	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	Low	2024-07-19T15:14:06+08:00	2068-01-15T02:05:55+08:00	Allow and Log
☐	6	ImAACPTRZR	Link	DDI2	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	Medium	2024-07-19T15:14:06+08:00	2065-05-21T10:39:21+08:00	Deny and Log
☐	7	MIQLFkwSN	Node	NOZOMI1	10.10.10.1	Medium	2024-07-19T15:14:06+08:00	2057-01-11T03:12:31+08:00	Allow and Log
☐	8	MUEPkaZBu	Node	DDI1	10.10.10.1	High	2024-07-19T15:14:06+08:00	2053-03-29T04:10:07+08:00	Allow and Log
☐	9	zsdXOoeSB	Node	DDI1	10.10.10.1	High	2024-07-19T15:14:06+08:00	2062-09-01T11:28:54+08:00	Deny and Log
☐	10	ndRlHlshF	Node	DDI2	10.10.10.1	High	2024-07-19T15:14:06+08:00	2064-06-04T07:24:56+08:00	Allow and Log
☐	11	LzZYyulRq	Node	NOZOMI1	10.10.10.1	High	2024-07-19T15:14:06+08:00	Never Expires	Allow and Log
☐	12	TkZzEniDmu	Link	NOZOMI1	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	Low	2024-07-19T15:14:06+08:00	2075-06-06T15:15:09+08:00	Deny and Log
☐	13	FloaxKRGJN	Node	DDI2	10.10.10.1	Medium	2024-07-19T15:14:06+08:00	2066-01-30T13:44:31+08:00	Allow and Log
☐	14	UQVMlBIMC	Link	DDI2	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	Medium	2024-07-19T15:14:06+08:00	2054-12-21T00:09:20+08:00	Deny and Log
☐	15	gjmYyYdgA	Link	NOZOMI1	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	High	2024-07-19T15:14:06+08:00	2040-09-07T12:08:12+08:00	Allow and Log

2. Enable the [Suspicious Objects] function by clicking the toggle.

! You cannot enable Suspicious Objects if:

- The Edge device is not connected to EdgeOne.
- The Edge device is not managed within an Edge group in EdgeOne.
- The [Suspicious Objects] group setting is not enabled for the Edge group to which the Edge device belongs.
- The [Dispatch Suspicious Objects to Edge Devices] option in EdgeOne ([Nodes] > [Suspicious Objects] > [Suspicious Object Settings]) is not enabled.

3. Select a mode, **Monitor Mode** or **Prevention Mode**, for the function.

4. To change the action (listed in the [Action] column) of a suspicious object, check the checkbox of an item and select an action, **Deny and Log** or **Allow and Log**, for when the pattern is matched.

Suspicious Object Rule List

Suspicious Object is prioritized over Policy Enforcement during traffic examination if this feature is enabled.

☒ Deny and Log ☒ Allow and Log 2 selected

All Types All Sources All Risk Levels All Actions

No.	ID	Type	Source	Object Contents	Risk Level	
☒	1	dtSYGGWeh	Link	NOZOMI1	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	Medium
☒	2	yYYapdqQzN	Link	DDI2	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	High
☐	3	vZQlyLzqpl	Link	NOZOMI1	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	Low
☐	4	XphZcqORag	Link	DDI1	Source IP=10.10.10.1, Source Port=Any, Destination IP=192.168.1.1, Destination Port=Any, Protocol=6	Medium

5. Click [Save].

The following table summarizes the settings.

Suspicious Object Operation Mode	Action Performed
Monitor Mode	• Detects network node or network link, but does not block communication related to the network node or network link. • Generates logs.
Prevention Mode	• Blocks network node or network link. • Generates logs.



The Suspicious Object is of higher priority than the Policy Enforcement.

7.3 CPSDR Policies

Create and manage network detection policy rules for early identification and prediction of anomalous network behaviors.

- [CPSDR Rules](#)
- [CPSDR Allow Lists & Deny Lists](#)

7.3.1 CPSDR Rules

TXOne CPSDR (Cyber-Physical System Detection and Response) is designed to identify and predict anomalous network behaviors through a comprehensive analysis of traffic patterns, rather than just inspecting individual packets for abnormalities like traditional IPS features. Unlike IPS, which scrutinizes individual packets within a traffic flow for any anomalies, CPSDR assesses the relationships between sequential traffic flows.

For example, if the system observes Traffic Type A followed by Traffic Type B, and this sequence is determined to be suspicious based on established behavioral patterns, the system will then predict the impact severity of this network behavior. This method allows CPSDR to provide a holistic view of network behavior and calculate impact severities more effectively.

You can manage the CPSDR network detection policy rules on the [CPSDR Rules] page or go to [CPSDR Allow Lists & Deny Lists] for advanced settings. If network traffic matches a specific rule, a log is generated on the [CPSDR Logs] page.



For more information about CPSDR allow lists and deny lists, see [CPSDR Allow Lists & Deny Lists](#).



For more information about CPSDR logs, see [CPSDR Logs](#).

7.3.1.1 Viewing and Configuring CPSDR Rules

Procedure

1. Go to [Security] > [CPSDR Policies] > [CPSDR Rules].

Security > CPSDR Policies > CPSDR Rules

CPSDR Settings

TXOne CPSDR (Cyber-Physical System Detection and Response) is designed to identify and predict anomalous network behaviors. You can manage network detection policy rules on this page or go to [CPSDR Allow Lists & Deny Lists] for advanced settings. If network traffic matches a specific rule, a log is generated on the [CPSDR Logs] page.

☒ CPSDR

Total Number of Items: 32 1 / 2 < > [icon]

All Detection Statuses* All Rule IDs All TIDs All Tactic IDs All Impact Severities All Confidence Levels [Search]

No.	Detection	Rule ID	Rule Name	TIDs	Tactic IDs	Impact Severity	Confidence Level	Action
1	☒	3010005	BabyShark Agent Pattern	T0869, T1071, T1071.001	TA0011, TA0101	High	Medium	[icon]
2	☒	3010006	Cobalt Strike Malicious Profile	T0869, T1071, T1071.001	TA0011, TA0101	High	Medium	[icon]
3	☒	3010008	WannaCry Killswitch Domain	T0869, T1071, T1071.001	TA0011, TA0101	High	Medium	[icon]
4	☒	3010010	Possible Lateral Tool Transfer via SMB	T1570	TA0008	Medium	Medium	[icon]
5	☒	3010013	Remote System Discovery Via RPC	T0846, T1016	TA0007, TA0102	Low	Low	[icon]
6	☒	3010014	Execution Via WMI	T1047	TA0002	High	Medium	[icon]
7	☒	3010015	Scheduled Task/Job Via At	T1053, T1053.002	TA0002	Medium	Medium	[icon]
8	☒	3010016	File Dir Discovery Via RPC	T1083	TA0007	Low	Low	[icon]
9	☒	3010017	Account Discovery Via RPC	T1087, T1087.002	TA0007	Medium	Medium	[icon]
10	☒	3010018	Network Share Discovery Via RPC	T1135	TA0007	Low	Low	[icon]
11	☒	3010019	Execution Via System Services	T1569, T1569.002	TA0002	High	Medium	[icon]
12	☒	3010020	Remote Schedule Task Lateral Movement via TaskSchedulerService	T1053, T1053.005	TA0008	Medium	Low	[icon]

2. Enable the [CPSDR] function by clicking the toggle.

The following table describes the CPSDR rule information.

Item	Description
No.	The ordinal number of the CPSDR rule.
Detection	The detection status of the CPSDR rule. Use the toggles to enable/disable rules and click [Save] on the [CPSDR Rules] page.
Rule ID	The ID number of the CPSDR rule. Click a specific rule ID to view details of a CPSDR rule. The [CPSDR Rule Details] window will appear.

CPSDR Rule Details

Detection

☒

Rule ID

3010002

Rule Name

MALWARE-C2 Trojan.Win32.AsukaStealer.A HTTP Connection (test2)

Description

Trojan.Win32.AsukaStealer.A is a Trojan that targets the Windows platform. The malware collects system information such as OS information, time zone, display size, ram, running applications, browser data and more. The collected data is sent to the attacker using a HTTP POST request.

Impact Severity

Low

Confidence Level

Low

TIDs

T1045, T1045.002

Tactic IDs

TA0043

Keywords

-

Created On

2024-03-04

Last Updated

2024-03-18

OK

Cancel

- Detection:** The detection status of the CPSDR rule. **Use the toggle to enable/disable this rule, click [OK] on the window, then click [Save] on the [CPSDR Rules] page.**
- Rule ID:** The ID number of the CPSDR rule.

Item	Description
	• Rule Name: The name of the CPSDR rule regarding the cyber attack. • Description: A brief overview of this type of cyber attack. • Impact Severity: The severity level (Low, Medium, High) of the cyber attack assessed and predicted by the system. • Confidence Level: The likelihood (Low, Medium, High) of the cyber attack assessed and predicted by the system. • TIDs: MITRE ID information. • Tactic IDs: The ID numbers of the cyber tactics. • Keywords: The keywords for searching this CPSDR rule. • Created On: The date when this CPSDR rule was initially created. • Last Updated: The date of the most recent update to the rule's details.
Rule Name	The name of the CPSDR rule regarding the cyber attack.
TIDs	MITRE ID information.
Tactic IDs	The ID numbers of the cyber tactics.
Impact Severity	The severity level (Low, Medium, High) of the cyber attack assessed and predicted by the system.
Confidence Level	The likelihood (Low, Medium, High) of the cyber attack assessed and predicted by the system.
Action	Click the  button to create a CPSDR allow list or deny list profile.  For more information about creating CPSDR allow lists and deny lists, see Configuring CPSDR Allow Lists and Deny Lists.

1. Click [Save].

7.3.2 CPSDR Allow Lists & Deny Lists

You can configure allow lists, deny lists, and threshold settings to enhance CPSDR detection accuracy. The Edge device responds to suspicious network behaviors based on the deny list. The CPSDR detection threshold (disabled by default) is designed to block network behaviors matching both CPSDR rules and the specified risk level. When all three are configured, the Edge device examines traffic in this order: Allow Lists > Deny Lists > CPSDR Detection Threshold.

7.3.2.1 Configuring CPSDR Allow Lists and Deny Lists

Procedure

1. Configure the required objects and profiles.
 - IP Objects

 For more information, see [Configuring IP Objects](#).

2. Go to [Security] > [CPSDR Policies] > [CPSDR Allow Lists & Deny Lists]. Then switch to the [Allow Lists] or [Deny Lists] tab.

Security > CPSDR Policies > CPSDR Allow Lists & Deny Lists

CPSDR Allow Lists & Deny Lists

You can configure allow lists, deny lists, and threshold settings to enhance CPSDR detection accuracy. The Edge device responds to suspicious network behaviors based on the alert list. The CPSDR detection threshold (disabled by default) is designed to block network behaviors matching both CPSDR rules and the specified risk level. When all three are configured, the Edge device examines traffic in this order: Allow Lists > Deny Lists > CPSDR Detection Threshold.

Allow Lists (3) | Deny Lists (1) | Threshold Settings

[+ Create](#) Total Number of Items: 3 (Max: 512) 1 / 1

No.	Name	Rule IDs	TIDs	Tactic IDs	Source IP / Object	Source IP / Object Info	Destination IP / Object	Destination IP / Object Info	
1	CPSDR_Allow_List_1	3010005, 3010010, 3010023	-	-	Any	Any	Any	Any	A
2	CPSDR_Allow_List_2	-	T0069, T1569 002, T1102 003	-	User Defined	192.168.2.56	Any	Any	A
3	CPSDR_Allow_List_3	-	-	TA0007, TA0101, TA0107	Any	Any	Any	Any	2

3. Do one of the following.

- Click [Create] to create an allow list or deny list. The [Create CPSDR Allow List] or [Create CPSDR Deny List] window will appear.
- Click an allow list or deny list name to edit settings. The [Edit CPSDR Allow List] or [Edit CPSDR Deny List] window will appear.

Create CPSDR Allow List

Name*

Description

Network Behavior

Source IP / Object

Destination IP / Object

VLAN IDs

4. Configure the following settings.

Item	Description
Name	Type a name for this CPSDR allow list/deny list.
Description	(Optional) Type a description.

- Network Behavior:** Select **Rule IDs**, **TIDs**, or **Tactic IDs**, then select one or multiple IDs from the drop-down list.

Item	Description
Source IP / Object	Select one of the following for the source IP addresses: - Any - Single IP - IP Range

Item	Description
	• IP Subnet • IP Object  If you select [IP Object], you can select an IP object that has been created beforehand on the [IP Objects] page or click the  button to create a new IP object.
Destination IP / Object	Select one of the following for the destination IP addresses: • Any • Single IP • IP Range • IP Subnet • IP Object
VLAN IDs	Select either one of the following for the VLAN IDs: • Any • Custom: Specify at least one VLAN ID for this protocol.  A maximum of five VLAN IDs can be applied to one policy rule.

5. Click [Save].

7.3.2.2 Configuring CPSDR Detection Threshold Settings

Procedure

1. Go to [Security] > [CPSDR Policies] > [CPSDR Allow Lists & Deny Lists]. Then switch to the [Threshold Settings] tab.

Security > CPSDR Policies > CPSDR Allow Lists & Deny Lists

CPSDR Allow Lists & Deny Lists

You can configure allow lists, deny lists, and threshold settings to enhance CPSDR detection accuracy. The Edge device responds to suspicious network behaviors based on the alert list. The CPSDR detection threshold (disabled by default) is designed to block network behaviors matching both CPSDR rules and the specified risk level. When all three are configured, the Edge device examines traffic in this order: Allow Lists > Deny Lists > CPSDR Detection Threshold.

Allow Lists (3)

Deny Lists (1)

Threshold Settings

When the CPSDR detection threshold is enabled, the Edge device uses the configured threshold (risk level) and CPSDR rules to examine network traffic and block any packet reaching the threshold.

☐ CPSDR Detection Threshold

Risk Level ⓘ Critical **High** Medium

Action Deny and Log

The action (deny and log) is triggered when the Edge device detects a network behavior that meets or exceeds the configured risk level. (Note: The effectiveness of traffic blocking may vary depending on the connection conditions (e.g., long vs. short connections) of Edge devices.)

Save

Cancel

2. Click the [CPSDR Detection Threshold] toggle to enable the feature.
3. Select a threshold (risk level), **Critical**, **High**, or **Medium**.
4. Click [Save].



The CPSDR action (deny and log) is triggered when the Edge device detects a network behavior that meets or exceeds the configured risk level.

8 Patterns

This chapter describes how to view the pattern information and import security and CPSDR patterns to the Edge series device.

8.1 Pattern Updates

Manage and update patterns for up-to-date threat prevention.

- [Security Patterns](#)
- [CPSDR Patterns](#)

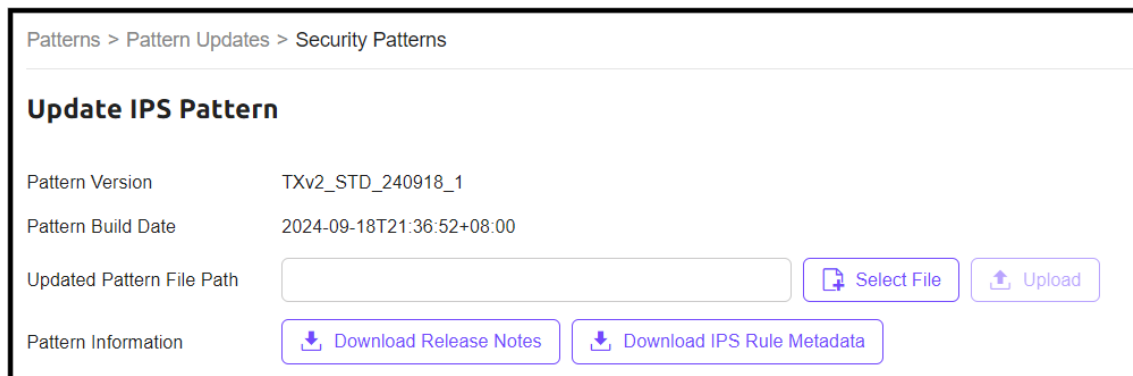
8.1.1 Security Patterns

Security Patterns contain DPI (Deep Packet Inspection) patterns (a.k.a. IPS patterns) with signatures to enable intrusion prevention features on the device. The intrusion prevention feature detects and prevents behaviors related to network intrusion attempts or targeted attacks at the network level.

8.1.1.1 Viewing IPS Pattern Information

Procedure

1. Go to [Patterns] > [Pattern Updates] > [Security Patterns].



Patterns > Pattern Updates > Security Patterns

Update IPS Pattern

Pattern Version: TXv2_STD_240918_1

Pattern Build Date: 2024-09-18T21:36:52+08:00

Updated Pattern File Path: [Select File](#) [Upload](#)

Pattern Information: [Download Release Notes](#) [Download IPS Rule Metadata](#)

2. In the [Update IPS Pattern] pane, you will see the pattern version, pattern build date, file path of the updated pattern, release notes, and IPS rule metadata.



Before updating the IPS pattern, ensure that you have activated the Edge device software license.

8.1.1.2 Updating IPS Patterns

Procedure

1. Go to [Patterns] > [Pattern Updates] > [Security Patterns].

Patterns > Pattern Updates > Security Patterns

Update IPS Pattern

Pattern Version	TXv2_STD_240918_1		
Pattern Build Date	2024-09-18T21:36:52+08:00		
Updated Pattern File Path		Select File	Upload
Pattern Information	Download Release Notes	Download IPS Rule Metadata	

2. In the [Update IPS Pattern] pane, click [Select File].

3. Select a pattern file from your local disk.

4. Click [Upload].

 Various versions of the pattern files can be downloaded at the MyTXOne Portal at <https://my.txone.com/>.

8.1.1.3 Downloading Pattern Release Notes and IPS Rule Metadata

Procedure

1. Go to [Patterns] > [Pattern Updates] > [Security Patterns].

Patterns > Pattern Updates > Security Patterns

Update IPS Pattern

Pattern Version	TXv2_STD_240918_1		
Pattern Build Date	2024-09-18T21:36:52+08:00		
Updated Pattern File Path		Select File	Upload
Pattern Information	Download Release Notes	Download IPS Rule Metadata	

2. Click the [Download Release Notes] button to download and view detailed pattern release information.

3. Click the [Download IPS Rule Metadata] button to download and view the IPS rule metadata.

8.1.2 CPSPDR Patterns

CPSPDR (Cyber-Physical System Detection and Response) patterns are specifically designed for the TXOne CPSPDR feature, which the Edge device employs to identify and predict anomalous network behaviors.

 For more information about the CPSPDR feature, see [CPSPDR Policies](#).

8.1.2.1 Viewing CPSPDR Pattern Information

Procedure

1. Go to [Patterns] > [Pattern Updates] > [CPSPDR Patterns].

Patterns > Pattern Updates > CPSDR Patterns

Update CPSDR Pattern

Pattern Version	TX_CPSDR_ND_STD_240925_33		
Pattern Build Date	2024-10-17T10:21:01+08:00		
Updated Pattern File Path		Select File	Upload
Pattern Information	Download Release Notes	Download CPSDR Rule Metadata	

2. In the [Update CPSDR Pattern] pane, you will see the pattern version, pattern build date, file path of the updated pattern, release notes, and CPSDR rule metadata.



Before updating the CPSDR pattern, ensure that you have activated the Edge device software license.

8.1.2.2 Updating CPSDR Patterns

Procedure

1. Go to [Patterns] > [Pattern Updates] > [CPSDR Patterns].

Patterns > Pattern Updates > CPSDR Patterns

Update CPSDR Pattern

Pattern Version	TX_CPSDR_ND_STD_240925_33		
Pattern Build Date	2024-10-17T10:21:01+08:00		
Updated Pattern File Path		Select File	Upload
Pattern Information	Download Release Notes	Download CPSDR Rule Metadata	

2. In the [Update CPSDR Pattern] pane, click [Select File].

3. Select a pattern file from your local disk.

4. Click [Upload].



Various versions of the pattern files can be downloaded at the MyTXOne Portal at <https://my.txone.com/>.

8.1.2.3 Downloading Pattern Release Notes and CPSDR Rule Metadata

Procedure

1. Go to [Patterns] > [Pattern Updates] > [CPSDR Patterns].

Patterns > Pattern Updates > CPSDR Patterns

Update CPSDR Pattern

Pattern Version	TX_CPSDR_ND_STD_240925_33		
Pattern Build Date	2024-10-17T10:21:01+08:00		
Updated Pattern File Path		Select File	Upload
Pattern Information	Download Release Notes Download CPSDR Rule Metadata		

2. Click the [Download Release Notes] button to download and view detailed pattern release information.
3. Click the [Download CPSDR Rule Metadata] button to download and view the CPSDR rule metadata.

9 Applications

This chapter describes how to configure and use additional applications and tools for advanced network management and control.

9.1 Backup Storage

Configure settings for storing device configuration backup files in a compatible USB disk.

- [USB Disk Storage](#)

9.1.1 USB Disk Storage

The Edge device supports backing up configurations and loading patterns, configurations, and firmware from a compatible USB disk. On the [USB Disk Storage] page, you can configure the relevant USB backup settings to meet your specific needs.

9.1.1.1 Configuring USB Application Settings

Procedure

1. Go to [Applications] > [Backup Storage] > [USB Disk Storage].

Applications > Backup Storage > USB Disk Storage

☒ USB Application Settings

USB Application Settings

USB Disk Status	Connected
USB Disk Capacity	8.00 GB
Free Capacity	4.70 GB

Advanced USB Application Settings

☒ Enable adaptive configuration backup to USB disk

Automatically back up administration settings to a USB disk (Zero Configuration).

Backup Behavior

☒ Back up configurations regularly

Every 6 Hours ▾

☐ Back up configurations only when they are changed ⓘ

Most Recent Backup

2024-07-19T17:02:27+08:00

Save

Cancel

2. Click the [USB Application Settings] toggle to enable the USB backup storage.
3. Once the USB Application function is enabled and a compatible USB disk is plugged in, you can view the status and information about the disk capacity and remaining free space.



If the USB Application function is disabled, the USB port on the front panel will not be active and cannot be used.



For more information about the supported USB functions, see [Extended Functions via USB Devices](#).

9.1.1.2 Configuring Advanced USB Application Settings

Procedure

1. Go to [Applications] > [Backup Storage] > [USB Disk Storage].

Applications > Backup Storage > USB Disk Storage

☒ USB Application Settings

USB Application Settings

USB Disk Status	Connected
USB Disk Capacity	8.00 GB
Free Capacity	4.70 GB

Advanced USB Application Settings

☒ Enable adaptive configuration backup to USB disk

Automatically back up administration settings to a USB disk (Zero Configuration).

Backup Behavior

☒ Back up configurations regularly

Every 6 Hours

☐ Back up configurations only when they are changed

?

Most Recent Backup

2024-07-19T17:02:27+08:00

Save

Cancel

2. Click the [Enable adaptive configuration backup to a USB disk] toggle to allow backups of configurations to a compatible USB disk.

3. Select one of the following backup behavior options:

- Back up configurations regularly:** If this option is selected, further specify the backup interval:
 - Every 6 Hours
 - Every 12 Hours
 - Every 24 Hours
 - Every 48 Hours
 - Every 72 Hours
- Back up configurations only when they are changed:** If this option is selected, the Edge device creates backed-up configuration files whenever settings are updated in the web console.



The advanced USB application settings are configurable only when the [USB Application Settings] option is enabled.

9.2 Cyber Forensic Tools

Utilize forensic tools for cybersecurity analysis.

- [IPS Event Packet Capture](#)
- [Captured Packet Downloads](#)
- [Pcap File Server Settings](#)

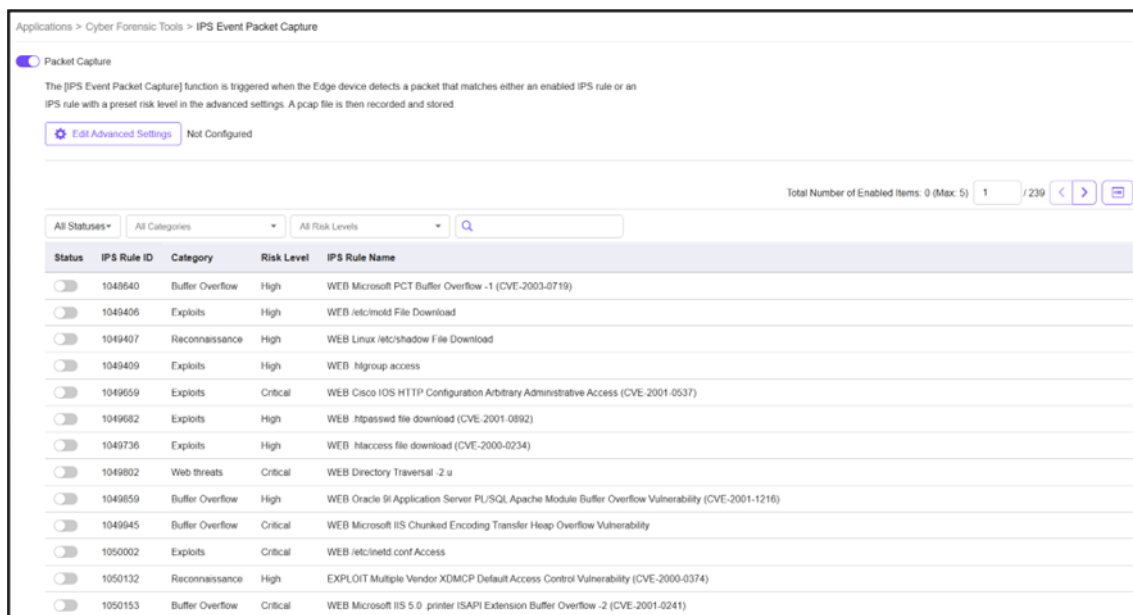
9.2.1 IPS Event Packet Capture

The IPS Event Packet Capture feature enables you to capture packets that trigger IPS events for detailed analysis. This capability allows precise control over packet capture settings for each IPS event rule, aiding support teams in quickly addressing false positive/false negative matching of IPS rules in the security module.

9.2.1.1 Enabling IPS Event Packet Capture

Procedure

1. Go to [Applications] > [Cyber Forensic Tools] > [IPS Event Packet Capture].



Applications > Cyber Forensic Tools > IPS Event Packet Capture

☒ Packet Capture

The [IPS Event Packet Capture] function is triggered when the Edge device detects a packet that matches either an enabled IPS rule or an IPS rule with a preset risk level in the advanced settings. A pcap file is then recorded and stored.

[Edit Advanced Settings](#) Not Configured

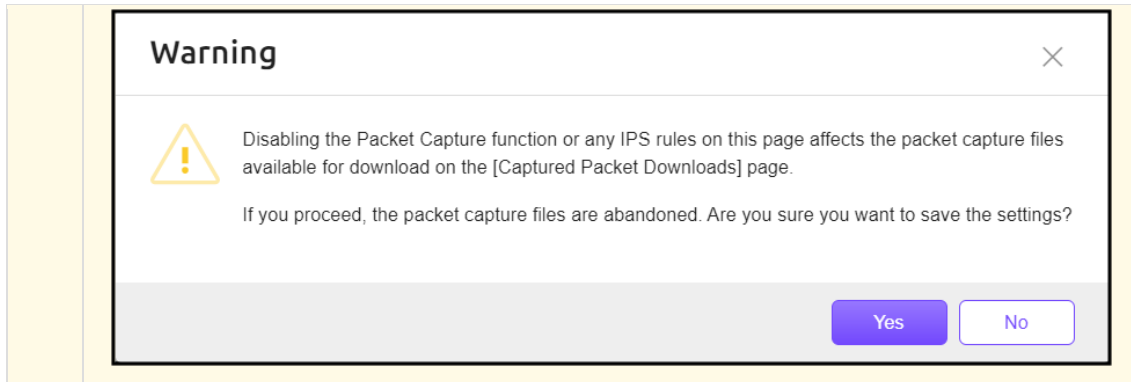
Total Number of Enabled Items: 0 (Max: 5) 1 / 239

Status	IPS Rule ID	Category	Risk Level	IPS Rule Name
☐	1048640	Buffer Overflow	High	WEB Microsoft PCT Buffer Overflow -1 (CVE-2003-0719)
☐	1049406	Exploits	High	WEB /etc/inetd File Download
☐	1049407	Reconnaissance	High	WEB Linux /etc/shadow File Download
☐	1049409	Exploits	High	WEB /tmp/.X000 access
☐	1049659	Exploits	Critical	WEB Cisco IOS HTTP Configuration Arbitrary Administrative Access (CVE-2001-0537)
☐	1049682	Exploits	High	WEB /tmp/.X000 file download (CVE-2001-0892)
☐	1049736	Exploits	High	WEB /tmp/.X000 file download (CVE-2000-0234)
☐	1049802	Web threats	Critical	WEB Directory Traversal - /..
☐	1049859	Buffer Overflow	High	WEB Oracle 9i Application Server PL/SQL Apache Module Buffer Overflow Vulnerability (CVE-2001-1216)
☐	1049945	Buffer Overflow	Critical	WEB Microsoft IIS Chunked Encoding Transfer Heap Overflow Vulnerability
☐	1050002	Exploits	Critical	WEB /etc/inetd.conf Access
☐	1050132	Reconnaissance	High	EXPLOIT Multiple Vendor XDMCP Default Access Control Vulnerability (CVE-2000-0374)
☐	1050153	Buffer Overflow	Critical	WEB Microsoft IIS 5.0 printer ISAPI Extension Buffer Overflow -2 (CVE-2001-0241)

2. Enable the Packet Capture function by clicking the [Packet Capture] toggle.



Disabling the previously enabled Packet Capture function will cause previously downloaded packet captures to be deleted.



3. After the function is enabled, you can see the entire IPS rule list. Select and enable one or multiple rules for IPS event packet capture.

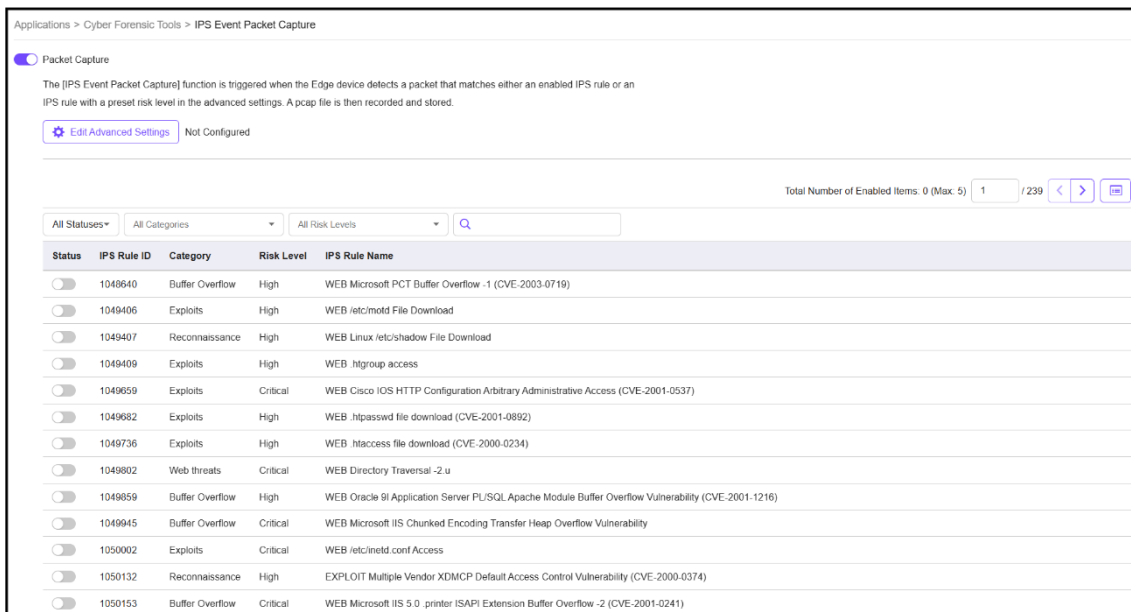
1. You can select up to five rules for the IPS Event Packet Capture feature.
2. The Edge device will save IPS event packets when the selected IPS rules are triggered. Only the last 10 packets of each IPS rule are saved. Older event packets will be overwritten.

4. Click [Save].

9.2.1.2 Configuring IPS Event Packet Capture Advanced Settings

Procedure

1. Go to [Applications] > [Cyber Forensic Tools] > [IPS Event Packet Capture].



2. Enable the Packet Capture function by clicking the [Packet Capture] toggle.

3. Click [Edit Advanced Settings]. The [Edit Advanced Settings] window will appear.

Edit Advanced Settings



The Edge device detects packets using IPS rules with the selected risk level below. If a match is found, a pcap file is recorded and stored on the device.

Risk Level

☐ Critical
 ☐ High

OK

Cancel

4. Select one or multiple risk levels. When the Edge device detects packets matching IPS rules with the specified risk levels, it will record and store the relevant pcap files.
5. Click [OK] on [Edit Advanced Settings] window, then click [Save] on the packet capture page.

9.2.2 Captured Packet Downloads

When the Edge device captures IPS event packets, you can download the captured packets on the [Captured Packet Downloads] page.

9.2.2.1 Downloading Captured IPS Event Packets

Procedure

1. Go to [Applications] > [Cyber Forensic Tools] > [Captured Packet Downloads].

Applications > Cyber Forensic Tools > Captured Packet Downloads

Clear Captured Packets

Total Number of Items: 5

1 / 1

< > [List Icon] [Refresh Icon]

All Categories

All Risk Levels

Q

No.	IPS Rule ID	Category	Risk Level	IPS Rule Name	Last Updated	Actions
1	1048640	Buffer Overflow	High	WEB Microsoft PCT Buffer Overflow -1 (CVE-2003-0719)	2025-05-07T03:38:39Z	
2	1049406	Exploits	High	WEB /etc/motd File Download	2025-05-07T03:38:28Z	
3	1049407	Reconnaissance	High	WEB Linux /etc/shadow File Download	2025-05-07T03:38:39Z	
4	1049409	Exploits	High	WEB /tmp/.X11-unix/.X11-unix access	2025-05-07T03:38:39Z	
5	1049556	Exploits	Medium	WEB Microsoft IIS 5.1 Frontpage /psrvadm.exe File Download	2025-05-07T03:38:39Z	

2. Click the download buttons  to download the zipped packet files to your local disk.



The download list is refreshed every 10 seconds. To get the latest update, click the refresh button .

9.2.2.2 Clearing Captured IPS Event Packets

Procedure

1. Go to [Applications] > [Cyber Forensic Tools] > [Captured Packet Downloads].

Applications > Cyber Forensic Tools > Captured Packet Downloads

[Clear Captured Packets](#) Total Number of Items: 5 / 1

All Categories All Risk Levels Q

No.	IPS Rule ID	Category	Risk Level	IPS Rule Name	Last Updated	Actions
1	1048640	Buffer Overflow	High	WEB Microsoft PCT Buffer Overflow -1 (CVE-2003-0719)	2025-05-07T03:38:39Z	Download
2	1049406	Exploits	High	WEB /etc/mold File Download	2025-05-07T03:38:28Z	Download
3	1049407	Reconnaissance	High	WEB Linux /etc/shadow File Download	2025-05-07T03:38:39Z	Download
4	1049409	Exploits	High	WEB .htgroup access	2025-05-07T03:38:39Z	Download
5	1049556	Exploits	Medium	WEB Microsoft IIS 5.1 Frontpage fprsvadm.exe File Download	2025-05-07T03:38:39Z	Download

2. Click [Clear Captured Packets].

9.2.3 Pcap File Server Settings

In addition to downloading IPS event packets to your local disk, you can also transmit these packet files to a designated remote server using the SMB protocol. On the [Pcap File Server Settings] page, configure the necessary settings to facilitate the transmission of packet files to the server.

9.2.3.1 Viewing Pcap File Server Information

Procedure

1. Go to [Applications] > [Cyber Forensic Tools] > [Pcap File Server Settings].

Applications > Cyber Forensic Tools > Pcap File Server Settings

Pcap File Server Information

SMB Service Status: ON

Server Address: smb:\\192.168.5.24:445\\fileserv\\pkt 

Last Sent Time: -

Pcap File Server Settings

☒ SMB Service

Recorded Event Source* ☒ IPS Event Packet Capture ☒ CPSDR Event Packet Capture

IP Address / Hostname*

Destination Folder* 

Port* 

Username*

Password* 

Sending Interval 

 Test Connection

2. In the [Pcap File Server Information] pane, you will see the SMB service status, server address (click the  button to copy the address), and last packet file sent time.

9.2.3.2 Configuring Pcap File Server Settings

Procedure

1. Go to [Applications] > [Cyber Forensic Tools] > [Pcap File Server Settings].

Applications > Cyber Forensic Tools > Pcap File Server Settings

Pcap File Server Information

SMB Service Status: ON

Server Address: smb:\\192.168.5.24:445\\fileservpkt 

Last Sent Time: -

Pcap File Server Settings

☒ SMB Service

Recorded Event Source* ☒ IPS Event Packet Capture ☒ CPSDR Event Packet Capture

IP Address / Hostname*

Destination Folder* 

Port* 

Username*

Password* 

Sending Interval 

 Test Connection

2. In the [Pcap File Server Settings] pane, configure the following settings.

Item	Description
SMB Service	Enable the SMB service by clicking the toggle.
Recorded Event Source	Select one or multiple sources of the packet files to be sent to the remote server.
IP Address / Hostname	Input the IP address or hostname of the server where the packet files will be sent.
Destination Folder	Input the destination folder on the server where the packet files will be sent.
Port	Input the port number for initializing an SMB connection. (Default: 445)

Item	Description
Username	Input the username for initializing an SMB connection.
Password	Input the password for initializing an SMB connection.
Sending Interval	Select a packet file sending interval, 30 Minutes , 1 Hour , 6 Hours , 12 Hours , or 24 Hours , from the drop-down list.

3. Click [Save].

9.3 Diagnostics

Troubleshoot root causes for device failures.

- [Hardware Diagnostics](#)

9.3.1 Hardware Diagnostics

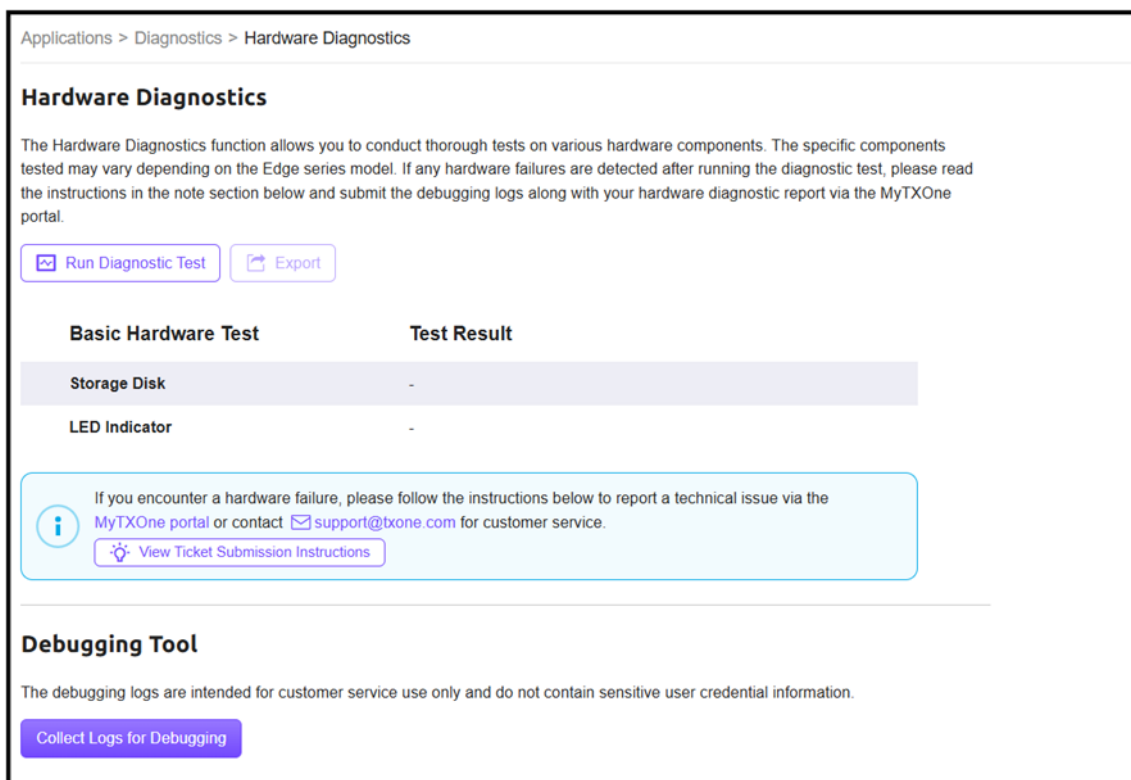
On the [Hardware Diagnostics] page, you can conduct thorough tests on various hardware components. The specific components tested may vary depending on the Edge series model.

If any hardware failures are detected after running the diagnostic test, please report the issue by visiting <https://my.txone.com/> or contacting support@txone.com.

9.3.1.1 Running Diagnostic Tests

Procedure

1. Go to [Admin] > [Diagnostics] > [Hardware Diagnostics].



The screenshot shows the 'Hardware Diagnostics' page. At the top, there is a breadcrumb trail: 'Applications > Diagnostics > Hardware Diagnostics'. Below this, the title 'Hardware Diagnostics' is followed by a descriptive paragraph. Two buttons, 'Run Diagnostic Test' and 'Export', are visible. A table titled 'Basic Hardware Test' shows the 'Test Result' for 'Storage Disk' and 'LED Indicator', both marked as '-'. Below the table, there is a light blue information box with an 'i' icon, containing text about reporting hardware failures and a link to 'View Ticket Submission Instructions'. At the bottom, the 'Debugging Tool' section includes a disclaimer and a 'Collect Logs for Debugging' button.

2. Click [Run Diagnostic Test]. Please wait about 10 to 20 seconds for the test result.

9.3.1.2 Viewing Diagnostic Results

Procedure

1. Go to [Admin] > [Diagnostics] > [Hardware Diagnostics].

Applications > Diagnostics > Hardware Diagnostics

Hardware Diagnostics

The Hardware Diagnostics function allows you to conduct thorough tests on various hardware components. The specific components tested may vary depending on the Edge series model. If any hardware failures are detected after running the diagnostic test, please read the instructions in the note section below and submit the debugging logs along with your hardware diagnostic report via the MyTXOne portal.




Basic Hardware Test	Test Result
Storage Disk	-
LED Indicator	-



If you encounter a hardware failure, please follow the instructions below to report a technical issue via the MyTXOne portal or contact support@txone.com for customer service.



Debugging Tool

The debugging logs are intended for customer service use only and do not contain sensitive user credential information.

Collect Logs for Debugging

2. On the [Hardware Diagnostics] page, you can view the result of the most recent diagnostic test.

Basic Hardware Test	Test Result	Last Updated: 2024-07-22T11:58:57+08:00
Storage Disk	 Storage disk failure detected	
LED Indicator	 LED Indicator failure detected (PWR1, PWR2 indicators)	

9.3.1.3 Exporting Diagnostic Reports

Procedure

1. Go to [Admin] > [Diagnostics] > [Hardware Diagnostics].

Applications > Diagnostics > Hardware Diagnostics

Hardware Diagnostics

The Hardware Diagnostics function allows you to conduct thorough tests on various hardware components. The specific components tested may vary depending on the Edge series model. If any hardware failures are detected after running the diagnostic test, please read the instructions in the note section below and submit the debugging logs along with your hardware diagnostic report via the MyTXOne portal.




Basic Hardware Test	Test Result	Last Updated: 2025-05-07T09:37:18Z
Storage Disk	✓ Passed	
LED Indicator	✗ LED Indicator failure detected (Test1, Test2 indicators)	



If you encounter a hardware failure, please follow the instructions below to report a technical issue via the MyTXOne portal or contact support@txone.com for customer service.



Debugging Tool

The debugging logs are intended for customer service use only and do not contain sensitive user credential information.

Collect Logs for Debugging

2. Click [Export] to download a PDF file containing the result of the most recent diagnostic test.

9.3.1.4 Collecting Debugging Logs

Procedure

1. Go to [Admin] > [Diagnostics] > [Hardware Diagnostics].

Applications > Diagnostics > Hardware Diagnostics

Hardware Diagnostics

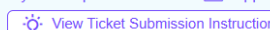
The Hardware Diagnostics function allows you to conduct thorough tests on various hardware components. The specific components tested may vary depending on the Edge series model. If any hardware failures are detected after running the diagnostic test, please read the instructions in the note section below and submit the debugging logs along with your hardware diagnostic report via the MyTXOne portal.




Basic Hardware Test	Test Result
Storage Disk	-
LED Indicator	-



If you encounter a hardware failure, please follow the instructions below to report a technical issue via the MyTXOne portal or contact support@txone.com for customer service.



Debugging Tool

The debugging logs are intended for customer service use only and do not contain sensitive user credential information.

Collect Logs for Debugging

2. In the [Debugging Tool] pane, click [Collect Logs for Debugging]. A tgz file will be downloaded to your local storage.

10 Logs

This chapter describes the security and system logs you can view in the management console.

10.1 Security Logs

Review and analyze logs for security incidents.

- [Cyber Security Logs](#)
- [Policy Enforcement Logs](#)
- [Protocol Filter Logs](#)
- [Domain Name Filter Logs](#)
- [File Filter Logs](#)
- [Suspicious Object Logs](#)
- [CPSDR Logs](#)

10.1.1 Cyber Security Logs

Cyber Security logs contain details about cybersecurity events detected by both Intrusion Prevention and Denial of Service Prevention features.

10.1.1.1 Viewing Cyber Security Logs

Procedure

1. Go to [Logs] > [Security Logs] > [Cyber Security Logs].

The following table describes the Cyber Security log information.

Field	Description
Time	The time when the log entry was created.
Function Type	The function type of the log, either Cyber Security or System Security.
Protocol Layer	The layer of OSI models that the matched signature belongs to.
Policy Rule Name	The name related to the matched policy rule triggering this log.
Profile Name	The name of the IPS profile triggering this log.
Event ID	The ID of the matched signature.
TIDs	MITRE TID Information
Security Category	The category of the matched signature.
Security Severity	The severity level assigned to the matched signature.
Security Rule Name	The name of the matched signature.
Direction	The flow direction of the connection.
Interface	The physical port interface which receives the packet.

Field	Description
Attacker	The IP address of the host device which initiated the cyber incident.
Victim	The IP address of the host device which was attacked in the cyber incident.
Source MAC Address	The source MAC address of the packet.
Source IP Address	The source IP address of the packet.
Source Asset Hostname	The hostname of the asset from which the packet originated.
Source Asset Alias Name	The user-defined alias name of the asset from which the packet originated.
Source Asset Type	The type of the asset from which the packet originated.
Source Port	The source port of the packet, if protocol is TCP/UDP. The ICMP type of the packet, if protocol is ICMP
Destination MAC address	The destination MAC address of the packet.
Destination IP Address	The destination IP address of the packet.
Destination Asset Hostname	The hostname of the asset to which the packet was sent.
Destination Asset Alias Name	The user-defined alias name of the asset to which the packet was sent.
Destination Asset Type	The type of the asset to which the packet was sent.
Destination Port	The destination port of the packet if the protocol is TCP/UDP. The ICMP code of the packet if the protocol is ICMP.
VLAN ID	The VLAN ID of the packet.
Ethernet Type	The Ethernet type of the packet.
IP Protocol Name	The IP protocol name of the packet.
Extra Information	Extra information provided with the cyber security log.
Preset Action	The action configured in the policy rule settings.
Enforced Action	The actual action executed by the device.
Count	The number of detected network packets within the detection period after the detection threshold is reached.

2. To query logs, select a time period, **Last 1 Hour**, **Last 24 Hours**, **Last 7 Days**, **Last 30 Days** or **Custom Time Range** from the drop-down list.

Custom Time Range ▾

+ Add Filters

Search

Last 1 Hour

Last 24 Hours

Last 7 Days

Last 30 Days

✓ Custom Time Range

2024-07-19

17:41:14

to

2024-07-19

17:41:14

<

July 2024

>

Sun	Mon	Tue	Wed	Thu	Fri	Sat
30	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31	1	2	3
4	5	6	7	8	9	10

<

July 2024

>

Sun	Mon	Tue	Wed	Thu	Fri	Sat
30	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31	1	2	3
4	5	6	7	8	9	10

Apply

Cancel

3. Click the [Add Filters] button and the following [Add Filters] window will appear.

Add Filters

×

Add up to 10 filters to search for specific logs. In each filter you can include up to 10 search values.

Field

Function Type ▾

Operator

is ▾

+

Value*

Cyber Security ×

System Security

×

Apply

Cancel

4. Select parameters from the [Field] and [Operator] drop-down lists.

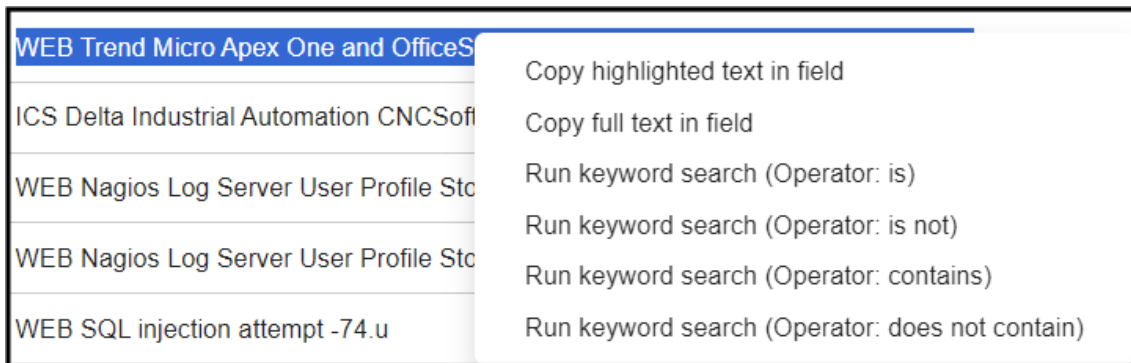
5. Select/Input up to 10 search values in the [Value] field.

6. To add more filters, click the add button **+**.

7. Click [Apply] to apply these filters for querying logs. Afterwards, the filters that you just applied will be listed.
8. Click the [Search] button to query logs.
9. Click the [Clear Filters] button to remove all the applied search filters.
10. To export a CSV file of your current search results, click the  button.
11. To show/hide certain type of information and adjust the column order, click the  button.

 For more information, see [Viewing in Table View Mode](#).

12. Right click on a log entry and a pop-up menu will appear. You can take one of the following actions.



10.1.2 Policy Enforcement Logs

Policy Enforcement logs are generated by the [Policy Enforcement] feature without [Protocol Filter] being enabled, i.e., the action of the Policy Enforcement rule is either [Allow] or [Deny]. The Protocol Filter profiles are not used in policy rules.

10.1.2.1 Viewing Policy Enforcement Logs

Procedure

1. Go to [Logs] > [Security Logs] > [Policy Enforcement Logs].

The following table describes the Policy Enforcement log information.

Field	Description
Time	The time when the log entry was created.
Protocol Layer	The layer of OSI models that the Policy Enforcement rule belongs to.
Policy Rule Name	The name related to the matched policy rule triggering this log.
Direction	The flow direction of the connection.
Interface	The physical port interface which receives the packet.
Source MAC Address	The source MAC address of the packet.

Field	Description
Source IP Address	The source IP address of the packet.
Source Asset Hostname	The hostname of the asset from which the packet originated.
Source Asset Alias Name	The user-defined alias name of the asset from which the packet originated.
Source Asset Type	The type of the asset from which the packet originated.
Source Port	The source port, if protocol is TCP/UDP. The ICMP type, if protocol is ICMP.
Destination MAC Address	The destination MAC address of the packet.
Destination IP Address	The destination IP address of the packet.
Destination Asset Hostname	The hostname of the asset to which the packet was sent.
Destination Asset Alias Name	The user-defined alias name of the asset to which the packet was sent.
Destination Asset Type	The type of the asset to which the packet was sent.
Destination Port	The destination port, if protocol is TCP/UDP. The ICMP code, if protocol is ICMP.
VLAN ID	The VLAN ID of the packet.
Ethernet Type	The Ethernet type of the packet.
IP Protocol Name	The IP protocol name of the packet.
Preset Action	The action configured in the policy rule settings.
Enforced Action	The actual action executed by the device.

 To query logs, see [Viewing Cyber Security Logs](#).

10.1.3 Protocol Filter Logs

Protocol Filter logs contain details about network traffic detected by the [Protocol Filter] feature. The [Protocol Filter] is an advanced setting option that can be configured under the [Policy Enforcement] settings.

10.1.3.1 Viewing Protocol Filter Logs

Procedure

1. Go to [Logs] > [Security Logs] > [Protocol Filter Logs].

The following table describes the Protocol Filter log information.

Field	Description
Time	The time when the log entry was created.
Protocol Layer	The layer of OSI models that the Policy Enforcement rule belongs to.
Policy Rule Name	The name related to the matched policy rule triggering this log.
Profile Name	The name of the Protocol Filter profile triggering this log.
Direction	The flow direction of the connection.
Interface	The physical port interface which receives the packet.
Source MAC Address	The source MAC address of the packet.
Source IP Address	The source IP address of the packet.
Source Asset Hostname	The hostname of the asset from which the packet originated.
Source Asset Alias Name	The user-defined alias name of the asset from which the packet originated.
Source Asset Type	The type of the asset from which the packet originated.
Source Port	The source port, if protocol is TCP/UDP. The ICMP type, if protocol is ICMP.
Destination MAC address	The destination MAC address of the packet.
Destination IP Address	The destination IP address of the packet.
Destination Asset Hostname	The hostname of the asset to which the packet was sent.
Destination Asset Alias Name	The user-defined alias name of the asset to which the packet was sent.
Destination Asset Type	The type of the asset to which the packet was sent.
Destination Port	The destination port, if protocol is TCP/UDP. The ICMP code, if protocol is ICMP.
VLAN ID	The VLAN ID of the packet.
Ethernet Type	The Ethernet type of the packet.
IP Protocol Name	The IP protocol name of the packet.
L7 Protocol Name	The layer 7 protocol name of the connection. The term layer 7 refers to the one defined in the OSI (Open Systems Interconnection) model.
Cmd / Fun No.	The command or function number that triggered the log.
Extra Information	Extra information provided with the log.
Preset Action	The action configured in the policy rule settings.
Enforced Action	The actual action executed by the device.

Field	Description
Count	The number of detected network packets.

 To query logs, see [Viewing Cyber Security Logs](#).

10.1.4 Domain Name Filter Logs

Domain Name Filter logs are generated by the [Domain Name Filter Rule] feature with the action being either [Allow] or [Deny].

10.1.4.1 Viewing Domain Name Filter Logs

Procedure

1. Go to [Logs] > [Security Logs] > [Domain Name Filter Logs].

The following table describes the Domain Name Filter log information.

Field	Description
Time	The time when the log entry was created.
Policy Rule Name	The name of the Domain Name Filter rule triggering this log.
Interface	The physical port interface which receives the packet.
Source MAC Address	The source MAC address of the packet.
Source IP Address	The source IP address of the packet.
Source Asset Hostname	The hostname of the asset from which the packet originated.
Source Asset Alias Name	The user-defined alias name of the asset from which the packet originated.
Source Asset Type	The type of the asset from which the packet originated.
Source Port	The source port, if protocol is TCP/UDP. The ICMP type, if protocol is ICMP.
Destination MAC Address	The destination MAC address of the packet.
Destination IP Address	The destination IP address of the packet.
Destination Asset Hostname	The hostname of the asset to which the packet was sent.
Destination Asset Alias Name	The user-defined alias name of the asset to which the packet was sent.
Destination Asset Type	The type of the asset to which the packet was sent.
Destination Port	The destination port, if protocol is TCP/UDP. The ICMP code, if protocol is ICMP.
VLAN ID	The VLAN ID of the packet.
Domain Name	The domain name in the DNS query.
Preset Action	The action configured in the Domain Name Filter rule settings.

Field	Description
Enforced Action	The actual action executed by the device.

 To query logs, see [Viewing Cyber Security Logs](#).

10.1.5 File Filter Logs

File Filter logs contain details about network traffic detected by the [File Filter] feature. The [File Filter] is an advanced setting option that can be configured under the [Policy Enforcement] settings.

10.1.5.1 Viewing File Filter Logs

Procedure

1. Go to [Logs] > [Security Logs] > [File Filter Logs].

The following table describes the File Filter log information.

Field	Description
Time	The time when the log entry was created.
Policy Rule Name	The name related to the matched policy rule triggering this log.
Profile Name	The name of the File Filter profile triggering this log.
Direction	The flow direction of the connection.
Interface	The physical port interface which receives the packet.
Source MAC Address	The source MAC address of the packet.
Source IP Address	The source IP address of the packet.
Source Asset Hostname	The hostname of the asset from which the packet originated.
Source Asset Alias Name	The user-defined alias name of the asset from which the packet originated.
Source Asset Type	The type of the asset from which the packet originated.
Source Port	The source port, if the protocol is TCP/UDP.
Destination MAC address	The destination MAC address of the packet.
Destination IP Address	The destination IP address of the packet.
Destination Asset Hostname	The hostname of the asset to which the packet was sent.
Destination Asset Alias Name	The user-defined alias name of the asset to which the packet was sent.
Destination Asset Type	The type of the asset to which the packet was sent.
Destination Port	The destination port, if protocol is TCP/UDP.
VLAN ID	The VLAN ID of the packet.

Field	Description
L7 Protocol Name	The layer 7 protocol name of the connection. The term layer 7 refers to the one defined in the OSI (Open Systems Interconnection) model.
Extra Information	Extra information provided with the file filter log.
Preset Action	The action configured in the policy rule settings.
Enforced Action	The actual action executed by the device.

 To query logs, see [Viewing Cyber Security Logs](#).

10.1.6 Suspicious Object Logs

Suspicious Object logs contain details about network traffic detected by the [Suspicious Objects] feature.

10.1.6.1 Viewing Suspicious Object Logs

Procedure

1. Go to [Logs] > [Security Logs] > [Suspicious Object Logs].

The following table describes the Suspicious Object log information.

Field	Description
Time	The time when the log entry was created.
ID	The hash ID of the matched suspicious object.
Type	The suspicious object type, which is node type or link type.
Source	The source of suspicious object.
Risk Level	The threat level of suspicious object.
Expiration Time	The expiration time of the suspicious object. When the expiration time reaches, the suspicious object will be deleted.
Direction	The flow direction of the connection.
Interface	The physical port interface which receives the suspicious object.
Source MAC Address	The source MAC address of the connection.
Source IP Address	The source IP address of the connection.
Source Asset Hostname	The hostname of the asset from which the suspicious object originated.
Source Asset Type	The type of the asset from which the suspicious object originated.
Source Port	The source port of the connection.
Destination MAC address	The destination MAC address of the connection.

Field	Description
Destination IP Address	The destination IP address of the connection.
Destination Asset Hostname	The hostname of the asset to which the suspicious object was sent.
Destination Asset Type	The type of the asset to which the suspicious object was sent.
Destination Port	The destination port of the connection.
VLAN ID	The VLAN ID of the connection.
Ethernet Type	The Ethernet type of the connection.
IP Protocol Name	The IP protocol name of the connection.
Preset Action	The action configured in the Suspicious Object settings.
Enforced Action	The actual action executed by the device.
Count	The number of detected suspicious object within the detection threshold.

 To query logs, see [Viewing Cyber Security Logs](#).

10.1.7 CPSDR Logs

CPSDR logs contain details about network traffic detected by the CPSDR feature with the action being either [Allow] or [Deny].

10.1.7.1 Viewing CPSDR Logs

Procedure

1. Go to [Logs] > [Security Logs] > [CPSDR Logs].

The following table describes the CPSDR log information.

Field	Description
Time	The time when the log entry was created.
CPSDR Rule ID	The rule ID of the CPSDR rule triggering this log.
CPSDR Rule Name	The name of the CPSDR rule triggering this log.
TIDs	MITRE TID Information
Tactic IDs	The ID numbers of the cyber tactics.
Impact Severity	The severity level of the cyber incident according to the CPSDR rule.
Confidence Level	The likelihood of the cyber incident according to the CPSDR rule.
Risk Level	The risk level calculated by the system based on the impact severity and confidence level.
Interface	The physical port interface which receives the packet.

Field	Description
Attacker	The IP address of the host device which initiated the cyber incident.
Victim	The IP address of the host device which was attacked in the cyber incident.
Source MAC Address	The source MAC address of the packet.
Source IP Address	The source IP address of the packet.
Source Asset Hostname	The hostname of the asset from which the packet originated.
Source Asset Alias Name	The user-defined alias name of the asset from which the packet originated.
Source Asset Type	The type of the asset from which the packet originated.
Source Port	The source port, if protocol is TCP/UDP. The ICMP type, if protocol is ICMP.
Destination MAC Address	The destination MAC address of the packet.
Destination IP Address	The destination IP address of the packet.
Destination Asset Hostname	The hostname of the asset to which the packet was sent.
Destination Asset Alias Name	The user-defined alias name of the asset to which the packet was sent.
Destination Asset Type	The type of the asset to which the packet was sent.
Destination Port	The destination port, if protocol is TCP/UDP. The ICMP code, if protocol is ICMP.
VLAN ID	The VLAN ID of the packet.
Ethernet Type	The Ethernet type of the packet.
IP Protocol Name	The IP protocol name of the packet.
Extra Information	Extra information provided with the log.
Source	The origin of the log based on the following criteria. • CPSDR Rule: The rule triggering this log is not included in an allow list or a deny list. In addition, it does not reach the risk level threshold defined in the CPSDR detection threshold settings. • Deny List: The rule triggering this log is included in a deny list. • Threshold Settings: The rule triggering this log reaches or exceeds the risk level threshold defined in the CPSDR detection threshold settings.  When a rule in an allow list is hit, no log entry is generated.
Preset Action	The action based on the log source.

Field	Description
	Allow: When the source of the rule triggering this log is CPSDR Rule. Deny: When the source of the rule triggering this log is Deny List or Threshold Settings.
Enforced Action	The actual action executed by the device, corresponding to the preset action.
Count	The number of detected network packets.

 To query logs, see [Viewing Cyber Security Logs](#).

10.2 Application Logs (Only in Gateway Mode)

Review and analyze logs collected from applications.

- [VPN Logs](#)

10.2.1 VPN Logs

VPN logs contain details about the network traffic passing through your VPN.

10.2.1.1 Viewing VPN Logs

Procedure

- Go to [Logs] > [Application Logs] > [VPN Logs].

The following table describes the VPN log information.

Field	Description
Time	The time when the log entry was created.
Profile Name	The name of the VPN profile triggering this log.
VPN Type	The type of the VPN that the network traffic passes through.
Source IP Address	The source IP address of the connection.
Source Port	The source port of the connection.
Destination IP Address	The destination IP address of the connection.
Destination Port	The destination port of the connection.
IP Protocol Name	The IP protocol name of the connection.
Connection Status	The status of the connection.

 To query logs, see [Viewing Cyber Security Logs](#).

10.3 System Logs

Review and analyze logs for system activities and operational events.

- [Asset Detection Logs](#)
- [System Event Logs](#)
- [Audit Logs](#)

10.3.1 Asset Detection Logs

Asset detection logs contain details about the system status changes of the managed assets.

10.3.1.1 Viewing Asset Detection Logs

Procedure

1. Go to [Logs] > [System Logs] > [Asset Detection Logs].

The following table describes the asset detection log information.

Field	Description
Time	The time when the log entry was created.
Event Type	The log event description.
Interface	The physical port interface which receives the asset information.
Asset MAC Address	The MAC address of the asset.
Asset IP Address	The IP address of the asset.
Asset Hostname	The hostname of the asset.
Asset Type	The type of the asset.

 To query logs, see [Viewing Cyber Security Logs](#).

10.3.2 System Event Logs

System event logs contain details about system events on the device.

10.3.2.1 Viewing System Event Logs

Procedure

1. Go to [Logs] > [System Logs] > [System Event Logs].

The following table describes the system event log information.

Field	Description
Time	The time when the log entry was created.
Severity	The severity level of the logs.
Message	The log event description.

 To query logs, see [Viewing Cyber Security Logs](#).

10.3.3 Audit Logs

Audit logs contain details about user access, configuration changes, and other events that occur when the device is operating.

 To view audit logs, please log in with an auditor account.

10.3.3.1 Viewing Audit Logs

Procedure

1. Go to [Logs] > [System Logs] > [Audit Logs].

The following table describes the audit log information.

Field	Description
Time	The time when the log entry was created.
User ID	The user account used to execute the task.
User IP Address	The IP address of the host used to access the management console.
Severity	The severity level of the logs.
Message	The log event description.

 To query logs, see [Viewing Cyber Security Logs](#).

11 Admin

This chapter describes the available administrative settings for the Edge device.



Log in to the management console with an admin account to access the Admin tab.

11.1 Accounts

Manage account-related settings, including role-based account management, account logout timer, and third-party account authentication methods.

- [Account Management](#)
- [Logout Timer](#)
- [Auth Services](#)

11.1.1 Account Management

This system uses role-based administration to grant and control access to the management console. Use this feature to assign specific management console privileges to the accounts and present them with only the tools and permissions necessary to perform specific tasks. Each account is assigned a specific role. A role defines the level of access to the management console. Users log on to the management console by using custom user accounts.

The following table outlines the tasks available on the [Account Management] page.

Task	Description
Create accounts	Click [Create] to create a new user account.  For more information, see Configuring User Accounts .
Delete existing accounts	Select existing user accounts and click [Delete].
Edit existing accounts	Click the name of an existing user account to view or modify the current account settings.  For more information, see Configuring User Accounts .
Configure password policy	Click the [Configure Password Policy] button to adjust password complexity settings.  For more information, see Configuring Password Policy .



The maximum number of user accounts on an Edge web console is 32.

11.1.1.1 User Roles

The following table describes the permission matrix for user roles.

		User Role			
Tab	Action	Admin	Operator	Viewer	Auditor
Visibility	View	Yes	Yes	Yes	Yes (Can only view the [System Dashboard] page)
	All operations	Yes	Yes	Yes (Cannot create profiles)	No
Device	View	Yes	Yes	No	No
	All operations	Yes	No	No	No
VPN (Only in Gateway Mode)	View	Yes	Yes	No	No
	All operations	Yes	No	No	No
Objects & Profiles	View	Yes	Yes	No	No
	All operations	Yes	Yes	No	No
Security	View	Yes	Yes	No	No
	All operations	Yes	Yes	No	No
Patterns	View	Yes	Yes	No	No
	All operations	Yes	Yes	No	No
Applications	View	Yes	Yes	Yes (Can only view the [Captured Packet Downloads] page)	No
	All operations	Yes	Yes	No	No
Logs - excluding Audit Logs	View	Yes	Yes	Yes	No
Logs - Audit Logs	View	No	No	No	Yes
Admin	View	Yes	No	No	No
	All operations	Yes	No	No	No

11.1.1.2 Built-in Admin Account Login Credential

The following table lists the default login credential of the built-in admin account.

Built-in Default Account ID	User Role	Default Password
admin	Admin	txone



The built-in admin account cannot be deleted from the system. Ensure that the password of the built-in admin account is changed when you first set up the device.

11.1.1.3 Account Input Format

Input format validation will apply to the account management form text fields. The following table describes the format restrictions on user input.

Field	Length	Format	Reserved Name
ID	1-32 characters	- Letters a-z & A-Z - Numbers 0-9 - Special characters: - periods [.] - underscores [_]  - Leading and trailing characters are not special characters. - Successive special characters are not allowed.	- admin - administrator - root
Name	1-32 characters	- Letters a-z & A-Z - Numbers 0-9 - Special characters: - periods [.] - underscores [_] - white spaces []  Single spaces are not allowed.	
Description	0-64 characters	- Letters a-z & A-Z - Numbers 0-9 - Special characters: - periods [.] - underscores [_] - white spaces [] - parentheses [() & []] - hyphens [-]	

11.1.1.4 Account ID/Password Reset

In some specific situations, for security reasons, users are required to reset their IDs or passwords the next time they log in.

User Role	Scenario	
	First Time Login	Password Changed by Admin

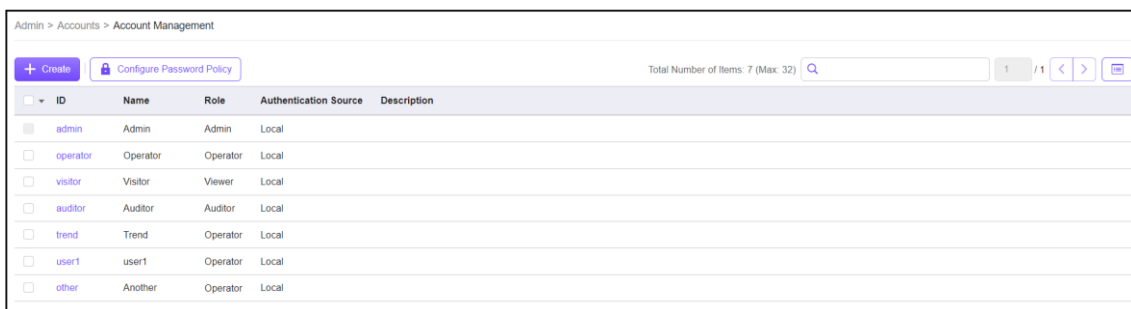
User Role	Scenario	
Admin	Reset ID and Password	N/A
Operator	Reset Password	Reset Password
Viewer	Reset Password	Reset Password
Auditor	Reset Password	Reset Password

11.1.1.5 Configuring User Accounts

When you log in to the web console using an administrator account (admin), you can create new user accounts to access the system or edit account settings.

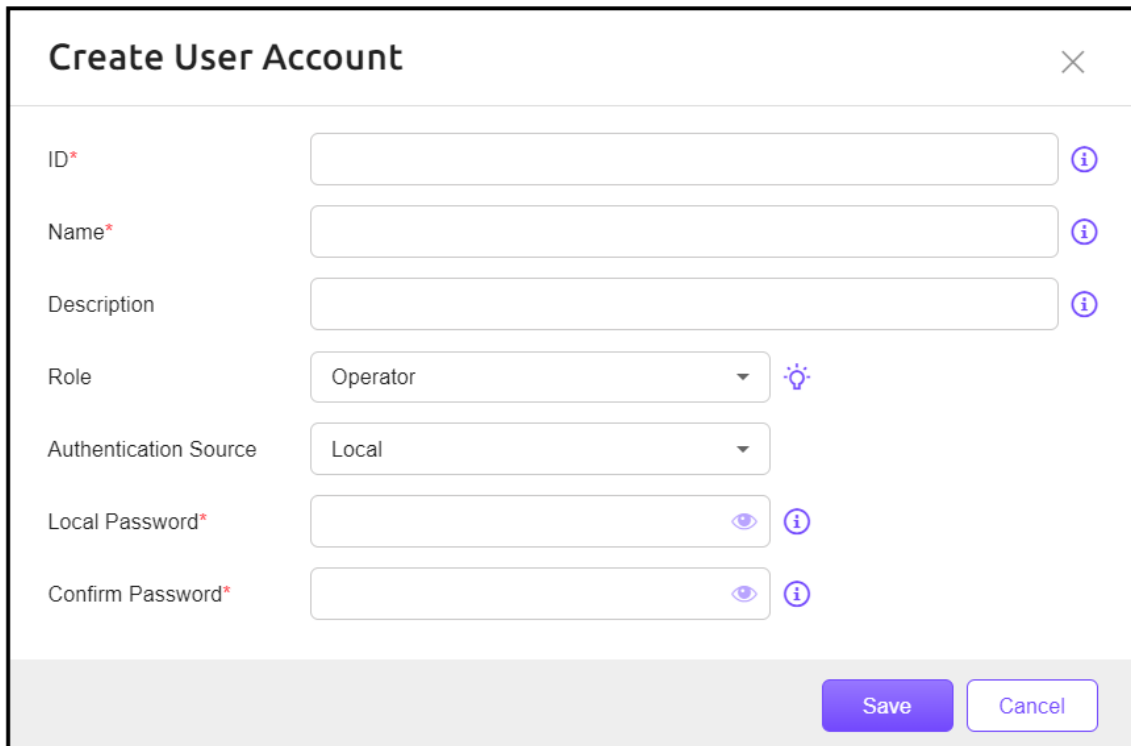
Procedure

1. Go to [Admin] > [Accounts] > [Account Management].



2. Do one of the following.

- Click [Create] to create a new user account. The [Create User Account] window will appear.
- Click a user ID to edit settings. The [Edit User Account] window will appear.



3. Configure the account settings.

Field	Description
ID	Input a user ID to log on to the management console.
Name	Input the name of the user for this account.
Description	Add a description for this account.
Role	Select a user role, Operator , Viewer , or Auditor , for this account.  For more information, see User Roles.
Authentication Source	Select an authentication source, Local or External Authentication Server . If you select External Authentication Server, further select an authentication protocol, TACACS+ , RADIUS , or SAML SSO , for this account.  For more information, see Auth Services.
Local Password	(Only for "Local" authentication source) Input an account password.
Confirm password	(Only for "Local" authentication source) Re-type the account password to confirm.

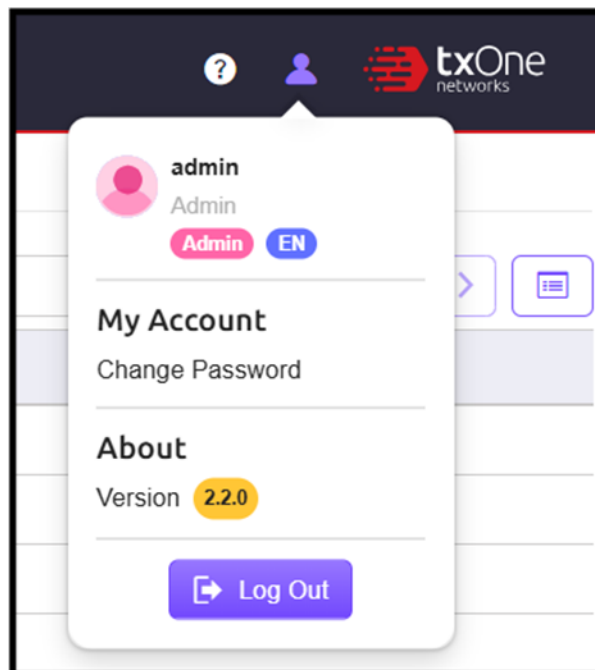
4. Click [Save].

11.1.1.6 Changing Account ID/Password

Procedure

- **Method 1: Access [Change Password] Window** (only for changing password)

1. On the web console banner, click your account  at the top right corner.



2. Click [Change Password]. The [Change Password] window will appear.

Change Password

Current Password*

.....

New Password*

.....

Confirm Password*

.....

Save

Cancel

3. Modify the password settings.

- Current password
- New password
- Confirm password

4. Click [Save].

- **Method 2: Access [Account Management] Page** (Only accessible with an admin account)

1. Go to [Admin] > [Accounts] > [Account Management].

Admin > Accounts > Account Management

+ Create

Configure Password Policy

Total Number of Items: 7 (Max: 32)

1 / 1

ID	Name	Role	Authentication Source	Description
☒ admin	Admin	Admin	Local	
☐ operator	Operator	Operator	Local	
☐ visitor	Visitor	Viewer	Local	
☐ auditor	Auditor	Auditor	Local	
☐ trend	Trend	Operator	Local	
☐ user1	user1	Operator	Local	
☐ other	Another	Operator	Local	

2. Click an ID in the [ID] column. The [Edit User Account] window will appear.

Edit User Account

ID*

Name*

Description

Role

Admin

Authentication Source

Local

Local Password

Confirm Password

Save

Cancel

3. Modify the ID/password settings.

- ID
- Local Password (new password)
- Confirm Password

4. Click [Save].

11.1.1.7 Configuring Password Policy

By configuring the password policy, users may enhance web console access security.

Configure password complexity settings to enforce strong passwords. For example, you can specify that users must create strong passwords that contain a combination of both upper-case and lower-case letters, numbers, and symbols, and which are at least eight characters in length.



When a user submits a new password, the password policy determines whether the password meets your company's established requirements. If strong passwords are required, strict password policies may sometimes increase costs to an organization when users select passwords that are too difficult to remember. Users call the help desk when they forget their passwords, or keep passwords in easily accessible locations that would increase their vulnerability to threats. When establishing a password policy, balance the necessity of highly secured password with the utility of easily-recalling password to make the policy easy for users to follow.

Procedure

1. Go to [Admin] > [Accounts] > [Account Management].
2. Click the [Configure Password Policy] button. The [Configure Password Policy] window will appear.

Configure Password Policy

Minimum password length*
(8 - 32)

☐ Must not include user account ID

☐ Must not include user account name

☐ Must include at least one uppercase letter (A-Z)

☐ Must include at least one lowercase letter (a-z)

☐ Must include at least one number (0-9)

☐ Must include at least one non-alphanumeric character (~!@#\$%^&* _+=`|()\{}[]:;'"<>.,?/)

☐ Must not be the same as the last password

Save

Cancel

3. Select one or multiple options that meet your required password policy.

4. Click [Save].

11.1.2 Logout Timer

On the [Logout Timer] page, you may set a specific time to automatically terminate login sessions after users have been idle for a specified time.

11.1.2.1 Configuring Logout Timer

Procedure

1. Go to [Admin] > [Accounts] > [Logout Timer].

Admin > Accounts > Logout Timer

Logout Timer Setting

Log off idle users after
i

Save

Cancel

2. Select a session timeout value, **10 minutes** (default), **15 minutes**, **30 minutes** or **60 minutes**, from the drop-down list.

3. Click [Save].



The logout timer setting applies to both Edge device web console and CLI.

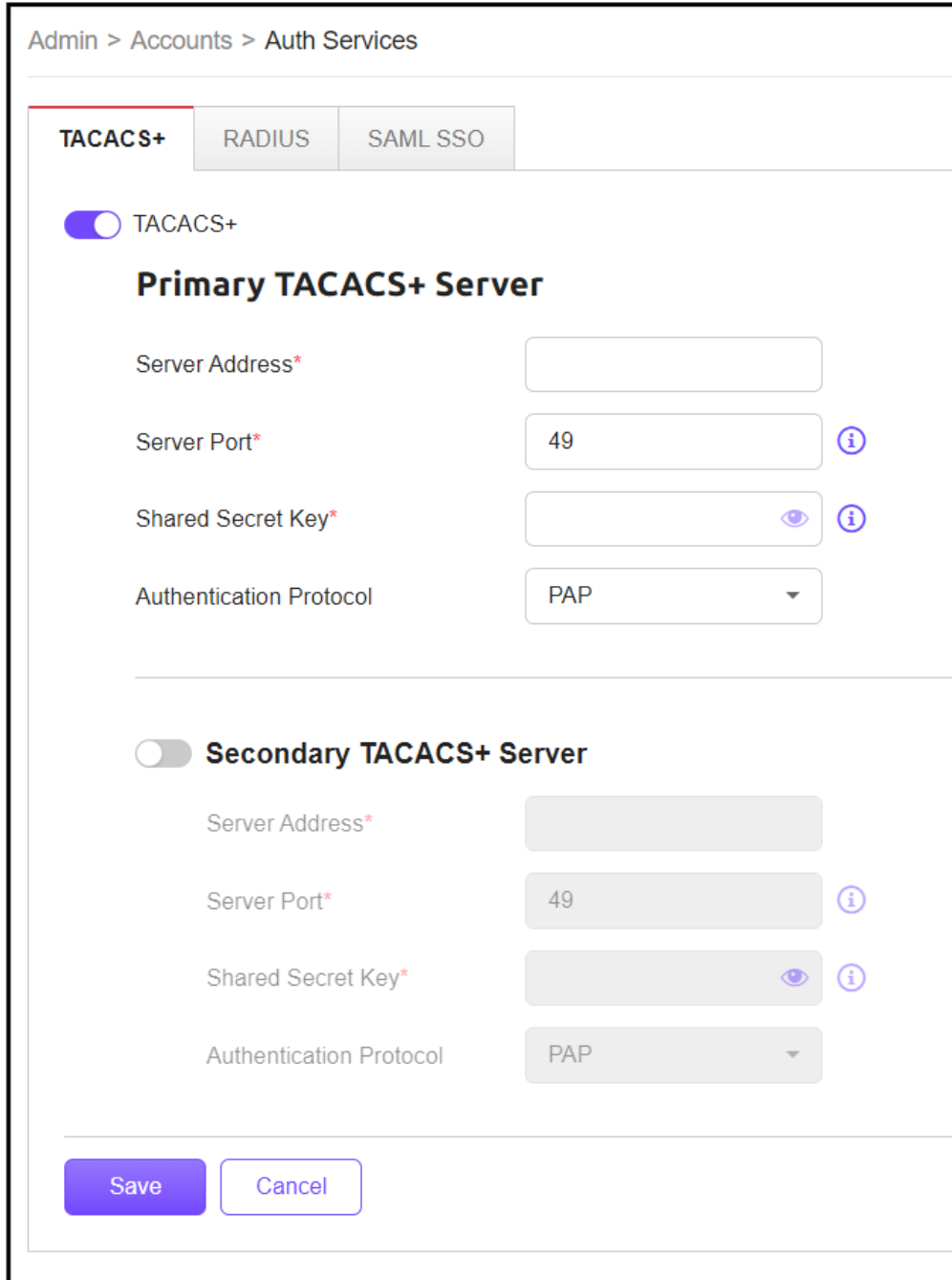
11.1.3 Auth Services

On the [Auth Services] page, you may configure the settings of authentication servers (TACACS+, RADIUS, and SAML SSO) for the Edge device's external authentication service.

11.1.3.1 Configuring TACACS+

Procedure

1. Go to [Admin] > [Accounts] > [Auth Services].



Admin > Accounts > Auth Services

TACACS+ RADIUS SAML SSO

☒ TACACS+

Primary TACACS+ Server

Server Address*

Server Port* ⓘ

Shared Secret Key* ⓘ ⓘ

Authentication Protocol

☐ **Secondary TACACS+ Server**

Server Address*

Server Port* ⓘ

Shared Secret Key* ⓘ ⓘ

Authentication Protocol

2. Under the [TACACS+] tab, enable TACACS+ and configure the following settings for the primary TACACS+ server.

- Server Address
- Server Port (Default Port: 49)
- Share Secret Key (Maximum Length: 64 characters)

- Authentication Protocol (MS-CHAPv1, CHAP, PAP or ASCII)
3. (Optional) Enable the secondary TACACS+ server and configure its settings if necessary.
 4. Click [Save].

 For more information about TACACS+, see the document RFC 8907.

11.1.3.2 Configuring RADIUS

Procedure

1. Go to [Admin] > [Accounts] > [Auth Services].

Admin > Accounts > Auth Services

TACACS+ **RADIUS** SAML SSO

☒ RADIUS

Primary RADIUS Server

Server IP / Name*

Server Port*

Shared Secret Key*

Authentication Protocol

1812

MS-CHAPv2

☐ Secondary RADIUS Server

Server IP / Name*

Server Port*

Shared Secret Key*

Authentication Protocol

1812

MS-CHAPv2

Save

Cancel

2. Under the [RADIUS] tab, enable RADIUS server and configure the following settings for the primary RADIUS server.

- Server IP/Name
- Server Port (Default Port: 1812)

- Share Secret Key (Maximum Length: 64 characters)
 - Authentication Protocol (MS-CHAPv2, MS-CHAPv1, CHAP, or PAP)
3. (Optional) Enable the secondary RADIUS server and configure its settings if necessary.
 4. Click [Save].



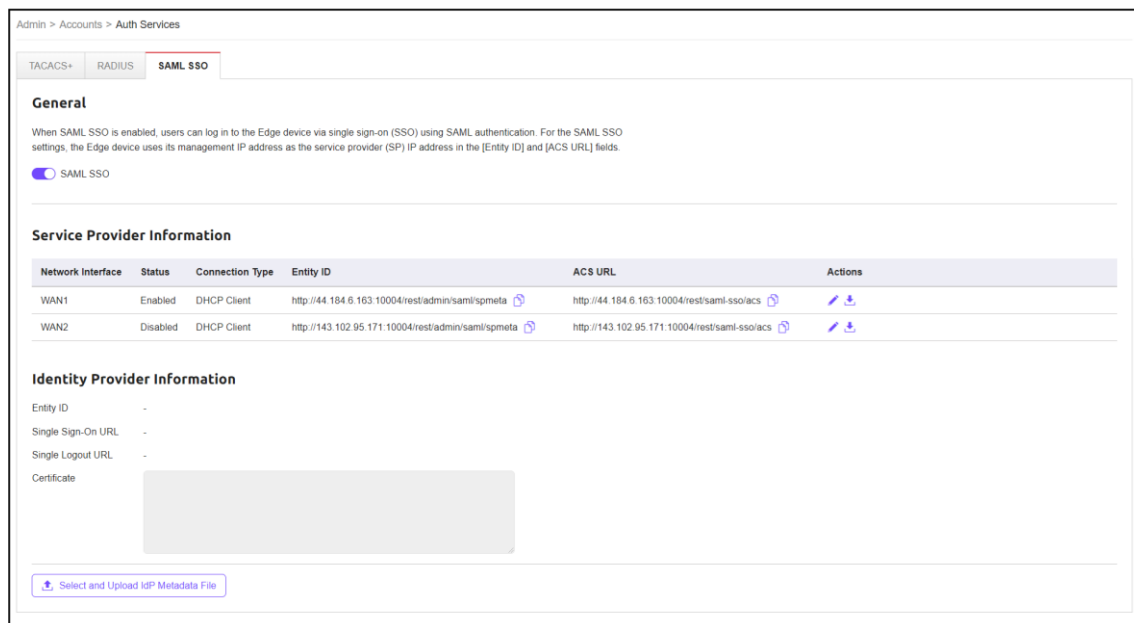
For more information about RADIUS, see the document RFC 2865.

11.1.3.3 Configuring SAML SSO

SAML SSO (Single Sign-On) allows users to log on to the Edge device after the Edge device metadata file with the external identity provider information is provided. The metadata file indicates the data that provide information about the basic configuration data for authentication and authorization.

Procedure

1. Go to [Admin] > [Accounts] > [Auth Services].
2. Under the [SAML SSO] tab, enable SAML SSO login by clicking the toggle.



Admin > Accounts > Auth Services

TACACS+ RADIUS **SAML SSO**

General

When SAML SSO is enabled, users can log in to the Edge device via single sign-on (SSO) using SAML authentication. For the SAML SSO settings, the Edge device uses its management IP address as the service provider (SP) IP address in the [Entity ID] and [ACS URL] fields.

☒ SAML SSO

Service Provider Information

Network Interface	Status	Connection Type	Entity ID	ACS URL	Actions
WAN1	Enabled	DHCP Client	http://44.184.6.163:10004/rest/admin/saml/spmeta	http://44.184.6.163:10004/rest/saml-ss0/acs	📄 ✎ 📄
WAN2	Disabled	DHCP Client	http://143.102.95.171:10004/rest/admin/saml/spmeta	http://143.102.95.171:10004/rest/saml-ss0/acs	📄 ✎ 📄

Identity Provider Information

Entity ID -

Single Sign-On URL -

Single Logout URL -

Certificate

[📄 Select and Upload IdP Metadata File](#)

3. In the [Service Provider Information] pane, the following information about the WAN1 and WAN2 network interfaces is shown in the table.

- Status
- Connection Type
- Entity ID
- ACS URL

4. Click the [📄](#) button to copy information, the [✎](#) button to edit service provider information, or the [📄](#) button to download the service provider metadata file in XML format.

5. Go to the external identity provider server and import the metadata file. An integrated metadata file will be generated.

6. In the [Identity Provider Information] pane, the following information will be automatically shown on the screen.

- Entity ID

- Single Sign-On URL
- Single Logout URL
- Certificate

7. After confirming the information above, click the [Select and Upload IdP Metadata File] button to import the integrated authentication and authorization data to the Edge device.

8. Click [Save].



For more information about SAML SSO, see the document RFC 7522.

11.2 Notifications

Configure notification settings.

- [SNMP](#)
- [Syslog](#)

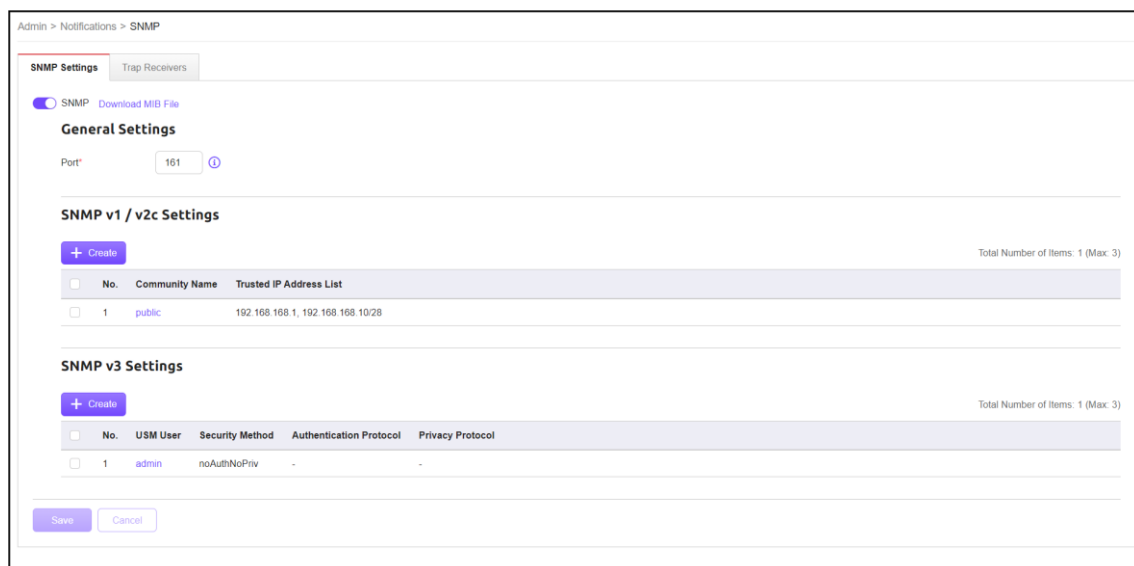
11.2.1 SNMP

The Simple Network Management Protocol is a protocol used for exchanging management information between an Edge series device and its management console. The device supports SNMPv1/v2c and a more secure v3, as well as SNMP traps.

11.2.1.1 Configuring SNMP General Settings

Procedure

1. Go to [Admin] > [Notifications] > [SNMP].



2. Under the [SNMP Settings] tab, enable the SNMP function by clicking the toggle.

3. In the [General Settings] pane, you can change the SNMP port. The default port is 161.

4. To download the Edge device MIB file, click the [Download MIB file] link.

11.2.1.2 Configuring SNMP v1/v2c Communities

Procedure

1. Go to [Admin] > [Notifications] > [SNMP]. Then switch to the [SNMP Settings] tab.

2. In the [SNMP v1/v2c Settings] pane, do one of the following.

- Click [Create] to create an SNMP v1/v2c community. The [Create Community] window will appear.
- Click an SNMP v1/v2c community name. The [Edit Community] window will appear.

Create Community

Community Name*

i

Trusted IP Address List

i

(Max: 3)

No.1

Single IP

+

OK

Cancel

3. Configure the following settings.

Item	Description
Community Name	Input an SNMP v1/v2c community name.
Trusted IP Address List	Specify an IP address or an IP subnet. Click the add button  to add another entry, or the remove button  to remove an entry.

4. Click [OK].

5. Click [Save] on the [SNMP] page.

11.2.1.3 Configuring SNMP v3 USM Users

Procedure

1. Go to [Admin] > [Notifications] > [SNMP]. Then switch to the [SNMP Settings] tab.
2. In the [SNMP v3 Settings] pane, do one of the following.

- Click [Create] to create an SNMP v3 USM user. The [Create USM User] window will appear.
- Click an SNMP v3 USM username. The [Edit USM User] window will appear.

Create USM User

USM User*

Security Method

No Authentication or Privacy

OK

Cancel

3. Configure the following settings.

Item	Description
USM User	Input an SNMP v3 USM username.
Security Method	Select one of the following options. • No Authentication or Privacy • Authentication without Privacy: Further select an appropriate authentication protocol (SHA or MD5) and enter an authentication key. • Authentication with Privacy: Further select an appropriate authentication protocol (SHA or MD5), enter an authentication key, select an appropriate privacy protocol (AES or DES), and enter a privacy key.

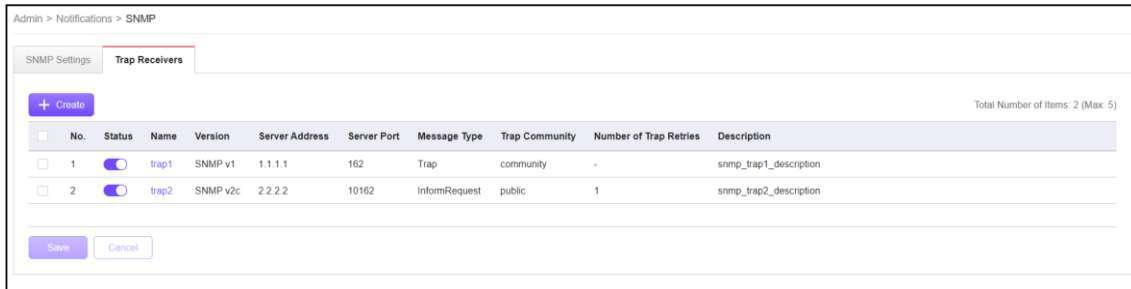
4. Click [OK].

5. Click [Save] on the [SNMP] page.

11.2.1.4 Configuring SNMP Trap Receivers

Procedure

1. Go to [Admin] > [Notifications] > [SNMP].



Admin > Notifications > SNMP

SNMP Settings **Trap Receivers**

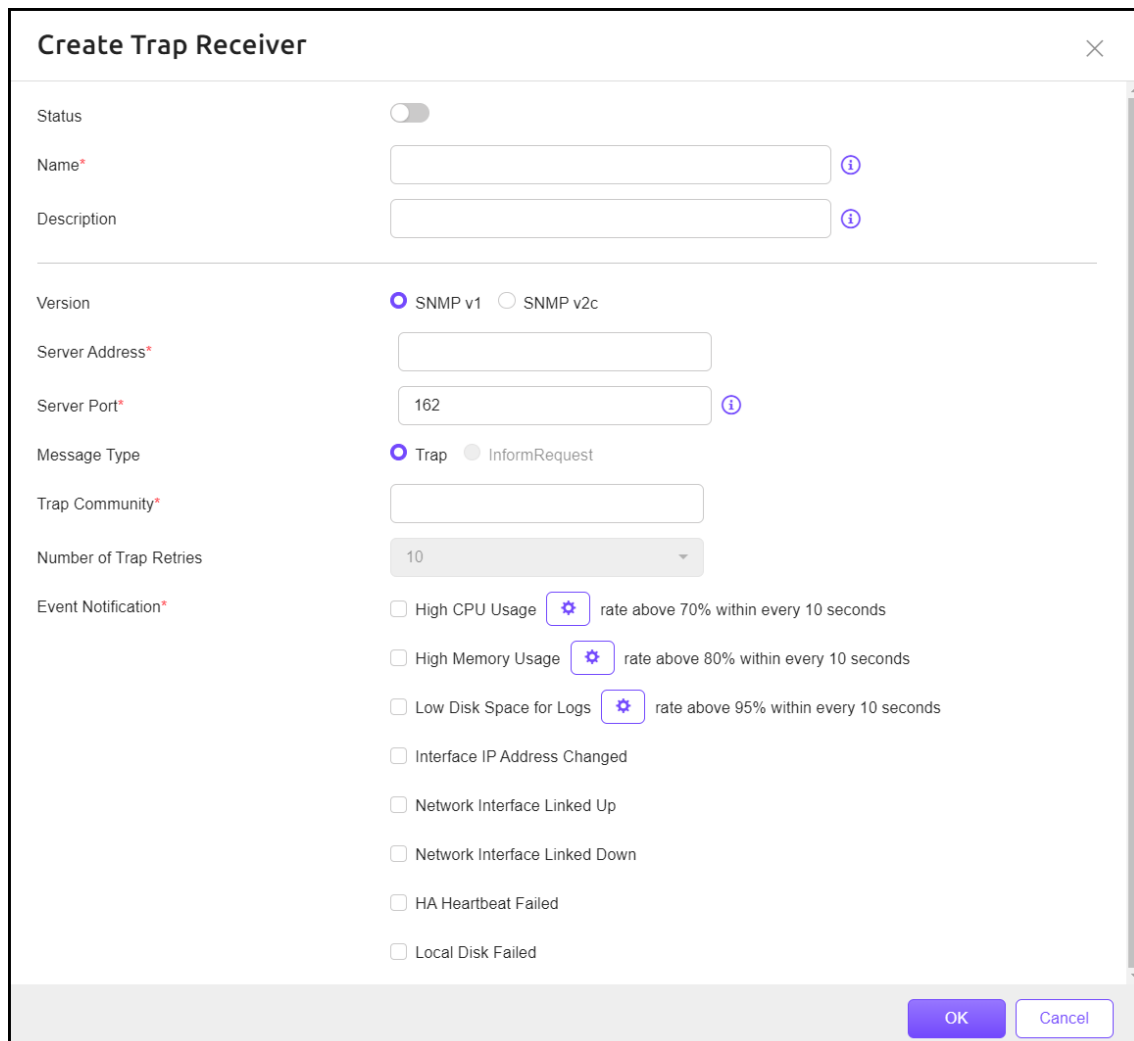
[+ Create](#) Total Number of Items: 2 (Max: 5)

No.	Status	Name	Version	Server Address	Server Port	Message Type	Trap Community	Number of Trap Retries	Description
1	☒	trap1	SNMP v1	1.1.1.1	162	Trap	community	-	snmp_trap1_description
2	☒	trap2	SNMP v2c	2.2.2.2	10162	InformRequest	public	1	snmp_trap2_description

[Save](#) [Cancel](#)

2. Under the [Trap Receivers] tab, do one of the following.

- Click [Create] to create a new trap receiver. The [Create Trap Receiver] window will appear.
- Click a trap receiver item. The [Edit Trap Receiver] window will appear.



Create Trap Receiver ×

Status ☐

Name* i

Description i

Version ☒ SNMP v1 ☐ SNMP v2c

Server Address*

Server Port* i

Message Type ☒ Trap ☐ InformRequest

Trap Community*

Number of Trap Retries

Event Notification*

☐ High CPU Usage i rate above 70% within every 10 seconds

☐ High Memory Usage i rate above 80% within every 10 seconds

☐ Low Disk Space for Logs i rate above 95% within every 10 seconds

☐ Interface IP Address Changed

☐ Network Interface Linked Up

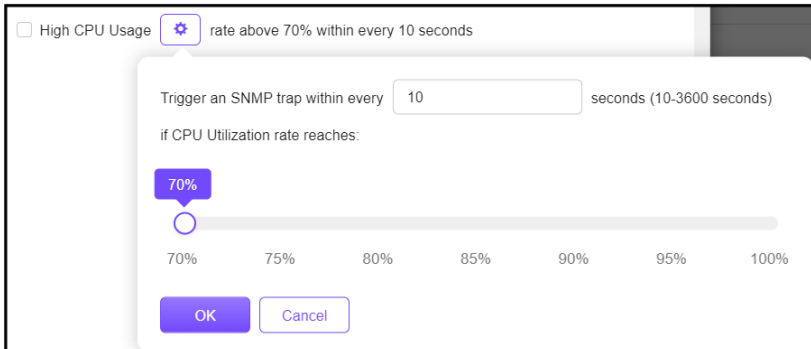
☐ Network Interface Linked Down

☐ HA Heartbeat Failed

☐ Local Disk Failed

[OK](#) [Cancel](#)

3. Configure the following trap receiver settings.

Item	Description
Status	Enable this trap receiver by clicking the toggle.
Name	Type a name for this trap receiver.
Description	(Optional) Type a description.
Version	Select an SNMP version, SNMP v1 or SNMP v2c .
Server Address	Input a server IP address.
Server Port	Input a server port. The default server port is 162.
Message Type	Select a message type, Trap or informRequest .
Trap Community	Input a trap community name. The default name is PUBLIC.
Number of Trap Retries	Set the number of trap retries ranging from 1 to 10.
Event Notification	Select what will trigger an event notification. For High CPU Usage, High Memory Usage, and Low Disk Space for Logs, click the  button to adjust the threshold for triggering the notification.  Due to differences in CPU operating temperature ranges between commercial-grade and industrial-grade Edge products, the SNMP trap setting for CPU temperature is not interchangeable. As a result, when a user imports a configuration file from one product type to another, the CPU temperature setting reverts to its default value (based on the specific Edge model) rather than applying the value from the imported file. 

4. Click [OK].

5. Click [Save] on the [SNMP] page.

11.2.2 Syslog

The system of this Edge series device maintains Syslog events that provide summaries of security and system events. Common Event Format (CEF) and Log Event Extended Format (LEEF) syslog messages are used in the device.

Configure the Syslog settings to enable the Edge device to send the Syslog to a Syslog server.

11.2.2.1 Configuring Syslog Settings

Procedure

1. Go to [Admin] > [Notifications] > [Syslog].

Admin > Notifications > Syslog

[+ Create](#) Total Number of Items: 2 (Max: 10) 1 / 1

☐	No.	Status	Name	Transmission Status	Server Address	Port	Protocol	Format	Facility Level	Log Output	Description
☐	1		Syslog_1	-	192.168.3.46	601	TCP	CEF	Local 0	Cyber Security Logs, Policy Enforcement Logs, Protocol Filter Logs	
☐	2		Syslog_2	-	10.35.1.64	601	TCP	CEF	Local 1	Asset Detection Logs, System Logs, Audit Logs	

2. Do one of the following.

- Click [Create] to create a Syslog profile. The [Create Syslog Settings] window will appear.
- Click a profile name to edit settings. The [Edit Syslog Settings] window will appear.

Create Syslog Settings

Profile Status

Name*

Description

Syslog Settings

Server Address*

Port*

601

Protocol*

☒ TCP
☐ UDP
☐ TLS

Format*

☒ CEF
☐ LEEF

Facility Level

Local 0

Log Output*

Security Logs

☐ Cyber Security Logs
Security Severity: All, Enforced Action: All

☐ Policy Enforcement Logs
Enforced Action: All

☐ Protocol Filter Logs
Enforced Action: All

☐ Domain Name Filter Logs
Enforced Action: All

☐ File Filter Logs
Enforced Action: All

☐ Suspicious Object Logs
Risk Level: All, Enforced Action: All

☐ CPSDR Logs
Impact Severity: All, Risk Level: All, Enforced Action: All

Application Logs

☐ VPN Logs
Connection Status: All

System Logs

☐ Asset Detection Logs
Event Type: All

☐ System Event Logs
Severity: All

☐ Audit Logs
Severity: All

Save

Cancel

3. Configure the following settings.

Field	Description
Status	Click the toggle to enable the Syslog settings configured in this profile.
Name	Type a name for this Syslog profile.
Description	(Optional) Type a description.

Admin – 231

- Syslog Settings

Field	Description
Server Address	Type the IP address of the Syslog server.
Port	Type the port number.
Protocol	Select a protocol for the communication.
Format	Select a syslog format, either CEF or LEEF.
Facility Level	Select a facility level to determine the source and priority of the logs.
Log Output	Select the types of logs to send. To configure detailed log filter settings for log output, click the  button.  For Cyber Security, Suspicious Object, and CPSPDR logs, you can configure the severity/risk levels of the output logs. For more information about Syslog severity and risk levels, see <i>Syslog Severity Levels</i>.

1. Click [Save].

11.2.2.2 Syslog Severity Levels

The Syslog severity level specifies the type of messages to be sent to a Syslog server.

Level	Severity	Description
0	Emergency	Complete system failure Take immediate action.
1	Alert	Primary system failure Take immediate action.
2	Critical	Urgent failure Take immediate action.
3	Error	Non-urgent failure Resolve issues quickly.
4	Warning	Error pending Take action to avoid errors.
5	Notice	Unusual events Immediate action is not required.
6	Informational	Normal operational messages useful for reporting, measuring throughput, and other purposes No action is required.
7	Debugging	Useful information when debugging the application.

Level	Severity	Description
		 Setting the debugging level can lead to a large amount of Syslog traffic in a busy network. Use with caution.

11.2.2.3 Syslog Severity Level Mapping Table

The following table summarizes the logs of Policy Enforcement/Protocol Filter/Cybersecurity and their equivalent Syslog severity levels.

Policy Enforcement/ Protocol Filter Action	Cyber Security Severity Level	Syslog Severity Level
		0 - Emergency
	Critical	1 – Alert
	High	2 – Critical
		3 – Error
Deny	Medium	4 - Warning
	Low	5 - Notice
Allow	Informational	6 - Informational
		7 - Debugging

11.3 Licensing & Updates

Manage product licenses and firmware updates.

- [License Management](#)
- [Firmware Updates & Management](#)

11.3.1 License Management

To fully access all functionalities of the Edge device, including pattern updates, you must first activate a valid Edge device software license. On the [License Management] page, you can activate your Edge device license using either a license key or a license file, and view detailed information about your license.

To activate a license, you can either enter a license key or upload a license file. If an Internet connection is unavailable, please upload your license file for activation.

11.3.1.1 Viewing License Information

Procedure

1. Go to [Admin] > [Licensing & Updates] > [License Management].

Admin > Licensing & Updates > License Management

License Management

A valid Edge device license is required for updating patterns via the Edge device console. Please activate the Edge device license before updating patterns under [Patterns] > [Pattern Updates].

License Key XXXXXXXXXXXX-XXXX-XXXX-XXXX-XXXX

Expiration Date 2026-12-22T08:00:00+08:00

To activate a license, you can either enter a license key or upload a license file. If Internet connection is unavailable, please upload your license file for activation.



License Key

- ✓ License Key
- ✓ Internet Connection



License File

- ✓ License File (.txt)

2. On the [License Management] page, you will see the license key and expiration date of the activated license.

11.3.1.2 Activating/Renewing License with License Key

Procedure

1. Go to [Admin] > [Licensing & Updates] > [License Management].

Admin > Licensing & Updates > License Management

License Management

A valid Edge device license is required for updating patterns via the Edge device console. Please activate the Edge device license before updating patterns under [Patterns] > [Pattern Updates].

License Key XXXXXXXXXXXX-XXXX-XXXX-XXXX-XXXX

Expiration Date 2026-12-22T08:00:00+08:00

To activate a license, you can either enter a license key or upload a license file. If Internet connection is unavailable, please upload your license file for activation.



License Key

- ✓ License Key
- ✓ Internet Connection



License File

- ✓ License File (.txt)

2. Select [License Key] and input a valid Edge device license key.
3. Click [Verify].
4. Verify the license information shown and click [Activate].



License Key

- ✓ License Key
- ✓ Internet Connection



License File

- ✓ License File (.txt)

License Key

Expiration Date 2026-12-22T08:00:00+08:00

Do you want to activate this license?



Internet connectivity is required for activating/renewing a license using a license key.

11.3.1.3 Activating/Renewing License with License File

Procedure

1. Go to [Admin] > [Licensing & Updates] > [License Management].

Admin > Licensing & Updates > License Management

License Management

A valid Edge device license is required for updating patterns via the Edge device console. Please activate the Edge device license before updating patterns under [Patterns] > [Pattern Updates].

License Key XXXXXXXXXXXX-XXXX-XXXX-XXXX

Expiration Date 2026-12-22T08:00:00+08:00

To activate a license, you can either enter a license key or upload a license file. If Internet connection is unavailable, please upload your license file for activation.



License Key

- ✓ License Key
- ✓ Internet Connection



License File

- ✓ License File (.txt)

2. Select [License File], and then click [Get License File] to connect to the TXOne License File Management server. (This step requires an Internet connection.)

License File Management 

 License File

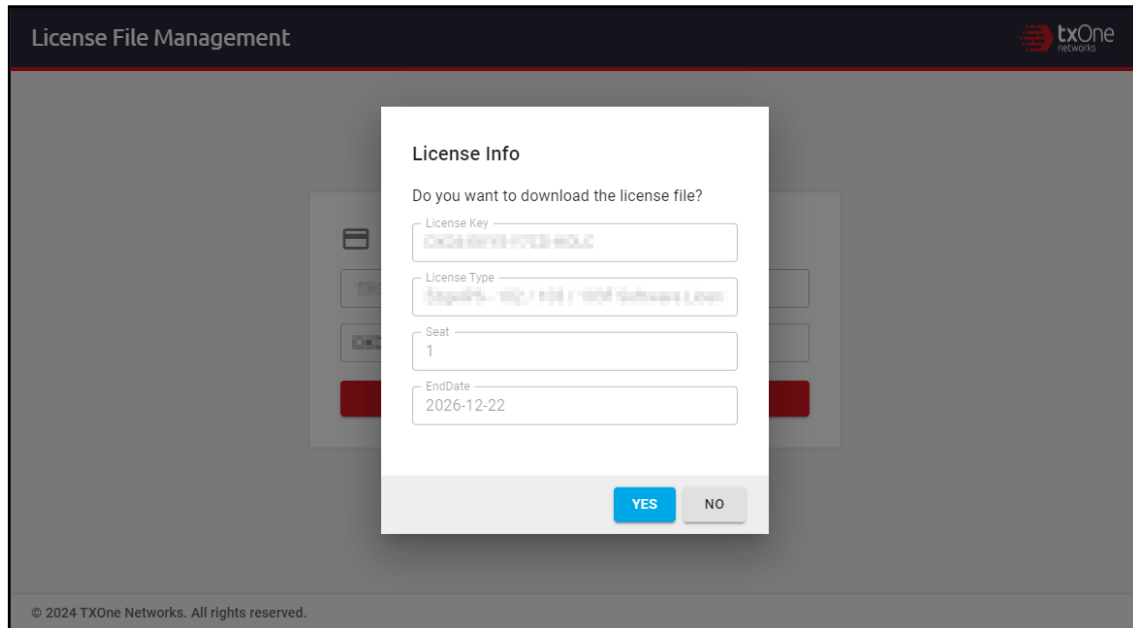
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1. The Edge device serial number is provided on the license file management server.
2. If you do not have an Internet connection, you can click [Copy Download Link] to copy the link of the TXOne License File Management server, and then open the link to download the license file when your Internet connection is restored. The

license file generated by the download link is only valid when it is applied to the same Edge device from which the link was generated.

3. Input your license key to download your license file.



The screenshot shows the 'License File Management' page with a 'License Info' dialog box. The dialog box contains the following fields:

- License Key:** A text input field containing the value 'CXC4-8878-PT28-HQJC'.
- License Type:** A dropdown menu with the selected value 'EdgeFire 1014 / 1014 / 1014 / 1014 / 1014 / 1014 / 1014 / 1014'.
- Seat:** A text input field containing the value '1'.
- EndDate:** A text input field containing the value '2026-12-22'.

At the bottom of the dialog box, there are two buttons: 'YES' (highlighted in blue) and 'NO'.

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4. Go back to the [License Management] page, and then click [Select File] to upload your license file.

5. Verify the license information shown and click [Activate].



License Key

- ✓ License Key
- ✓ Internet Connection



License File

- ✓ License File (.txt)

License Key

Expiration Date 2026-12-22T08:00:00+08:00

Do you want to activate this license?


 If you don't have a license file, please click [Get License File] to obtain one or [Copy Download Link] to request it later.

[Get License File](#)
[Copy Download Link](#)

11.3.2 Firmware Updates & Management

On the [Firmware Updates & Management] page, you can view the device firmware information, update the device firmware, and switch to the standby firmware.

11.3.2.1 Viewing Device Firmware Information

Procedure

1. Go to [Admin] > [Licensing & Updates] > [Firmware Updates & Management].

Admin > Licensing & Updates > Firmware Updates & Management

No.	Partition Name	Partition Status	Firmware Version	Firmware Build Time	Actions
1	p1	Running	2.1.6	2024-04-29T12:00:00+08:00	
2	p2	Standby	2.0.0	2019-05-09T08:00:00+08:00	 

2. The table on this page lists the two available firmware partitions and shows information including the partition name, partition status, firmware version, and firmware build time.

The device can have up to two firmwares to be installed. Each firmware is installed in its own separate partition. At any given point in time, one partition will have the status [Running], which indicates the currently running and active firmware. The other partition will have the status [Standby], which indicates an alternative or standby partition.

11.3.2.2 Updating Device Firmware

Procedure

1. Go to [Admin] > [Licensing & Updates] > [Firmware Updates & Management].

Admin > Licensing & Updates > Firmware Updates & Management

No.	Partition Name	Partition Status	Firmware Version	Firmware Build Time	Actions
1	p1	Running	2.1.6	2024-04-29T12:00:00+08:00	
2	p2	Standby	2.0.0	2019-05-09T08:00:00+08:00	

2. Click the [Update Firmware] button  to install firmware to the [Standby] partition. The [Update Firmware] window will appear.

Update Firmware

Firmware Information

Running Firmware2.0.0
Firmware Build Time2019-05-09T08:00:00+08:00

Firmware Update

Uploaded Firmware File

Select File

Upload

Close

During a firmware update, firmware will always be installed to the [Standby] partition. As such, the [Update Firmware] button is only available in the [Standby] partition row.

3. On the [Update Firmware] window, click [Select File] to select a firmware file from your local device, and then click [Upload] to install the firmware to the standby partition.
4. Click [Close].

Various versions of the firmware files can be downloaded at the MyTXOne portal at <https://my.txone.com/>.

11.3.2.3 Switching to Standby Firmware

To boot into an updated firmware or to revert to a previous firmware, you would need to boot into the [Standby] partition and load the firmware from it.

Procedure

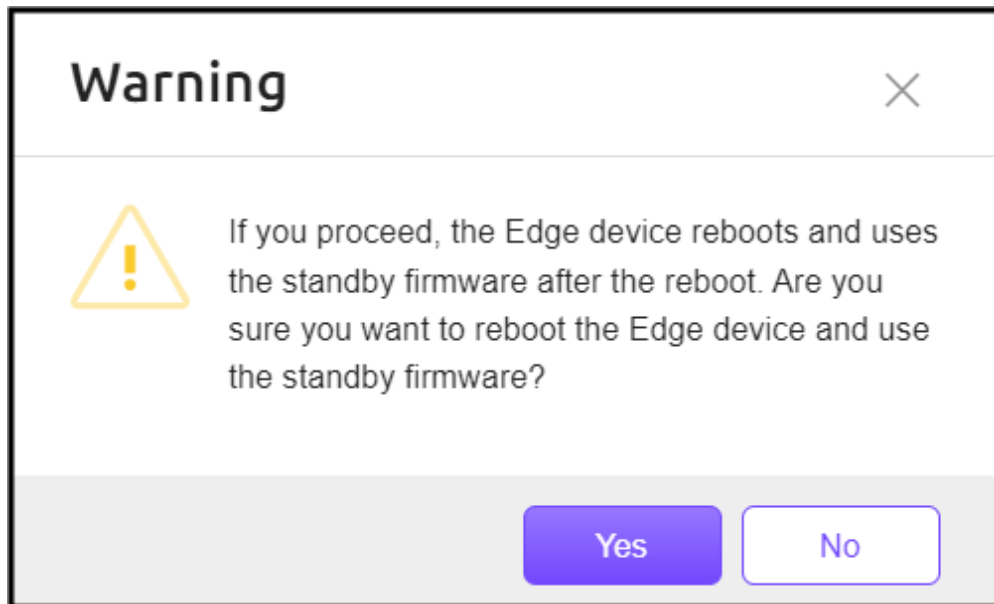
1. Go to [Admin] > [Licensing & Updates] > [Firmware Updates & Management].

Admin > Licensing & Updates > Firmware Updates & Management

No.	Partition Name	Partition Status	Firmware Version	Firmware Build Time	Actions
1	p1	Running	2.1.6	2024-04-29T12:00:00+08:00	
2	p2	Standby	2.0.0	2019-05-09T08:00:00+08:00	↔

2. Click the [Switch to Standby Firmware] button [↔](#) in the [Standby] partition row.

3. The below warning will be shown to the user. Click [Yes] to proceed with rebooting into the [Standby] partition and making it the [Running] partition.



11.4 System

Configure administrative settings to optimize device operations.

- [System Management](#)
- [Edge Device Security](#)
- [EdgeOne Connection Settings](#)
- [Language Package](#)
- [API Management](#)
- [SSL Certificate](#)
- [Backup/Restore](#)
- [System Time](#)
- [System Reboot](#)

11.4.1 System Management

On the [System Management] page, you can do the following:

- Configure the hostname and location information of the device.

- Choose the protocols and ports that can be used to manage the device.
 - Configure the IP addresses that are allowed to access these protocols.
- Allow pings to the management interface.

11.4.1.1 Configuring Device Hostname and Location

Procedure

1. Go to [Admin] > [System] > [System Management].

Admin > System > System Management

Edge Device Information

Hostname*

IEF-local

Location

Example: Zone1, Network-1

OT Layer

L3

Management Methods

☒ HTTP*

10004

☐ HTTPS

443

☐ SSH

22

☐ Telnet

23

☐ Allow access from WAN

☐ Access Control List

Allowed IP Address 1

Allowed IP Address 2

Allowed IP Address 3

Allowed IP Address 4

Management Interface Ping Settings

☒ Allow ping to management interface

Save

Cancel

2. In the [Edge Device Information] pane, input a hostname and location, and select an OT layer for the device.

3. Click [Save].

11.4.1.2 Configuring Management Protocols and Ports

Procedure

1. Go to [Admin] > [System] > [System Management].

Admin > System > System Management

Edge Device Information

Hostname* ⓘ

Location ⓘ
Example: Zone1, Network-1

OT Layer

Management Methods

☒ HTTP* ⓘ

☐ HTTPS ⓘ

☐ SSH ⓘ

☐ Telnet ⓘ

☐ Allow access from WAN

☐ Access Control List

Allowed IP Address 1

Allowed IP Address 2

Allowed IP Address 3

Allowed IP Address 4

Management Interface Ping Settings

☒ Allow ping to management interface

2. In the [Management Methods] pane:

- Select the protocols that are allowed to be used.
- Input port numbers for the protocols.



The HTTP and HTTPS protocols are used for connecting to the web management console. The SSH and Telnet protocols are used for connecting to the CLI commands.

3. Click [Save].

11.4.1.3 Configuring Access Control List and Access from WAN Interface

Procedure

1. Go to [Admin] > [System] > [System Management].

Admin > System > System Management

Edge Device Information

Hostname* ⓘ

Location ⓘ
Example: Zone1, Network-1

OT Layer

Management Methods

☒ HTTP* ⓘ

☐ HTTPS ⓘ

☐ SSH ⓘ

☐ Telnet ⓘ

☐ Allow access from WAN
 ☐ Access Control List

Allowed IP Address 1

Allowed IP Address 2

Allowed IP Address 3

Allowed IP Address 4

Management Interface Ping Settings

☒ Allow ping to management interface

2. Click the [Allow access from WAN] toggle to enable or disable access from the WAN interfaces and the [Access Control List] toggle to enable or disable access control of the management clients.

3. List the IP addresses that are allowed for accessing the device.

4. If the connection between the Edge device and your network is not stable or is disconnected, enable the toggle to ping to the management Interface.

Management Interface Ping Settings

☒ Allow ping to management interface

5. Click [Save].

11.4.2 Edge Device Security

The feature, System Security, is used to protect the Edge device itself against file vulnerabilities, buffer overflow, denial of service attacks, exploits, etc. by applying pre-defined IPS profiles and configuring the thresholds of the denial of service rules.

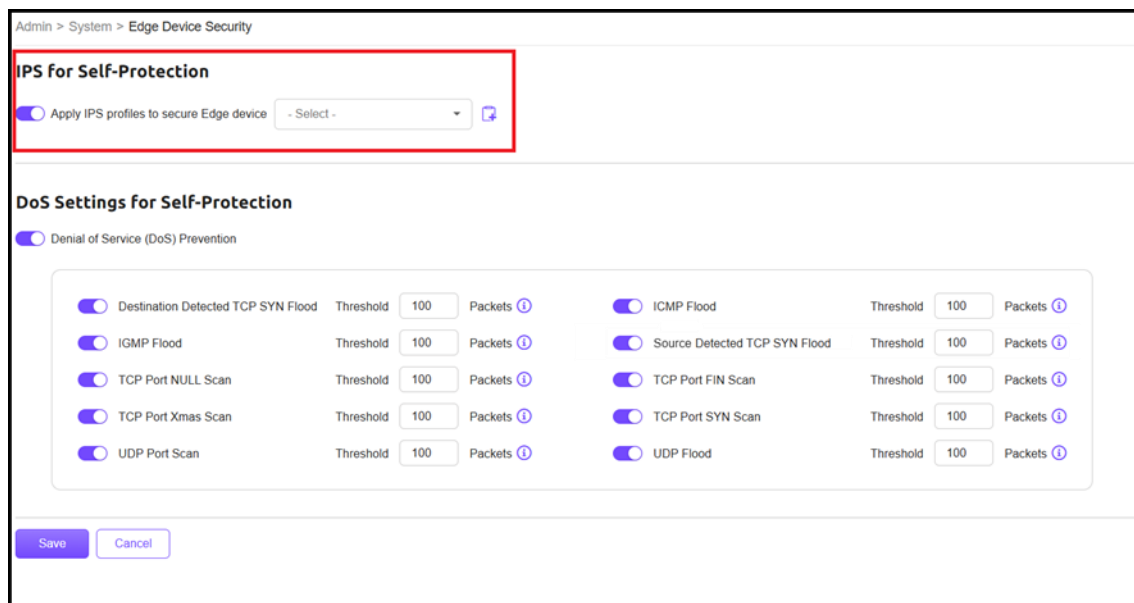
11.4.2.1 Applying IPS Profile for Self-Protection

Procedure

1. Configure the required IPS profile.

 For more information, see [Configuring IPS Profiles](#).

2. Go to [Admin] > [System] > [Edge Device Security].



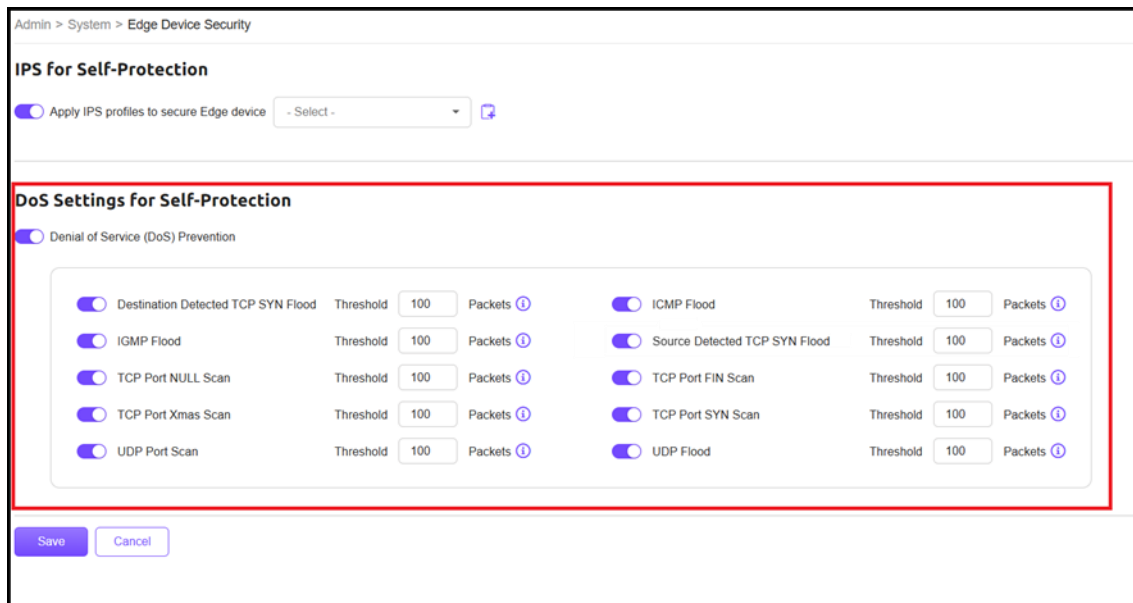
3. In the [IPS for Self-Protection] pane, enable [Apply IPS profiles to secure Edge device] by clicking the toggle.
4. Select an IPS profile from the drop-down list or click the  button to create a profile.
5. Click [Save].

11.4.2.2 Configuring DoS Prevention for Self-Protection

Procedure

1. Go to [Admin] > [System] > [Edge Device Security].

2. In the [DoS Settings for Self-Protection] pane, enable the [Denial of Service (DoS) Prevention] function by clicking the toggle.



3. Configure the thresholds of the denial of service rules.
4. Click [Save].

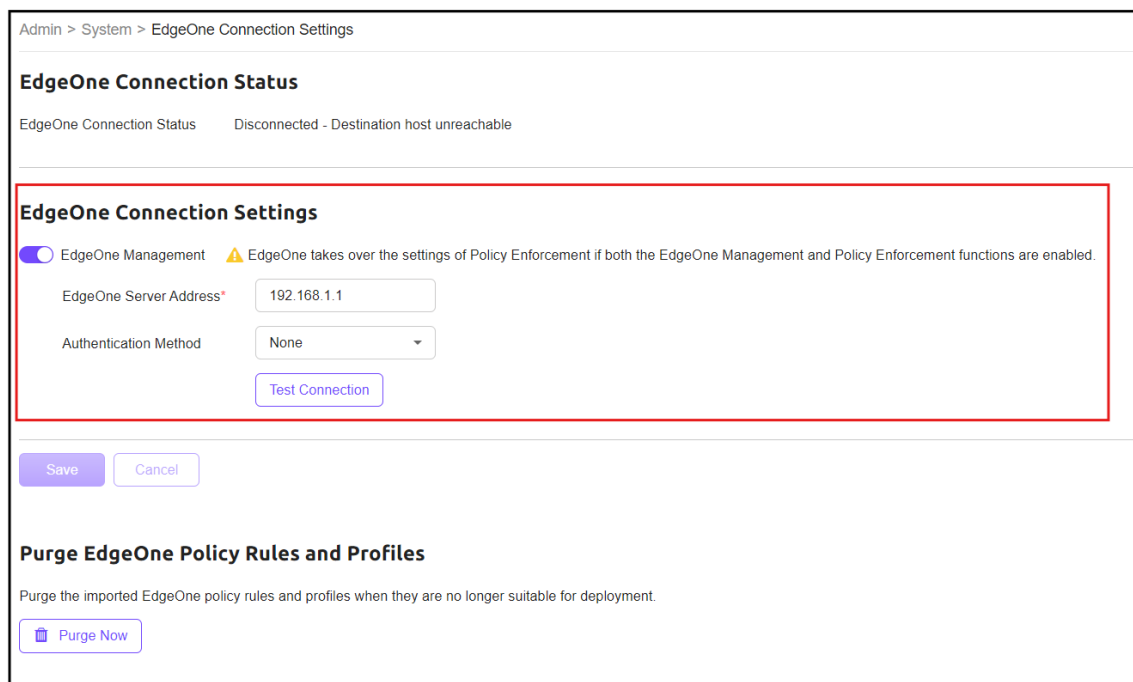
11.4.3 EdgeOne Connection Settings

This Edge series device can be managed by TXOne EdgeOne, a centralized management system. Use this tab to register your device to EdgeOne.

11.4.3.1 Enabling EdgeOne Management

Procedure

1. Go to [Admin] > [System] > [EdgeOne Connection Settings].



2. In the [EdgeOne Connection Settings] pane, enable [EdgeOne Management] by clicking the toggle.

3. Configure the following connection settings.

Item	Description
EdgeOne Server Address	Input the IP address of the EdgeOne server.
Authentication Method	Select None or Passcode from the drop-down list. To test connectivity between the Edge device and EdgeOne, click the [Test Connection] button.  If you have selected passcode as the authentication method in the EdgeOne system, please select "Passcode" as the authentication method here accordingly and input the passcode specified in the EdgeOne system for onboarding. The passcode is used only for first-time onboarding to EdgeOne.

4. Click [Save].

 The following settings/functions cannot be configured/performed on the device management console when the Edge device is connected to EdgeOne.

- Cyber Security
- Policy Enforcement
- Packet Capture
- File Exclusion
- Suspicious Objects
- Pattern Update
- Delete EdgeOne Rules & Profiles

11.4.3.2 Purging EdgeOne Policy Rules and Profiles

Procedure

1. Go to [Admin] > [System] > [EdgeOne Connection Settings].

Admin > System > EdgeOne Connection Settings

EdgeOne Connection Status

EdgeOne Connection Status Disconnected - Destination host unreachable

EdgeOne Connection Settings

☒ EdgeOne Management ⚠ EdgeOne takes over the settings of Policy Enforcement if both the EdgeOne Management and Policy Enforcement functions are enabled.

EdgeOne Server Address* 192.168.1.1

Authentication Method None ▼

[Test Connection](#)

[Save](#) [Cancel](#)

Purge EdgeOne Policy Rules and Profiles

Purge the imported EdgeOne policy rules and profiles when they are no longer suitable for deployment.

[Purge Now](#)

2. In the [Purge EdgeOne Policy Rules and Profiles] pane, click the [Purge Now] button to delete objects, profiles, and Policy Enforcement rules synced from EdgeOne.

11.4.4 Language Package

You can import language packages on the [Language Package] page. To change the displayed language, select your preferred language on the login page. The system retains only the most recently uploaded package. To add support for a new language, import a language package containing the necessary data for your target language. Note that updating the firmware with major or minor version changes, or resetting the system, reverts the language settings to English.

11.4.4.1 Importing Language Packages

Procedure

1. Go to [Admin] > [System] > [Language Package].

Admin > System > Language Package

This page is for importing language packages. To change the displayed language, select your preferred language on the login page. The system retains only the most recently uploaded package. To add support for a new language, import a language package containing the necessary data for your target language. Note that updating the firmware with major or minor version changes, or resetting the system, reverts the language settings to English.

Language Package Version 2.1.2

Release Date 2024-07-08T09:35:09+08:00

[Import Language Package](#) Total Number of Items: 1

No.	Language	Language Version
1	English (EN)	2.1.27

2. Click [Import Language Package] and select a language package file (ACF file) from your local disk. The [Import Language Package] window will appear.

Import Language Package

×

Language Package Version

2.1.2

Release Date

2024-07-08T18:25:00+08:00

Description

English (EN), Language Version: 2.1.27
日本語 (JP), Language Version: 2.1.27

Import

Cancel

3. Click [Import].

11.4.5 API Management

The API Key and API Secret are the software-level credentials for external systems to access the device via the RESTful APIs. It's a common practice to provide separate API Key/Secret for each API client. The API Key Management function allows the administrator to create and manage up to 10 API Keys.



This API key management function is provided by request. To utilize this function, please contact local sales representative for more information.

11.4.5.1 Creating API Clients

Procedure

1. Go to [Admin] > [System] > [API Management].

Admin > System > API Management

+ Create

Total Number of Items: 2 (Max: 10)

☐	No.	Status	Name	API Key	Cert. Fingerprint	Expiration Date	Trusted IP Address	Created At	Last Used	Action	Description
☐	1	☒	API_1	M2FYzhYzMI	-	-	Any	2024-07-22T10:44:14+08:00	-	Reset	
☐	2	☒	API_2	-	504c30c4448b1b50c31700c176a8f15a296d0c10008a411204a081c40f8a2710a338	2034-07-14	Any	2024-07-22T10:44:57+08:00	-		

2. Click the [Create] button to create a new API client.

Create API Client

Status ☒

Name*

Description

Authentication Method

Use API Key ▾

Trusted IP Address

Any ▾

Generate

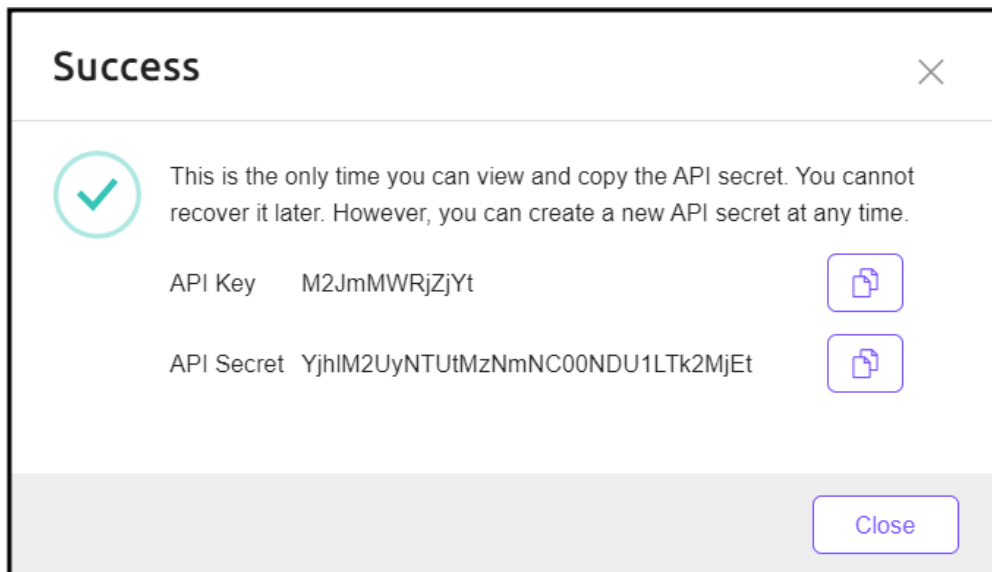
Cancel

3. Configure the following settings.

Item	Description
Status	Click the toggle to enable this API client.
Name	Type a name for this API client profile.
Description	(Optional) Type a description.
Authentication Method	Select an authentication method for accessing this API client, either Use API Key or Use Certificate. Use API Key: Authenticate using a randomly generated API key and secret. Use Certificate: Authenticate using a user-defined authentication credential embedded within a certificate. If this option is selected, click [Upload] to import a certificate from your local disk.
Trusted IP Address	Select Any or Custom . If you select Custom , specify an IP address. The system will reject API requests if the client's IP address does not match the given trusted IP address.

4. Click [Generate].

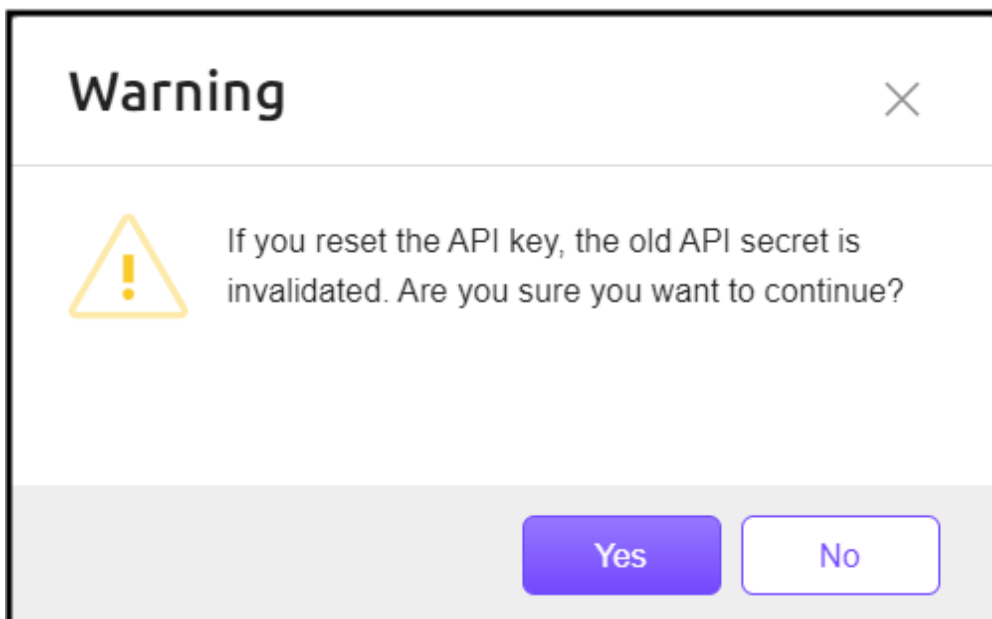
5. If the authentication method is **Use API Key**, a new API key and API secret will be generated and displayed on the screen. Click the copy button  to copy the API key and API secret.



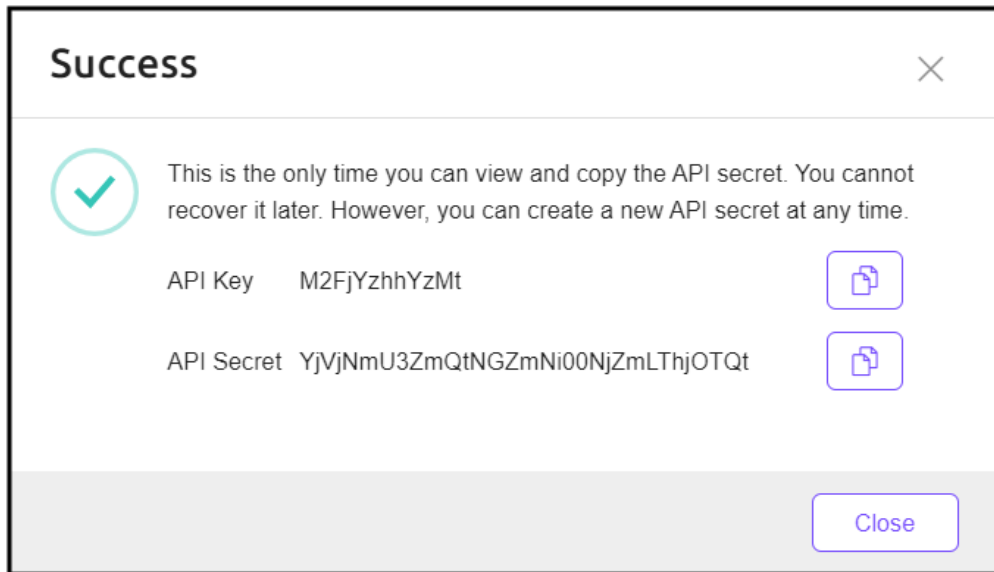
11.4.5.2 Resetting API Clients

Procedure

1. Go to [Admin] > [System] > [API Management].
2. Click [Reset] for the selected API client. A warning window will appear.



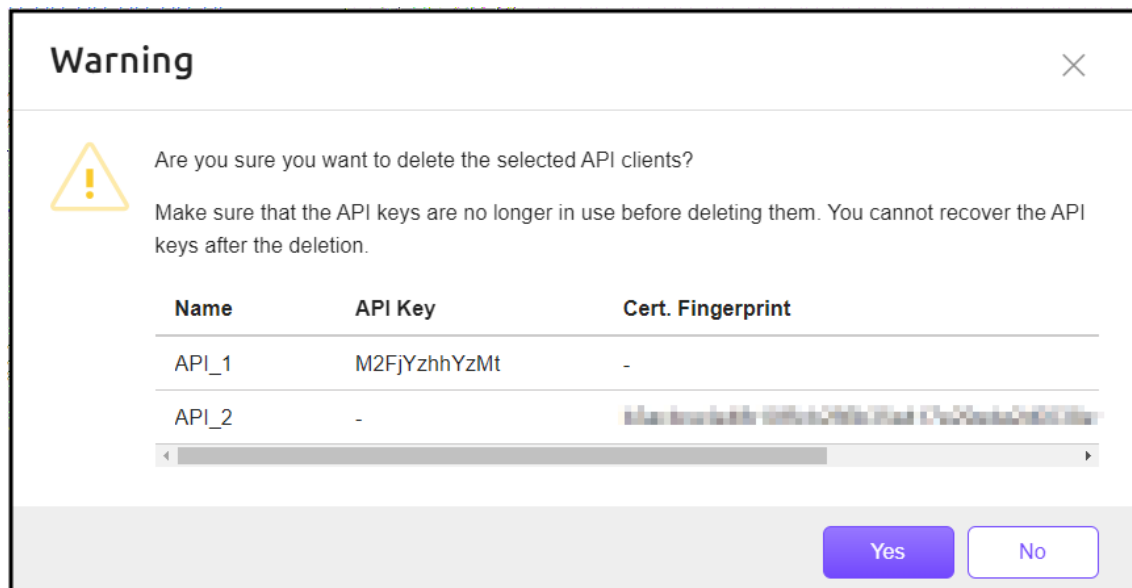
3. Click [Yes]. Please note that the old API secret will be invalidated then.
4. A new API key and API secret will be generated and displayed on the screen. Click the copy button  to copy the API key and API secret and paste them onto the API client.



11.4.5.3 Deleting API Clients

Procedure

1. Go to [Admin] > [System] > [API Management].
2. Select the API clients to be deleted.
3. Click the [Delete] button. A confirm window will appear.



4. Click [Yes].

11.4.6 SSL Certificate

The Edge device uses the HTTPS protocol to encrypt web traffic between the user's web browser and the Edge device web server. By default, the Edge device web server uses an auto-generated and self-signed SSL certificate for the HTTPS connections. This chapter introduces how to change the SSL certificate.

11.4.6.1 Importing SSL Certificates

Procedure

1. Go to [Admin] > [System] > [SSL Certificate]. If the Edge device is currently using its default certificate (auto-generated and self-signed), you will see the following page. The information shown on the page is the default Edge device certificate information.

Admin > System > SSL Certificate

Certificate	CN=Self-Signed CA,OU=TXOne,O=TXOne Networks,C=TW
Issuer	CN=Self-Signed CA,OU=TXOne,O=TXOne Networks,C=TW
Expiration	2034-07-17T15:14:06+08:00

Import Certificate

2. Click [Import Certificate]. The [Import Certificate] window will appear.

Import Certificate



You must restart the virtual instance to activate an imported certificate. After importing a certificate, please close the current web page, reopen the browser, and access the Edge device console again.

Certificate*	Select File
Private Key*	Select File
Passphrase	

Import and Restart Cancel

3. Import your certificate file by clicking the [Select File] button next to the [Certificate] field.
4. Import the private key (PEM-encoded PKCS#1 format) for the certificate file by clicking the [Select File] button next to the [Private Key] field.
5. Input the passphrase of the private key if the private key is encrypted.
6. Click [Import and Restart].
7. Close the current web page, reopen the browser, and access the Edge device web console again.

11.4.6.2 Replacing SSL Certificates

Procedure

1. Go to [Admin] > [System] > [SSL Certificate]. If the Edge device is currently using an imported certificate, you will see the following page. The information shown on the page is the current Edge device certificate information.

Admin > System > SSL Certificate

Certificate	CN=TPS Certificate,OU=TPS,O=TXOne Networks,L=Taipei,C=TW
Issuer	CN=TXOne Certificate Authority,OU=MgmtPlane,O=TXOne Networks,L=Taipei,C=TW
Expiration	2033-10-23T16:00:54+08:00

Delete Certificate
Replace Certificate

2. Click [Replace Certificate]. The [Replace Certificate] window will appear.

Replace Certificate ×

 You must restart the virtual instance to replace the current certificate. Please close the current web page, reopen the browser, and access the Edge device console again after importing a new certificate.

Certificate*	Select File
Private Key*	Select File
Passphrase	

Import and Restart
Cancel

3. Import your certificate file by clicking the [Select File] button next to the [Certificate] field.
4. Import the private key (PEM-encoded PKCS#1 format) for the certificate file by clicking the [Select File] button next to the [Private Key] field.
5. Input the passphrase of the private key if the private key is encrypted.
6. Click [Import and Restart].
7. Close the current web page, reopen the browser, and access the Edge device web console again.

11.4.6.3 Verifying SSL Certificates

After adding a new certificate to the Edge device web console, you can verify whether the certificate is valid.

Procedure

1. Log in to the Edge device web console with the Chrome browser.
2. Go to the Three Dots Menu > More Tools > Developer Tools.
3. Go to the [Security] Tab. It will give you a security overview.
4. In the [Security Overview] pane, click the [View certificate] button, and you will see the certificate details of the Edge device web console.

11.4.6.4 Deleting SSL Certificates

Procedure

1. Go to [Admin] > [System] > [SSL Certificate].

Admin > System > SSL Certificate

Certificate	CN=TPS Certificate,OU=TPS,O=TXOne Networks,L=Taipei,C=TW
Issuer	CN=TXOne Certificate Authority,OU=MgmtPlane,O=TXOne Networks,L=Taipei,C=TW
Expiration	2033-10-23T16:00:54+08:00

 Delete Certificate
  Replace Certificate

2. Click [Delete Certificate].

3. Close the current web page, reopen the browser, and access the Edge device web console again.

11.4.7 Backup/Restore

You can export settings from the management console to back up the configuration of your Edge series device. If a system failure occurs, you can restore the settings by importing the configuration file that you previously backed up.

We recommend the following:

- Backing up the current configuration before each import operation.
- Performing the backup operation when the device is idle. Importing and exporting configuration settings affect the performance of your device.

11.4.7.1 Configuring Backup Settings

Procedure

1. Go to [Admin] > [System] > [Backup/Restore].

Admin > System > Backup / Restore

Back Up Configuration

Periodic Backup: **OFF**

Backup Server Address: -

Last Backup Time: -

 Edit Backup Settings

Back up Edge device policies and administration settings to a file on your computer.

 Back Up Now

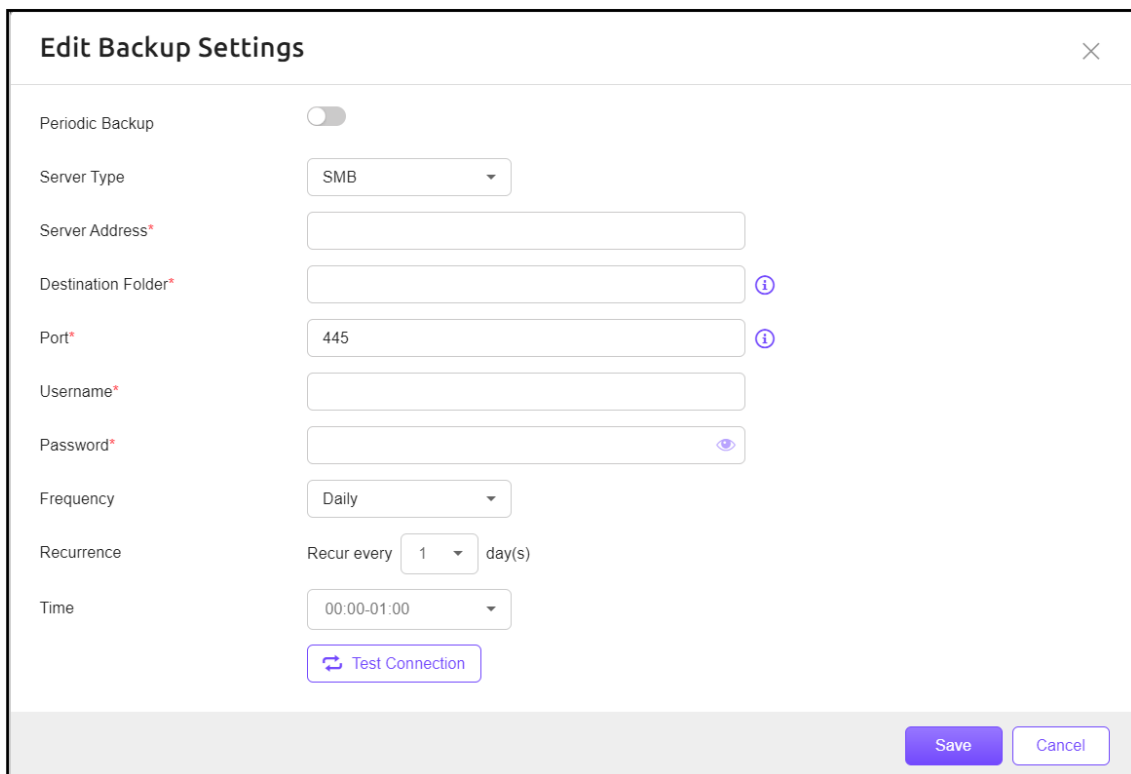
Restore Configuration

Restore the Edge device configuration from a backup file. We recommend that you back up your current configuration before replacing it.

Note: By restoring the configuration to a previous state, the current configuration settings are replaced.

 Select File

2. Click [Edit Backup Settings]. The [Edit Backup Settings] window will appear.



3. Configure the following settings.

Item	Description
Periodic Backup	Click the toggle to enable or disable backing up configuration to a remote server periodically.
Server Type	Select a protocol, SMB or SFTP , for sending backed-up files to a remote server.
Server Address	Input the IP address of the server where the backed-up files will be sent.
Destination Folder	Input the destination folder on the server where the backed-up files will be sent.
Port	Input the port number for initializing an SMB or SFTP connection. (Default: 445)
Username	Input the username for initializing an SMB or SFTP connection.
Password	Input the password for initializing an SMB or SFTP connection.
Frequency	Select a backup frequency, Daily or Weekly , from the drop-down list.
Recurrence	Specify the exact days when the backups should run.
Time	Specify the exact times when the backups should run.

4. Click [Save].

11.4.7.2 Backing Up Device Configuration

Procedure

1. Go to [Admin] > [System] > [Backup/Restore].

Admin > System > Backup / Restore

Back Up Configuration

Periodic Backup: **OFF**

Backup Server Address: -

Last Backup Time: -

 Edit Backup Settings

Back up Edge device policies and administration settings to a file on your computer.

 Back Up Now

Restore Configuration

Restore the Edge device configuration from a backup file. We recommend that you back up your current configuration before replacing it.

Note: By restoring the configuration to a previous state, the current configuration settings are replaced.

 Select File

2. Click [Back Up Now]. A configuration backup file will automatically be saved to your computer.

11.4.7.3 Restoring Device Configuration

Procedure

1. Go to [Admin] > [System] > [Backup/Restore].

Admin > System > Backup / Restore

Back Up Configuration

Periodic Backup: **OFF**

Backup Server Address: -

Last Backup Time: -

 Edit Backup Settings

Back up Edge device policies and administration settings to a file on your computer.

 Back Up Now

Restore Configuration

Restore the Edge device configuration from a backup file. We recommend that you back up your current configuration before replacing it.

Note: By restoring the configuration to a previous state, the current configuration settings are replaced.

 Select File

2. In the [Restore Configuration] pane, click [Select File], then import a configuration file from your local disk. All services will restart. It can take some time to restart services after applying imported settings and rules.

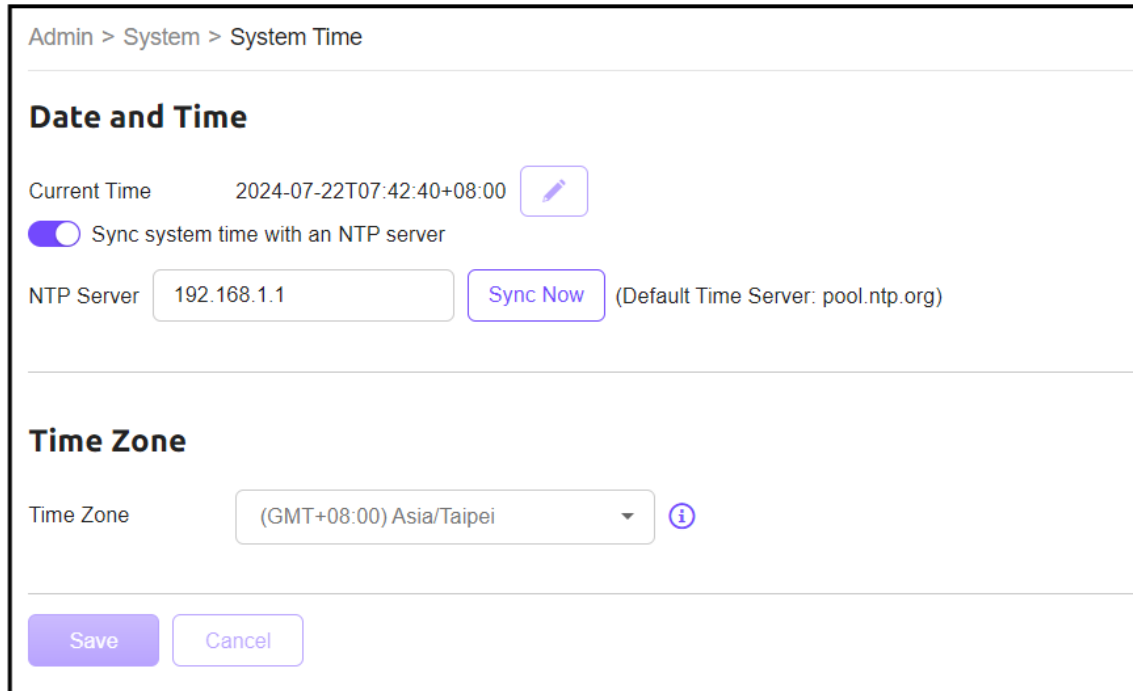
11.4.8 System Time

Network Time Protocol (NTP) synchronizes computer system clocks across the Internet. Configure NTP settings to synchronize the server clock with an NTP server, or manually set the system time.

11.4.8.1 Configuring System Time

Procedure

1. Go to [Admin] > [System] > [System Time].



2. In the [Date and Time] pane, you can choose to synchronize the system time with an NTP server or set the system time manually.

- Synchronize the system time with an NTP server:
 - a. Enable the [Sync system time with an NTP server] setting.
 - b. Specify the domain name or IP address of the NTP server.
 - c. Click [Sync Now].
- Set the system time manually:
 - a. Click the edit button .
 - b. Set the date and time.
 - c. Click [Apply].

3. From the [Time Zone] drop-down list, select a time zone.

4. Click [Save].



EdgeOne system synchronizes the system time with its managed instances.

11.4.9 System Reboot

Access the [System Reboot] page to reboot the Edge device.

11.4.9.1 Rebooting Edge Device

Procedure

1. Go to [Admin] > [System] > [System Reboot].
2. Click [Reboot Now] to reboot the Edge device.

Admin > System > System Reboot

Use the current configuration and reboot the Edge device now.

Reboot Now

12 Extended Functions via USB Devices

This chapter describes the use of USB devices with the Edge series device for extended or supported functions.

After a USB device is plugged in, you can perform the following tasks.

- [Back up configuration](#)
- [Load pattern from USB device](#)
- [Load configuration from USB device](#)
- [Load firmware from USB device](#)



Given that this feature allows anyone to perform the above tasks, the physical security of your Edge series device must be considered carefully.



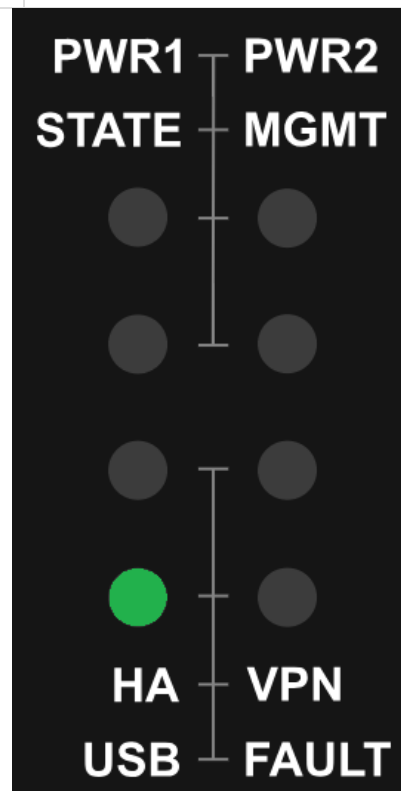
Only USB devices formatted with the FAT32 file system are supported.

12.1 Backing Up Configuration

Procedure

1. Plug a USB disk device into the USB port on the front panel of your Edge series device.
2. Upon successful detection of the USB disk device, the corresponding LED as listed below will change to a steady green. Relevant system logs can also be checked to confirm that a USB disk device was detected when inserted. This state is referred to as the "Default Action" state.

State/Action	LED	LED Color/State
Default State – USB Disk Plugged in	USB	Green – Steady



3. The function of the reset button will also change until the USB device is unplugged. When a USB device is plugged in, the reset button will not serve the as the reboot/factory reset button. It will instead serve as a button to cycle through a set of possible actions that may be taken during this time.

4. Press the reset button for more than 10 seconds to back up configuration. The corresponding LED as listed below will blink.

State/Action	LED	LED Color/State
Back Up Configuration	USB	Green – Blinking once every 0.5 seconds

5. Once the reset button is released, the LED will be restored to the default state (as shown in the table in step 2).

6. If any error occurs, the corresponding LED as listed below will indicate it.

State/Action	LED	LED Color/State
Error	FAULT	Red – Blinking once every second



An error can only be cleared if:

1. the reset button is pressed once more (LEDs return to default states with no action selected), or
2. the USB disk is unplugged.

7. Relevant system logs can be checked to verify if the configuration backup was completed successfully or not.

8. Unplug the USB disk, after which the LEDs will return to their states prior to the USB disk device being plugged in (USB LED off), and relevant system logs will show that the USB disk has been unplugged.



Since configuration files are usually not very large, this process may finish quickly.

12.2 Loading Pattern from Disk

A DPI pattern file may be easily and quickly loaded from a USB disk device. This function allows for a floor operator to update/restore the pattern file on the physical floor of the ICS environment without the need of a client computer to log in to the device.

Procedure

1. Create a folder “**TXOne**” and a subfolder “**pattern**” in the TXOne folder on a USB disk.

2. Save a pattern file you would like to load onto your Edge series device on your USB disk under the path **“/TXOne/pattern/”**. Assuming the filename of the pattern file is “pattern.acf”, the file path on the USB disk would be **“/TXOne/pattern/pattern.acf”**.



Saving pattern files under other paths or incorrect folder names will cause the file to not be detected during the pattern load process.

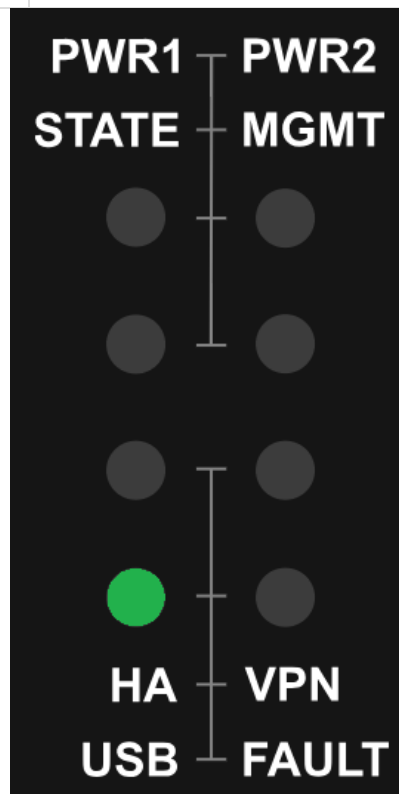


1. Folder names are case-insensitive. If multiple pattern files exist in the folder, the newest one will be used.
2. Various versions of the pattern files can be downloaded at the MyTXOne portal at <https://my.txone.com/>.

3. Plug your USB disk into the USB port on the front panel of your Edge series device.

4. Upon successful detection of the USB disk, the corresponding LED as listed below will change to a steady green. Relevant system logs can also be checked to confirm that a USB disk was detected when inserted. This state is referred to as the “Default Action” state.

State/Action	LED	LED Color/State
Default State – USB Disk Plugged in	USB	Green – Steady



5. The function of the reset button will also change until the USB disk is unplugged. When a USB disk is plugged in, the reset button will not serve as the reboot/factory reset button. It will instead serve as a button to cycle through a set of possible actions that may be taken during this time.

6. Give the reset button one short press to toggle to the “Load Pattern from USB Disk” action. The LEDs will change to the following colors/states.

State/Action	LED 1	LED 1 Color/State	LED 2	LED 2 Color/State
Toggle to Pattern Load	USB	Green – Steady	STATE	Green – Blinking once every second

7. The action must be confirmed within 10 seconds by holding down the reset button for more than 3 seconds. After the reset button is pressed more than 3 seconds, the LEDs will change to the following colors/states.

State/Action	LED 1	LED 1 Color/State	LED 2	LED 2 Color/State
Pattern Load Action Confirmed	USB	Green – Steady	STATE	Green – Blinking once every 0.5 seconds

8. Once the reset button is released, pattern load will be in progress, and the LEDs will change to the following colors/states.

State/Action	LED 1	LED 1 Color/State	LED 2	LED 2 Color/State
Pattern Load in Progress	USB	Green – Blinking once every 0.5 seconds	STATE	Green – Blinking once every 0.5 seconds



When the pattern load is in progress on EdgeFire, the color of the IPS/IDS LED stays the same as it was before the reset button was released. For example, if the IPS/IDS LED was green when the reset button was released, it will remain green during the pattern load. The same is true for an amber IPS/ID LED; it will remain amber throughout the pattern load.

9. If any error occurs, the corresponding LED as listed below will indicate it.

State/Action	LED	LED Color/State
Error	FAULT	Red – Blinking once every second



An error can only be cleared if:

1. the reset button is pressed once more (LEDs return to default states with no action selected), or
2. the USB disk is unplugged.

10. Relevant system logs can be checked to verify if the pattern load was completed successfully or not. If so, the LEDs will be restored to the default state (as shown in the table in step 4).

11. Unplug the USB disk, after which the LEDs will return to their states prior to the USB disk device being plugged in (USB LED off), and relevant system logs will show that the USB disk has been unplugged.

12.3 Loading Configuration from USB Device

A configuration file may be easily and quickly loaded from a USB disk device. This function allows for a floor operator to update/restore the configuration on the physical floor of the ICS environment without the need of a client computer to log in to the device.

Procedure

1. Create a folder “**TXOne**” and a subfolder “**config**” in the TXOne folder on a USB disk.
2. Save a configuration file you would like to load onto your Edge series device on your USB disk under the path “**/TXOne/config/**”. Assuming the filename of the configuration file is “**config.acf**”, the file path on the USB disk would be “**/TXOne/config/config.acf**”.
3. Plug your USB disk into the USB port on the front panel of your Edge series device.



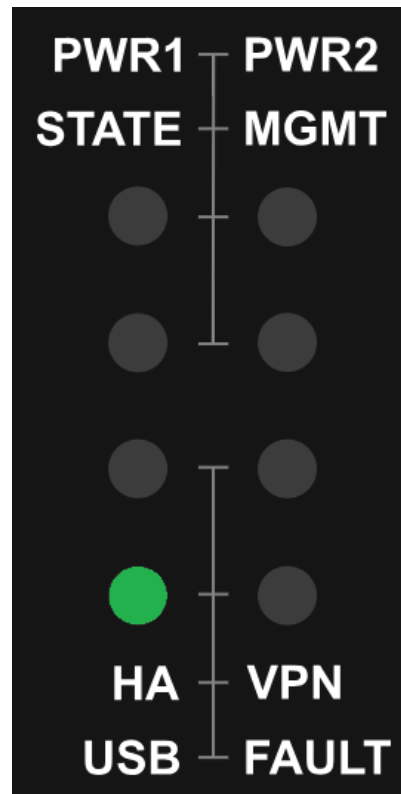
Saving configuration files under other paths or incorrect folder names will cause the file to not be detected during the configuration load process.



Folder names are case-insensitive. If multiple configuration files exist in the folder, the newest one will be used.

4. Upon successful detection of the USB disk, the corresponding LED as listed below will change to a steady green. Relevant system logs can also be checked to confirm that a USB disk was detected when inserted. This state is referred to as the “Default Action” state.

State/Action	LED	LED Color/State
Default State – USB Disk Plugged in	USB	Green – Steady



5. The function of the reset button will also change until the USB disk is unplugged. When a USB disk is plugged in, the reset button will not serve the as the reboot/factory reset button. It will instead serve as a button to cycle through a set of possible actions that may be taken during this time.

6. Give the reset button two short presses to toggle to the “Load Configuration from USB Disk” action. The LEDs will change to the following colors/states.

State/Action	LED 1	LED 1 Color/State	LED 2	LED 2 Color/State
Toggle to Configuration Load	USB	Green – Steady	MGMT	Green – Blinking once every second

7. The action must be confirmed within 10 seconds by holding down the reset button for more than 3 seconds. After the reset button is pressed more than 3 seconds, the LEDs will change to the following colors/states.

State/Action	LED 1	LED 1 Color/State	LED 2	LED 2 Color/State
Configuration Load Action Confirmed	USB	Green – Steady	MGMT	Green – Blinking once every 0.5 seconds

8. Once the reset button is released, the configuration load will be in progress, and the LEDs will change to the following colors/states.

State/Action	LED 1	LED 1 Color/State	LED 2	LED 2 Color/State
Configuration Load in Progress	USB	Green – Blinking once every 0.5 seconds	MGMT	Green – Blinking once every 0.5 seconds

9. If any error occurs, the corresponding LED as listed below will indicate it.

State/Action	LED	LED Color/State
Error	FAULT	Red – Blinking once every second



An error can only be cleared if:

1. the reset button is pressed once more (LEDs return to default states with no action selected), or
2. the USB disk is unplugged.

10. Relevant system logs can be checked to verify if the configuration load was completed successfully or not. If so, the LEDs will be restored to the default state (as shown in the table in step 4).

11. Unplug the USB disk, after which the LEDs will return to their states prior to the USB disk device being plugged in (USB LED off), and relevant system logs will show that the USB disk has been unplugged.

12.4 Loading Firmware from USB Device

Device firmware may be easily and quickly loaded from a USB disk device. This function allows for a floor operator to upgrade/change the firmware of a device on the physical floor of the ICS environment without the need of a client computer to log in to the device.

Procedure

1. Create a folder “**TXOne**” and a subfolder “**firmware**” in the TXOne folder on a USB disk.
2. Save a firmware file you would like to load onto your Edge series device on your USB disk under the path “**/TXOne/firmware/**”. Assuming the filename of the firmware file is “firmware.acf”, the file path on the USB disk would be “**/TXOne/firmware/firmware.acf**”.



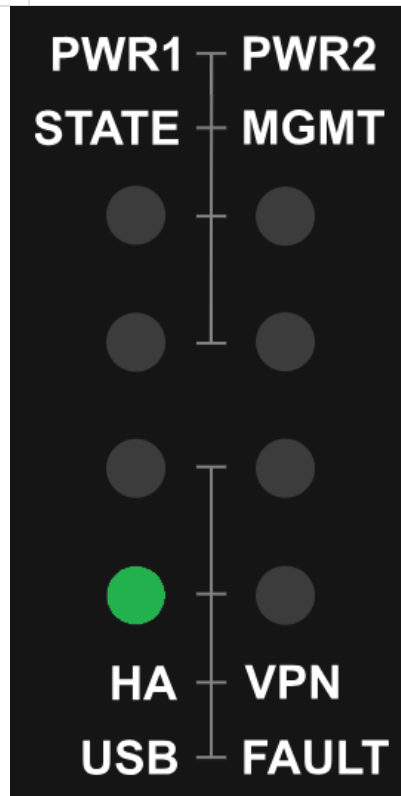
Saving firmware files under other paths or incorrect folder names will cause the file to not be detected during the firmware load process.



1. Folder names are case-insensitive. If multiple firmware files exist in the folder, the newest one will be used.
2. Various versions of the firmware files can be downloaded at the MyTXOne portal at <https://my.txone.com/>.

3. Plug your USB disk into the USB port on the front panel of your Edge series device.
4. Upon successful detection of the USB disk, the corresponding LED as listed below will change to a steady green. Relevant system logs can also be checked to confirm that a USB disk was detected when inserted. This state is referred to as the “Default Action” state.

State/Action	LED	LED Color/State
Default State – USB Disk Plugged in	USB	Green – Steady



5. The function of the reset button will also change until the USB disk is unplugged. When a USB disk is plugged in, the reset button will not serve the as the reboot/factory reset button. It will instead serve as a button to cycle through a set of possible actions that may be taken during this time.

6. Give the reset button three short presses to toggle to the “Load Firmware from USB Disk” action. The LEDs will change to the following colors/states.

State/Action	LED 1	LED 1 Color/State	LED 2	LED 2 Color/State	LED 3	LED 3 Color/State
Toggle to Firmware Load	USB	Green – Steady	STATE	Green – Blinking once every second	MGMT	Green – Blinking once every second

7. The action must be confirmed within 10 seconds by holding down the reset button for more than 3 seconds. After the reset button is pressed more than 3 seconds, the LEDs will change to the following colors/states.

State/Action	LED 1	LED 1 Color/State	LED 2	LED 2 Color/State	LED 3	LED 3 Color/State
Firmware Load Action Confirmed	USB	Green – Steady	STATE	Green – Blinking once every 0.5 seconds	MGMT	Green – Blinking once every 0.5 seconds

8. Once the reset button is released, the firmware load will be in progress, and the LEDs will change to the following colors/states.

State/ Action	LED 1	LED 1 Color/State	LED 2	LED 2 Color/State	LED 3	LED 3 Color/State
Firmware Load in Progress	USB	Green – Blinking once every 0.5 seconds	STATE	Green – Blinking once every 0.5 seconds	MGMT	Green – Blinking once every 0.5 seconds

9. If any error occurs, the corresponding LED as listed below will indicate it.

State/Action	LED	LED Color/State
Error	FAULT	Red – Blinking once every second

	An error can only be cleared if: 1. the reset button is pressed once more (LEDs return to default states with no action selected), or 2. the USB disk is unplugged.
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10. Relevant system logs can be checked to verify if the firmware load was completed successfully or not. If so, the LEDs will be restored to the default state (as shown in the table in step 4).

11. Unplug the USB disk, after which the LEDs will return to their states prior to the USB disk device being plugged in (USB LED off), and relevant system logs will show that the USB disk has been unplugged.

13 Technical Support

This chapter includes the following topics.

- [Troubleshooting Resources](#)
- [Contacting TXOne](#)
- [Other Resources](#)

13.1 Troubleshooting Resources

Before contacting technical support, consider visiting the following TXOne online resources.

13.1.1 Help Center

TXOne Help Center provides 24/7 online resources that contain the most up-to-date information about both common and unusual problems.

To access the TXOne Help Center, click the  icon in the top right corner of the console or visit <https://help.txone.com/>, where you can learn more about:

- News & Releases
- Technical Advisory
- Security Advisory

13.1.2 Threat Encyclopedia

Most malware today consists of blended threats, which combine two or more technologies, to bypass computer security protocols. TXOne combats this complex malware with products that create a custom defense strategy. The Threat Encyclopedia provides a comprehensive list of names and symptoms for various blended threats, including known malware, spam, malicious URLs, and known vulnerabilities.

Visit <https://www.encyclopedia.txone.com/> to learn more about:

- Malware and malicious mobile code currently active or "in the wild"
- Correlated threat information pages to form a complete web attack story
- Internet threat advisories about targeted attacks and security threats
- Web attack and online trend information
- Weekly malware reports

13.2 Contacting TXOne

In the United States, TXOne representatives are available by phone or email:

Address	TXOne Networks, Incorporated 222 West Las Colinas Boulevard, Irving, Texas 75062, U.S.A.
Phone	Phone: +1 (817) 569-8900 Toll-free: (888) 762-8736
Website	https://www.txone.com/

Email Address

- Contact us: <https://www.txone.com/contact/> or mail to support@txone.com

13.2.1 Expediting Support Call

To accelerate the speed to reply and solve the problem, ensure that you have the following details available:

- Steps to reproduce the problem
- Appliance or network information
- Computer brand and model, as well as any additional connected hardware or devices
- Amount of memory and free hard disk space
- Operating system and service pack version
- Version of the installed agent
- Serial number or activation code
- Detailed description of install environment
- Exact text of any error message received

13.3 Other Resources

In addition to solutions and support, there are many other helpful resources available online to stay up to date, learn about innovations, and be aware of the latest security trends.

13.3.1 MyTXOne Portal

MyTXOne portal (<https://my.txone.com/>) is designed to consolidate everything you need all through a single pane of glass, aiming to maximize your efficiency when using TXOne products.

By visiting our MyTXOne portal, you are able to:

- Manage your TXOne product licenses, including license activation and license renewal, all at one place;
- 3. Download the firmware and pattern files for your security appliances' regular updates and enhancement;
- Access our product documentation, such as user's guides, quick start guides, and release notes, enabling you to use our products efficiently and effectively;
- Read our white papers, case studies, security reports, etc. to gain insights into OT security and stay ahead of the curve;
- Submit support case tickets to resolve technical issues or other concerns.

13.3.2 Documentation Feedback

TXOne always seeks to improve its documentation. If you have questions, comments, or suggestions about this or any other TXOne document, please mail to support@txone.com.

14 Appendix A: Terms and Acronyms

The following table lists the terms and acronyms used in this document.

Term/Acronym	Definition
ALG	Application Layer Gateway
CEF	Common Event Format
CIDR	Classless Inter-Domain Routing
CPSDR	Cyber-Physical System Detection and Response
DPI	Deep Packet Inspection
EWS	Engineering Workstation
HMI	Human-Machine Interface
ICS	Industrial Control System
IT	Information Technology
NAT	Network Address Translation
OT	Operational Technology
PLC	Programmable Logic Controller
SCADA	Supervisory Control and Data Acquisition